

**SUNDRY NOTICES AND REPORTS ON WELLS - FORM 4**

INDUSTRIAL COMMISSION OF NORTH DAKOTA

OIL AND GAS DIVISION

600 EAST BOULEVARD DEPT 405

BISMARCK, ND 58505-0840

SFN 5749 (09-2006)

Well File No.

**28652**

Received

DEC 09 2016

PLEASE READ INSTRUCTIONS BEFORE FILLING OUT FORM.

PLEASE SUBMIT THE ORIGINAL AND ONE COPY.

|                                                                                                                                                 |                                                |
|-------------------------------------------------------------------------------------------------------------------------------------------------|------------------------------------------------|
| <input type="checkbox"/> Notice of Intent                                                                                                       | Approximate Start Date                         |
| <input checked="" type="checkbox"/> Report of Work Done                                                                                         | Date Work Completed<br><b>November 3, 2016</b> |
| <input type="checkbox"/> Notice of Intent to Begin a Workover Project that may Qualify for a Tax Exemption Pursuant to NDCC Section 57-51.1-03. | Approximate Start Date                         |

|                                               |                                                              |
|-----------------------------------------------|--------------------------------------------------------------|
| <input type="checkbox"/> Drilling Prognosis   | <input type="checkbox"/> Spill Report                        |
| <input type="checkbox"/> Redrilling or Repair | <input type="checkbox"/> Shooting                            |
| <input type="checkbox"/> Casing or Liner      | <input type="checkbox"/> Acidizing                           |
| <input type="checkbox"/> Plug Well            | <input type="checkbox"/> Fracture Treatment                  |
| <input type="checkbox"/> Supplemental History | <input checked="" type="checkbox"/> Change Production Method |
| <input type="checkbox"/> Temporarily Abandon  | <input type="checkbox"/> Reclamation                         |
| <input type="checkbox"/> Other                | <b>Well is now on pump</b>                                   |

|                                                             |                       |                           |                      |                          |                       |
|-------------------------------------------------------------|-----------------------|---------------------------|----------------------|--------------------------|-----------------------|
| Well Name and Number<br><b>Kline Federal 5300 41-18 10B</b> |                       |                           |                      |                          |                       |
| Footages<br><b>500 F S L 237 F W L</b>                      |                       | Qtr-Qtr<br><b>SWSW</b>    | Section<br><b>18</b> | Township<br><b>153 N</b> | Range<br><b>100 W</b> |
| Field<br><b>Baker</b>                                       | Pool<br><b>Bakken</b> | County<br><b>McKenzie</b> |                      |                          |                       |

| 24-HOUR PRODUCTION RATE |      |       |      |
|-------------------------|------|-------|------|
| Before                  |      | After |      |
| Oil                     | Bbls | Oil   | Bbls |
| Water                   | Bbls | Water | Bbls |
| Gas                     | MCF  | Gas   | MCF  |

|                       |      |       |          |
|-----------------------|------|-------|----------|
| Name of Contractor(s) |      |       |          |
| Address               | City | State | Zip Code |

**DETAILS OF WORK**

Effective 11/03/2016 the above referenced well was converted to rod pump.

End of Tubing: 2-7/8" L-80 tubing @ 10122.38'

Pump: 2-1/2" x 2.0" x 24' insert pump @ 9995.42'

|                                                     |                                         |                                         |  |
|-----------------------------------------------------|-----------------------------------------|-----------------------------------------|--|
| Company<br><b>Oasis Petroleum North America LLC</b> |                                         | Telephone Number<br><b>281-404-9436</b> |  |
| Address<br><b>1001 Fannin, Suite 1500</b>           |                                         |                                         |  |
| City<br><b>Houston</b>                              | State<br><b>TX</b>                      | Zip Code<br><b>77002</b>                |  |
| Signature<br>                                       | Printed Name<br><b>Jennifer Swenson</b> |                                         |  |
| Title<br><b>Regulatory Specialist</b>               | Date<br><b>December 7, 2016</b>         |                                         |  |
| Email Address<br><b>jswenson@oasispetroleum.com</b> |                                         |                                         |  |

| FOR STATE USE ONLY                                    |                                   |
|-------------------------------------------------------|-----------------------------------|
| <input checked="" type="checkbox"/> Received          | <input type="checkbox"/> Approved |
| Date<br><b>1-16-2017</b>                              |                                   |
| By<br>                                                |                                   |
| Title<br><b>TAYLOR ROTH</b><br>Engineering Technician |                                   |

# Oasis Petroleum North America

## Kline Federal 5300 41-18 10B



### Vertical Section View

**Surface Location:**  
500' FSL & 237' FWL  
SW SW Sec. 18, T153N, R100W  
McKenzie County, ND

**Bottom Hole Location**  
254.45' S & 9968.39' E  
of surface location or approx.  
754.45' FSL & 283.76' FEL  
SE SE Sec. 17 T153N, R100W

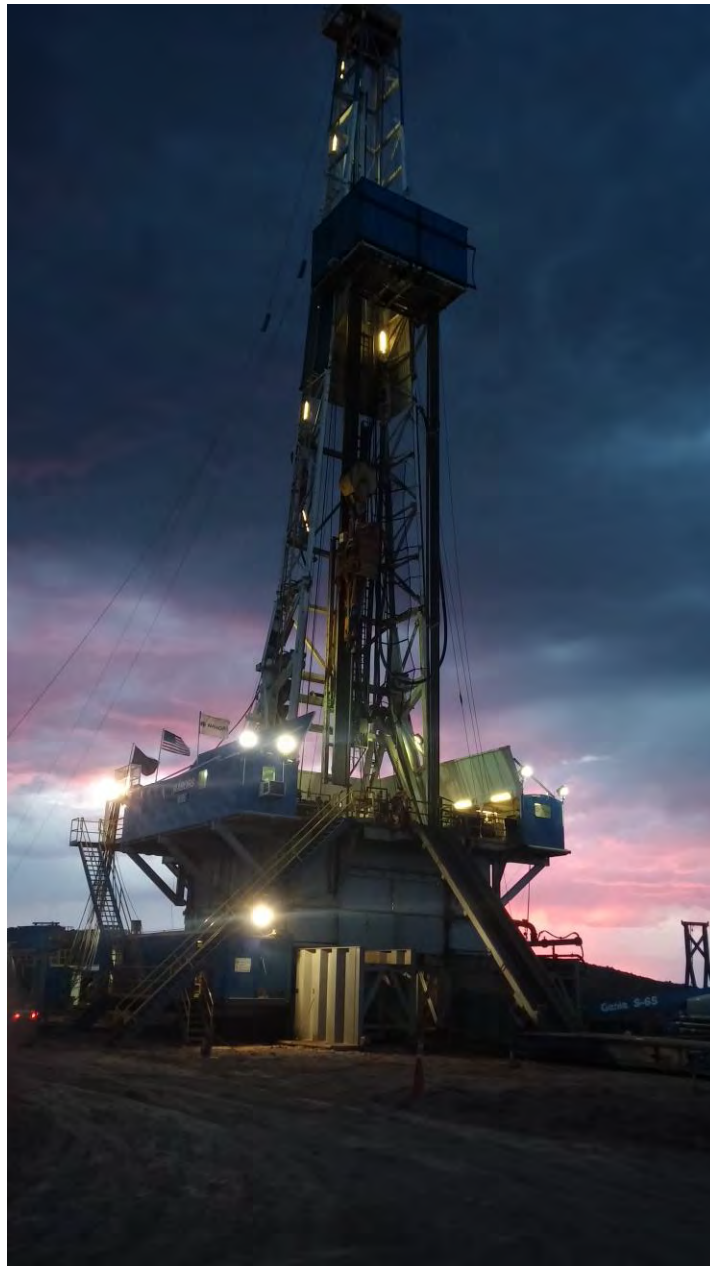


Services Performed For:  
Brendan Hargrove  
Oasis Petroleum North America  
1001 Fannin Suite 202  
Houston, TX 77002

Onsite Geology Performed by:  
Annika Tostengard, Timber Solberg  
RPM Geologic  
[geology@rpmconsultinginc.com](mailto:geology@rpmconsultinginc.com)  
(303) 595-7625

# WELL EVALUATION

Oasis Petroleum  
Kline Federal 5300 41-18 10B  
SW SW Sec. 18, T153N, R100W  
Williams County, North Dakota





## **Synopsis**

Oasis Petroleum *Kline Federal 5300 41-18 10B* [SW SW Sec. 18 T153N R100W] is located ~6 miles Northwest of the town of Williston, North Dakota. *Kline Federal 5300 41-18 10B* is situated within the Baker Field. A single lateral leg trending 90° was proposed to be drilled from the SW SW corner of Section 18 to the SE SE corner of Section 17 targeting the porous sandstone and silty sandstone of the **Middle Bakken**.

## **Control Wells**

Two completed wells were used as the control offsets to the *Kline Federal 5300 41-18 10B*.

The primary offset considered was the *Kline Federal 5300 41-18 11T2* [SW SW Sec. 18 T153N, R100W] which is a horizontal well drilled by Oasis Petroleum. *Kline Federal 5300 41-18 11T2* is located approximately 33 feet south of *Kline Federal 5300 41-18 10B*.

The other offset considered was the *Kline Federal 5300 41-18 12TX* [SW SW Sec. 18 T153N, R100W] which is a horizontal well drilled by Oasis Petroleum. *Kline Federal 5300 41-18 12TX* is located approximately 66 feet south of *Kline Federal 5300 41-18 10B*.

During the curve, the gamma ray that was produced was constantly compared to the *Kline Federal 5300 41-18 11T2* and the *Kline Federal 5300 41-18 12TX* wells. Gamma ray was used to help determine a proper landing depth.

To assist in the choosing of the landing target, an isopach table (Table 1) was constructed to measure the distance of formation tops to the target depth determined from the *Kline Federal 5300 41-18 11T2* and *Kline Federal 5300 41-18 12TX* offset wells. Comparison of gamma ray markers showed the most consistent isopach for up-hole formations to be the **False Bakken**.



| <b><i>Control Well</i></b> |                                                                             |                  |                  |                       |
|----------------------------|-----------------------------------------------------------------------------|------------------|------------------|-----------------------|
| <b>Operator:</b>           | <b>Oasis Petroleum</b>                                                      |                  |                  |                       |
| <b>Well Name:</b>          | <b><i>Kline Federal 5300 41-18 11T2</i></b>                                 |                  |                  |                       |
| <b>Location:</b>           | SW SW Sec. 18 T153N, R100W<br>McKenzie Co., ND<br>Approximately 25 ft South |                  |                  |                       |
| <b>Elevation:</b>          | KB: 2,082'                                                                  |                  |                  |                       |
| <b>Formation/<br/>Zone</b> | <b>Est. TVD</b>                                                             | <b>MSL Datum</b> | <b>Thickness</b> | <b>Dist to target</b> |
| Kibbey Lime                | 8,382'                                                                      | -6,300'          | 183'             | 2,368'                |
| Charles Salt               | 8,565'                                                                      | -6,483'          | 642'             | 2,185'                |
| Base Last Salt             | 9,207'                                                                      | -7,125'          | 223'             | 1,543'                |
| Mission Canyon             | 9,430'                                                                      | -7,348'          | 567'             | 1,320'                |
| Lodgepole                  | 9,997'                                                                      | -7,915'          | 708'             | 753'                  |
| False Bakken               | 10,705'                                                                     | -8,623'          | 7'               | 45'                   |
| Upper Bakken Shale         | 10,712'                                                                     | -8,630'          | 18'              | 38'                   |
| Middle Bakken              | 10,730'                                                                     | -8,648'          | 40'              | 20'                   |

| <b><i>Control Well</i></b> |                                                                               |                  |                  |                       |
|----------------------------|-------------------------------------------------------------------------------|------------------|------------------|-----------------------|
| <b>Operator:</b>           | <b>Oasis Petroleum</b>                                                        |                  |                  |                       |
| <b>Well Name:</b>          | <b><i>Kline Federal 5300 41-18 12TX</i></b>                                   |                  |                  |                       |
| <b>Location:</b>           | SW SW Sec. 18 T153N, R100W<br>McKenzie Co., ND<br>Approximately 50 feet South |                  |                  |                       |
| <b>Elevation:</b>          | KB: 2,082'                                                                    |                  |                  |                       |
| <b>Formation/Zone</b>      | <b>Est. TVD</b>                                                               | <b>MSL Datum</b> | <b>Thickness</b> | <b>Dist to target</b> |
| Kibbey Lime                | 8,384                                                                         | -6,302'          | 172'             | 2,361'                |
| Charles Salt               | 8,556                                                                         | -6,474'          | 650'             | 2,189'                |
| Base Last Salt             | 9,206                                                                         | -7,124'          | 222'             | 1,539'                |
| Mission Canyon             | 9,428                                                                         | -7,346'          | 547'             | 1,317'                |
| Lodgepole                  | 9,975                                                                         | -7,893'          | 729'             | 770'                  |
| False Bakken               | 10,704                                                                        | -8,622'          | 9'               | 41'                   |
| Upper Bakken Shale         | 10,713                                                                        | -8,631'          | 17'              | 32'                   |
| Middle Bakken              | 10,730                                                                        | -8,648'          | 27'              | 15'                   |

**Table 1:** Distance from Formation tops in the control wells were utilized to determine an initial drilling target.

# **Geologic Assessment**

## **Methods**

Geologic supervision of *Kline Federal 5300 41-18 10B* was provided by experienced RPM well site geologists. Gas and chromatograph levels were measured using NOV WellSight Gas Detection real time gas detector and chromatograph system. The WellSight gas detection system uses non-dispersive infrared and chemical sensor gas detection to evaluate gases liberated from the formation by the drill bit and carried to the surface by the drilling fluid.

The gas detector was interfaced with a RigWatch electronic data recorder system. RigWatch provided drill rate, on-off bottom and pump strokes to the gas detector and received total gas information from the gas detector for viewing on location and remotely.

Under the direction of RPM well site geologists, rig crews were instructed to catch lagged drill cutting samples at 30' intervals in the vertical hole and curve build from 8,350' to 11,175' MD and 50' samples from 11,175' to TD (20,475').

Sampled drill cuttings were examined wet and dry under a binocular microscope using both plain (broad spectrum) and transmitted light. Cuttings were evaluated for hydrocarbon "cut" by immersion in Acetone and inspection under a UV fluoroscope. 10% hydrochloric acid and alizarin red were used to determine the calcareous and dolomitic content of rocks and cementing.

RPM Geologic, Inc. (RPM) well site geologists also closely examined MWD gamma-ray information and penetration rates to aid in steering decisions and dip degree estimations.

## **Vertical Operations**

### **Overview**

The *Kline Federal 5300 41-18 10B* was spud for surface drilling on 9 September 2014 utilizing Nabors 486 as the drilling contractor. Prior to commencement of RPM mud logging services, a 17 1/2" hole was drilled with fresh water to depth of 2,125' and isolated with 13 3/8" 54# J-55 casing cemented from 2,083' to surface. To ensure hole stability a 12 1/4" hole was drilled to a depth of 6,140' and 40# 8 5/8 casing was set at 6,125'.

RPM well site geologists began logging the vertical interval at 8,350'.

Two PDC bit were used to drill out of secondary casing casing to vertical TD. PDC bit #1 (8 3/4" Varel) was used to drill from 2,140' to 9,9500' and averaged an ROP of 62 ft/hr in 61.5 hours of use. A second Varel PDC bit was used to continue drilling to a vertical TD of 10,240' MD in 4.5 hours with an average penetration rate of 64.4 ft/hr

Diesel invert drilling fluid with a mud weight ranging from 9.80-9.90 ppg was used for the remainder of the vertical hole and in the curve builds sections. The vertical interval was drilled to a kick off point (KOP) of 10,450', at which point vertical operations were ceased.

### **Lithology**

The top of the **Kibbey Lime** [Mississippian Madison Group] was picked at 8,376' (-6,294'), 6' high to the *Kline Federal 5300 41-18 11T2*. Samples from this interval (Figure 1) were described as:

LIMESTONE: wackestone, light gray to medium gray, light gray brown to medium gray brown, occasional off white friable to firm, crystalline to microcrystalline



*Figure 1: Photograph of Limestone seen in sample from the **Kibby Limestone** formation.*

The top of the **Charles Salt** [Mississippian Madison Group] was logged at 8,566' (-6,484'), 1' low to the *Kline Federal 5300 41-18 11T2*. The **Base Last Salt** was drilled at 9,206' (-7,123'), 2' high to the *Kline Federal 5300 41-18 11T2*. Samples from this interval (Figure 2) were described as:

SALT: translucent, trace off white, crystalline, firm, crystalline texture, no visible porosity, no visible oil stain

LIMESTONE: wackestone, light gray to medium gray, light gray brown to medium gray brown, occasional off white friable to firm, crystalline to microcrystalline

ANHYDRITE: off white, cream, soft to firm, amorphous texture, no visible porosity





*Figure 2: Photograph of Salt seen in sample from the Charles formation.*

The top of the Mission Canyon formation of the Madison Group [Mississippian] was reached at 9,428' (-7,346'), 2' high to the *Kline Federal 5300 41-18 11T2*. Samples from this interval (Figure 3) were described as:

LIMESTONE: wackestone, light gray to medium gray, light gray brown to medium gray brown, occasional off white friable to firm, crystalline to microcrystalline



*Figure 3: Photograph of limestone from the Mission Canyon formation.*

The top of the **Lodgepole** formation of the Madison Group (Mississippian) was logged at 9,980' (-7,898'), 17' high to the *Kline Federal 5300 41-18 11T2*. Samples from this interval (Figure 4) were described as:

ARGILLACEOUS LIMESTONE: wackestone, medium gray brown to light gray brown, firm, crystalline texture, trace sparry calcareous, carbonaceous, possible intercrystalline porosity, possible slightly light brown spotty oil stain



*Figure 4: Photograph of argillaceous limestone from **Lodgepole** formation.*

## **Directional Operations**

Gyrodata provided equipment and personnel for MWD teams, while RPM Consulting provided directional personnel. RPM geologists worked closely with RPM directional as well as Gyrodata MWD teams to formulate steering decisions to maximize the footage of borehole in the pay zone.

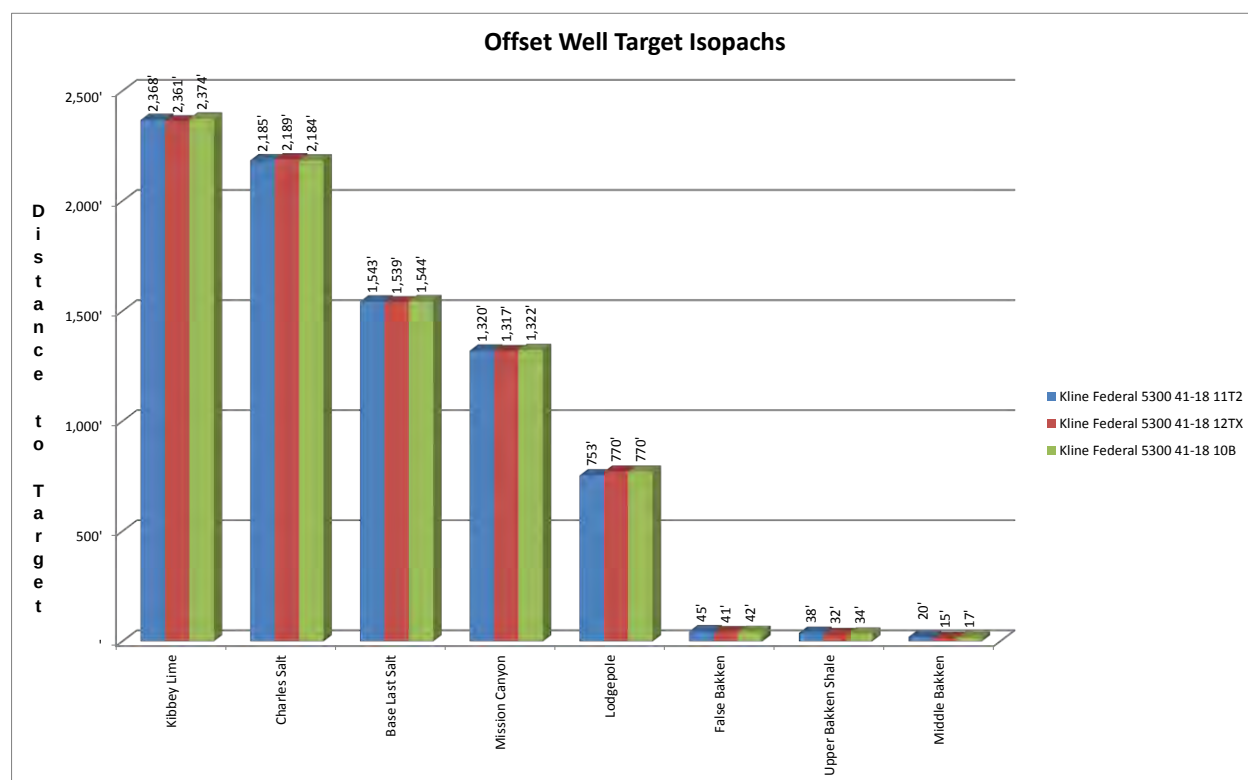
## **Curve Build**

### **Overview**

Two PDC bits were used to build the curve section. Bit #3 was an 8 3/4" Security bit used to drill the curve from a measured depth of 10,240' to 10,859' in 17.5 hours, averaging 34.50 ft/hr. Bit #4 (8 3/4" Security) was used to drill the curve from a measured depth of 10,859' to 11,175' in

10.5 hours, averaging 30.1 ft/hr. 7" 32# P-110 casing was set to 11,037' MD upon completion of the curve.

A Chart was constructed for the curve (Chart 1) that compared the isopach of certain easily recognizable gamma markers from the **Kibbey Lime** to **Middle Bakken** on the control well *Kline Federal 5300 41-18 11T2* and *Kline Federal 5300 41-18 12TX* to the *Kline Federal 5300 41-18 10B* as marker picks were made. This chart shows the **False Bakken** top being very close in thickness to target based on the offset wells, and was the main point used to make changes to the target landing depth during the curve. The curve was completed at a measured depth of 11,175' and TVD of 10,752', ~19 feet below the base of the **Upper Bakken Shale**.



**Chart 1:** Comparing distances from gamma markers on the offset wells to markers in gamma in the curve.

## **Lithology**

The top of the "**False Bakken**" was drilled at a measured depth of 10,864' (TVD 10,708') (-8,626') 3' low to the *Kline Federal 5300 41-18 11T2*.

The *Upper Shale* of the **Bakken** formation [Mississippian – Devonian] was drilled at 10,890' MD (10,716' TVD) (-8,634') with sample returns of black, carbonaceous, and petroliferous shale (Figure 5) characterized by gamma ray values in excess of 255 API counts. The *Upper Shale* was 4' low to the *Kline Federal 5300 41-18 11T2*. Samples were described as:

SHALE: black, firm, earthy to waxy texture, pyrite, carbonaceous, platy, visible fracture porosity





*Figure 5: Photograph sample from the Upper Bakken Shale.*

The *Middle Member* of the **Bakken** formation was penetrated at 10,954' MD, (10,733' TVD) (-8,651'), 3' low to the *Kline Federal 5300 41-18 11T2*. The *Middle Member* contained layers of sandstone and silty sandstones with varying amounts of porosity and oil stain (Figure 6). Samples from the *Middle Member* were described as:

SILTY SANDSTONE: common salt and pepper, occasional light gray to medium gray, visible white, very fine grained, soft to firm, moderately to well sorted, rounded to sub rounded, moderately to poorly cemented, visible calcareous fluid fractures, possible fracture porosity, common intergranular porosity, visible light brown spotty oil stain



*Figure 6: Photograph of sandstone and silty sandstone from the middle member of the Bakken formation.*



## **Lateral**

### **Overview**

Drilling fluid consisting of open system salt water brine (9.6-9.7 ppg) was used while drilling the lateral section up to the TD of 20,475'. Four 6" PDC bits were used to drill the lateral.

Bit #5 (Varel VM613P2) was a PDC bit used to drill the start of the lateral from 11,175' to 13,813' and averaged an ROP of 57.30 over 46 hours of use. Bit #6 (Reed) was a PDC bit used to drill from 13,813' to 16,761'; it was used for 69 hours at an average drilling rate of 42.70 feet per hour. Bit #7 (Smith MDSi613) was a PDC bit used to drill from 16,761' to 19,731', and was used for 91 hours and averaged an ROP of 42.70 ft/hr. Bit #8 (Smith MDSi613) was a PDC bit used to drill from 19,731' to TD @ 20,475'

The Upper Bakken Shale was encountered at the measured depth 15,220'. The decision was made to move the bit back to 15,070' MD and sidetrack. Time drilling began using the 5-5 4-4 3-3 2-2 1-1 method and once that was concluded at 15,102' MD the sidetrack was drilled to forward.

The Lower Bakken Shale was encountered at the measured depth 18,550'. The decision was made to move the bit back to 18,000' MD and sidetrack. Time drilling began using the 5-5 4-4 3-3 2-2 1-1 method and once that was concluded at 18,021' MD the sidetrack was drilled to TD.

Drilling operations were concluded at 10:48 on December 31, 2014 at a bottom hole location of 10,630.79' S & 1,109.25' E of surface location or approximately 754.45' FSL & 283.76' FEL of SE SE Sec. 22 T153N, R100W.

### **Lithology**

The target Zone of the **Middle Bakken** consisted of a zone 9 to 19 feet below the **Upper Bakken Shale**. This zone consists of a silty sandstone with very intergranular porosity, moderate gas, and visible oil stain (Figure 6). Samples from the target Zone were generally described as:

SILTY SANDSTONE: light to medium gray, common light brown to tan, common white to off white, very fine grained, friable to firm, moderately to well sorted, sub rounded to sub angular, moderately to poorly cemented, visible calcareous fluid fractures, trace disseminated pyrite, possible fracture porosity, common intergranular porosity, trace light brown spotty oil stain

### **Gas and hydrocarbon Shows**

It was found that the gas and hydrocarbon shows were elevated through the entire lateral with an average background gas of ~3500 units and common peaks of between 7000u and 8000u.

The NOV gas chromatograph showed components of all C1-C4 present through the entire lateral. Sample cuttings throughout the lateral showed oil stain and a fast white diffuse cut when submerged in Acetone solution and observed with a UV fluoroscope. Trace oil was observed in the sample wash water.

## **Summary**

- 1) The *Kline Federal 5300 41-18 10B* was initially spud for surface drilling on September 9th 2014 in the Indian Hill field of McKenzie County, North Dakota. The well was spud by Nabors 486 and the vertical hole was drilled to a total depth (KOP) of 10,240'.
- 2) A mud program consisted of diesel invert (9.8-9.9 ppg) after surface casing through the curve build sections. The drilling mud was successful in maintaining stable hole conditions and minimizing washout through the salt intervals. Brine water with a weight of 9.6-9.7 ppg was used during lateral drilling operations to maintain hydrostatic balance until a TD of 20,475'.
- 3) Two 8 ¾" Varel PDC bit was used to drill out of surface casing to a KOP of 10,240'. Two 8 ¾" PDC bits were used to drill the curve to 11,175' MD. Four 6" PDC bits were used to drill the lateral to 20,475' TD.
- 4) The lateral was targeting the **Middle Member** of the **Bakken Formation** member ranging from about 9' and 19' below the base of the **Upper Bakken Shale**
- 5) Gas levels between 3000 and 4000 units were maintained with peaks over 8000 units through the lateral drilling operations while in the **Middle Member**.
- 6) The **Middle Member** of the **Bakken Formation** target consisted of one observed gross lithology. The **Middle Member** was observed to be a silty sandstone with lots of disseminated pyrite and apparent intergranular porosity with visible oil stain.
- 7) Trace to moderate oil stain was observed in zone associated with a fast, white, diffuse cut. Trace oil washed out of the samples into the wash water when they were cleaned.
- 8) Drilling operations were concluded at 14:00 on July 8<sup>th</sup>, 2014 at a bottom hole location of 254.45' south & 9968.39' east of surface location or approximately 754.45' FSL & 283.76' FEL of SE SE Sec. 22, T153N-R100W.
- 9) Oasis Petroleum's *Kline Federal 5300 41-18 10B* awaits completion operations to determine its ultimate production potential.

Respectfully submitted,  
Annika Tostengard  
RPM Geologic LLC



# Well Information

**Operator:** Oasis Petroleum North America

**Address:** 1001 Fannin Suite 202  
Houston, TX 77002

**Well Name:** Kline Federal 5300 41-18 10B

**Field/ Prospect:** Baker

**Elevation:** GL: 2,057'  
KB: 2,082'

**Spud Date:** 9-Sep-14

**API #:** 33-053-06026

**NDIC Well File #:** 28652

**Surface Location:** SW SW Sec. 18 T153N, R100W

**Footages:** 500' FSL & 237' FWL

**County, State:** McKenzie Co., ND

**Basin:** Williston

**Well Type:** Horizontal Bakken

**Contractor:** Nabors 486

**Toolpushers:** Logan Penhollow, Tyler Schultz

**Field Supervisors:** Mark Lawlar/Travis Handran

**Directional Drilling** RPM Consulting  
Mark Lawlar/Travis Handran  
Eric Nielssen

**Chemical Company** Halliburton

**Mud Engineer** Tim Vaira

**H<sub>2</sub>S MONITORING:** NOV WellSight Gas Watch

**MWD** Gyro Data  
Robert Porter, Kasey (curve)

**Wellsite Geologist** Annika Tostengard  
Timber Solberg

**Rock Sampling:** 30' from 8,350' to 10,860' MD  
50' 11,175' MD to 20,475' (TD)

**Prospect Geologist** Brendan Hargrove

**Gas Detector** NOV WellSight Gas Watch

**Sample Examination:** Binocular microscope & fluoroscope

**Sample Cuts:** Acetone

**Horizontal Target** Middle Bakken

## Key Offset Wells:

|                  |                                      |                            |                  |
|------------------|--------------------------------------|----------------------------|------------------|
| Oasis Petroleum. | <b>Kline Federal 5300 41-18 11T2</b> | SW SW Sec. 18 T153N, R100W | McKenzie Co., ND |
| Oasis Petroleum  | <b>Kline Federal 5300 41-18 12TX</b> | SW SW Sec. 18 T153N, R100W | McKenzie Co., ND |

**Pumps:** #1 & #2: HHF 1600 Triplex - 12" Stroke length Output: 0.0997 bbl/stk

**Mud Type:** Diesel invert mud 2,140'-11,125' (Curve TD)  
Salt Water Brine, 10,860' - 20,475' (TD)

**Casing:** Surface: 13 3/8" 54# J-55 @ 2,083'  
Intermediate: 9 5/8" 40# HCL80 @ 6,125', 7" 32# P-110 @ 11,037'

**Hole Size:** 17 1/2 to 2,125', 12 1/4" to 6,140' MD, 8 3/4" to 11,125' MD 6" to 20,475'(TD)

**Total Drilling Days:** 21

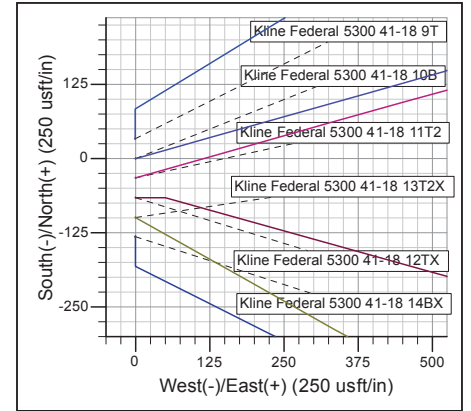
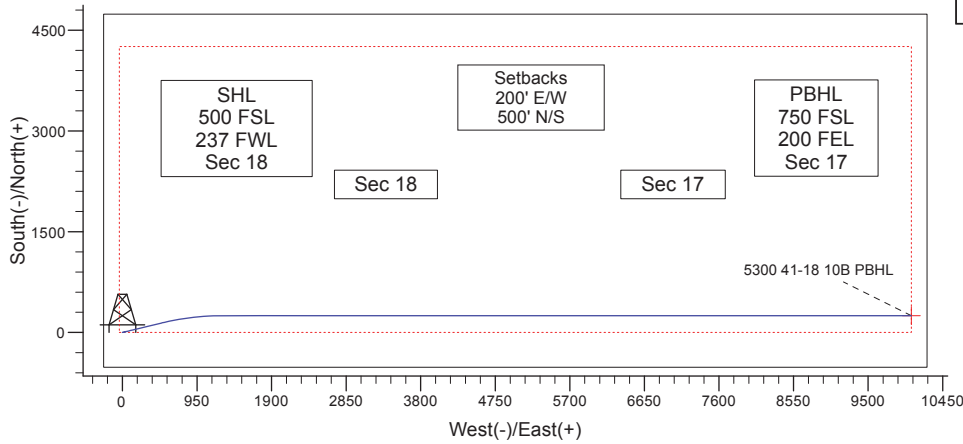
|                                       |                                |                                          |
|---------------------------------------|--------------------------------|------------------------------------------|
| <b><u>Horizontal Target:</u></b>      | <i>Middle Bakken</i>           | <b><u>BOTTOM HOLE LOCATION:</u></b>      |
|                                       |                                | <i>254.45' S &amp; 9968.39' E</i>        |
| <b><u>Kick-Off Point / Date:</u></b>  | <i>10,240' on 15 Nov, 2014</i> | <i>of surface location or approx. or</i> |
|                                       |                                | <i>754.45' FSL &amp; 283.76' FEL</i>     |
| <b><u>Total Depth/ Date:</u></b>      | <i>20,475' 31 Dec, 2014</i>    | <i>SE SE Sec. 22, T153N, R100W</i>       |
| <b><u>Ending Vertical Section</u></b> | <i>9968.4</i>                  |                                          |
| <b><u>Ending Azimuth</u></b>          | <i>90.81</i>                   |                                          |
| <b><u>Status of Well:</u></b>         | <i>Complete</i>                |                                          |



Project: Indian Hills  
 Site: 153N-100W-17/18  
 Well: Kline Federal 5300 41-18 10B  
 Wellbore: Kline Federal 5300 41-18 10B  
 Design: Design #1

#### WELL DETAILS: Kline Federal 5300 41-18 10B

Ground Level: 2057.0  
 Northing: 405202.44 Easting: 1209970.87 Latitude: 48° 4' 8.200 N Longitude: 103° 36' 11.950 W



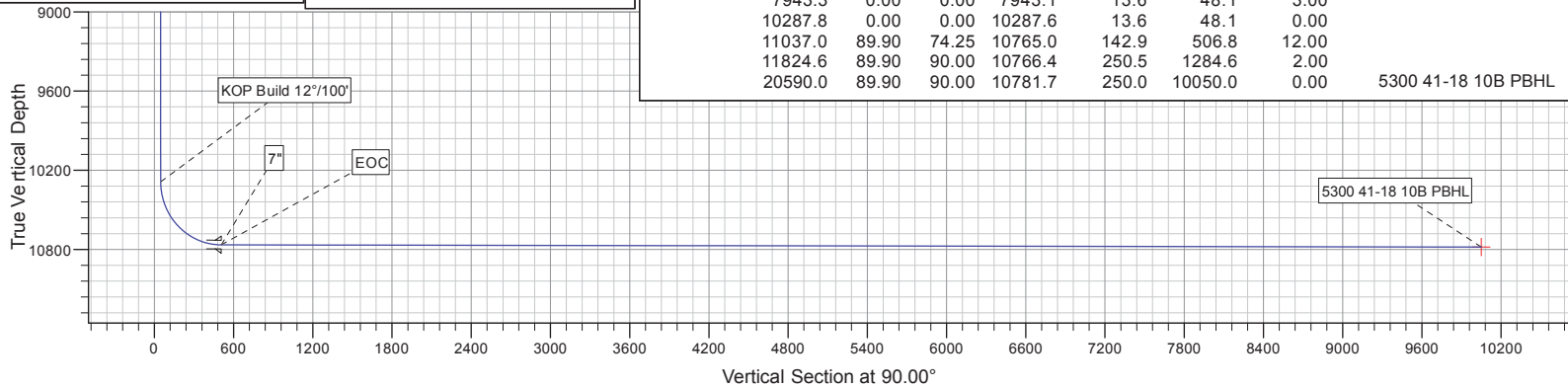
| CASING DETAILS |         |      |      |
|----------------|---------|------|------|
| TVD            | MD      | Name | Size |
| 10765.0        | 11037.0 | 7"   | 7    |



Azimuths to True North  
 Magnetic North: 8.27°  
 Magnetic Field  
 Strength: 56438.3snT  
 Dip Angle: 72.99°  
 Date: 5/19/2014  
 Model: IGRF2010

#### SECTION DETAILS

| MD      | Inc   | Azi   | TVD     | +N/-S | +E/-W   | Dleg  | Target              |
|---------|-------|-------|---------|-------|---------|-------|---------------------|
| 0.0     | 0.00  | 0.00  | 0.0     | 0.0   | 0.0     | 0.00  |                     |
| 2200.0  | 0.00  | 0.00  | 2200.0  | 0.0   | 0.0     | 0.00  |                     |
| 2216.7  | 0.50  | 74.25 | 2216.7  | 0.0   | 0.1     | 3.00  |                     |
| 7926.7  | 0.50  | 74.25 | 7926.4  | 13.5  | 48.0    | 0.00  |                     |
| 7943.3  | 0.00  | 0.00  | 7943.1  | 13.6  | 48.1    | 3.00  |                     |
| 10287.8 | 0.00  | 0.00  | 10287.6 | 13.6  | 48.1    | 0.00  |                     |
| 11037.0 | 89.90 | 74.25 | 10765.0 | 142.9 | 506.8   | 12.00 |                     |
| 11824.6 | 89.90 | 90.00 | 10766.4 | 250.5 | 1284.6  | 2.00  |                     |
| 20590.0 | 89.90 | 90.00 | 10781.7 | 250.0 | 10050.0 | 0.00  | 5300 41-18 10B PBHL |



## **Daily Activity**

| <b>Day</b> | <b>Date 2014</b> | <b>Depth<br/>0600<br/>Hrs</b> | <b>24 Hr<br/>Footage</b> | <b>Bit #</b> | <b>WOB<br/>(Klbs) Rotate</b> | <b>WOB<br/>(Klbs) Slide</b> | <b>RPM<br/>(RT)</b> | <b>Pump<br/>Pressure</b> | <b>SPM 1</b> | <b>SPM 2</b> | <b>GPM</b> | <b>24 Hr Activity</b>                                                                                                                                                                                                                       | <b>Formation</b> |
|------------|------------------|-------------------------------|--------------------------|--------------|------------------------------|-----------------------------|---------------------|--------------------------|--------------|--------------|------------|---------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|------------------|
| 1          | 9-Sep            | 569'                          | 489'                     | 1            | 5                            | 10                          | 40                  | 0                        | 0            | 0            | 0          | circulate and condition, rig up cementers, primary cementing, circulate and condition, primary cementing second stage, rig down, install wellheads, pressure test, nipple down bops, rig released at 2200 hours, rig down, skid rig, rig up | Surface          |
| 2          | 10-Sep           | 2,125'                        | 1,556'                   | 1            | 5                            | 12                          | 60                  | 0                        | 0            | 0            | 0          | Drill 596'-1,865' TOOHP for MWD, change out BHA, TIH, drill f/ 1,865 - 2,125, TOOHP, rig up casers                                                                                                                                          | Piere            |
| 3          | 11-Sep           | 2,125'                        | 0'                       | 1            | 0                            | 0                           | 0                   | 0                        | 0            | 0            | 0          | run casing, rig up cementers, pump cement, rig down cementers, weld on wellhead                                                                                                                                                             | Dakota           |
| 4          | 7-Nov            | 2,125'                        | 0'                       | 2            | 0                            | 0                           | 0                   | 0                        | 0            | 0            | 0          | nipple up BOP, TIH, drill cement                                                                                                                                                                                                            | Opeche           |
| 5          | 9-Nov            | 5,832'                        | 3,707'                   | 2            | 5                            | 15                          | 65                  | 2100                     | 75           | 75           | 520        | drill 2,125'-5,832'                                                                                                                                                                                                                         | Tyler            |
| 6          | 10-Nov           | 6,140'                        | 308'                     | 2            | 5                            | 15                          | 47                  | 2900                     | 75           | 75           | 520        | Drill f/ 5,832'-6,140', TOOHP, rig up casers, run casing, rig up cement, pump cement, pressure test, rig down                                                                                                                               | Tyler            |
| 7          | 11-Nov           | 7,630'                        | 1,490'                   | 3            | 38                           | 0                           | 68                  | 3200                     | 73           | 73           | 611        | P/U BHA, trip in hole, drill shoe, drill f/ 6140-7630                                                                                                                                                                                       | Charles          |
| 8          | 12-Nov           | 8,950'                        | 1,320'                   | 3            | 25                           | 35                          | 55                  | 2750                     | 150          |              | 520        | Drill f/7,630' - 8,950'                                                                                                                                                                                                                     | Lodgepole        |



## **Daily Activity**

| <b>Day</b> | <b>Date 2014</b> | <b>Depth<br/>0600<br/>Hrs</b> | <b>24 Hr<br/>Footage</b> | <b>Bit #</b> | <b>WOB<br/>(Klbs) Rotate</b> | <b>WOB<br/>(Klbs) Slide</b> | <b>RPM<br/>(RT)</b> | <b>Pump<br/>Pressure</b> | <b>SPM 1</b> | <b>SPM 2</b> | <b>GPM</b> | <b>24 Hr Activity</b>                                                                                | <b>Formation</b> |
|------------|------------------|-------------------------------|--------------------------|--------------|------------------------------|-----------------------------|---------------------|--------------------------|--------------|--------------|------------|------------------------------------------------------------------------------------------------------|------------------|
| 9          | 13-Nov           | 9,859'                        | 909'                     | 3            | 10                           | 40                          | 50                  | 3300                     | 150          |              | 525        | Drill f/ 8,950'-9,859'                                                                               | Lodgepole        |
| 10         | 14-Nov           | 10,240'                       | 381'                     | 3,4          | 10                           | 45                          | 50                  | 2800                     | 150          |              | 527        | Drill f/ 9,859-9,950, TOOH, inspect BHA, P/U new BHA, TIH, ream salts, TIH, drill 9,950'-10,240, TOH | Lodgepole        |
| 11         | 15-Nov           | 10,561'                       | 321'                     | 4,5          | 10                           | 40                          | 50                  | 2800                     | 150          |              | 527        | TOOH, run logs, P/U curve assembly, TIH, Drill f/ 10,240'-10,583'                                    | Middle Bakken    |
| 12         | 16-Nov           | 10,859'                       | 298'                     | 5,6          | 10                           | 35                          | 25                  | 3400                     |              | 150          | 510        | drill f/ 10,583-10,859, TOOH wet, inspect and changeout BHA                                          | Middle Bakken    |
| 13         | 17-Nov           | 11,082'                       | 223'                     | 6            | 10                           | 25                          | 25                  | 1450                     | 98           | 145          | 0          | TIH, ream salts, TIH, drill f/ 10,859-11,082'                                                        | Middle Bakken    |
| 14         | 18-Nov           | 11,175'                       | 93'                      | 6            | 10                           | 25                          | 26                  | 2577                     | 140          | 0            | 492        | Drill f/ 11,082-11,175, TOOH, slip n cut, L/D BHA, rig up casers, run 7" intermediate casing         | Middle Bakken    |
| 15         | 19-Nov           | 11,175'                       | 0'                       |              |                              |                             |                     |                          |              |              |            | Run 7" intermediate Casing                                                                           | Middle Bakken    |
| 16         | 20-Nov           | 11,175'                       | 0'                       | 6            |                              |                             |                     |                          |              |              |            | rig down casing crew, rig up cementers, pump cement, set casing slips, prepare to skid rig           | Middle Bakken    |
| 17         | 18-Dec           | 11,175'                       | 0'                       |              |                              |                             |                     |                          |              |              |            | skid rig back, NU BOP, pressure test                                                                 | Bkken            |

## **Daily Activity**

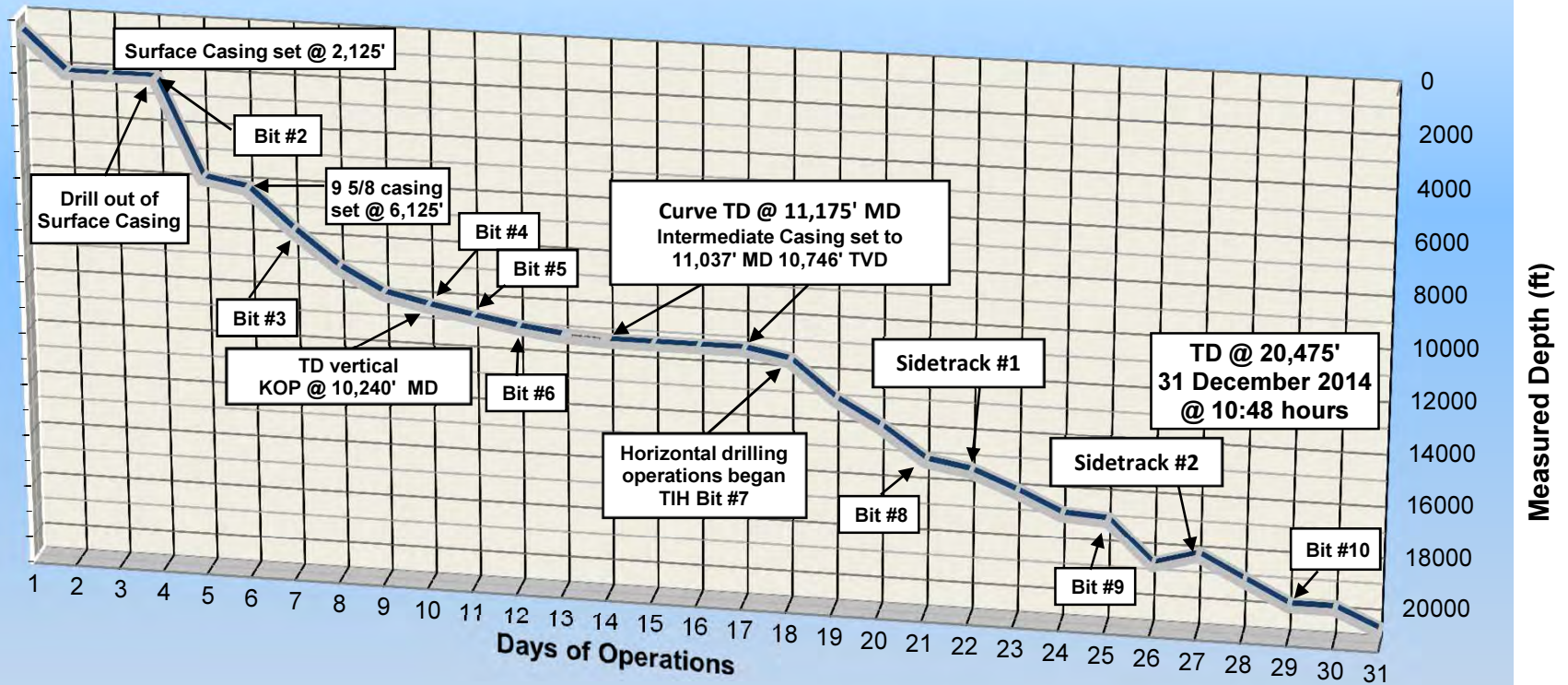
| <b>Day</b> | <b>Date 2014</b> | <b>Depth<br/>0600<br/>Hrs</b> | <b>24 Hr<br/>Footage</b> | <b>Bit #</b> | <b>WOB<br/>(Klbs) Rotate</b> | <b>WOB<br/>(Klbs)<br/>Slide</b> | <b>RPM<br/>(RT)</b> | <b>Pump<br/>Pressure</b> | <b>SPM 1</b> | <b>SPM 2</b> | <b>GPM</b> | <b>24 Hr Activity</b>                                                                     | <b>Formation</b> |
|------------|------------------|-------------------------------|--------------------------|--------------|------------------------------|---------------------------------|---------------------|--------------------------|--------------|--------------|------------|-------------------------------------------------------------------------------------------|------------------|
| 18         | 19-Dec           | 11,493'                       | 318'                     | 7            | 5                            | 15                              | 45                  | 2100                     | 75           | 75           | 263        | P/U BHA, TIH, drill f/ 11,175' to 11,493'                                                 | Middle Bakken    |
| 19         | 20-Dec           | 12,866'                       | 1,373'                   | 7            | 10                           | 18                              | 45                  | 860                      | 30           | 40           | 245        | drill f/ 11,493- 12,866'                                                                  | Middle Bakken    |
| 20         | 21-Dec           | 13,813'                       | 947'                     | 7,8          | 12                           | 25                              | 45                  | 3158                     | 75           | 0            | 263        | Drill f/ 12.866'-13,813, TOOH for motor, C/O BHA, TIH                                     | Middle Bakken    |
| 21         | 22-Dec           | 15,008'                       | 1,195'                   | 8            | 8                            | 20                              | 40                  | 3200                     | 75           | 0            | 263        | TIH, drill f/ 13,813'-15,008'                                                             | Middle Bakken    |
| 22         | 23-Dec           | 15,303'                       | 295'                     | 8            | 15                           | 18                              | 45                  | 2835                     | 75           | 0            | 281        | Drill f/ 15,008-15303, confirm shale strike, back up 15,024, time drill                   | Middle Bakken    |
| 23         | 24-Dec           | 15,984'                       | 681'                     | 8            | 20                           | 25                              | 30                  | 3736                     | 0            | 75           | 263        | time drill, , drill f/ 15,098'-15,984'                                                    | Middle Bakken    |
| 24         | 25-Dec           | 16,761'                       | 777'                     | 8            | 15                           | 22                              | 30                  | 3100                     | 0            | 75           | 264        | Drill 15,984' - 16,761', TOOH for motor                                                   | Middle Bakken    |
| 25         | 26-Dec           | 16,865'                       | 104'                     | 8,9          | 12                           | 20                              | 40                  | 3100                     | 0            | 75           | 264        | TOOH, C/O BHA, TIH, drill f/ 16,761' - 16,865                                             | Middle Bakken    |
| 26         | 27-Dec           | 18,500'                       | 1,635'                   | 9            | 12                           | 20                              | 45                  | 3200                     | 0            | 75           | 263        | Drill f/16,865 - 18,500'                                                                  | Middle Bakken    |
| 27         | 28-Dec           | 18,009'                       | -491'                    | 9            | 15                           | 101                             | 45                  | 3200                     | 0            | 75           | 263        | Drill f/ 18,500-18,627, circ Bottoms up, TOH to sidetrack Time drill from 17,936 - 18,010 | Middle Bakken    |
| 28         | 29-Dec           | 18,877'                       | 868'                     | 9            | 12                           | 20                              | 40                  | 3200                     | 0            | 75           | 281        | Time Drill 18,010 -18,021'. Drill f/ 18,021-18,877'                                       | Middle Bakken    |
| 29         | 30-Dec           | 19,731'                       | 854'                     | 10           | 12                           | 20                              | 40                  | 3200                     | 0            | 75           | 281        | Drill f 18,877'-19,731' TOOH                                                              | Middle Bakken    |
| 30         | 31-Dec           | 19,731'                       | 0'                       | 10           | -                            | -                               | -                   | -                        | -            | -            |            | TOOH, L/D BHA, P/U new BHA TIH, orintate to slide                                         | Middle Bakken    |
| 31         | 1-Jan            | 20,475'                       | 744'                     | -            | -                            | -                               | -                   | -                        | -            | -            |            | Drill 19,731'-20,475' TOOH, ream, prepare for completions                                 | Middle Bakken    |



## Daily Progress

Oasis Petroleum  
Kline Federal 5300 41-18 10B

Spud 09 September 2014



## **Bit Record**

| <i>Bit #</i> | <i>Size</i> | <i>Make</i> | <i>Model</i> | <i>Serial #</i> | <i>Jets</i> | <i>Depth In</i> | <i>Depth Out</i> | <i>Footage</i> | <i>Hours</i> | <i>Mean ROP<br/>(ft/hr)</i> | <i>Accum.<br/>Hours</i> |
|--------------|-------------|-------------|--------------|-----------------|-------------|-----------------|------------------|----------------|--------------|-----------------------------|-------------------------|
| 1            | 17 1/2      | Nov         | TFR6195      | E163630         | 12x18       | 0'              | 2,125'           | 2,125'         | 21           | 101.20                      | 21.0                    |
| 2            | 12 1/4      | Varel       | Varel        | 4007781         | 6x20        | 2,125'          | 6,140'           | 4,000'         | 26           | 153.80                      | 47.00                   |
| 3            | 8 3/4       | Varel       | Varel        | 4006758         | 6x16        | 6,140'          | 9,950'           | 3810           | 61.5         | 62.00                       | 108.5                   |
| 4            | 8 3/4       | Varel       | Varel        | 4006379         | 6x16        | 9,950'          | 10,240'          | 290            | 4.5          | 64.40                       | 113.0                   |
| 5            | 8 3/4       | Halliburton | Sec          | 12560298        | 5x18        | 10,240'         | 10,859'          | 619            | 17.5         | 35.40                       | 130.5                   |
| 6            | 8 3/4       | Halliburton | Sec          | 12487646        | 5x18        | 10,859'         | 11,175'          | 316            | 10.5         | 30.10                       | 141.0                   |
| 7            | 6           | Varel       | Varel        | 4007155         | 6x18        | 11,175'         | 13,813'          | 2638           | 46           | 57.30                       | 187.0                   |
| 8            | 6           | Reed        | Reed         | A208139         | 6x19        | 13,813'         | 16,761'          | 2948           | 69           | 42.70                       | 256.0                   |
| 9            | 6           | Smith       | MDSi613      | JK0375          | 6X18        | 16,761'         | 19,731'          | 3461           | 91           | 42.70                       | 347.0                   |
| 10           | 6           | Smith       | MDSi613      | JJ035           | 6x18        | 19,731'         | 20,475'          | 744            | 16           | 46.50                       | 363.0                   |

| DRILLING PLAN                                                                                                                     |  |                              |  |                             |                         |                                                                                                                                  |  |                                        |  |
|-----------------------------------------------------------------------------------------------------------------------------------|--|------------------------------|--|-----------------------------|-------------------------|----------------------------------------------------------------------------------------------------------------------------------|--|----------------------------------------|--|
| OPERATOR                                                                                                                          |  | Oasis Petroleum              |  |                             |                         | COUNTY/STATE                                                                                                                     |  | McKenzie Co., ND                       |  |
| WELL NAME                                                                                                                         |  | Kline Federal 5300 41-18 10B |  |                             |                         | RIG                                                                                                                              |  | 0                                      |  |
| WELL TYPE                                                                                                                         |  | Horizontal Middle Bakken     |  |                             |                         |                                                                                                                                  |  |                                        |  |
| LOCATION                                                                                                                          |  | SWNW 18-153N-100W            |  |                             |                         | Surface Location (survey plat):                                                                                                  |  | 500' fsl 237' fwl                      |  |
| EST. T.D.                                                                                                                         |  | 20,590'                      |  |                             |                         | GROUND ELEV:                                                                                                                     |  | 2057 Finished Pad Elev. Sub Height: 25 |  |
| TOTAL LATERA                                                                                                                      |  | 9,553'                       |  |                             |                         | KB ELEV:                                                                                                                         |  | 2082                                   |  |
| PROGNOSIS:                                                                                                                        |  | Based on 2,082' KB(est)      |  |                             |                         | LOGS:                                                                                                                            |  | Type Interval                          |  |
| MARKER                                                                                                                            |  | DEPTH (Surf Loc)             |  | DATUM (Surf Loc)            |                         | OH Logs: Triple Combo KOP to Kibby (or min run of 1800' whichever is greater); GR/Res to BSC; GR to surf; CND through the Dakota |  |                                        |  |
|                                                                                                                                   |  |                              |  |                             |                         | CBL/GR: Above top of cement/GR to base of casing                                                                                 |  |                                        |  |
|                                                                                                                                   |  |                              |  |                             |                         | MWD GR: KOP to lateral TD                                                                                                        |  |                                        |  |
| Pierre                                                                                                                            |  | NDIC MAP                     |  | 1,972 110                   |                         | DEVIATION:                                                                                                                       |  |                                        |  |
| Greenhorn                                                                                                                         |  |                              |  | 4,646 (2,564)               |                         | Surf: 3 deg. max., 1 deg / 100'; srvy every 500'                                                                                 |  |                                        |  |
| Mowry                                                                                                                             |  |                              |  | 5,057 (2,975)               |                         | Prod: 5 deg. max., 1 deg / 100'; srvy every 100'                                                                                 |  |                                        |  |
| Dakota                                                                                                                            |  |                              |  | 5,471 (3,389)               |                         |                                                                                                                                  |  |                                        |  |
| Rierdon                                                                                                                           |  |                              |  | 6,486 (4,404)               |                         |                                                                                                                                  |  |                                        |  |
| Dunham Salt                                                                                                                       |  |                              |  | 6,814 (4,732)               |                         |                                                                                                                                  |  |                                        |  |
| Dunham Salt Base                                                                                                                  |  |                              |  | 6,929 (4,847)               |                         |                                                                                                                                  |  |                                        |  |
| Spearfish                                                                                                                         |  |                              |  | 7,024 (4,942)               |                         |                                                                                                                                  |  |                                        |  |
| Pine Salt                                                                                                                         |  |                              |  | 7,283 (5,201)               |                         |                                                                                                                                  |  |                                        |  |
| Pine Salt Base                                                                                                                    |  |                              |  | 7,318 (5,236)               |                         |                                                                                                                                  |  |                                        |  |
| Opeche Salt                                                                                                                       |  |                              |  | 7,374 (5,292)               |                         |                                                                                                                                  |  |                                        |  |
| Opeche Salt Base                                                                                                                  |  |                              |  | 7,403 (5,321)               |                         | DST'S:                                                                                                                           |  |                                        |  |
| Broom Creek (Top of Minnelusa Gp.)                                                                                                |  |                              |  | 7,605 (5,523)               |                         | None planned                                                                                                                     |  |                                        |  |
| Amsden                                                                                                                            |  |                              |  | 7,684 (5,602)               |                         |                                                                                                                                  |  |                                        |  |
| Tyler                                                                                                                             |  |                              |  | 7,853 (5,771)               |                         |                                                                                                                                  |  |                                        |  |
| Otter (Base of Minnelusa Gp.)                                                                                                     |  |                              |  | 8,047 (5,965)               |                         |                                                                                                                                  |  |                                        |  |
| Kibbey Lime                                                                                                                       |  |                              |  | 8,390 (6,308)               |                         | CORES:                                                                                                                           |  |                                        |  |
| Charles Salt                                                                                                                      |  |                              |  | 8,542 (6,460)               |                         | None planned                                                                                                                     |  |                                        |  |
| UB                                                                                                                                |  |                              |  | 9,159 (7,077)               |                         |                                                                                                                                  |  |                                        |  |
| Base Last Salt                                                                                                                    |  |                              |  | 9,238 (7,156)               |                         |                                                                                                                                  |  |                                        |  |
| Ratcliffe                                                                                                                         |  |                              |  | 9,301 (7,219)               |                         |                                                                                                                                  |  |                                        |  |
| Mission Canyon                                                                                                                    |  |                              |  | 9,454 (7,372)               |                         | MUDLOGGING:                                                                                                                      |  |                                        |  |
| Lodgepole                                                                                                                         |  |                              |  | 10,022 (7,940)              |                         | Two-Man: 8,342'                                                                                                                  |  |                                        |  |
| Lodgepole Fracture Zone                                                                                                           |  |                              |  | 10,209 (8,127)              |                         | ~200' above the Charles (Kibbey) to Casing point; Casing point to TD                                                             |  |                                        |  |
| False Bakken                                                                                                                      |  |                              |  | 10,726 (8,644)              |                         | 30' samples at direction of wellsite geologist; 10' through target @ curve land                                                  |  |                                        |  |
| Upper Bakken                                                                                                                      |  |                              |  | 10,735 (8,653)              |                         |                                                                                                                                  |  |                                        |  |
| Middle Bakken                                                                                                                     |  |                              |  | 10,751 (8,669)              |                         |                                                                                                                                  |  |                                        |  |
| Middle Bakken Sand Target                                                                                                         |  |                              |  | 10,760 (8,678)              |                         |                                                                                                                                  |  |                                        |  |
| Base Middle Bakken Sand Target                                                                                                    |  |                              |  | 10,770 (8,688)              |                         |                                                                                                                                  |  |                                        |  |
| Lower Bakken                                                                                                                      |  |                              |  | 10,785 (8,703)              |                         | BOP:                                                                                                                             |  |                                        |  |
| Three Forks                                                                                                                       |  |                              |  | 10,813 (8,731)              |                         | 11" 5000 psi blind, pipe & annular                                                                                               |  |                                        |  |
| Dip Rate:                                                                                                                         |  | -0.1                         |  |                             |                         |                                                                                                                                  |  |                                        |  |
| Max. Anticipated BHP:                                                                                                             |  | 4673                         |  |                             |                         | Surface Formation: Glacial till                                                                                                  |  |                                        |  |
| MUD:                                                                                                                              |  | Interval                     |  | Type                        |                         | WT                                                                                                                               |  | Vis                                    |  |
| Surface:                                                                                                                          |  | 0' -                         |  | 2,072' FW/Gel - Lime Sweeps |                         | 8.4-9.0                                                                                                                          |  | 28-32                                  |  |
| Intermediate:                                                                                                                     |  | 2,072' -                     |  | 11,037' Invert              |                         | 9.5-10.4                                                                                                                         |  | 40-50                                  |  |
| Lateral:                                                                                                                          |  | 11,037' -                    |  | 20,590' Salt Water          |                         | 9.8-10.2                                                                                                                         |  | 28-32                                  |  |
| CASING:                                                                                                                           |  | Size                         |  | Wt ppf                      |                         | Hole                                                                                                                             |  | Depth                                  |  |
| Surface:                                                                                                                          |  | 9-5/8"                       |  | 36#                         |                         | 13-1/2"                                                                                                                          |  | 2,072'                                 |  |
| Intermediate:                                                                                                                     |  | 7"                           |  | 29/32#                      |                         | 8-3/4"                                                                                                                           |  | 11,037'                                |  |
| Production Liner:                                                                                                                 |  | 4.5"                         |  | 11.6#                       |                         | 6"                                                                                                                               |  | 20,590'                                |  |
|                                                                                                                                   |  |                              |  |                             |                         |                                                                                                                                  |  | TOL @ 10,238'                          |  |
| PROBABLE PLUGS, IF REQ'D:                                                                                                         |  |                              |  |                             |                         |                                                                                                                                  |  |                                        |  |
| OTHER:                                                                                                                            |  | MD                           |  | TVD                         |                         | FNL/FSL                                                                                                                          |  | FEL/FWL                                |  |
| Surface:                                                                                                                          |  | 2,072                        |  | 2,072                       |                         | 500' FSL                                                                                                                         |  | 237' FWL                               |  |
| KOP:                                                                                                                              |  | 10,288'                      |  | 10,288'                     |                         | 514' FSL                                                                                                                         |  | 285' FWL                               |  |
| EOC:                                                                                                                              |  | 11,037'                      |  | 10,765'                     |                         | 643' FSL                                                                                                                         |  | 744' FWL                               |  |
| Casing Point:                                                                                                                     |  | 11,037'                      |  | 10,765'                     |                         | 643' FSL                                                                                                                         |  | 744' FWL                               |  |
| Middle Bakken Lateral TD:                                                                                                         |  | 20,590'                      |  | 10,782'                     |                         | 750' FSL                                                                                                                         |  | 200' FEL                               |  |
|                                                                                                                                   |  |                              |  |                             |                         |                                                                                                                                  |  | S-T-R                                  |  |
|                                                                                                                                   |  |                              |  |                             |                         |                                                                                                                                  |  | AZI                                    |  |
|                                                                                                                                   |  |                              |  |                             |                         |                                                                                                                                  |  | Survey Company:                        |  |
|                                                                                                                                   |  |                              |  |                             |                         |                                                                                                                                  |  | Build Rate: 12 deg / 100'              |  |
| Comments:                                                                                                                         |  |                              |  |                             |                         |                                                                                                                                  |  |                                        |  |
| Request a Sundry for an Open Hole Log Waiver                                                                                      |  |                              |  |                             |                         |                                                                                                                                  |  |                                        |  |
| Exception well: Oasis Petroleum's Kline Federal 5300 11-18H (153N 100W 18 NW NW)                                                  |  |                              |  |                             |                         |                                                                                                                                  |  |                                        |  |
| 35 packers, 35 sleeves, no frac string                                                                                            |  |                              |  |                             |                         |                                                                                                                                  |  |                                        |  |
| Oasis Petroleum does not use Diesel Fuel, as defined by the US EPA in the list below, in our hydraulic fracture operations.       |  |                              |  |                             |                         |                                                                                                                                  |  |                                        |  |
| 68334-30-5 (Primary Name: Fuels, diesel) 68476-34-6 (Primary Name: Fuels, diesel, No. 2)68476-30-2 (Primary Name: Fuel oil No. 2) |  |                              |  |                             |                         |                                                                                                                                  |  |                                        |  |
| 68476-31-3 (Primary Name: Fuel oil, No. 4)8008-20-6 (Primary Name: Kerosene)                                                      |  |                              |  |                             |                         |                                                                                                                                  |  |                                        |  |
|                                                |  |                              |  |                             |                         |                                                                                                                                  |  |                                        |  |
| Geology: M. Steed 4/3/2014                                                                                                        |  |                              |  |                             | Engineering: hbadar rpm |                                                                                                                                  |  |                                        |  |



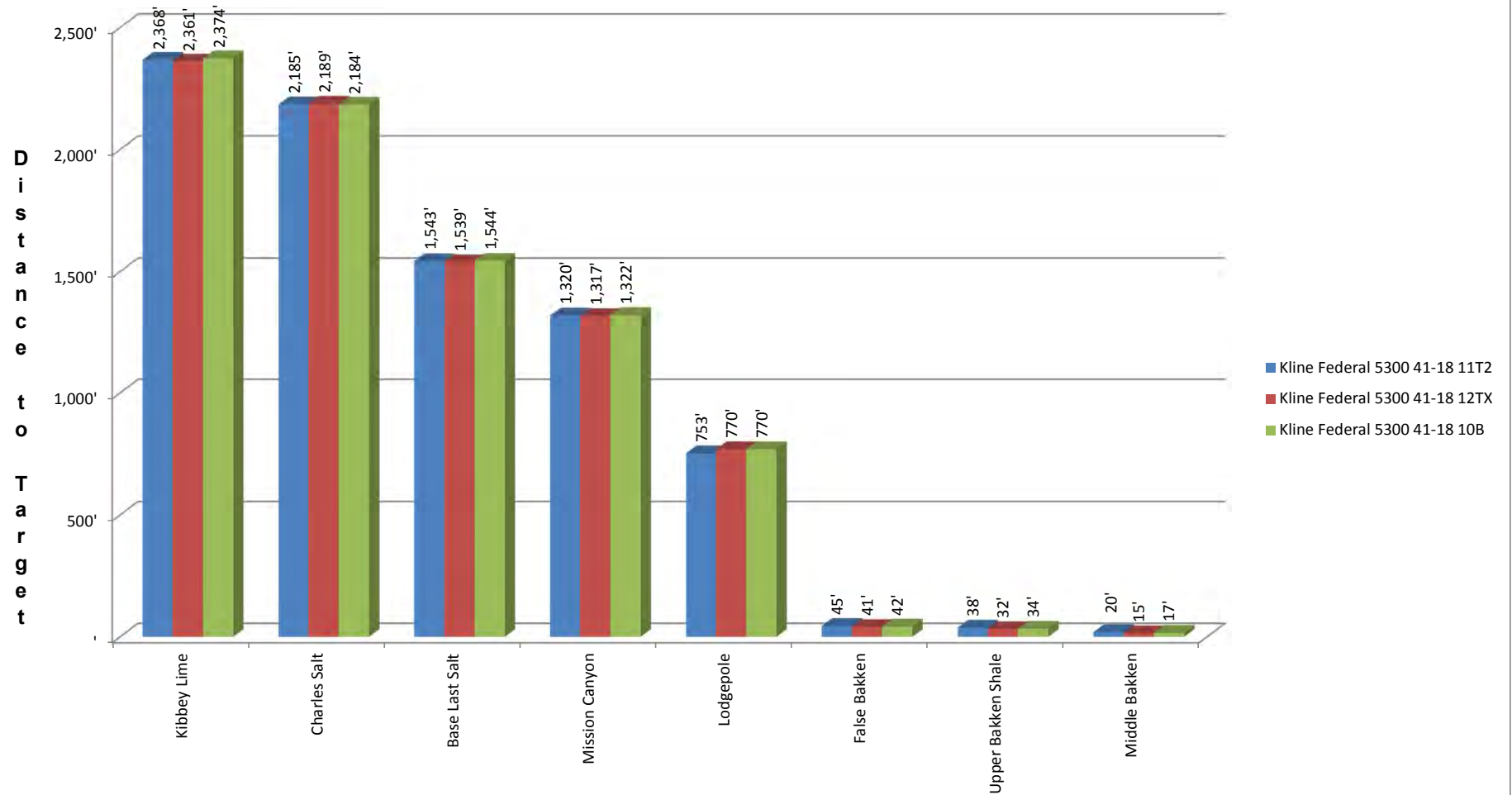
## **Formation Tops**

| <b>Operator:</b><br><b>Well Name:</b><br><b>Location:</b><br><br><b>Elevation:</b> | <b>Subject Well:</b>                                                                                                                                                                             |                            |                                    |                            |                           |                  |                                                        |
|------------------------------------------------------------------------------------|--------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|----------------------------|------------------------------------|----------------------------|---------------------------|------------------|--------------------------------------------------------|
|                                                                                    | <b>Oasis Petroleum North America</b><br><b><i>Kline Federal 5300 41-18 10B</i></b><br>533' FSL & 237' FWL<br>SW SW Sec. 18 T153N, R100W<br>GL: 2,057'   Sub: 25'                      KB: 2,082' |                            |                                    |                            |                           |                  |                                                        |
| <b>Formation/<br/>Zone</b>                                                         | <b>Prog.<br/>Top</b>                                                                                                                                                                             | <b>Prog.<br/>MSL Datum</b> | <b>Est. MD<br/>Top<br/>(Gamma)</b> | <b>TVD<br/>Top (Gamma)</b> | <b>Est. MLS<br/>Datum</b> | <b>Thickness</b> | <b>Dip To<br/><i>Kline Federal 5300 41-18 11T2</i></b> |
| Kibbey Lime                                                                        | 8,380                                                                                                                                                                                            | -6,298'                    | 8,376'                             | 8,376'                     | -6,294'                   | 190'             | 6' High                                                |
| Charles Salt                                                                       | 8,560                                                                                                                                                                                            | -6,478'                    | 8,566'                             | 8,566'                     | -6,484'                   | 640'             | 1' Low                                                 |
| Base Last Salt                                                                     | 9,207                                                                                                                                                                                            | -7,125'                    | 9,206'                             | 9,206'                     | -7,124'                   | 222'             | 1' High                                                |
| Mission Canyon                                                                     | 9,429                                                                                                                                                                                            | -7,347'                    | 9,428'                             | 9,428'                     | -7,346'                   | 552'             | 2' High                                                |
| Lodgepole                                                                          | 9,975                                                                                                                                                                                            | -7,893'                    | 9,980'                             | 9,980'                     | -7,898'                   | 728'             | 3' Low                                                 |
| False Bakken                                                                       | 10,707                                                                                                                                                                                           | -8,625'                    | 10,864'                            | 10,708'                    | -8,626'                   | 8'               | 4' High                                                |
| Upper Bakken Shale                                                                 | 10,717                                                                                                                                                                                           | -8,635'                    | 10,890'                            | 10,716'                    | -8,634'                   | 17'              | 3' High                                                |
| Middle Bakken                                                                      | 10,734                                                                                                                                                                                           | -8,652'                    | 10,954'                            | 10,733'                    | -8,651'                   | -                | 2' High                                                |

## **Control Well**

|                            |                                                                             |                  |                  |                       |                                                                               |                  |                  |                       |
|----------------------------|-----------------------------------------------------------------------------|------------------|------------------|-----------------------|-------------------------------------------------------------------------------|------------------|------------------|-----------------------|
| <b>Operator:</b>           | <b>Oasis Petroleum</b>                                                      |                  |                  |                       | <b>Oasis Petroleum</b>                                                        |                  |                  |                       |
| <b>Well Name:</b>          | <b><i>Kline Federal 5300 41-18 11T2</i></b>                                 |                  |                  |                       | <b><i>Kline Federal 5300 41-18 12TX</i></b>                                   |                  |                  |                       |
| <b>Location:</b>           | SW SW Sec. 18 T153N, R100W<br>McKenzie Co., ND<br>Approximately 25 ft South |                  |                  |                       | SW SW Sec. 18 T153N, R100W<br>McKenzie Co., ND<br>Approximately 50 feet South |                  |                  |                       |
| <b>Elevation:</b>          | KB: 2,082'                                                                  |                  |                  |                       | KB: 2,082'                                                                    |                  |                  |                       |
| <b>Formation/<br/>Zone</b> | <b>Est. TVD</b>                                                             | <b>MSL Datum</b> | <b>Thickness</b> | <b>Dist to target</b> | <b>Est. TVD</b>                                                               | <b>MSL Datum</b> | <b>Thickness</b> | <b>Dist to target</b> |
| Kibbey Lime                | 8,382'                                                                      | -6,300'          | 183'             | 2,368'                | 8,384                                                                         | -6,302'          | 172'             | 2,361'                |
| Charles Salt               | 8,565'                                                                      | -6,483'          | 642'             | 2,185'                | 8,556                                                                         | -6,474'          | 650'             | 2,189'                |
| Base Last Salt             | 9,207'                                                                      | -7,125'          | 223'             | 1,543'                | 9,206                                                                         | -7,124'          | 222'             | 1,539'                |
| Mission Canyon             | 9,430'                                                                      | -7,348'          | 567'             | 1,320'                | 9,428                                                                         | -7,346'          | 547'             | 1,317'                |
| Lodgepole                  | 9,997'                                                                      | -7,915'          | 708'             | 753'                  | 9,975                                                                         | -7,893'          | 729'             | 770'                  |
| False Bakken               | 10,705'                                                                     | -8,623'          | 7'               | 45'                   | 10,704                                                                        | -8,622'          | 9'               | 41'                   |
| Upper Bakken Shale         | 10,712'                                                                     | -8,630'          | 18'              | 38'                   | 10,713                                                                        | -8,631'          | 17'              | 32'                   |
| Middle Bakken              | 10,730'                                                                     | -8,648'          | 20'              | 20'                   | 10,730                                                                        | -8,648'          | 27'              | 15'                   |
| Target                     | 10,750'                                                                     | -8,668'          |                  | '                     | 10,745'                                                                       | -8,663'          |                  | '                     |

## Offset Well Target Isopachs



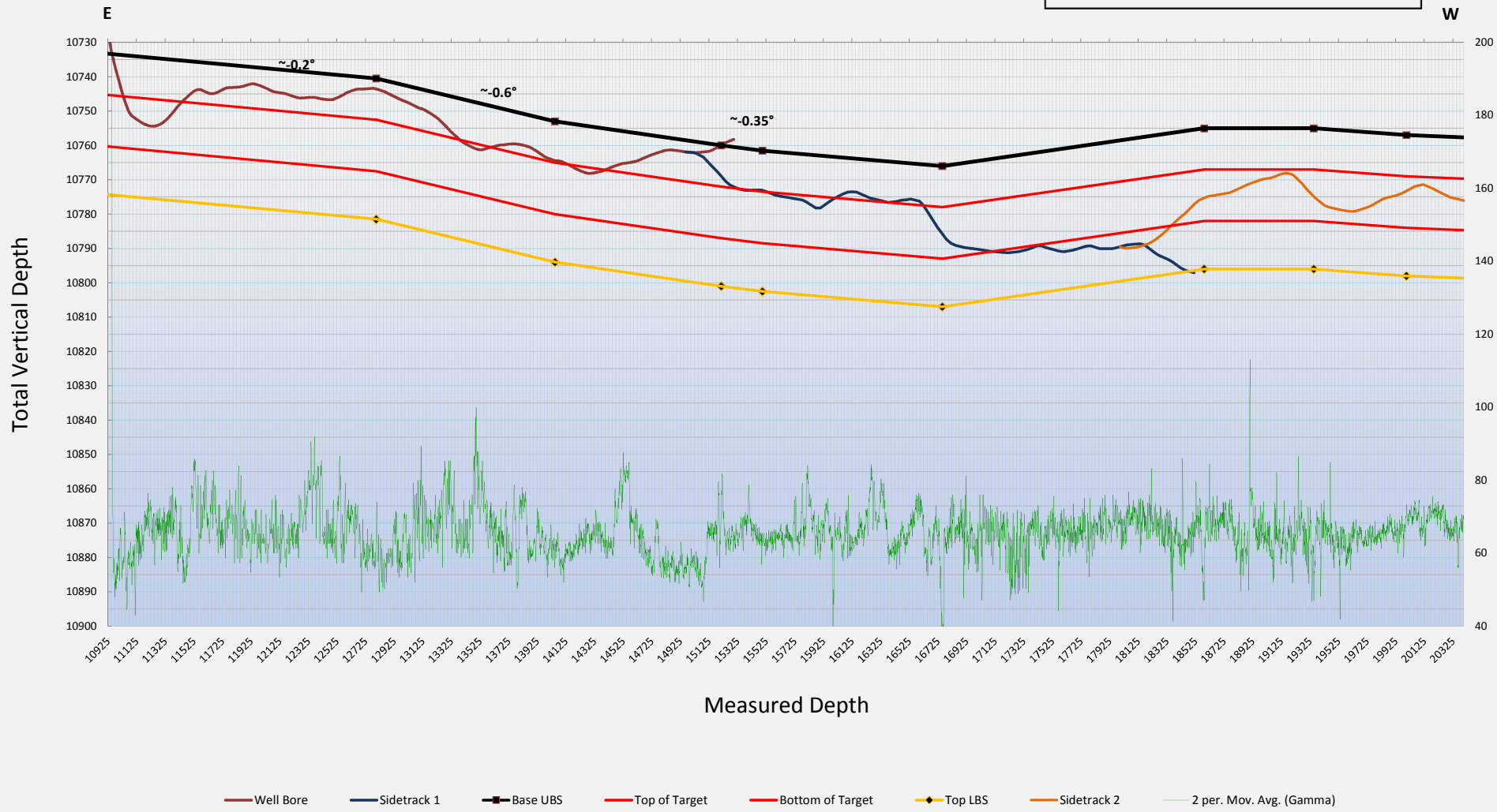




Surface: 500' FSL & 237' FEL  
SW SW Sec. 18 T153N, R100W  
McKenzie County, ND

Kline Federal 5300 41-18 10B  
Gamma Cross Section

Bottom Hole Location:  
254.45' S & 9968.39' E  
of surface location or approx.  
754.45' FSL & 283.76' FEL  
SE SE Sec. 17 T153N, R100W





The diagram illustrates the flow of water from a reservoir through a series of gates and a culvert into a river. The flow is indicated by a red arrow pointing from left to right.

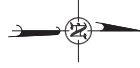
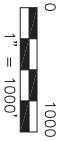
**McKenzie County, ND**

SE SE Sec. 17 T153N, R100W





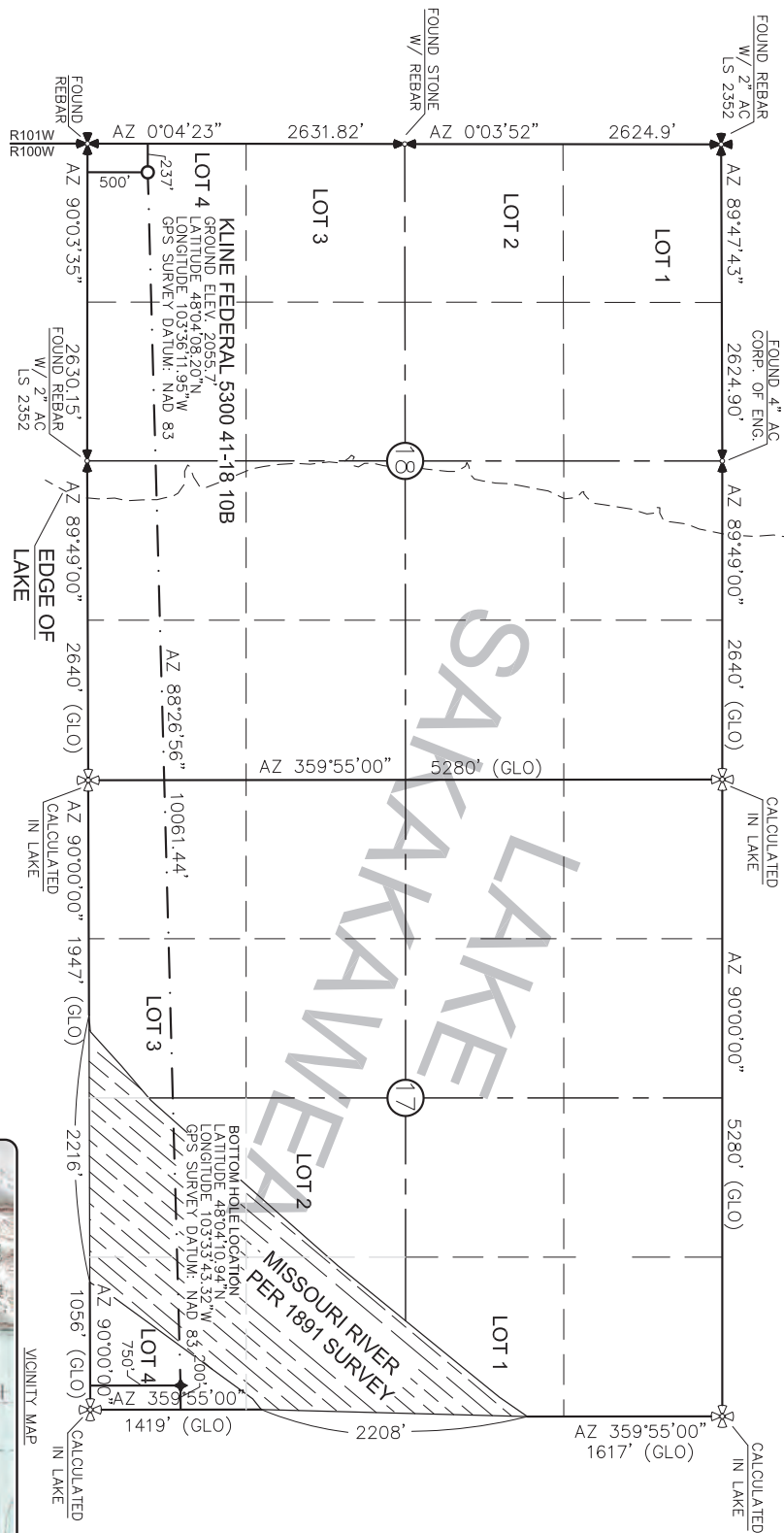
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THIS SURVEY AND PLAT IS BEING PROVIDED AT THE REQUEST OF ERIC BAYES OF OASIS PETROLEUM, I CERTIFY THAT THIS PLAT CORRECTLY REPRESENTS WORK PERFORMED BY ME OR UNDER MY SUPERVISION AND IS TRUE AND CORRECT TO THE BEST OF MY KNOWLEDGE AND BELIEF.

ROBERT L. PROCVIE 2884LS



**WELL LOCATION PLAT**  
OASIS PETROLEUM NORTH AMERICA, LLC  
1001 FANNIN, SUITE 1500, HOUSTON, TX 77002  
"KLINE FEDERAL 5300 41-18 10B"  
500 FEET FROM SOUTH LINE AND 237 FEET FROM WEST LINE  
SECTION 18, T153N, R100W, 5th P.M., MCKENZIE COUNTY, NORTH DAKOTA

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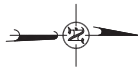
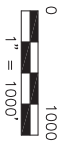


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Other offices in Minnesota, North Dakota and South Dakota

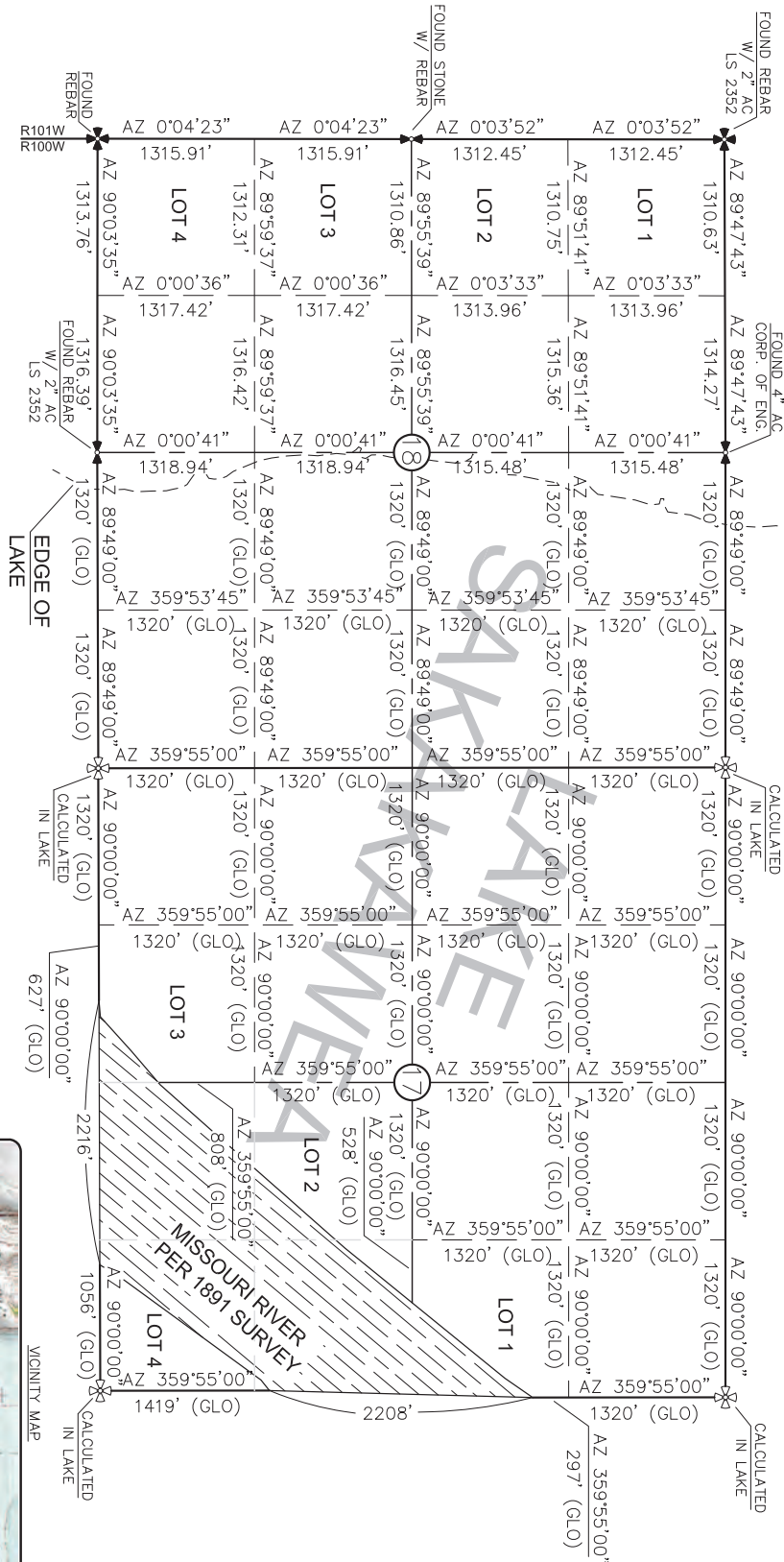
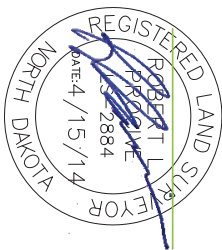
OASIS PETROLEUM NORTH AMERICA, LLC  
WELL LOCATION PLAT  
SECTION 18, T153N, R100W  
MCKENZIE COUNTY, NORTH DAKOTA  
Drawn By: B.H.H. Project No.: S14-09-081.04  
Checked By: B.L.P. Date: APRIL 2014

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 WAS 1891. THE CORNERS FOUND ARE AS  
 INDICATED AND ALL OTHERS ARE COMPUTED FROM  
 THOSE CORNERS FOUND AND BASED ON G.L.O.  
 DATA. THE MAPPING ANGLE FOR THIS AREA IS  
 APPROXIMATELY 0°03'.



**SECTION BREAKDOWN**  
 OASIS PETROLEUM NORTH AMERICA, LLC  
 1001 FANNIN, SUITE 1500, HOUSTON, TX 77002  
 KLINE FEDERAL 5300 41-18 108"  
 500 FEET FROM SOUTH LINE AND 237 FEET FROM WEST LINE  
 SECTIONS 17 & 18, T153N, R100W, 5th P.M., MCKENZIE COUNTY, NORTH DAKOTA

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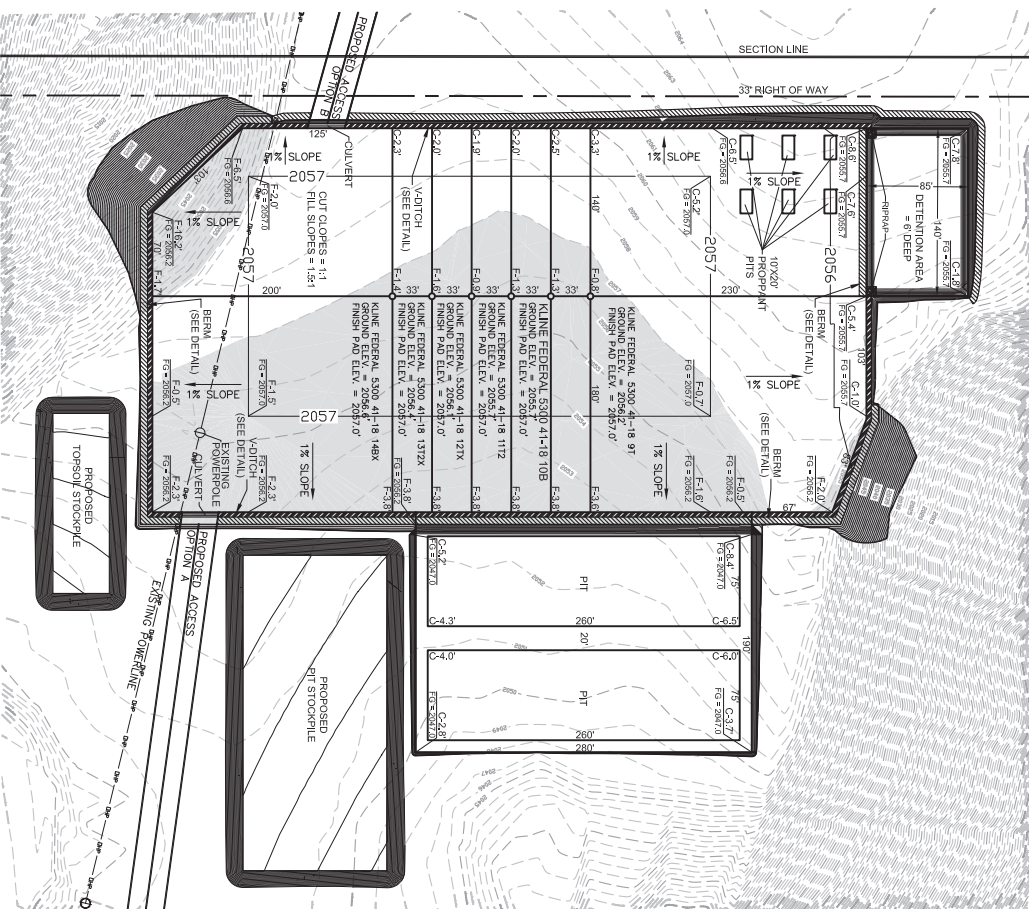
OASIS PETROLEUM NORTH AMERICA, LLC  
 SECTION BREAKDOWN  
 SECTIONS 17 & 18, T153N, R100W  
 MCKENZIE COUNTY, NORTH DAKOTA

Drawn By: B.H.H. Project No.: S14-09-001.04  
 Checked By: R.L.P. Date: APRIL 2014

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 KLINE FEDERAL 5300 41-18 108"  
 500 FEET FROM SOUTH LINE AND 237 FEET FROM WEST LINE  
 SECTIONS 17 & 18, T153N, R100W, 5th P.M., MCKENZIE COUNTY, NORTH DAKOTA

PAD LAYOUT  
OASIS PETROLEUM NORTH AMERICA, LLC  
1001 FANNIN, SUITE 1500, HOUSTON, TX 77002  
KLINE FEDERAL 5300 41-18 108"  
SECTION 18, T155N, R100W, S01 P.M., MCKENZIE COUNTY, NORTH DAKOTA  
500 FEET FROM SOUTH LINE AND 297 FEET FROM WEST LINE

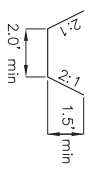


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NOTE: Pad dimensions shown are to outside area, the location and berm dimensions are to the outside of the pad dimensions.



V-DITCH DETAIL



Proposed Contours  
Original Contours



NOTE: All utilities shown are preliminary only, a complete utility investigation is recommended before construction.



3/8

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P: (281) 416-1111  
F: (281) 416-1112  
WWW.INTERSTATEENGINEERING.COM

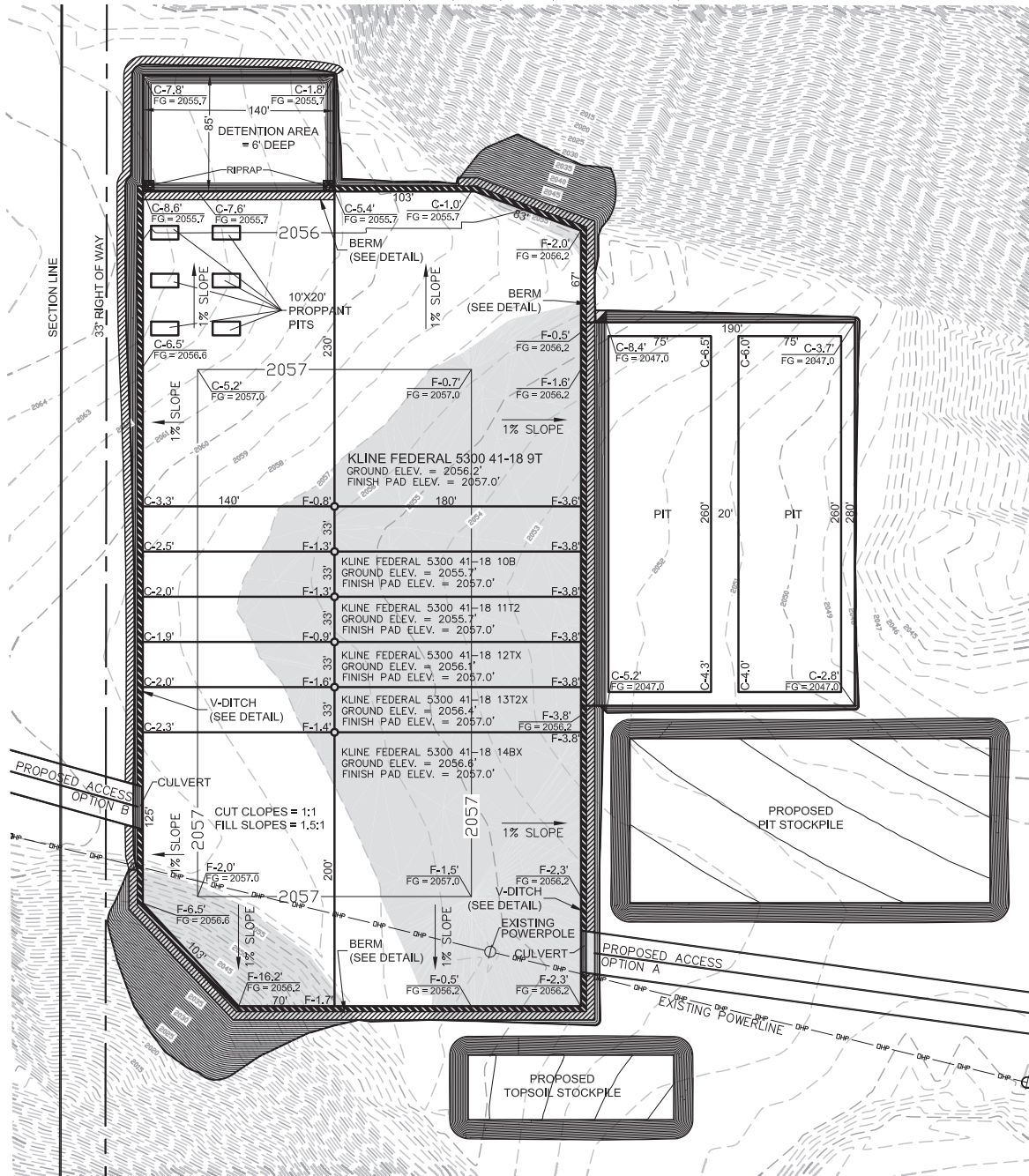
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F: (409) 533-5418  
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DATE: 4/15/14  
BY: R.



**PAD LAYOUT**  
 OASIS PETROLEUM NORTH AMERICA, LLC  
 1001 FANNIN, SUITE 1500, HOUSTON, TX 77002  
 KLINE FEDERAL 5300 41-18 9T  
 533 FEET FROM SOUTH LINE AND 237 FEET FROM WEST LINE  
 SECTION 18, T153N, R100W, 5th P.M., MCKENZIE COUNTY, NORTH DAKOTA

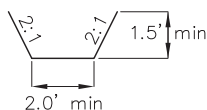


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NOTE: Pad dimensions shown are to usable area, the v-ditch and berm areas shall be built to the outside of the pad dimensions.



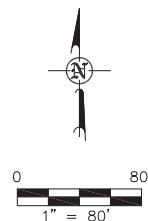
**V-DITCH DETAIL**



- Proposed Contours
- Original Contours
- BERM
- DITCH

NOTE: All utilities shown are preliminary only, a complete utilities location is recommended before construction.

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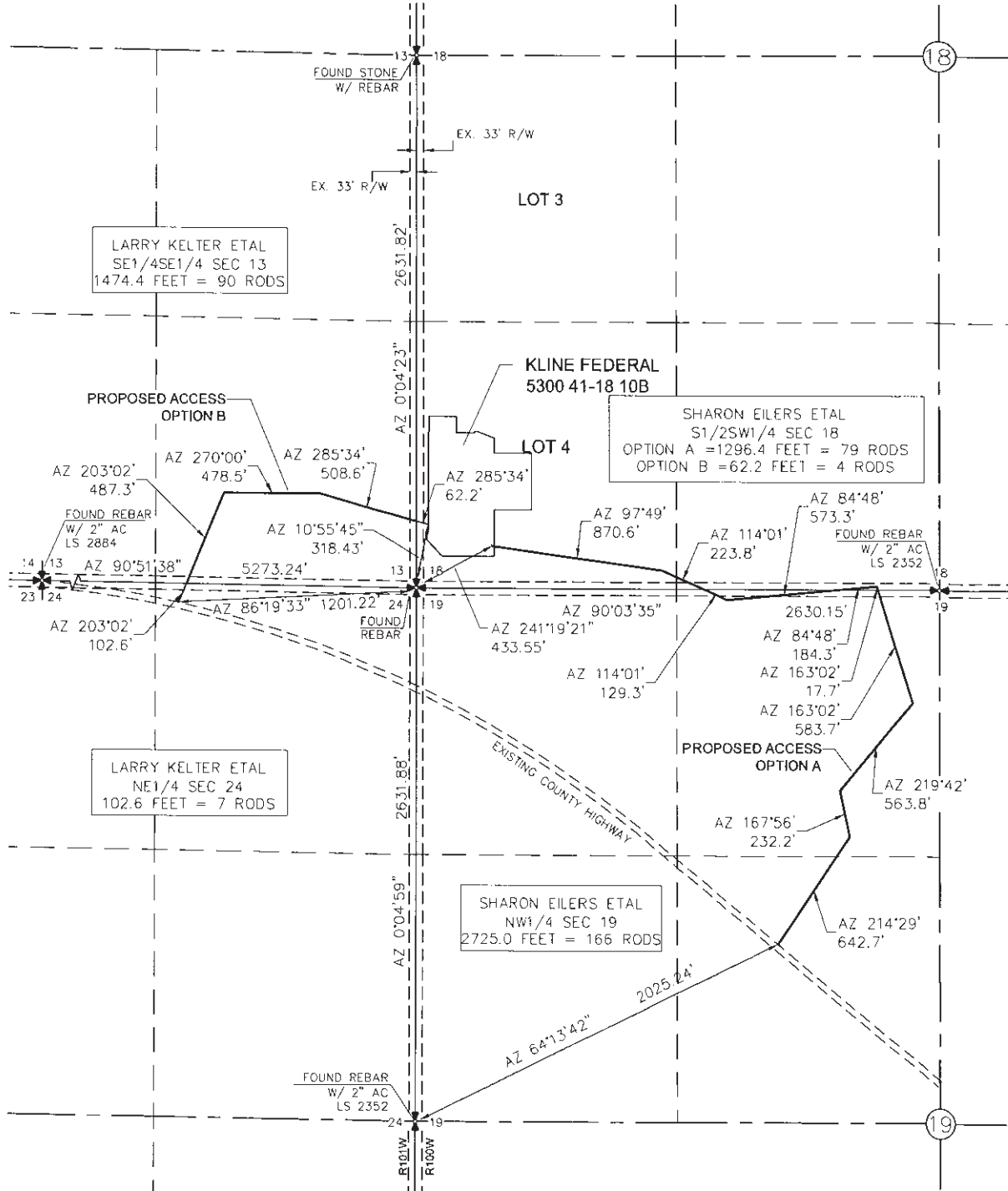
OASIS PETROLEUM NORTH AMERICA, LLC  
 PAD LAYOUT  
 SECTION 18, T153N, R100W  
 MCKENZIE COUNTY, NORTH DAKOTA

Drawn By: B.H.H. Project No.: S14-09-081.05  
 Checked By: D.D.K. Date: APRIL 2014

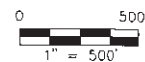
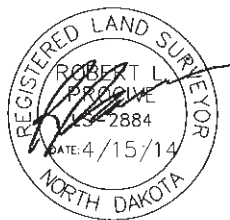
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**ACCESS APPROACH**  
 OASIS PETROLEUM NORTH AMERICA, LLC  
 1001 FANNIN, SUITE 1500, HOUSTON, TX 77002  
 "KLINE FEDERAL 5300 41-18 10B"  
 500 FEET FROM SOUTH LINE AND 237 FEET FROM WEST LINE  
 SECTION 18, T153N, R100W, 5th P.M., MCKENZIE COUNTY, NORTH DAKOTA



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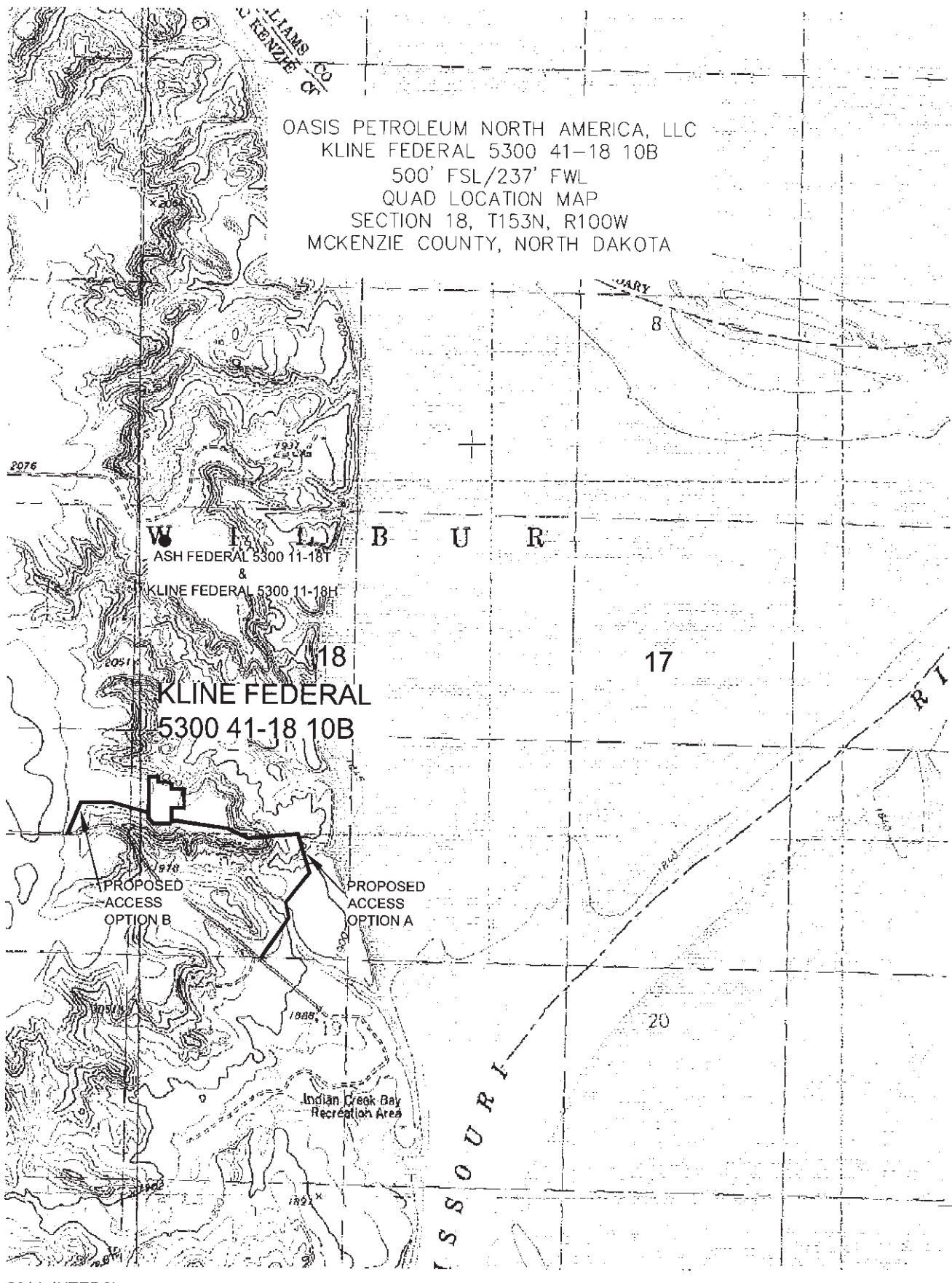
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Drawn, Plotted & Mailed: North Dakota and South Dakota

OASIS PETROLEUM NORTH AMERICA, LLC  
 ACCESS APPROACH  
 SECTION 18, T153N, R100W  
 MCKENZIE COUNTY, NORTH DAKOTA  
 Drawn By: B.H.L.  
 Checked By: B.H.L.  
 Project No.: 515-09-08124  
 Date: APRIL 2014

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OASIS PETROLEUM NORTH AMERICA, LLC  
QUAD LOCATION MAP  
SECTION 18, T153N, R100W  
MCKENZIE COUNTY, NORTH DAKOTA

Drawn By: B.J.H. Project No.: S14-09-081.04  
Checked By: R.L.P. Date: APRIL 2014

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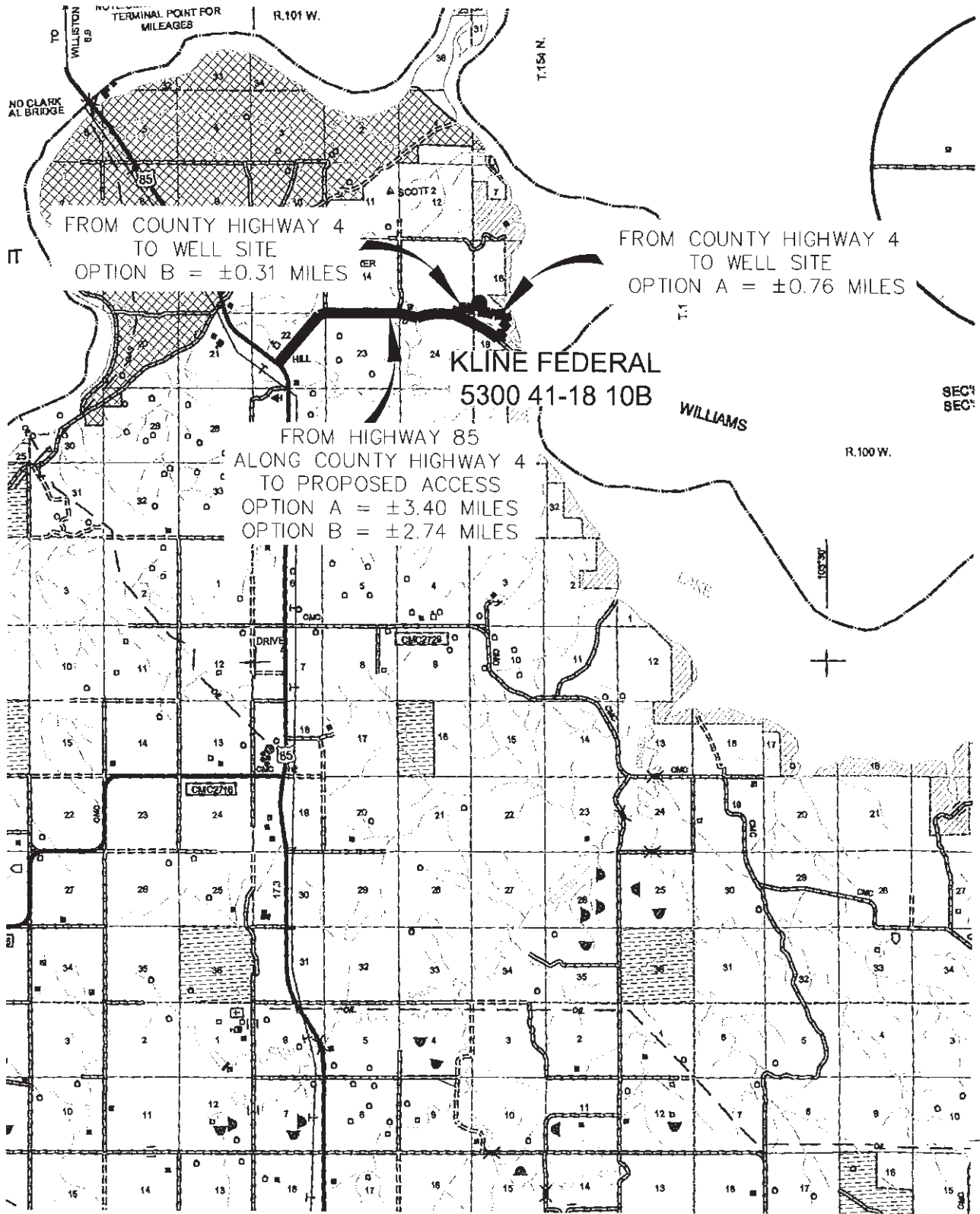


# COUNTY ROAD MAP

OASIS PETROLEUM NORTH AMERICA, LLC  
1001 FANNIN, SUITE 1500, HOUSTON, TX 77002

"KLINE FEDERAL 5300 41-18 10B"

500 FEET FROM SOUTH LINE AND 237 FEET FROM WEST LINE  
SECTION 18, T153N, R100W, 5th P.M., MCKENZIE COUNTY, NORTH DAKOTA



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SCALE: 1" = 2 MILE

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OASIS PETROLEUM NORTH AMERICA, LLC  
COUNTY ROAD MAP  
SECTION 18, T153N, R100W  
MCKENZIE COUNTY, NORTH DAKOTA

Drawn By: B.H.H. Project No.: S14-09-081.04  
Checked By: R.L.P. Date: APRIL 2014

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|                          |                              |
|--------------------------|------------------------------|
| <b>Operator:</b>         | <b>Oasis Petroleum</b>       |
| Well :                   | Kline Federal 5300 41-18 10B |
| MWD Providers            | GyroData                     |
| Directional Supervision: | RPM                          |

|           |     |      |       |           |          |        |    |
|-----------|-----|------|-------|-----------|----------|--------|----|
| Section:  | 18  | QQ:  | SW SW | County:   | McKenzie | State: | ND |
| Township: | 153 | N/S: | N     | Footages: | 500      | FN/SL: | S  |
| Range:    | 100 | E/W: | W     |           | 237      | FE/WL: | W  |

|                         |           |
|-------------------------|-----------|
| Vertical Section Plane: | <b>90</b> |
|-------------------------|-----------|

| #          | MD              | Inc.        | Azim.        | T.V.D.          | Ver. Sect.   | Coordinates  |              | DLS         |
|------------|-----------------|-------------|--------------|-----------------|--------------|--------------|--------------|-------------|
|            |                 |             |              |                 |              | +N / -S      | +E / -W      |             |
| <b>Tie</b> | <b>10187.00</b> | <b>0.66</b> | <b>66.22</b> | <b>10186.59</b> | <b>43.31</b> | <b>15.00</b> | <b>43.31</b> | <b>0.22</b> |
| 1          | 10217.00        | 0.44        | 40.99        | 10216.59        | 43.54        | 15.16        | 43.54        | 1.07        |
| 2          | 10248.00        | 2.68        | 70.70        | 10247.58        | 44.31        | 15.49        | 44.31        | 7.45        |
| 3          | 10279.00        | 7.78        | 74.22        | 10278.44        | 47.01        | 16.30        | 47.01        | 16.48       |
| 4          | 10311.00        | 13.05       | 74.83        | 10309.90        | 52.59        | 17.83        | 52.59        | 16.47       |
| 5          | 10343.00        | 18.46       | 74.57        | 10340.68        | 60.96        | 20.13        | 60.96        | 16.91       |
| 6          | 10375.00        | 20.40       | 71.84        | 10370.86        | 71.15        | 23.21        | 71.15        | 6.69        |
| 7          | 10406.00        | 22.11       | 70.88        | 10399.75        | 81.79        | 26.81        | 81.79        | 5.63        |
| 8          | 10438.00        | 24.53       | 70.79        | 10429.14        | 93.76        | 30.97        | 93.76        | 7.56        |
| 9          | 10470.00        | 25.93       | 73.60        | 10458.08        | 106.74       | 35.13        | 106.74       | 5.76        |
| 10         | 10502.00        | 30.11       | 78.00        | 10486.33        | 121.31       | 38.78        | 121.31       | 14.56       |
| 11         | 10533.00        | 33.54       | 79.67        | 10512.66        | 137.35       | 41.93        | 137.35       | 11.42       |
| 12         | 10565.00        | 37.14       | 79.40        | 10538.76        | 155.55       | 45.29        | 155.55       | 11.26       |
| 13         | 10597.00        | 41.14       | 75.09        | 10563.58        | 175.23       | 49.78        | 175.23       | 15.11       |
| 14         | 10628.00        | 45.36       | 73.69        | 10586.16        | 195.68       | 55.50        | 195.68       | 13.96       |
| 15         | 10660.00        | 49.85       | 71.84        | 10607.73        | 218.24       | 62.52        | 218.24       | 14.67       |
| 16         | 10691.00        | 53.49       | 73.69        | 10626.95        | 241.46       | 69.71        | 241.46       | 12.64       |
| 17         | 10723.00        | 57.10       | 76.06        | 10645.17        | 266.85       | 76.56        | 266.85       | 12.82       |
| 18         | 10754.00        | 60.40       | 77.82        | 10661.25        | 292.67       | 82.54        | 292.67       | 11.70       |
| 19         | 10785.00        | 64.04       | 77.73        | 10675.70        | 319.47       | 88.35        | 319.47       | 11.74       |
| 20         | 10816.00        | 67.91       | 77.91        | 10688.32        | 347.14       | 94.32        | 347.14       | 12.50       |
| 21         | 10848.00        | 68.31       | 78.52        | 10700.25        | 376.20       | 100.39       | 376.20       | 2.17        |
| 22         | 10879.00        | 68.35       | 78.08        | 10711.69        | 404.41       | 106.23       | 404.41       | 1.33        |
| 23         | 10910.00        | 71.74       | 76.76        | 10722.27        | 432.85       | 112.58       | 432.85       | 11.64       |
| 24         | 10942.00        | 77.23       | 76.41        | 10730.83        | 462.83       | 119.73       | 462.83       | 17.19       |
| 25         | 10973.00        | 81.05       | 75.62        | 10736.67        | 492.37       | 127.09       | 492.37       | 12.57       |
| 26         | 11005.00        | 81.49       | 73.60        | 10741.53        | 522.86       | 135.48       | 522.86       | 6.39        |
| 27         | 11036.00        | 81.54       | 73.69        | 10746.10        | 552.28       | 144.11       | 552.28       | 0.33        |
| 28         | 11068.00        | 85.01       | 73.07        | 10749.85        | 582.73       | 153.20       | 582.73       | 11.01       |
| 29         | 11078.00        | 86.42       | 73.25        | 10750.59        | 592.27       | 156.09       | 592.27       | 14.21       |
| 30         | 11098.00        | 88.09       | 71.84        | 10751.55        | 611.33       | 162.08       | 611.33       | 10.92       |
| 31         | 11166.00        | 88.70       | 71.38        | 10753.46        | 675.83       | 183.53       | 675.83       | 1.12        |

|    |          |       |       |          |         |        |         |       |
|----|----------|-------|-------|----------|---------|--------|---------|-------|
| 32 | 11197.00 | 89.05 | 71.21 | 10754.06 | 705.18  | 193.47 | 705.18  | 1.26  |
| 33 | 11229.00 | 89.71 | 70.68 | 10754.41 | 735.43  | 203.91 | 735.43  | 2.65  |
| 34 | 11260.00 | 90.59 | 71.91 | 10754.33 | 764.79  | 213.86 | 764.79  | 4.88  |
| 35 | 11291.00 | 91.30 | 73.84 | 10753.82 | 794.41  | 222.98 | 794.41  | 6.63  |
| 36 | 11322.00 | 91.96 | 75.07 | 10752.94 | 824.26  | 231.29 | 824.26  | 4.50  |
| 37 | 11353.00 | 92.70 | 76.39 | 10751.68 | 854.28  | 238.92 | 854.28  | 4.88  |
| 38 | 11384.00 | 92.88 | 78.50 | 10750.17 | 884.50  | 245.65 | 884.50  | 6.82  |
| 39 | 11415.00 | 92.88 | 79.38 | 10748.61 | 914.89  | 251.59 | 914.89  | 2.84  |
| 40 | 11446.00 | 92.35 | 80.88 | 10747.20 | 945.40  | 256.90 | 945.40  | 5.13  |
| 41 | 11477.00 | 92.09 | 82.72 | 10745.99 | 976.06  | 261.32 | 976.06  | 5.99  |
| 42 | 11508.00 | 92.18 | 82.72 | 10744.84 | 1006.78 | 265.24 | 1006.78 | 0.29  |
| 43 | 11539.00 | 91.12 | 86.06 | 10743.95 | 1037.62 | 268.27 | 1037.62 | 11.30 |
| 44 | 11570.00 | 89.63 | 86.32 | 10743.74 | 1068.55 | 270.33 | 1068.55 | 4.88  |
| 45 | 11602.00 | 88.53 | 88.76 | 10744.26 | 1100.51 | 271.70 | 1100.51 | 8.36  |
| 46 | 11633.00 | 89.36 | 87.03 | 10744.83 | 1131.48 | 272.84 | 1131.48 | 6.19  |
| 47 | 11664.00 | 90.46 | 88.35 | 10744.88 | 1162.46 | 274.09 | 1162.46 | 5.54  |
| 48 | 11695.00 | 91.08 | 88.70 | 10744.46 | 1193.44 | 274.89 | 1193.44 | 2.30  |
| 49 | 11726.00 | 91.56 | 88.79 | 10743.75 | 1224.43 | 275.57 | 1224.43 | 1.58  |
| 50 | 11758.00 | 90.29 | 90.02 | 10743.23 | 1256.42 | 275.90 | 1256.42 | 5.52  |
| 51 | 11820.00 | 90.02 | 90.81 | 10743.06 | 1318.42 | 275.45 | 1318.42 | 1.35  |
| 52 | 11851.00 | 90.42 | 90.28 | 10742.94 | 1349.42 | 275.16 | 1349.42 | 2.14  |
| 53 | 11882.00 | 90.73 | 90.10 | 10742.63 | 1380.42 | 275.05 | 1380.42 | 1.16  |
| 54 | 11913.00 | 90.77 | 90.37 | 10742.23 | 1411.41 | 274.93 | 1411.41 | 0.88  |
| 55 | 11945.00 | 89.85 | 90.72 | 10742.05 | 1443.41 | 274.62 | 1443.41 | 3.08  |
| 56 | 12007.00 | 88.66 | 91.42 | 10742.86 | 1505.39 | 273.46 | 1505.39 | 2.23  |
| 57 | 12069.00 | 89.05 | 90.19 | 10744.10 | 1567.37 | 272.59 | 1567.37 | 2.08  |
| 58 | 12100.00 | 89.54 | 90.02 | 10744.48 | 1598.37 | 272.54 | 1598.37 | 1.67  |
| 59 | 12163.00 | 89.71 | 87.64 | 10744.89 | 1661.35 | 273.82 | 1661.35 | 3.79  |
| 60 | 12194.00 | 88.70 | 88.26 | 10745.32 | 1692.33 | 274.93 | 1692.33 | 3.82  |
| 61 | 12256.00 | 89.80 | 87.03 | 10746.13 | 1754.27 | 277.48 | 1754.27 | 2.66  |
| 62 | 12287.00 | 90.11 | 89.14 | 10746.16 | 1785.25 | 278.52 | 1785.25 | 6.88  |
| 63 | 12349.00 | 90.20 | 88.87 | 10745.99 | 1847.24 | 279.59 | 1847.24 | 0.46  |
| 64 | 12380.00 | 89.49 | 88.26 | 10746.07 | 1878.23 | 280.37 | 1878.23 | 3.02  |
| 65 | 12411.00 | 89.36 | 89.84 | 10746.39 | 1909.22 | 280.88 | 1909.22 | 5.11  |
| 66 | 12473.00 | 90.07 | 88.35 | 10746.69 | 1971.21 | 281.86 | 1971.21 | 2.66  |
| 67 | 12505.00 | 90.51 | 89.31 | 10746.53 | 2003.20 | 282.51 | 2003.20 | 3.30  |
| 68 | 12567.00 | 91.69 | 89.23 | 10745.34 | 2065.19 | 283.30 | 2065.19 | 1.91  |
| 69 | 12599.00 | 91.08 | 91.16 | 10744.57 | 2097.18 | 283.20 | 2097.18 | 6.32  |
| 70 | 12662.00 | 90.55 | 91.95 | 10743.67 | 2160.14 | 281.49 | 2160.14 | 1.51  |
| 71 | 12693.00 | 89.67 | 93.44 | 10743.61 | 2191.11 | 280.03 | 2191.11 | 5.58  |
| 72 | 12756.00 | 90.64 | 93.00 | 10743.44 | 2254.01 | 276.49 | 2254.01 | 1.69  |
| 73 | 12788.00 | 89.41 | 93.71 | 10743.43 | 2285.95 | 274.62 | 2285.95 | 4.44  |
| 74 | 12851.00 | 89.14 | 93.18 | 10744.23 | 2348.83 | 270.83 | 2348.83 | 0.94  |
| 75 | 12883.00 | 88.62 | 93.80 | 10744.85 | 2380.77 | 268.88 | 2380.77 | 2.53  |

|     |          |       |       |          |         |        |         |      |
|-----|----------|-------|-------|----------|---------|--------|---------|------|
| 76  | 12945.00 | 89.05 | 93.62 | 10746.11 | 2442.63 | 264.87 | 2442.63 | 0.75 |
| 77  | 13040.00 | 88.79 | 92.48 | 10747.90 | 2537.47 | 259.82 | 2537.47 | 1.23 |
| 78  | 13103.00 | 88.97 | 91.42 | 10749.13 | 2600.42 | 257.68 | 2600.42 | 1.71 |
| 79  | 13135.00 | 89.41 | 91.25 | 10749.59 | 2632.41 | 256.93 | 2632.41 | 1.47 |
| 80  | 13229.00 | 87.56 | 91.25 | 10752.07 | 2726.35 | 254.88 | 2726.35 | 1.97 |
| 81  | 13324.00 | 87.69 | 90.63 | 10756.01 | 2821.26 | 253.32 | 2821.26 | 0.67 |
| 82  | 13355.00 | 87.74 | 90.46 | 10757.24 | 2852.23 | 253.03 | 2852.23 | 0.57 |
| 83  | 13387.00 | 88.00 | 90.28 | 10758.43 | 2884.21 | 252.82 | 2884.21 | 0.99 |
| 84  | 13418.00 | 88.57 | 89.31 | 10759.36 | 2915.19 | 252.93 | 2915.19 | 3.63 |
| 85  | 13481.00 | 88.97 | 89.40 | 10760.71 | 2978.17 | 253.64 | 2978.17 | 0.65 |
| 86  | 13512.00 | 89.41 | 89.93 | 10761.15 | 3009.17 | 253.82 | 3009.17 | 2.22 |
| 87  | 13544.00 | 90.29 | 88.87 | 10761.24 | 3041.17 | 254.16 | 3041.17 | 4.31 |
| 88  | 13607.00 | 91.08 | 88.87 | 10760.48 | 3104.15 | 255.40 | 3104.15 | 1.25 |
| 89  | 13670.00 | 90.02 | 88.52 | 10759.88 | 3167.13 | 256.84 | 3167.13 | 1.77 |
| 90  | 13700.00 | 90.33 | 89.14 | 10759.79 | 3197.12 | 257.45 | 3197.12 | 2.31 |
| 91  | 13732.00 | 90.42 | 88.61 | 10759.58 | 3229.12 | 258.08 | 3229.12 | 1.68 |
| 92  | 13763.00 | 89.71 | 89.49 | 10759.54 | 3260.11 | 258.59 | 3260.11 | 3.65 |
| 93  | 13795.00 | 89.85 | 90.46 | 10759.66 | 3292.11 | 258.60 | 3292.11 | 3.06 |
| 94  | 13858.00 | 89.05 | 91.86 | 10760.27 | 3355.09 | 257.33 | 3355.09 | 2.56 |
| 95  | 13890.00 | 88.75 | 91.25 | 10760.88 | 3387.08 | 256.46 | 3387.08 | 2.12 |
| 96  | 13953.00 | 88.22 | 89.49 | 10762.55 | 3450.05 | 256.05 | 3450.05 | 2.92 |
| 97  | 13984.00 | 88.53 | 88.87 | 10763.43 | 3481.03 | 256.50 | 3481.03 | 2.24 |
| 98  | 14047.00 | 89.76 | 91.77 | 10764.37 | 3544.02 | 256.15 | 3544.02 | 5.00 |
| 99  | 14079.00 | 89.63 | 89.93 | 10764.54 | 3576.01 | 255.67 | 3576.01 | 5.76 |
| 100 | 14110.00 | 88.92 | 88.79 | 10764.93 | 3607.01 | 256.02 | 3607.01 | 4.33 |
| 101 | 14173.00 | 88.66 | 87.38 | 10766.26 | 3669.96 | 258.12 | 3669.96 | 2.28 |
| 102 | 14205.00 | 88.79 | 89.75 | 10766.97 | 3701.94 | 258.92 | 3701.94 | 7.42 |
| 103 | 14268.00 | 89.32 | 88.87 | 10768.01 | 3764.92 | 259.68 | 3764.92 | 1.63 |
| 104 | 14299.00 | 90.24 | 89.84 | 10768.13 | 3795.92 | 260.03 | 3795.92 | 4.31 |
| 105 | 14362.00 | 90.59 | 89.93 | 10767.68 | 3858.92 | 260.16 | 3858.92 | 0.57 |
| 106 | 14393.00 | 90.90 | 92.30 | 10767.27 | 3889.91 | 259.55 | 3889.91 | 7.71 |
| 107 | 14457.00 | 91.08 | 90.72 | 10766.17 | 3953.87 | 257.87 | 3953.87 | 2.48 |
| 108 | 14520.00 | 90.51 | 90.90 | 10765.29 | 4016.86 | 256.98 | 4016.86 | 0.95 |
| 109 | 14551.00 | 90.11 | 89.75 | 10765.12 | 4047.86 | 256.80 | 4047.86 | 3.93 |
| 110 | 14614.00 | 90.77 | 89.75 | 10764.64 | 4110.86 | 257.08 | 4110.86 | 1.05 |
| 111 | 14646.00 | 91.12 | 89.93 | 10764.11 | 4142.85 | 257.17 | 4142.85 | 1.23 |
| 112 | 14709.00 | 90.90 | 89.75 | 10763.00 | 4205.84 | 257.34 | 4205.84 | 0.45 |
| 113 | 14740.00 | 90.90 | 88.52 | 10762.52 | 4236.83 | 257.81 | 4236.83 | 3.97 |
| 114 | 14803.00 | 90.66 | 89.40 | 10761.66 | 4299.82 | 258.95 | 4299.82 | 1.45 |
| 115 | 14835.00 | 90.33 | 87.91 | 10761.38 | 4331.81 | 259.70 | 4331.81 | 4.77 |
| 116 | 14866.00 | 89.85 | 87.99 | 10761.33 | 4362.79 | 260.81 | 4362.79 | 1.57 |
| 117 | 14929.00 | 89.58 | 87.12 | 10761.65 | 4425.73 | 263.50 | 4425.73 | 1.45 |
| 118 | 14961.00 | 89.54 | 89.75 | 10761.89 | 4457.71 | 264.37 | 4457.71 | 8.22 |
| 119 | 15024.00 | 90.07 | 90.98 | 10762.11 | 4520.71 | 263.97 | 4520.71 | 2.13 |

|      |          |       |       |          |         |        |         |      |
|------|----------|-------|-------|----------|---------|--------|---------|------|
| 120  | 15118.00 | 90.37 | 91.77 | 10761.75 | 4614.68 | 261.72 | 4614.68 | 0.90 |
| 121  | 15149.00 | 91.08 | 92.04 | 10761.35 | 4645.66 | 260.69 | 4645.66 | 2.45 |
| 122  | 15181.00 | 91.52 | 90.54 | 10760.63 | 4677.64 | 259.97 | 4677.64 | 4.88 |
| 123  | 15212.00 | 91.91 | 90.98 | 10759.70 | 4708.63 | 259.56 | 4708.63 | 1.90 |
| Proj | 15300.00 | 90.00 | 90.00 | 10758.23 | 4796.61 | 258.80 | 4796.61 | 2.44 |





|                                 |                                     |
|---------------------------------|-------------------------------------|
| <b>Operator:</b>                | <b>Oasis Petroleum</b>              |
| <b>Well :</b>                   | <b>Kline Federal 5300 41-18 10B</b> |
| <b>MWD Providers</b>            | <b>Ryan</b>                         |
| <b>Directional Supervision:</b> | <b>RPM</b>                          |

|           |     |      |       |           |          |        |    |
|-----------|-----|------|-------|-----------|----------|--------|----|
| Section:  | 18  | QQ:  | SW SW | County:   | McKenzie | State: | ND |
| Township: | 153 | N/S: | N     | Footages: | 500      | FN/SL: | S  |
| Range:    | 100 | E/W: | W     |           | 237      | FE/WL: | W  |

### Sidetrack #1 Surveys

Vertical Section Plane: **90**

| <i>Coordinates</i> |                 |              |              |                 |                |               |                |             |
|--------------------|-----------------|--------------|--------------|-----------------|----------------|---------------|----------------|-------------|
| #                  | MD              | Inc.         | Azm.         | T.V.D.          | Ver. Sect.     | +N / -S       | +E / -W        | DLS         |
| <b>Tie</b>         | <b>14961.00</b> | <b>89.54</b> | <b>89.75</b> | <b>10761.87</b> | <b>4457.76</b> | <b>264.58</b> | <b>4457.76</b> | <b>0.00</b> |
| 1                  | 15024.00        | 90.07        | 90.98        | 10762.08        | 4520.76        | 264.18        | 4520.76        | 2.13        |
| 2                  | 15087.00        | 87.56        | 92.39        | 10763.39        | 4583.71        | 262.33        | 4583.71        | 4.57        |
| 3                  | 15118.00        | 87.30        | 93.09        | 10764.78        | 4614.64        | 260.85        | 4614.64        | 2.41        |
| 4                  | 15149.00        | 87.69        | 92.92        | 10766.13        | 4645.57        | 259.22        | 4645.57        | 1.37        |
| 5                  | 15181.00        | 87.47        | 91.51        | 10767.48        | 4677.52        | 257.99        | 4677.52        | 4.46        |
| 6                  | 15212.00        | 87.16        | 91.77        | 10768.94        | 4708.47        | 257.10        | 4708.47        | 1.30        |
| 7                  | 15244.00        | 87.25        | 92.65        | 10770.50        | 4740.41        | 255.87        | 4740.41        | 2.76        |
| 8                  | 15275.00        | 88.75        | 91.77        | 10771.58        | 4771.37        | 254.67        | 4771.37        | 5.61        |
| 9                  | 15306.00        | 89.05        | 91.25        | 10772.17        | 4802.35        | 253.86        | 4802.35        | 1.94        |
| 10                 | 15338.00        | 89.10        | 92.04        | 10772.69        | 4834.33        | 252.94        | 4834.33        | 2.47        |
| 11                 | 15369.00        | 89.67        | 91.42        | 10773.02        | 4865.31        | 252.00        | 4865.31        | 2.72        |
| 12                 | 15401.00        | 90.02        | 91.25        | 10773.11        | 4897.31        | 251.26        | 4897.31        | 1.22        |
| 13                 | 15432.00        | 90.29        | 92.30        | 10773.03        | 4928.29        | 250.30        | 4928.29        | 3.50        |
| 14                 | 15495.00        | 89.71        | 89.75        | 10773.03        | 4991.27        | 249.17        | 4991.27        | 4.15        |
| 15                 | 15527.00        | 88.79        | 89.84        | 10773.44        | 5023.27        | 249.29        | 5023.27        | 2.89        |
| 16                 | 15590.00        | 89.32        | 89.49        | 10774.48        | 5086.26        | 249.65        | 5086.26        | 1.01        |
| 17                 | 15621.00        | 89.67        | 88.87        | 10774.76        | 5117.26        | 250.10        | 5117.26        | 2.30        |
| 18                 | 15684.00        | 89.54        | 88.61        | 10775.19        | 5180.24        | 251.48        | 5180.24        | 0.46        |
| 19                 | 15716.00        | 89.54        | 87.73        | 10775.45        | 5212.22        | 252.50        | 5212.22        | 2.75        |
| 20                 | 15779.00        | 89.67        | 88.26        | 10775.88        | 5275.18        | 254.71        | 5275.18        | 0.87        |
| 21                 | 15810.00        | 87.87        | 89.75        | 10776.55        | 5306.17        | 255.25        | 5306.17        | 7.54        |
| 22                 | 15873.00        | 89.05        | 90.54        | 10778.24        | 5369.14        | 255.09        | 5369.14        | 2.25        |
| 23                 | 15905.00        | 91.03        | 90.63        | 10778.22        | 5401.14        | 254.76        | 5401.14        | 6.19        |
| 24                 | 15968.00        | 92.35        | 90.46        | 10776.36        | 5464.11        | 254.16        | 5464.11        | 2.11        |
| 25                 | 16000.00        | 90.81        | 89.93        | 10775.48        | 5496.10        | 254.05        | 5496.10        | 5.09        |
| 26                 | 16063.00        | 91.65        | 89.31        | 10774.13        | 5559.08        | 254.47        | 5559.08        | 1.66        |
| 27                 | 16094.00        | 90.20        | 87.12        | 10773.62        | 5590.06        | 255.44        | 5590.06        | 8.47        |
| 28                 | 16126.00        | 90.29        | 88.17        | 10773.49        | 5622.03        | 256.75        | 5622.03        | 3.29        |
| 29                 | 16157.00        | 89.32        | 86.41        | 10773.59        | 5652.99        | 258.22        | 5652.99        | 6.48        |
| 30                 | 16188.00        | 88.62        | 88.17        | 10774.15        | 5683.95        | 259.68        | 5683.95        | 6.11        |
| 31                 | 16251.00        | 89.27        | 88.61        | 10775.31        | 5746.92        | 261.45        | 5746.92        | 1.25        |
| 32                 | 16346.00        | 89.58        | 88.08        | 10776.26        | 5841.87        | 264.19        | 5841.87        | 0.65        |
| 33                 | 16377.00        | 89.05        | 88.70        | 10776.63        | 5872.86        | 265.07        | 5872.86        | 2.63        |

|    |          |       |       |          |         |        |         |      |
|----|----------|-------|-------|----------|---------|--------|---------|------|
| 34 | 16440.00 | 91.52 | 88.87 | 10776.32 | 5935.84 | 266.40 | 5935.84 | 3.93 |
| 35 | 16472.00 | 89.85 | 90.02 | 10775.94 | 5967.83 | 266.71 | 5967.83 | 6.34 |
| 36 | 16503.00 | 90.46 | 91.25 | 10775.86 | 5998.83 | 266.37 | 5998.83 | 4.43 |
| 37 | 16535.00 | 90.42 | 90.63 | 10775.61 | 6030.82 | 265.84 | 6030.82 | 1.94 |
| 38 | 16598.00 | 88.44 | 91.60 | 10776.24 | 6093.80 | 264.62 | 6093.80 | 3.50 |
| 39 | 16629.00 | 86.15 | 91.33 | 10777.70 | 6124.76 | 263.83 | 6124.76 | 7.44 |
| 40 | 16724.00 | 86.37 | 91.33 | 10783.90 | 6219.53 | 261.63 | 6219.53 | 0.23 |
| 41 | 16787.00 | 87.21 | 92.92 | 10787.42 | 6282.38 | 259.29 | 6282.38 | 2.85 |
| 42 | 16818.00 | 88.75 | 92.04 | 10788.52 | 6313.33 | 257.95 | 6313.33 | 5.72 |
| 43 | 16851.00 | 89.14 | 91.33 | 10789.12 | 6346.31 | 256.98 | 6346.31 | 2.45 |
| 44 | 16914.00 | 89.71 | 91.25 | 10789.76 | 6409.30 | 255.56 | 6409.30 | 0.91 |
| 45 | 16945.00 | 89.67 | 90.54 | 10789.92 | 6440.29 | 255.08 | 6440.29 | 2.29 |
| 46 | 17008.00 | 89.71 | 89.75 | 10790.26 | 6503.29 | 254.92 | 6503.29 | 1.26 |
| 47 | 17103.00 | 89.54 | 88.87 | 10790.89 | 6598.28 | 256.06 | 6598.28 | 0.94 |
| 48 | 17197.00 | 90.07 | 88.52 | 10791.21 | 6692.25 | 258.21 | 6692.25 | 0.68 |
| 49 | 17228.00 | 89.89 | 88.17 | 10791.22 | 6723.24 | 259.10 | 6723.24 | 1.27 |
| 50 | 17291.00 | 90.64 | 88.26 | 10790.93 | 6786.21 | 261.06 | 6786.21 | 1.20 |
| 51 | 17323.00 | 90.46 | 89.93 | 10790.62 | 6818.20 | 261.57 | 6818.20 | 5.25 |
| 52 | 17354.00 | 90.73 | 90.28 | 10790.30 | 6849.20 | 261.51 | 6849.20 | 1.43 |
| 53 | 17386.00 | 90.95 | 90.02 | 10789.83 | 6881.20 | 261.43 | 6881.20 | 1.06 |
| 54 | 17418.00 | 91.08 | 90.10 | 10789.26 | 6913.19 | 261.39 | 6913.19 | 0.48 |
| 55 | 17449.00 | 89.01 | 90.90 | 10789.24 | 6944.19 | 261.12 | 6944.19 | 7.16 |
| 56 | 17480.00 | 89.14 | 91.07 | 10789.74 | 6975.18 | 260.59 | 6975.18 | 0.69 |
| 57 | 17575.00 | 89.58 | 90.10 | 10790.80 | 7070.17 | 259.62 | 7070.17 | 1.12 |
| 58 | 17607.00 | 89.89 | 89.66 | 10790.95 | 7102.17 | 259.69 | 7102.17 | 1.68 |
| 59 | 17669.00 | 90.90 | 89.68 | 10790.52 | 7164.16 | 260.04 | 7164.16 | 1.63 |
| 60 | 17701.00 | 90.46 | 90.37 | 10790.14 | 7196.16 | 260.03 | 7196.16 | 2.56 |
| 61 | 17764.00 | 91.03 | 89.84 | 10789.32 | 7259.16 | 259.92 | 7259.16 | 1.24 |
| 62 | 17796.00 | 89.19 | 89.84 | 10789.26 | 7291.15 | 260.00 | 7291.15 | 5.75 |
| 63 | 17858.00 | 89.27 | 89.40 | 10790.09 | 7353.15 | 260.42 | 7353.15 | 0.72 |
| 64 | 17952.00 | 90.81 | 89.14 | 10790.03 | 7447.14 | 261.61 | 7447.14 | 1.66 |
| 65 | 18046.00 | 90.46 | 89.23 | 10788.98 | 7541.12 | 262.95 | 7541.12 | 0.38 |
| 66 | 18078.00 | 90.02 | 89.66 | 10788.85 | 7573.12 | 263.26 | 7573.12 | 1.92 |
| 67 | 18140.00 | 90.37 | 89.49 | 10788.64 | 7635.12 | 263.72 | 7635.12 | 0.63 |
| 68 | 18172.00 | 88.18 | 89.66 | 10789.04 | 7667.11 | 263.96 | 7667.11 | 6.86 |
| 69 | 18235.00 | 88.18 | 89.49 | 10791.04 | 7730.08 | 264.42 | 7730.08 | 0.27 |
| 70 | 18266.00 | 88.53 | 89.49 | 10791.93 | 7761.06 | 264.70 | 7761.06 | 1.13 |
| 71 | 18329.00 | 89.36 | 89.49 | 10793.09 | 7824.05 | 265.26 | 7824.05 | 1.32 |
| 72 | 18360.00 | 88.09 | 90.19 | 10793.78 | 7855.04 | 265.35 | 7855.04 | 4.68 |
| 73 | 18423.00 | 88.48 | 89.93 | 10795.67 | 7918.01 | 265.28 | 7918.01 | 0.74 |
| 74 | 18454.00 | 88.97 | 89.58 | 10796.36 | 7949.01 | 265.41 | 7949.01 | 1.94 |
| 75 | 18516.00 | 89.71 | 89.49 | 10797.07 | 8011.00 | 265.92 | 8011.00 | 1.20 |
| 76 | 18551.00 | 88.04 | 90.28 | 10797.76 | 8045.99 | 265.99 | 8045.99 | 5.28 |
| 77 | 18629.00 | 90.00 | 90.00 | 10799.10 | 8123.97 | 265.80 | 8123.97 | 2.54 |



|                          |                              |
|--------------------------|------------------------------|
| <b>Operator:</b>         | <b>Oasis Petroleum</b>       |
| Well :                   | Kline Federal 5300 41-18 10B |
| MWD Providers            | Ryan                         |
| Directional Supervision: | RPM                          |

|           |     |      |       |           |          |        |    |
|-----------|-----|------|-------|-----------|----------|--------|----|
| Section:  | 18  | QQ:  | SW SW | County:   | McKenzie | State: | ND |
| Township: | 153 | N/S: | N     | Footages: | 500      | FN/SL: | S  |
| Range:    | 100 | E/W: | W     |           | 237      | FE/WL: | W  |

### Sidetrack #2 Surveys

Vertical Section Plane: **90**

| <i>Coordinates</i> |                 |              |              |                 |                |               |                |             |
|--------------------|-----------------|--------------|--------------|-----------------|----------------|---------------|----------------|-------------|
| #                  | MD              | Inc.         | Azm.         | T.V.D.          | Ver. Sect.     | +N / -S       | +E / -W        | DLS         |
| <b>Tie</b>         | <b>17858.00</b> | <b>89.27</b> | <b>89.40</b> | <b>10790.09</b> | <b>7353.09</b> | <b>258.85</b> | <b>7353.09</b> | <b>0.72</b> |
| 1                  | 17952.00        | 90.59        | 89.14        | 10790.20        | 7447.08        | 260.05        | 7447.08        | 1.43        |
| 2                  | 17984.00        | 90.81        | 88.87        | 10789.81        | 7479.07        | 260.60        | 7479.07        | 1.09        |
| 3                  | 18015.00        | 89.23        | 90.10        | 10789.80        | 7510.07        | 260.88        | 7510.07        | 6.46        |
| 4                  | 18046.00        | 90.07        | 90.46        | 10789.99        | 7541.07        | 260.73        | 7541.07        | 2.95        |
| 5                  | 18109.00        | 90.42        | 89.84        | 10789.72        | 7604.07        | 260.57        | 7604.07        | 1.13        |
| 6                  | 18140.00        | 90.55        | 90.37        | 10789.46        | 7635.07        | 260.51        | 7635.07        | 1.76        |
| 7                  | 18203.00        | 91.12        | 90.98        | 10788.54        | 7698.05        | 259.77        | 7698.05        | 1.33        |
| 8                  | 18235.00        | 91.65        | 90.98        | 10787.77        | 7730.04        | 259.22        | 7730.04        | 1.66        |
| 9                  | 18297.00        | 92.00        | 91.16        | 10785.79        | 7792.00        | 258.06        | 7792.00        | 0.63        |
| 10                 | 18329.00        | 92.40        | 90.90        | 10784.57        | 7823.97        | 257.49        | 7823.97        | 1.49        |
| 11                 | 18360.00        | 92.57        | 90.90        | 10783.22        | 7854.94        | 257.00        | 7854.94        | 0.55        |
| 12                 | 18423.00        | 91.78        | 90.98        | 10780.83        | 7917.88        | 255.97        | 7917.88        | 1.26        |
| 13                 | 18454.00        | 92.00        | 90.72        | 10779.81        | 7948.86        | 255.51        | 7948.86        | 1.10        |
| 14                 | 18517.00        | 92.84        | 90.63        | 10777.15        | 8011.80        | 254.77        | 8011.80        | 1.34        |
| 15                 | 18548.00        | 91.30        | 91.16        | 10776.03        | 8042.78        | 254.28        | 8042.78        | 5.25        |
| 16                 | 18611.00        | 90.81        | 91.60        | 10774.87        | 8105.75        | 252.77        | 8105.75        | 1.05        |
| 17                 | 18705.00        | 89.98        | 91.69        | 10774.22        | 8199.70        | 250.07        | 8199.70        | 0.89        |
| 18                 | 18768.00        | 90.90        | 91.07        | 10773.74        | 8262.68        | 248.55        | 8262.68        | 1.76        |
| 19                 | 18799.00        | 91.25        | 90.90        | 10773.16        | 8293.67        | 248.02        | 8293.67        | 1.26        |
| 20                 | 18893.00        | 90.95        | 90.81        | 10771.35        | 8387.65        | 246.62        | 8387.65        | 0.33        |
| 21                 | 18987.00        | 90.77        | 90.90        | 10769.94        | 8481.62        | 245.21        | 8481.62        | 0.21        |
| 22                 | 19018.00        | 90.15        | 91.69        | 10769.69        | 8512.61        | 244.51        | 8512.61        | 3.24        |
| 23                 | 19050.00        | 90.55        | 91.69        | 10769.50        | 8544.60        | 243.57        | 8544.60        | 1.25        |
| 24                 | 19081.00        | 90.95        | 91.42        | 10769.09        | 8575.59        | 242.73        | 8575.59        | 1.56        |
| 25                 | 19112.00        | 91.25        | 91.51        | 10768.49        | 8606.57        | 241.93        | 8606.57        | 1.01        |
| 26                 | 19144.00        | 89.98        | 91.33        | 10768.15        | 8638.56        | 241.14        | 8638.56        | 4.01        |
| 27                 | 19175.00        | 90.02        | 91.60        | 10768.15        | 8669.55        | 240.35        | 8669.55        | 0.88        |
| 28                 | 19207.00        | 88.53        | 91.33        | 10768.56        | 8701.53        | 239.53        | 8701.53        | 4.73        |
| 29                 | 19269.00        | 86.95        | 91.42        | 10771.00        | 8763.47        | 238.04        | 8763.47        | 2.55        |
| 30                 | 19300.00        | 87.25        | 91.51        | 10772.57        | 8794.42        | 237.25        | 8794.42        | 1.01        |
| 31                 | 19331.00        | 87.69        | 91.51        | 10773.94        | 8825.37        | 236.44        | 8825.37        | 1.42        |
| 32                 | 19363.00        | 87.56        | 91.25        | 10775.26        | 8857.34        | 235.67        | 8857.34        | 0.91        |
| 33                 | 19426.00        | 88.48        | 91.07        | 10777.44        | 8920.29        | 234.39        | 8920.29        | 1.49        |

|      |          |       |       |          |         |        |         |      |
|------|----------|-------|-------|----------|---------|--------|---------|------|
| 34   | 19457.00 | 89.54 | 90.10 | 10777.98 | 8951.28 | 234.08 | 8951.28 | 4.63 |
| 35   | 19520.00 | 89.41 | 88.08 | 10778.55 | 9014.27 | 235.08 | 9014.27 | 3.21 |
| 36   | 19551.00 | 89.41 | 87.29 | 10778.87 | 9045.24 | 236.33 | 9045.24 | 2.55 |
| 37   | 19614.00 | 89.93 | 87.03 | 10779.24 | 9108.16 | 239.45 | 9108.16 | 0.92 |
| 38   | 19646.00 | 90.42 | 87.29 | 10779.14 | 9140.12 | 241.04 | 9140.12 | 1.73 |
| 39   | 19740.00 | 91.16 | 86.32 | 10777.84 | 9233.96 | 246.27 | 9233.96 | 1.30 |
| 40   | 19771.00 | 91.38 | 86.41 | 10777.15 | 9264.89 | 248.24 | 9264.89 | 0.77 |
| 41   | 19834.00 | 91.38 | 85.71 | 10775.64 | 9327.73 | 252.57 | 9327.73 | 1.11 |
| 42   | 19865.00 | 90.33 | 88.17 | 10775.17 | 9358.68 | 254.22 | 9358.68 | 8.63 |
| 43   | 19928.00 | 90.73 | 87.47 | 10774.59 | 9421.63 | 256.62 | 9421.63 | 1.28 |
| 44   | 19960.00 | 90.99 | 88.96 | 10774.11 | 9453.61 | 257.61 | 9453.61 | 4.73 |
| 45   | 19991.00 | 91.30 | 87.73 | 10773.49 | 9484.59 | 258.51 | 9484.59 | 4.09 |
| 46   | 20022.00 | 91.30 | 88.96 | 10772.79 | 9515.57 | 259.40 | 9515.57 | 3.97 |
| 47   | 20054.00 | 91.52 | 87.91 | 10772.00 | 9547.54 | 260.28 | 9547.54 | 3.35 |
| 48   | 20117.00 | 89.63 | 89.63 | 10771.37 | 9610.52 | 261.63 | 9610.52 | 4.06 |
| 49   | 20180.00 | 88.35 | 91.77 | 10772.48 | 9673.50 | 260.86 | 9673.50 | 3.96 |
| 50   | 20211.00 | 88.62 | 90.63 | 10773.30 | 9704.48 | 260.21 | 9704.48 | 3.78 |
| 51   | 20305.00 | 89.19 | 92.13 | 10775.10 | 9798.44 | 257.95 | 9798.44 | 1.71 |
| 52   | 20400.00 | 89.67 | 90.81 | 10776.04 | 9893.40 | 255.51 | 9893.40 | 1.48 |
| PROJ | 20475.00 | 89.67 | 90.81 | 10776.47 | 9968.39 | 254.45 | 9968.39 | 0.00 |

## **Kline Federal 5300 41-18 10B LITHOLOGY**

Geologists caught lagged samples in 30' intervals in the vertical hole, from 8,350' to 10,240' MD; 30' intervals in the curve build from 10,860' to 11,175' MD; and 50' samples in the lateral from 11,175' to TD at 20,475'

Electric geophysical log, sample and/or MWD gamma ray markers and tops are included in the sample descriptions below for reference. Samples were examined wet and dry under a binocular microscope in approximately 30' to 50' intervals. Sample descriptions begin in the **Kibbey Formation**. The drilling fluid was diesel invert in the vertical and curve, and salt water brine in the lateral.

### **Drilling vertical hole in Tyler Formation**

8350-8376 SILTSTONE: reddish brown, occasional, gray to light gray, brown to light brown gray, soft, sub blocky, earthy texture, no visible porosity, no visible oil stain

### **Kibbey Lime**

**8,376' MD, 8,376' TVD (-6,294')**

8376-8410 SILTSTONE: brown, occasional, gray to light gray, brown to light brown gray, soft, sub blocky, earthy texture, no visible porosity, no visible oil stain; LIMESTONE: wackestone, light gray to medium gray, light gray brown to medium gray brown, occasional off white friable to firm, crystalline to microcrystalline

8410-8440 SILTY SANDSTONE: medium gray, firm to friable, rounded, well sorted, moderately cemented, calcite cement, possible intergranular porosity, no visible oil stain; SILTSTONE: reddish brown, occasional, gray to light gray, brown to light brown gray, soft, sub blocky, earthy texture, no visible porosity, no visible oil stain

8440-8470 SILTY SANDSTONE: medium gray, firm to friable, rounded, well sorted, moderately cemented, calcite cement, possible intergranular porosity, no visible oil stain; SILTSTONE: reddish brown, occasional, gray to light gray, brown to light brown gray, soft, sub blocky, earthy texture, no visible porosity, no visible oil stain

8470-8500 SILTY SANDSTONE: medium gray, firm to friable, rounded, well sorted, moderately cemented, calcite cement, possible intergranular porosity, no visible oil stain; SILTSTONE: reddish brown, occasional, gray to light gray, brown to light brown gray, soft, sub blocky, earthy texture, no visible porosity, no visible oil stain

8500-8530 SILTY SANDSTONE: medium gray, firm to friable, rounded, well sorted, moderately cemented, calcite cement, possible intergranular porosity, no visible oil stain; SILTSTONE: reddish brown, occasional, gray to light gray, brown to light brown gray, soft, sub blocky, earthy texture, no visible porosity, no visible oil stain

8530-8566 SILTY SANDSTONE: medium gray, firm to friable, rounded, well sorted, moderately cemented, calcite cement, possible intergranular porosity, no visible oil stain; SILTSTONE: reddish brown, occasional, gray to light gray, brown to light brown gray, soft, sub blocky, earthy texture, no visible porosity, no visible oil stain

8566-8590 SALT: translucent, trace off white, crystalline, firm, crystalline texture, no visible porosity, no visible oil stain

8590-8620 SALT: translucent, trace off white, crystalline, firm, crystalline texture, no visible porosity, no visible oil stain

8620-8650 SALT: translucent, trace off white, crystalline, firm, crystalline texture, no visible porosity, no visible oil stain, LIMESTONE: wackestone, light gray to medium gray, light gray brown to medium gray brown, occasional off white friable to firm, crystalline to microcrystalline

8650-8680 SALT: translucent, trace off white, crystalline, firm, crystalline texture, no visible porosity, no visible oil stain, LIMESTONE: wackestone, light gray to medium gray, light gray brown to medium gray brown, occasional off white friable to firm, crystalline to microcrystalline

8680-8710 SALT: translucent, trace off white, crystalline, firm, crystalline texture, no visible porosity, no visible oil stain

8710-8740 SALT: translucent, trace off white, crystalline, firm, crystalline texture, no visible porosity, no visible oil stain, LIMESTONE: wackestone, light gray to medium gray, light gray brown to medium gray brown, occasional off white friable to firm, crystalline to microcrystalline

8740-8770 SALT: translucent, trace off white, crystalline, firm, crystalline texture, no visible porosity, no visible oil stain, LIMESTONE: wackestone, light gray to medium gray, light gray brown to medium gray brown, occasional off white friable to firm, crystalline to microcrystalline

8770-8800 SALT: translucent, trace off white, crystalline, firm, crystalline texture, no visible porosity, no visible oil stain, LIMESTONE: wackestone, light gray to medium gray, light gray brown to medium gray brown, occasional off white friable to firm, crystalline to microcrystalline

8800-8830 SALT: translucent, trace off white, crystalline, firm, crystalline texture, no visible porosity, no visible oil stain, LIMESTONE: wackestone, light gray to medium gray, light gray brown to medium gray brown, occasional off white friable to firm, crystalline to microcrystalline

8830-8860 SALT: translucent, trace off white, crystalline, firm, crystalline texture, no visible porosity, no visible oil stain, LIMESTONE: wackestone, light gray to medium gray, light gray brown to medium gray brown, occasional off white friable to firm, crystalline to microcrystalline

8860-8890 SALT: translucent, trace off white, crystalline, firm, crystalline texture, no visible porosity, no visible oil stain, LIMESTONE: wackestone, light gray to medium gray, light gray brown to medium gray brown, occasional off white friable to firm, crystalline to microcrystalline

8890-8920 SALT: translucent, trace off white, crystalline, firm, crystalline texture, no visible porosity, no visible oil stain, LIMESTONE: wackestone, light gray to medium gray, light gray brown to medium gray brown, occasional off white friable to firm, crystalline to microcrystalline



8920-8950 SALT: translucent, trace off white, crystalline, firm, crystalline texture, no visible porosity, no visible oil stain

8950-8980 SALT: translucent, trace off white, crystalline, firm, crystalline texture, no visible porosity, no visible oil stain, LIMESTONE: wackestone, light gray to medium gray, light gray brown to medium gray brown, occasional off white friable to firm, crystalline to microcrystalline

8980-9010 SALT: translucent, trace off white, crystalline, firm, crystalline texture, no visible porosity, no visible oil stain, LIMESTONE: wackestone, light gray to medium gray, light gray brown to medium gray brown, occasional off white friable to firm, crystalline to microcrystalline

9010-9040 SALT: translucent, trace off white, crystalline, firm, crystalline texture, no visible porosity, no visible oil stain, LIMESTONE: wackestone, light gray to medium gray, light gray brown to medium gray brown, occasional off white friable to firm, crystalline to microcrystalline

9040-9070 SALT: translucent, trace off white, crystalline, firm, crystalline texture, no visible porosity, no visible oil stain, LIMESTONE: wackestone, light gray to medium gray, light gray brown to medium gray brown, occasional off white friable to firm, crystalline to microcrystalline

9070-9100 SALT: translucent, trace off white, crystalline, firm, crystalline texture, no visible porosity, no visible oil stain, LIMESTONE: wackestone, light gray to medium gray, light gray brown to medium gray brown, occasional off white friable to firm, crystalline to microcrystalline

9100-9130 SALT: translucent, trace off white, crystalline, firm, crystalline texture, no visible porosity, no visible oil stain, LIMESTONE: wackestone, light gray to medium gray, light gray brown to medium gray brown, occasional off white friable to firm, crystalline to microcrystalline

9130-9160 SALT: translucent, trace off white, crystalline, firm, crystalline texture, no visible porosity, no visible oil stain, LIMESTONE: wackestone, light gray to medium gray, light gray brown to medium gray brown, occasional off white friable to firm, crystalline to microcrystalline

9160-9206 SALT: translucent, trace off white, crystalline, firm, crystalline texture, no visible porosity, no visible oil stain, LIMESTONE: wackestone, light gray to medium gray, light gray brown to medium gray brown, occasional off white friable to firm, crystalline to microcrystalline

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***Base Charles Salt***

***9,206' MD, 9,206' TVD (-7,124')***

9160-9190 SALT: translucent, trace off white, crystalline, firm, crystalline texture, no visible porosity, no visible oil stain, LIMESTONE: wackestone, light gray to medium gray, light gray brown to medium gray brown, occasional off white friable to firm, crystalline to microcrystalline

9206-9220 LIMESTONE: wackestone, light gray to medium gray, light gray brown to medium gray brown, occasional off white friable to firm, crystalline to microcrystalline

9220-9250 LIMESTONE: wackestone, light gray to medium gray, light gray brown to medium gray brown, occasional off white friable to firm, crystalline to microcrystalline

9250-9280 LIMESTONE: wackestone, light gray to medium gray, light gray brown to medium gray brown, occasional off white friable to firm, crystalline to microcrystalline

9280-9310 LIMESTONE: wackestone, light gray to medium gray, light gray brown to medium gray brown, occasional off white friable to firm, crystalline to microcrystalline

9310-9340 LIMESTONE: wackestone, light gray to medium gray, light gray brown to medium gray brown, occasional off white friable to firm, crystalline to microcrystalline

9340-9370 LIMESTONE: wackestone, light gray to medium gray, light gray brown to medium gray brown, occasional off white friable to firm, crystalline to microcrystalline

9370-9400 LIMESTONE: wackestone, light gray to medium gray, light gray brown to medium gray brown, occasional off white friable to firm, crystalline to microcrystalline

9400-9428 LIMESTONE: wackestone, light gray to medium gray, light gray brown to medium gray brown, occasional off white friable to firm, crystalline to microcrystalline

### *Mission Canyon*

*9,428 MD, 9,428' TVD (-7,346')*

9428-9460 LIMESTONE: wackestone, light gray to medium gray, light gray brown to medium gray brown, occasional off white friable to firm, crystalline to microcrystalline

9460-9490 LIMESTONE: wackestone, light gray to medium gray, light gray brown to medium gray brown, occasional off white friable to firm, crystalline to microcrystalline

9490-9520 LIMESTONE: wackestone, light gray to medium gray, light gray brown to medium gray brown, occasional off white friable to firm, crystalline to microcrystalline

9520-9550 LIMESTONE: wackestone, light gray to medium gray, light gray brown to medium gray brown, occasional off white friable to firm, crystalline to microcrystalline

9550-9580 LIMESTONE: wackestone, light gray to medium gray, light gray brown to medium gray brown, occasional off white friable to firm, crystalline to microcrystalline

9580-9610 LIMESTONE: wackestone, light gray to medium gray, light gray brown to medium gray brown, occasional off white friable to firm, crystalline to microcrystalline

9610-9640 LIMESTONE: wackestone, light gray to medium gray, light gray brown to medium gray brown, occasional off white friable to firm, crystalline to microcrystalline

9640-9670 LIMESTONE: wackestone, light gray to medium gray, light gray brown to medium gray brown, occasional off white friable to firm, crystalline to microcrystalline

9670-9700 LIMESTONE: wackestone, light gray to medium gray, light gray brown to medium gray brown, occasional off white friable to firm, crystalline to microcrystalline

9700-9730 LIMESTONE: wackestone, light gray to medium gray, light gray brown to medium gray brown, occasional off white friable to firm, crystalline to microcrystalline

9730-9760 LIMESTONE: wackestone, light gray to medium gray, light gray brown to medium gray brown, occasional off white friable to firm, crystalline to microcrystalline

9760-9790 LIMESTONE: wackestone, light gray to medium gray, light gray brown to medium gray brown, occasional off white friable to firm, crystalline to microcrystalline

9790-9820 LIMESTONE: wackestone, light gray to medium gray, light gray brown to medium gray brown, occasional off white friable to firm, crystalline to microcrystalline

9820-9850 LIMESTONE: wackestone, light gray to medium gray, light gray brown to medium gray brown, occasional off white friable to firm, crystalline to microcrystalline

9850-9880 LIMESTONE: wackestone, light gray to medium gray, light gray brown to medium gray brown, occasional off white friable to firm, crystalline to microcrystalline

9880-9910 LIMESTONE: wackestone, light gray to medium gray, light gray brown to medium gray brown, occasional off white friable to firm, crystalline to microcrystalline

9910-9940 LIMESTONE: wackestone, light gray to medium gray, light gray brown to medium gray brown, occasional off white friable to firm, crystalline to microcrystalline

9940-9980 LIMESTONE: wackestone, light gray to medium gray, light gray brown to medium gray brown, occasional off white friable to firm, crystalline to microcrystalline

### Lodgepole

9,980 MD, 9,980' TVD (-7,898')

9880-9910 LIMESTONE: wackestone, light gray to medium gray, light gray brown to medium gray brown, occasional off white friable to firm, crystalline to microcrystalline

9910-9940 LIMESTONE: wackestone, light gray to medium gray, light gray brown to medium gray brown, occasional off white friable to firm, crystalline to microcrystalline

9940-9970 LIMESTONE: wackestone, light gray to medium gray, light gray brown to medium gray brown, occasional off white friable to firm, crystalline to microcrystalline

9980-10000 LIMESTONE: wackestone, light gray to medium gray, light gray brown to medium gray brown, occasional off white friable to firm, crystalline to microcrystalline

10000-10030 ARGILLACEOUS LIMESTONE: wackestone, medium gray brown to light gray brown, firm, crystalline texture, trace sparry calcareous, carbonaceous, possible intercrystalline porosity, possible slightly light brown spotty oil stain

10030-10060 ARGILLACEOUS LIMESTONE: wackestone, medium gray brown to light gray brown, firm, crystalline texture, trace sparry calcareous, carbonaceous, possible intercrystalline porosity, possible slightly light brown spotty oil stain

10060-10090 ARGILLACEOUS LIMESTONE: wackestone, medium gray brown to light gray brown, firm, crystalline texture, trace sparry calcareous, carbonaceous, possible intercrystalline porosity, possible slightly light brown spotty oil stain

10090-10120 ARGILLACEOUS LIMESTONE: wackestone, medium gray brown to light gray brown, firm, crystalline texture, trace sparry calcareous, carbonaceous, possible intercrystalline porosity, possible slightly light brown spotty oil stain

10120-10150 ARGILLACEOUS LIMESTONE: wackestone, medium gray brown to light gray brown, firm, crystalline texture, trace sparry calcareous, carbonaceous, possible intercrystalline porosity, possible slightly light brown spotty oil stain

10150-10180 ARGILLACEOUS LIMESTONE: wackestone, medium gray brown to light gray brown, firm, crystalline texture, trace sparry calcareous, carbonaceous, possible intercrystalline porosity, possible slightly light brown spotty oil stain

10180-10210 ARGILLACEOUS LIMESTONE: wackestone, medium gray brown to light gray brown, firm, crystalline texture, trace sparry calcareous, carbonaceous, possible intercrystalline porosity, possible slightly light brown spotty oil stain

10210-10240 ARGILLACEOUS LIMESTONE: wackestone, medium gray brown to light gray brown, firm, crystalline texture, trace sparry calcareous, carbonaceous, possible intercrystalline porosity, possible slightly light brown spotty oil stain

***10,240' Vertical TD reached @ 03:00 hours CST, 14 November 2014***

***Drilling curve build in the Lodgepole Formation***

10240-10270 ARGILLACEOUS LIMESTONE: wackestone, medium gray brown to light gray brown, firm, crystalline texture, trace sparry calcareous, carbonaceous, possible intercrystalline porosity, possible slightly light brown spotty oil stain

10270-10300 ARGILLACEOUS LIMESTONE: wackestone, medium gray brown to light gray brown, firm, crystalline texture, trace sparry calcareous, carbonaceous, possible intercrystalline porosity, possible slightly light brown spotty oil stain

10300-10330 ARGILLACEOUS LIMESTONE: wackestone, medium gray brown to light gray brown, firm, crystalline texture, trace sparry calcareous, carbonaceous, possible intercrystalline porosity, possible slightly light brown spotty oil stain

10330-10360 ARGILLACEOUS LIMESTONE: wackestone, medium gray brown to light gray brown, firm, crystalline texture, trace sparry calcareous, carbonaceous, possible intercrystalline porosity, possible slightly light brown spotty oil stain

10360-10390 ARGILLACEOUS LIMESTONE: wackestone, medium gray brown to light gray brown, firm, crystalline texture, trace sparry calcareous, carbonaceous, possible intercrystalline porosity, possible slightly light brown spotty oil stain

10390-10420 ARGILLACEOUS LIMESTONE: wackestone, medium gray brown to light gray brown, firm, crystalline texture, trace sparry calcareous, carbonaceous, possible intercrystalline porosity, possible slightly light brown spotty oil stain

10420-10450 ARGILLACEOUS LIMESTONE: wackestone, medium gray brown to light gray brown, firm, crystalline texture, trace sparry calcareous, carbonaceous, possible intercrystalline porosity, possible slightly light brown spotty oil stain

10450-10480 ARGILLACEOUS LIMESTONE: wackestone, medium gray brown to light gray brown, firm, crystalline texture, trace sparry calcareous, carbonaceous, possible intercrystalline porosity, possible slightly light brown spotty oil stain

10480-10510 ARGILLACEOUS LIMESTONE: wackestone, medium gray brown to light gray brown, firm, crystalline texture, trace sparry calcareous, carbonaceous, possible intercrystalline porosity, possible slightly light brown spotty oil stain

10510-10540 ARGILLACEOUS LIMESTONE: wackestone, medium gray brown to light gray brown, firm, crystalline texture, trace sparry calcareous, carbonaceous, possible intercrystalline porosity, possible slightly light brown spotty oil stain

10540-10570 ARGILLACEOUS LIMESTONE: wackestone, medium gray brown to light gray brown, firm, crystalline texture, trace sparry calcareous, carbonaceous, possible intercrystalline porosity, possible slightly light brown spotty oil stain

10570-10600 ARGILLACEOUS LIMESTONE: wackestone, medium gray brown to light gray brown, firm, crystalline texture, trace sparry calcareous, carbonaceous, possible intercrystalline porosity, possible slightly light brown spotty oil stain

10600-10630 ARGILLACEOUS LIMESTONE: wackestone, medium gray brown to light gray brown, firm, crystalline texture, trace sparry calcareous, carbonaceous, possible intercrystalline porosity, possible slightly light brown spotty oil stain

10630-10660 ARGILLACEOUS LIMESTONE: wackestone, medium gray brown to light gray brown, firm, crystalline texture, trace sparry calcareous, carbonaceous, possible intercrystalline porosity, possible slightly light brown spotty oil stain

10660-10690 ARGILLACEOUS LIMESTONE: wackestone, medium gray brown to light gray brown, firm, crystalline texture, trace sparry calcareous, carbonaceous, possible intercrystalline porosity, possible slightly light brown spotty oil stain

10690-10720 ARGILLACEOUS LIMESTONE: wackestone, medium gray brown to light gray brown, firm, crystalline texture, trace sparry calcareous, carbonaceous, possible intercrystalline porosity, possible slightly light brown spotty oil stain

10720-10750 ARGILLACEOUS LIMESTONE: wackestone, medium gray brown to light gray brown, firm, crystalline texture, trace sparry calcareous, carbonaceous, possible intercrystalline porosity, possible slightly light brown spotty oil stain

10750-10780 ARGILLACEOUS LIMESTONE: wackestone, medium gray brown to light gray brown, firm, crystalline texture, trace sparry calcareous, carbonaceous, possible intercrystalline porosity, possible slightly light brown spotty oil stain

10780-10810 ARGILLACEOUS LIMESTONE: wackestone, medium gray brown to light gray brown, firm, crystalline texture, trace sparry calcareous, carbonaceous, possible intercrystalline porosity, possible slightly light brown spotty oil stain

10810-10840 ARGILLACEOUS LIMESTONE: wackestone, medium gray brown to light gray brown, firm, crystalline texture, trace sparry calcareous, carbonaceous, possible intercrystalline porosity, possible slightly light brown spotty oil stain

10840-10864 SHALE: black, dark gray black, firm, sbblky, earthy, calcareous cement, petro, carbonaceous, abundt fine disseminated pyrite, fine laminated; LIMESTONE: wackestone, light gray to medium gray, light gray brown to medium gray brown, occasional off white friable to firm, crystalline to microcrystalline

### ***False Bakken***

***10,864' MD, 10,708' TVD (-8,626')***

10864-10890 SHALE: black, dark gray black, firm, sub blocky, earthy, calcareous cement, petro, carbonaceous, abundant fine disseminated pyrite, fine laminated

### ***Upper Bakken Shale***

***10,890' MD, 10,716' TVD (-8,634')***

10890-10954 SHALE: black, dark gray black, firm, sub blocky, earthy, calcareous cement, petro, carbonaceous, abundant fine disseminated pyrite, fine laminated

### ***Middle Bakken***

***10,954' MD, 10,733' TVD (-8,651')***

10930-10960 SILTY SANDSTONE: common salt and pepper, occasional light gray to medium gray, visible white, very fine grained, soft to firm, moderately to well sorted, rounded to sub rounded, moderately to poorly cemented, visible calcareous fluid fractures, possible fracture porosity, common intergranular porosity, visible light brown spotty oil stain

10960-10990 SILTY SANDSTONE: common salt and pepper, occasional light gray to medium gray, visible white, very fine grained, soft to firm, moderately to well sorted, rounded to sub rounded, moderately to poorly cemented, visible calcareous fluid fractures, possible fracture porosity, common intergranular porosity, visible light brown spotty oil stain

10990-11020 SILTY SANDSTONE: common salt and pepper, occasional light gray to medium gray, visible white, very fine grained, soft to firm, moderately to well sorted, rounded to sub rounded, moderately to poorly cemented, visible calcareous fluid fractures, possible fracture porosity, common intergranular porosity, visible light brown spotty oil stain



11020-11050 SILTY SANDSTONE: common salt and pepper, occasional light gray to medium gray, visible white, very fine grained, soft to firm, moderately to well sorted, rounded to sub rounded, moderately to poorly cemented, visible calcareous fluid fractures, possible fracture porosity, common intergranular porosity, visible light brown spotty oil stain

11050-11080 SILTY SANDSTONE: common salt and pepper, occasional light gray to medium gray, visible white, very fine grained, soft to firm, moderately to well sorted, rounded to sub rounded, moderately to poorly cemented, visible calcareous fluid fractures, possible fracture porosity, common intergranular porosity, visible light brown spotty oil stain

11080-11110 SILTY SANDSTONE: common salt and pepper, occasional light gray to medium gray, visible white, very fine grained, soft to firm, moderately to well sorted, rounded to sub rounded, moderately to poorly cemented, visible calcareous fluid fractures, possible fracture porosity, common intergranular porosity, visible light brown spotty oil stain

11110-11125 SILTY SANDSTONE: common salt and pepper, occasional light gray to medium gray, visible white, very fine grained, soft to firm, moderately to well sorted, rounded to sub rounded, moderately to poorly cemented, visible calcareous fluid fractures, possible fracture porosity, common intergranular porosity, visible light brown spotty oil stain

### **Curve TD of 11,125' @ 12:00 CST 17 November-2014**

#### ***Intermediate 7" Casing Set @ 11,185' MD***

11125-11175 SILTY SANDSTONE: common salt and pepper, occasional light gray to medium gray, visible white, very fine grained, soft to firm, moderately to well sorted, rounded to sub rounded, moderately to poorly cemented, visible calcareous fluid fractures, possible fracture porosity, common intergranular porosity, visible light brown spotty oil stain

11175-11200 SILTY SANDSTONE: common salt and pepper, occasional light gray to medium gray, visible white, very fine grained, soft to firm, moderately to well sorted, rounded to sub rounded, moderately to poorly cemented, visible calcareous fluid fractures, possible fracture porosity, common intergranular porosity, visible light brown spotty oil stain

11200-11250 SILTY SANDSTONE: common salt and pepper, occasional light gray to medium gray, visible white, very fine grained, soft to firm, moderately to well sorted, rounded to sub rounded, moderately to poorly cemented, visible calcareous fluid fractures, possible fracture porosity, common intergranular porosity, visible light brown spotty oil stain

11250-11300 SILTY SANDSTONE: salt and pepper, common light gray to medium gray, occasional light brown to tan, occasional visible white, very fine grained, soft to firm, moderately to well sorted, sub rounded to sub angular, moderately to poorly cemented, visible calcareous fluid fractures, possible fracture porosity, common intergranular porosity, visible light brown spotty oil stain

11300-11350 SILTY SANDSTONE: salt and pepper, common light gray to medium gray, occasional light brown to tan, occasional visible white, very fine grained, soft to firm, moderately to well sorted, sub rounded to sub angular, moderately to poorly cemented, visible calcareous fluid fractures, possible fracture porosity, common intergranular porosity, visible light brown spotty oil stain

11350-11400 SILTY SANDSTONE: salt and pepper, common light gray to medium gray, cc light brown to tan, occasional visible white, very fine grained, soft to firm, moderately to well sorted, sub rounded to sub angular, moderately to poorly cemented, visible calcareous fluid fractures, possible fracture porosity, common intergranular porosity, visible light brown spotty oil stain

11400-11450 SILTY SANDSTONE: salt and pepper, common light gray to medium gray, common white, rare tan, very fine grained, soft to firm, moderately to well sorted, sub rounded to sub angular, moderately to poorly cemented, visible calcareous fluid fractures, possible fracture porosity, common intergranular porosity, visible light brown spotty oil stain

11450-11500 SILTY SANDSTONE: salt and pepper, common light gray to medium gray, common white, rare tan, very fine grained, soft to firm, moderately to well sorted, sub rounded to sub angular, moderately to poorly cemented, visible calcareous fluid fractures, possible fracture porosity, common intergranular porosity, visible light brown spotty oil stain

11500-11550 SILTY SANDSTONE: salt and pepper, common light gray to medium gray, common white, rare tan, very fine grained, soft to firm, moderately to well sorted, sub rounded to sub angular, moderately to poorly cemented, visible calcareous fluid fractures, possible fracture porosity, common intergranular porosity, visible light brown spotty oil stain

11550-11600 SILTY SANDSTONE: salt and pepper, common light gray to medium gray, common white, rare tan, very fine grained, soft to firm, moderately to well sorted, sub rounded to sub angular, moderately to poorly cemented, visible calcareous fluid fractures, possible fracture porosity, common intergranular porosity, visible light brown spotty oil stain

11600-11650 SILTY SANDSTONE: salt and pepper, common light gray to medium gray, common white, rare tan, very fine grained, soft to firm, moderately to well sorted, sub rounded to sub angular, moderately to poorly cemented, visible calcareous fluid fractures, possible fracture porosity, common intergranular porosity, visible light brown spotty oil stain

11650-11700 SILTY SANDSTONE: salt and pepper, common light gray to medium gray, common white, rare tan, very fine grained, soft to firm, moderately to well sorted, sub rounded to sub angular, moderately to poorly cemented, visible calcareous fluid fractures, possible fracture porosity, common intergranular porosity, visible light brown spotty oil stain

11700-11750 SILTY SANDSTONE: salt and pepper, common light gray to medium gray, common white, rare tan, very fine grained, soft to firm, moderately to well sorted, sub rounded to sub angular, moderately to poorly cemented, visible calcareous fluid fractures, possible fracture porosity, common intergranular porosity, visible light brown spotty oil stain

11750-11800 SILTY SANDSTONE: salt and pepper, common light gray to medium gray, common white, rare tan, very fine grained, soft to firm, moderately to well sorted, sub rounded to sub angular, moderately to poorly cemented, visible calcareous fluid fractures, possible fracture porosity, common intergranular porosity, visible light brown spotty oil stain

11800-11850 SILTY SANDSTONE: salt and pepper, common light gray to medium gray, common white, rare tan, very fine grained, soft to firm, moderately to well sorted, sub rounded to sub angular, moderately to poorly cemented, visible calcareous fluid fractures, possible fracture porosity, common intergranular porosity, visible light brown spotty oil stain

11850-11900 SILTY SANDSTONE: salt and pepper, common light gray to medium gray, common white, rare tan, very fine grained, soft to firm, moderately to well sorted, sub rounded to sub angular, moderately to poorly cemented, visible calcareous fluid fractures, possible fracture porosity, common intergranular porosity, visible light brown spotty oil stain

11900-11950 SILTY SANDSTONE: salt and pepper, common light gray to medium gray, common white, rare tan, very fine grained, soft to firm, moderately to well sorted, sub rounded to sub angular, moderately to poorly cemented, visible calcareous fluid fractures, possible fracture porosity, common intergranular porosity, visible light brown spotty oil stain

11950-12000 SILTY SANDSTONE: light to medium gray, common light to medium brown, occasional white to off white, very fine grained, friable to firm, moderately to well sorted, sub rounded to sub angular, moderately to poorly cemented, visible calcareous fluid fractures, trace disseminated pyrite, possible fracture porosity, common intergranular porosity, trace light brown spotty oil stain

12000-12050 SILTY SANDSTONE: light to medium gray, common light to medium brown, occasional white to off white, very fine grained, friable to firm, moderately to well sorted, sub rounded to sub angular, moderately to poorly cemented, visible calcareous fluid fractures, trace disseminated pyrite, possible fracture porosity, common intergranular porosity, trace light brown spotty oil stain

12050-12100 SILTY SANDSTONE: light to medium gray, common medium brown, occasional light brown, occasional white to off white, very fine grained, friable to firm, moderately to well sorted, sub rounded to sub angular, moderately to poorly cemented, visible calcareous fluid fractures, rare crystalline pyrite, trace disseminated pyrite, possible fracture porosity, common intergranular porosity, trace light brown spotty oil stain

12100-12150 SILTY SANDSTONE: light to medium gray, common medium brown, occasional light brown, occasional white to off white, very fine grained, friable to firm, moderately to well sorted, sub rounded to sub angular, moderately to poorly cemented, visible calcareous fluid fractures, rare crystalline pyrite, trace disseminated pyrite, possible fracture porosity, common intergranular porosity, trace light brown spotty oil stain

12150-12200 SILTY SANDSTONE: medium brown, common light to medium gray, occasional white to off white, very fine grained, friable to firm, moderately to well sorted, sub rounded to sub angular, moderately to poorly cemented, visible calcareous fluid fractures, trace disseminated pyrite, possible fracture porosity, common intergranular porosity, trace light brown spotty oil stain

12200-12250 SILTY SANDSTONE: medium brown, common light to medium gray, occasional dark brown, rare white to off white, very fine grained, friable to firm, moderately to well sorted, sub rounded to sub angular, moderately to poorly cemented, visible calcareous fluid fractures, trace disseminated pyrite, possible fracture porosity, common intergranular porosity, trace light brown spotty oil stain

12250-12300 SILTY SANDSTONE: light to medium gray, common light to medium brown, rare white to off white, very fine grained, friable to firm, moderately to well sorted, sub rounded to sub angular, moderately to poorly cemented, visible calcareous fluid fractures, trace disseminated pyrite, possible fracture porosity, common intergranular porosity, trace light brown spotty oil stain

12300-12350 SILTY SANDSTONE: light to medium gray, common light to medium brown, occasional white to off white, very fine grained, friable to firm, moderately to well sorted, sub rounded to sub angular, moderately to poorly cemented, visible calcareous fluid fractures, trace disseminated pyrite, possible fracture porosity, common intergranular porosity, trace light brown spotty oil stain

12350-12400 SILTY SANDSTONE: light to medium gray, common light to medium brown, occasional white to off white, very fine grained, friable to firm, moderately to well sorted, sub rounded to sub angular, moderately to poorly cemented, visible calcareous fluid fractures, trace disseminated pyrite, possible fracture porosity, common intergranular porosity, trace light brown spotty oil stain

12400-12450 SILTY SANDSTONE: light to medium gray, common light to medium brown, occasional white to off white, very fine grained, friable to firm, moderately to well sorted, sub rounded to sub angular, moderately to poorly cemented, visible calcareous fluid fractures, trace disseminated pyrite, possible fracture porosity, common intergranular porosity, trace light brown spotty oil stain

12450-12500 SILTY SANDSTONE: light to medium gray, common light to medium brown, occasional white to off white, very fine grained, friable to firm, moderately to well sorted, sub rounded to sub angular, moderately to poorly cemented, visible calcareous fluid fractures, rare crystalline pyrite, trace disseminated pyrite, possible fracture porosity, common intergranular porosity, trace light brown spotty oil stain

12500-12550 SILTY SANDSTONE: light brown to tan, common light to medium gray, common white to off white, occasional medium brown, very fine grained, friable to firm, moderately to well sorted, sub rounded to sub angular, moderately to poorly cemented, visible calcareous fluid fractures, rare crystalline pyrite, trace disseminated pyrite, possible fracture porosity, common intergranular porosity, trace light brown spotty oil stain

12550-12600 SILTY SANDSTONE: light to medium gray, common light to medium brown, common white to off white, very fine grained, friable to firm, moderately to well sorted, sub rounded to sub angular, moderately to poorly cemented, visible calcareous fluid fractures, trace crystalline pyrite, trace disseminated pyrite, possible fracture porosity, common intergranular porosity, trace light brown spotty oil stain

12600-12650 SILTY SANDSTONE: light to medium gray, common light to medium brown, common white to off white, very fine grained, friable to firm, moderately to well sorted, sub rounded to sub angular, moderately to poorly cemented, visible calcareous fluid fractures, trace crystalline pyrite, trace disseminated pyrite, possible fracture porosity, common intergranular porosity, trace light brown spotty oil stain

12650-12700 SILTY SANDSTONE: light to medium gray, common light brown to tan, occasional white to off white, very fine grained, friable to firm, moderately to well sorted, sub rounded to sub angular, moderately to poorly cemented, visible calcareous fluid fractures, trace disseminated pyrite, possible fracture porosity, common intergranular porosity, trace light brown spotty oil stain

12700-12750 SILTY SANDSTONE: light to medium gray, common light brown to tan, common white to off white, very fine grained, friable to firm, moderately to well sorted, sub rounded to sub angular, moderately to poorly cemented, visible calcareous fluid fractures, trace disseminated pyrite, possible fracture porosity, common intergranular porosity, trace light brown spotty oil stain

13200-13250 SILTY SANDSTONE: light to medium gray, common light brown to tan, common white to off white, very fine grained, friable to firm, moderately to well sorted, sub rounded to sub angular, moderately to poorly cemented, visible calcareous fluid fractures, trace disseminated pyrite, possible fracture porosity, common intergranular porosity, trace light brown spotty oil stain

13700-13750 SILTY SANDSTONE: light to medium gray, common light brown to tan, common white to off white, very fine grained, friable to firm, moderately to well sorted, sub rounded to sub angular, moderately to poorly cemented, visible calcareous fluid fractures, trace disseminated pyrite, possible fracture porosity, common intergranular porosity, trace light brown spotty oil stain



13750-13800 SILTY SANDSTONE: light to medium gray, common light brown to tan, common white to off white, very fine grained, friable to firm, moderately to well sorted, sub rounded to sub angular, moderately to poorly cemented, visible calcareous fluid fractures, trace disseminated pyrite, possible fracture porosity, common intergranular porosity, trace light brown spotty oil stain

13800-13850 SILTY SANDSTONE: light to medium gray, common light brown to tan, common white to off white, very fine grained, friable to firm, moderately to well sorted, sub rounded to sub angular, moderately to poorly cemented, visible calcareous fluid fractures, trace disseminated pyrite, possible fracture porosity, common intergranular porosity, trace light brown spotty oil stain

13850-13900 SILTY SANDSTONE: light to medium gray, common light brown to tan, common white to off white, very fine grained, friable to firm, moderately to well sorted, sub rounded to sub angular, moderately to poorly cemented, visible calcareous fluid fractures, trace disseminated pyrite, possible fracture porosity, common intergranular porosity, trace light brown spotty oil stain

NOTE: Oil highly visible in sample and coming over shakers

13900-13950 SILTY SANDSTONE: light to medium gray, common light brown to tan, common white to off white, very fine grained, friable to firm, moderately to well sorted, sub rounded to sub angular, moderately to poorly cemented, visible calcareous fluid fractures, trace disseminated pyrite, possible fracture porosity, common intergranular porosity, trace light brown spotty oil stain

13950-14000 SILTY SANDSTONE: light to medium gray, common light brown to tan, common white to off white, very fine grained, friable to firm, moderately to well sorted, sub rounded to sub angular, moderately to poorly cemented, visible calcareous fluid fractures, trace disseminated pyrite, possible fracture porosity, common intergranular porosity, trace light brown spotty oil stain

14000-14050 SILTY SANDSTONE: light to medium gray, common light brown to tan, common white to off white, very fine grained, friable to firm, moderately to well sorted, sub rounded to sub angular, moderately to poorly cemented, visible calcareous fluid fractures, trace disseminated pyrite, possible fracture porosity, common intergranular porosity, trace light brown spotty oil stain

14050-14100 SILTY SANDSTONE: light to medium gray, common light brown to tan, common white to off white, very fine grained, friable to firm, moderately to well sorted, sub rounded to sub angular, moderately to poorly cemented, visible calcareous fluid fractures, trace disseminated pyrite, possible fracture porosity, common intergranular porosity, trace light brown spotty oil stain

14100-14150 SILTY SANDSTONE: light to medium gray, common light brown to tan, common white to off white, very fine grained, friable to firm, moderately to well sorted, sub rounded to sub angular, moderately to poorly cemented, visible calcareous fluid fractures, trace disseminated pyrite, possible fracture porosity, common intergranular porosity, trace light brown spotty oil stain

14150-14200 SILTY SANDSTONE: light to medium gray, common light brown to tan, common white to off white, very fine grained, friable to firm, moderately to well sorted, sub rounded to sub angular, moderately to poorly cemented, visible calcareous fluid fractures, trace disseminated pyrite, possible fracture porosity, common intergranular porosity, trace light brown spotty oil stain

14650-14700 SILTY SANDSTONE: light to medium gray, common light brown to tan, common white to off white, very fine grained, friable to firm, moderately to well sorted, sub rounded to sub angular, moderately to poorly cemented, visible calcareous fluid fractures, trace disseminated pyrite, possible fracture porosity, common intergranular porosity, trace light brown spotty oil stain

14700-14750 SILTY SANDSTONE: light to medium gray, common light brown to tan, common white to off white, very fine grained, friable to firm, moderately to well sorted, sub rounded to sub angular, moderately to poorly cemented, visible calcareous fluid fractures, trace disseminated pyrite, possible fracture porosity, common intergranular porosity, trace light brown spotty oil stain

14750-14800 SILTY SANDSTONE: light to medium gray, common light brown to tan, common white to off white, very fine grained, friable to firm, moderately to well sorted, sub rounded to sub angular, moderately to poorly cemented, visible calcareous fluid fractures, trace disseminated pyrite, possible fracture porosity, common intergranular porosity, trace light brown spotty oil stain

14800-14850 SILTY SANDSTONE: light to medium gray, common light brown to tan, common white to off white, very fine grained, friable to firm, moderately to well sorted, sub rounded to sub angular, moderately to poorly cemented, visible calcareous fluid fractures, trace disseminated pyrite, possible fracture porosity, common intergranular porosity, trace light brown spotty oil stain

14850-14900 SILTY SANDSTONE: light to medium gray, common light brown to tan, common white to off white, very fine grained, friable to firm, moderately to well sorted, sub rounded to sub angular, moderately to poorly cemented, visible calcareous fluid fractures, trace disseminated pyrite, possible fracture porosity, common intergranular porosity, trace light brown spotty oil stain

NOTE: Oil highly visible in sample and coming over shakers

14900-14950 SILTY SANDSTONE: light to medium gray, common light brown to tan, common white to off white, very fine grained, friable to firm, moderately to well sorted, sub rounded to sub angular, moderately to poorly cemented, visible calcareous fluid fractures, trace disseminated pyrite, possible fracture porosity, common intergranular porosity, trace light brown spotty oil stain

14950-15000 SILTY SANDSTONE: light to medium gray, common light brown to tan, common white to off white, very fine grained, friable to firm, moderately to well sorted, sub rounded to sub angular, moderately to poorly cemented, visible calcareous fluid fractures, trace disseminated pyrite, possible fracture porosity, common intergranular porosity, trace light brown spotty oil stain

15000-15050 SILTY SANDSTONE: light to medium gray, common light brown to tan, common white to off white, very fine grained, friable to firm, moderately to well sorted, sub rounded to sub angular, moderately to poorly cemented, visible calcareous fluid fractures, trace disseminated pyrite, possible fracture porosity, common intergranular porosity, trace light brown spotty oil stain

15050-15100 SILTY SANDSTONE: light to medium gray, common light brown to tan, common white to off white, very fine grained, friable to firm, moderately to well sorted, sub rounded to sub angular, moderately to poorly cemented, visible calcareous fluid fractures, trace disseminated pyrite, possible fracture porosity, common intergranular porosity, trace light brown spotty oil stain

15100-15150 SILTY SANDSTONE: light to medium gray, common light brown to tan, common white to off white, very fine grained, friable to firm, moderately to well sorted, sub rounded to sub angular, moderately to poorly cemented, visible calcareous fluid fractures, trace disseminated pyrite, possible fracture porosity, common intergranular porosity, trace light brown spotty oil stain

15600-15650 SILTY SANDSTONE: light to medium gray, common light brown to tan, common white to off white, very fine grained, friable to firm, moderately to well sorted, sub rounded to sub angular, moderately to poorly cemented, visible calcareous fluid fractures, trace disseminated pyrite, possible fracture porosity, common intergranular porosity, trace light brown spotty oil stain

16100-16150 SILTY SANDSTONE: light to medium gray, common light brown to tan, common white to off white, very fine grained, friable to firm, moderately to well sorted, sub rounded to sub angular, moderately to poorly cemented, visible calcareous fluid fractures, trace disseminated pyrite, possible fracture porosity, common intergranular porosity, trace light brown spotty oil stain

16600-16650 SILTY SANDSTONE: light to medium gray, common light brown to tan, common white to off white, very fine grained, friable to firm, moderately to well sorted, sub rounded to sub angular, moderately to poorly cemented, visible calcareous fluid fractures, trace disseminated pyrite, possible fracture porosity, common intergranular porosity, trace light brown spotty oil stain

16650-16700 SILTY SANDSTONE: light to medium gray, common light brown to tan, common white to off white, very fine grained, friable to firm, moderately to well sorted, sub rounded to sub angular, moderately to poorly cemented, visible calcareous fluid fractures, trace disseminated pyrite, possible fracture porosity, common intergranular porosity, trace light brown spotty oil stain

16700-16750 SILTY SANDSTONE: light to medium gray, common light brown to tan, common white to off white, very fine grained, friable to firm, moderately to well sorted, sub rounded to sub angular, moderately to poorly cemented, visible calcareous fluid fractures, trace disseminated pyrite, possible fracture porosity, common intergranular porosity, trace light brown spotty oil stain

16750-16800 SILTY SANDSTONE: light to medium gray, common light brown to tan, common white to off white, very fine grained, friable to firm, moderately to well sorted, sub rounded to sub angular, moderately to poorly cemented, visible calcareous fluid fractures, trace disseminated pyrite, possible fracture porosity, common intergranular porosity, trace light brown spotty oil stain

NOTE: Oil highly visible in sample and coming over shakers

16800-16850 SILTY SANDSTONE: light to medium gray, common light brown to tan, common white to off white, very fine grained, friable to firm, moderately to well sorted, sub rounded to sub angular, moderately to poorly cemented, visible calcareous fluid fractures, trace disseminated pyrite, possible fracture porosity, common intergranular porosity, trace light brown spotty oil stain

16850-16900 SILTY SANDSTONE: light to medium gray, common light brown to tan, common white to off white, very fine grained, friable to firm, moderately to well sorted, sub rounded to sub angular, moderately to poorly cemented, visible calcareous fluid fractures, trace disseminated pyrite, possible fracture porosity, common intergranular porosity, trace light brown spotty oil stain

16900-16950 SILTY SANDSTONE: light to medium gray, common light brown to tan, common white to off white, very fine grained, friable to firm, moderately to well sorted, sub rounded to sub angular, moderately to poorly cemented, visible calcareous fluid fractures, trace disseminated pyrite, possible fracture porosity, common intergranular porosity, trace light brown spotty oil stain

16950-17000 SILTY SANDSTONE: light to medium gray, common light brown to tan, common white to off white, very fine grained, friable to firm, moderately to well sorted, sub rounded to sub angular, moderately to poorly cemented, visible calcareous fluid fractures, trace disseminated pyrite, possible fracture porosity, common intergranular porosity, trace light brown spotty oil stain

17000-17050 SILTY SANDSTONE: light to medium gray, common light brown to tan, common white to off white, very fine grained, friable to firm, moderately to well sorted, sub rounded to sub angular, moderately to poorly cemented, visible calcareous fluid fractures, trace disseminated pyrite, possible fracture porosity, common intergranular porosity, trace light brown spotty oil stain

17050-17100 SILTY SANDSTONE: light to medium gray, common light brown to tan, common white to off white, very fine grained, friable to firm, moderately to well sorted, sub rounded to sub angular, moderately to poorly cemented, visible calcareous fluid fractures, trace disseminated pyrite, possible fracture porosity, common intergranular porosity, trace light brown spotty oil stain



17550-17600 SILTY SANDSTONE: light to medium gray, common light brown to tan, common white to off white, very fine grained, friable to firm, moderately to well sorted, sub rounded to sub angular, moderately to poorly cemented, visible calcareous fluid fractures, trace disseminated pyrite, possible fracture porosity, common intergranular porosity, trace light brown spotty oil stain

18050-18100 SILTY SANDSTONE: light to medium gray, common light brown to tan, common white to off white, very fine grained, friable to firm, moderately to well sorted, sub rounded to sub angular, moderately to poorly cemented, visible calcareous fluid fractures, trace disseminated pyrite, possible fracture porosity, common intergranular porosity, trace light brown spotty oil stain

18100-18150 SILTY SANDSTONE: light to medium gray, common light brown to tan, common white to off white, very fine grained, friable to firm, well sorted, sub angular, moderately to well cemented, visible calcareous fluid fractures, occasional disseminated pyrite, possible fracture porosity, common intergranular porosity, trace light brown spotty oil stain

18150-18200 SILTY SANDSTONE: light to medium gray, common light brown to tan, common white to off white, very fine grained, friable to firm, well sorted, sub angular, moderately to well cemented, visible calcareous fluid fractures, occasional disseminated pyrite, possible fracture porosity, common intergranular porosity, trace light brown spotty oil stain

18200-18250 SILTY SANDSTONE: light to medium gray, common light brown to tan, common white to off white, very fine grained, friable to firm, well sorted, sub angular, moderately to well cemented, visible calcareous fluid fractures, occasional disseminated pyrite, possible fracture porosity, common intergranular porosity, trace light brown spotty oil stain

18250-18300 SILTY SANDSTONE: light to medium gray, common light brown to tan, common white to off white, very fine grained, friable to firm, well sorted, sub angular, moderately to well cemented, visible calcareous fluid fractures, occasional disseminated pyrite, possible fracture porosity, common intergranular porosity, trace light brown spotty oil stain

18300-18350 SILTY SANDSTONE: light to medium gray, common light brown to tan, common white to off white, very fine grained, friable to firm, well sorted, sub angular, moderately to well cemented, visible calcareous fluid fractures, occasional disseminated pyrite, possible fracture porosity, common intergranular porosity, trace light brown spotty oil stain

18350-18400 SILTY SANDSTONE: light to medium gray, common light brown to tan, common white to off white, very fine grained, friable to firm, well sorted, sub angular, moderately to well cemented, visible calcareous fluid fractures, occasional disseminated pyrite, possible fracture porosity, common intergranular porosity, trace light brown spotty oil stain

18400-18450 SILTY SANDSTONE: light to medium gray, common light brown to tan, common white to off white, very fine grained, friable to firm, well sorted, sub angular, moderately to well cemented, visible calcareous fluid fractures, occasional disseminated pyrite, possible fracture porosity, common intergranular porosity, trace light brown spotty oil stain

18450-18500 SILTY SANDSTONE: light to medium gray, common light brown to tan, common white to off white, very fine grained, friable to firm, well sorted, sub angular, moderately to well cemented, visible calcareous fluid fractures, occasional disseminated pyrite, possible fracture porosity, common intergranular porosity, trace light brown spotty oil stain

18500-18550 SILTY SANDSTONE: light to medium gray, common light brown to tan, common white to off white, very fine grained, friable to firm, well sorted, sub angular, moderately to well cemented, visible calcareous fluid fractures, occasional disseminated pyrite, possible fracture porosity, common intergranular porosity, trace light brown spotty oil stain

18550-18600 SILTY SANDSTONE: light to medium gray, common light brown to tan, common white to off white, very fine grained, friable to firm, well sorted, sub angular, moderately to well cemented, visible calcareous fluid fractures, occasional disseminated pyrite, possible fracture porosity, common intergranular porosity, trace light brown spotty oil stain

18600-18650 SILTY SANDSTONE: light to medium gray, common light brown to tan, common white to off white, very fine grained, friable to firm, well sorted, sub angular, moderately to well cemented, visible calcareous fluid fractures, occasional disseminated pyrite, possible fracture porosity, common intergranular porosity, trace light brown spotty oil stain

18650-18700 SILTY SANDSTONE: light to medium gray, common light brown to tan, common white to off white, very fine grained, friable to firm, well sorted, sub angular, moderately to well cemented, visible calcareous fluid fractures, occasional disseminated pyrite, possible fracture porosity, common intergranular porosity, trace light brown spotty oil stain

18700-18750 SILTY SANDSTONE: light to medium gray, common light brown to tan, common white to off white, very fine grained, friable to firm, well sorted, sub angular, moderately to well cemented, visible calcareous fluid fractures, occasional disseminated pyrite, possible fracture porosity, common intergranular porosity, trace light brown spotty oil stain

18750-18800 SILTY SANDSTONE: light to medium gray, common light brown to tan, common white to off white, very fine grained, friable to firm, well sorted, sub angular, moderately to well cemented, visible calcareous fluid fractures, occasional disseminated pyrite, possible fracture porosity, common intergranular porosity, trace light brown spotty oil stain

18800-18850 SILTY SANDSTONE: light to medium gray, common light brown to tan, common white to off white, very fine grained, friable to firm, moderately to well sorted, sub angular, moderately to well cemented, visible calcareous fluid fractures, common disseminated pyrite, possible fracture porosity, common intergranular porosity, trace light brown spotty oil stain

18850-18900 SILTY SANDSTONE: light to medium gray, common light brown to tan, common white to off white, very fine grained, friable to firm, moderately to well sorted, sub angular, moderately to well cemented, visible calcareous fluid fractures, common disseminated pyrite, possible fracture porosity, common intergranular porosity, trace light brown spotty oil stain

18900-18950 SILTY SANDSTONE: light to medium gray, common light brown to tan, common white to off white, very fine grained, friable to firm, moderately to well sorted, sub angular, moderately to well cemented, visible calcareous fluid fractures, common disseminated pyrite, possible fracture porosity, common intergranular porosity, trace light brown spotty oil stain

18950-19000 SILTY SANDSTONE: light to medium gray, common light brown to tan, common white to off white, very fine grained, friable to firm, moderately to well sorted, sub angular to angular, moderately to well cemented, visible calcareous fluid fractures, common disseminated pyrite, possible fracture porosity, common intergranular porosity, trace light brown spotty oil stain

19000-19050 SILTY SANDSTONE: light to medium gray, common light brown to tan, common white to off white, very fine grained, friable to firm, moderately to well sorted, sub angular to angular, moderately to well cemented, visible calcareous fluid fractures, common disseminated pyrite, possible fracture porosity, common intergranular porosity, trace light brown spotty oil stain

19050-19100 SILTY SANDSTONE: light to medium gray, common light brown to tan, common white to off white, very fine grained, friable to firm, moderately to well sorted, sub angular to angular, moderately to well cemented, visible calcareous fluid fractures, common disseminated pyrite, possible fracture porosity, common intergranular porosity, trace light brown spotty oil stain

19550-19600 SILTY SANDSTONE: light to medium gray, common white to off white, occasional light brown to tan very fine grained, friable to firm, moderately to well sorted, sub angular to angular, moderately to well cemented, visible calcareous fluid fractures, common disseminated pyrite, possible fracture porosity, common intergranular porosity, trace light brown spotty oil stain

19600-19650 SILTY SANDSTONE: light to medium gray, common white to off white, occasional light brown to tan very fine grained, friable to firm, moderately to well sorted, sub angular to angular, moderately to well cemented, visible calcareous fluid fractures, common disseminated pyrite, possible fracture porosity, common intergranular porosity, trace light brown spotty oil stain

19650-19700 SILTY SANDSTONE: light to medium gray, common white to off white, occasional light brown to tan very fine grained, friable to firm, moderately to well sorted, sub angular to angular, moderately to well cemented, visible calcareous fluid fractures, common disseminated pyrite, possible fracture porosity, common intergranular porosity, trace light brown spotty oil stain

19700-19750 SILTY SANDSTONE: light to medium gray, common white to off white, occasional light brown to tan very fine grained, friable to firm, moderately to well sorted, sub angular to angular, moderately to well cemented, visible calcareous fluid fractures, common disseminated pyrite, possible fracture porosity, common intergranular porosity, trace light brown spotty oil stain

19750-19800 SILTY SANDSTONE: light to medium gray, common white to off white, occasional light brown to tan very fine grained, friable to firm, moderately to well sorted, sub angular to angular, moderately to well cemented, visible calcareous fluid fractures, common disseminated pyrite, possible fracture porosity, common intergranular porosity, trace light brown spotty oil stain

19800-19850 SILTY SANDSTONE: light to medium gray, common white to off white, occasional light brown to tan very fine grained, friable to firm, moderately to well sorted, sub angular to angular, moderately to well cemented, visible calcareous fluid fractures, common disseminated pyrite, possible fracture porosity, common intergranular porosity, trace light brown spotty oil stain

19850-19900 SILTY SANDSTONE: light to medium gray, common white to off white, occasional light brown to tan very fine grained, friable to firm, moderately to well sorted, sub angular, moderately to well cemented, visible calcareous fluid fractures, common disseminated pyrite, possible fracture porosity, common intergranular porosity, trace light brown spotty oil stain

19900-19950 SILTY SANDSTONE: light to medium gray, common white to off white, occasional light brown to tan very fine grained, friable to firm, moderately to well sorted, sub angular, moderately to well cemented, visible calcareous fluid fractures, common disseminated pyrite, possible fracture porosity, common intergranular porosity, trace light brown spotty oil stain

19950-20000 SILTY SANDSTONE: light to medium gray, common white to off white, occasional light brown to tan very fine grained, friable to firm, moderately to well sorted, sub angular, moderately to well cemented, visible calcareous fluid fractures, common disseminated pyrite, possible fracture porosity, common intergranular porosity, trace light brown spotty oil stain

20000-20050 SILTY SANDSTONE: light to medium gray, common white to off white, occasional light brown to tan very fine grained, friable to firm, moderately to well sorted, sub angular to angular, moderately to well cemented, visible calcareous fluid fractures, common disseminated pyrite, possible fracture porosity, common intergranular porosity, trace light brown spotty oil stain

20050-20100 SILTY SANDSTONE: light to medium gray, common white to off white, occasional light brown to tan very fine grained, friable to firm, moderately to well sorted, sub angular to angular, moderately to well cemented, visible calcareous fluid fractures, common disseminated pyrite, possible fracture porosity, common intergranular porosity, trace light brown spotty oil stain

20100-20150 SILTY SANDSTONE: light to medium gray, common white to off white, occasional light brown to tan very fine grained, friable to firm, moderately to well sorted, sub angular to angular, moderately to well cemented, visible calcareous fluid fractures, common disseminated pyrite, possible fracture porosity, common intergranular porosity, trace light brown spotty oil stain

20150-20200 SILTY SANDSTONE: light to medium gray, common white to off white, occasional light brown to tan very fine grained, friable to firm, moderately to well sorted, sub angular to angular, moderately to well cemented, visible calcareous fluid fractures, common disseminated pyrite, possible fracture porosity, common intergranular porosity, trace light brown spotty oil stain

20200-20250 SILTY SANDSTONE: light to medium gray, common white to off white, occasional light brown to tan very fine grained, friable to firm, moderately to well sorted, sub angular to angular, moderately to well cemented, visible calcareous fluid fractures, common disseminated pyrite, possible fracture porosity, common intergranular porosity, trace light brown spotty oil stain

20250-20300 SILTY SANDSTONE: light to medium gray, common white to off white, occasional light brown to tan very fine grained, friable to firm, moderately to well sorted, sub angular to angular, moderately to well cemented, visible calcareous fluid fractures, common disseminated pyrite, possible fracture porosity, common intergranular porosity, trace light brown spotty oil stain

20300-20350 SILTY SANDSTONE: light to medium gray, common white to off white, occasional light brown to tan very fine grained, friable to firm, moderately to well sorted, sub angular to angular, moderately to well cemented, visible calcareous fluid fractures, common disseminated pyrite, possible fracture porosity, common intergranular porosity, trace light brown spotty oil stain

20350-20400 SILTY SANDSTONE: light to medium gray, common white to off white, occasional light brown to tan very fine grained, friable to firm, moderately to well sorted, sub angular to angular, moderately to well cemented, visible calcareous fluid fractures, common disseminated pyrite, possible fracture porosity, common intergranular porosity, trace light brown spotty oil stain

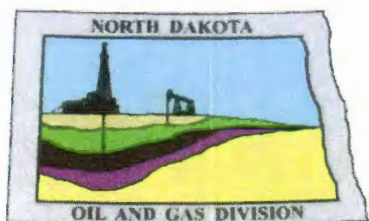
20400-20450 SILTY SANDSTONE: light to medium gray, common white to off white, occasional light brown to tan very fine grained, friable to firm, moderately to well sorted, sub angular to angular, moderately to well cemented, visible calcareous fluid fractures, common disseminated pyrite, possible fracture porosity, common intergranular porosity, trace light brown spotty oil stain

20450-20475 SILTY SANDSTONE: light to medium gray, common white to off white, occasional light brown to tan very fine grained, friable to firm, moderately to well sorted, sub angular to angular, moderately to well cemented, visible calcareous fluid fractures, common disseminated pyrite, possible fracture porosity, common intergranular porosity, trace light brown spotty oil stain

*20,475' TD reached @ 10:48 hours CST, 31 December 2014*







# Oil and Gas Division

Lynn D. Helms - Director

Bruce E. Hicks - Assistant Director

Department of Mineral Resources

Lynn D. Helms - Director

North Dakota Industrial Commission

www.dmr.nd.gov/oilgas

28652

January 8, 2016

Michael Kukuk  
OASIS PETRO NO AMER  
1001 FANNIN ST. SUITE 1500  
HOUSTON, TX 77002

RE: See Attached List of Wells

*List Reference WF# 28652*

Dear Michael Kukuk:

We have not received the geological reports on the referenced well list. Please submit one paper copy and one digital pdf file of this report for each well.

If you have any questions, please contact Richard Suggs at (701) 328-8020.

Sincerely,

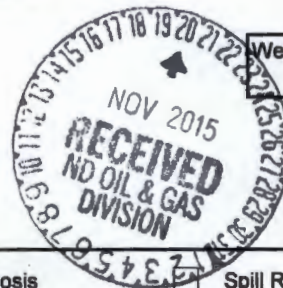
A handwritten signature in blue ink, appearing to read "Taylor Roth".

Taylor Roth  
Engineering Technician



# SUNDRY NOTICES AND REPORTS ON WELLS - FORM 4

INDUSTRIAL COMMISSION OF NORTH DAKOTA  
OIL AND GAS DIVISION  
600 EAST BOULEVARD DEPT 405  
BISMARCK, ND 58505-0840  
SFN 5749 (09-2008)



Well File No.  
**28652**

PLEASE READ INSTRUCTIONS BEFORE FILLING OUT FORM.  
PLEASE SUBMIT THE ORIGINAL AND ONE COPY.

☐ Notice of Intent

Approximate Start Date

☒ Report of Work Done

Date Work Completed  
**November 13, 2015**

☐ Notice of Intent to Begin a Workover Project that may Qualify  
for a Tax Exemption Pursuant to NDCC Section 57-51.1-03.

Approximate Start Date

☐ Drilling Prognosis

☐ Redrilling or Repair

☐ Casing or Liner

☐ Plug Well

☐ Supplemental History

☐ Temporarily Abandon

☐ Other

☐ Spill Report

☐ Shooting

☐ Acidizing

☐ Fracture Treatment

☒ Change Production Method

☐ Reclamation

**Well is now on pump**

Well Name and Number

**Kline Federal 5300 41-18 10B**

Footages

**500 F S L**

**237 F W L**

Qtr-Qtr

**SWSW**

Section

**18**

Township

**153 N**

Range

**100 W**

Field

**Baker**

Pool

**Bakken**

County

**McKenzie**

## 24-HOUR PRODUCTION RATE

| Before |      | After |      |
|--------|------|-------|------|
| Oil    | Bbls | Oil   | Bbls |
| Water  | Bbls | Water | Bbls |
| Gas    | MCF  | Gas   | MCF  |

Name of Contractor(s)

Address

City

State

Zip Code

## DETAILS OF WORK

Effective 11/13/2015 the above referenced well is on pump.

End of Tubing: 2-7/8" L-80 tubing @ 10150'

Pump: ESP @ 10114'

Company  
**Oasis Petroleum North America LLC**

Telephone Number  
**281-404-9436**

Address  
**1001 Fannin, Suite 1500**

City  
**Houston**

State  
**TX**

Zip Code  
**77002**

Signature

Printed Name  
**Jennifer Swenson**

Title  
**Regulatory Specialist**

Date  
**November 19, 2015**

Email Address  
**jswenson@oasispetroleum.com**

## FOR STATE USE ONLY

|                                              |                                   |
|----------------------------------------------|-----------------------------------|
| <input checked="" type="checkbox"/> Received | <input type="checkbox"/> Approved |
| Date                                         | <b>11-30-2015</b>                 |
| By                                           | <b>Original Signed By:</b>        |
|                                              | <b>TAYLOR ROTH</b>                |
| Title                                        | <b>Engineering Technician</b>     |



**WELL COMPLETION OR RECOMPLETION REPORT - FORM**

INDUSTRIAL COMMISSION OF NORTH DAKOTA  
OIL AND GAS DIVISION  
600 EAST BOULEVARD DEPT 405  
BISMARCK, ND 58505-0840  
SFN 2468 (04-2010)



Well File No.  
**28652**

PLEASE READ INSTRUCTIONS BEFORE FILLING OUT FORM.

PLEASE SUBMIT THE ORIGINAL AND ONE COPY.

|                                                             |                                   |                                                                                                                                    |                                        |
|-------------------------------------------------------------|-----------------------------------|------------------------------------------------------------------------------------------------------------------------------------|----------------------------------------|
| Designate Type of Completion                                |                                   |                                                                                                                                    |                                        |
| <input checked="" type="checkbox"/> Oil Well                | <input type="checkbox"/> EOR Well | <input type="checkbox"/> Recompletion                                                                                              | <input type="checkbox"/> Deepened Well |
| <input type="checkbox"/> Gas Well                           | <input type="checkbox"/> SWD Well | <input type="checkbox"/> Water Supply Well                                                                                         | <input type="checkbox"/> Other:        |
| <input type="checkbox"/> Added Horizontal Leg               |                                   | <input type="checkbox"/> Extended Horizontal Leg                                                                                   |                                        |
| Well Name and Number<br><b>Kline Federal 5300 41-18 10B</b> |                                   | Spacing Unit Description<br><b>Sec. 17/18 T153N R100W</b>                                                                          |                                        |
| Operator<br><b>Oasis Petroleum North America</b>            |                                   | Telephone Number<br><b>(281) 404-9591</b>                                                                                          |                                        |
| Address<br><b>1001 Fannin, Suite 1500</b>                   |                                   | Field<br><b>Baker</b>                                                                                                              |                                        |
| City<br><b>Houston</b>                                      |                                   | Pool<br><b>Bakken</b>                                                                                                              |                                        |
| State<br><b>TX</b>                                          | Zip Code<br><b>77002</b>          | Permit Type<br><input type="checkbox"/> Wildcat <input checked="" type="checkbox"/> Development <input type="checkbox"/> Extension |                                        |

**LOCATION OF WELL**

|                                                                                                                     |                                           |                                                         |                      |                                  |                                      |                           |
|---------------------------------------------------------------------------------------------------------------------|-------------------------------------------|---------------------------------------------------------|----------------------|----------------------------------|--------------------------------------|---------------------------|
| At Surface<br><b>500 F S L</b>                                                                                      | <b>237 F W L</b>                          | Qtr-Qtr<br><b>SWSW</b>                                  | Section<br><b>18</b> | Township<br><b>153 N</b>         | Range<br><b>100 W</b>                | County<br><b>McKenzie</b> |
| Spud Date<br><b>September 8, 2014</b>                                                                               | Date TD Reached<br><b>January 1, 2015</b> | Drilling Contractor and Rig Number<br><b>Nabors 486</b> |                      | KB Elevation (Ft)<br><b>2082</b> | Graded Elevation (Ft)<br><b>2057</b> |                           |
| Type of Electric and Other Logs Run (See Instructions)<br><b>MWD/GR FROM KOP TO TD; CBL FROM INT. TD TO SURFACE</b> |                                           |                                                         |                      |                                  |                                      |                           |

**CASING & TUBULARS RECORD (Report all strings set in well)**

| Well Bore     | Type         | String Size (Inch) | Top Set (MD Ft) | Depth Set (MD Ft) | Hole Size (Inch) | Weight (Lbs/Ft) | Anchor Set (MD Ft) | Packer Set (MD Ft) | Sacks Cement | Top of Cement |
|---------------|--------------|--------------------|-----------------|-------------------|------------------|-----------------|--------------------|--------------------|--------------|---------------|
| Surface Hole  | Surface      | 13 3/8             | 0               | 2125              | 17 1/2           | 54.5            |                    |                    | 950          | 0             |
| Vertical Hole | Intermediate | 9 5/8              | 0               | 6125              | 13 1/2           | 36              |                    |                    | 1204         | 2400          |
| Vertical Hole | Intermediate | 7                  | 0               | 11125             | 8 3/4            | 32              |                    |                    | 842          |               |
| Lateral1      | Liner        | 4 1/2              | 10190           | 20460             | 6                | 13.5            |                    |                    | 510          | 10190         |
|               |              |                    |                 |                   |                  |                 |                    |                    |              |               |
|               |              |                    |                 |                   |                  |                 |                    |                    |              |               |
|               |              |                    |                 |                   |                  |                 |                    |                    |              |               |

**PERFORATION & OPEN HOLE INTERVALS**

| Well Bore | Well Bore TD Drillers Depth (MD Ft) | Completion Type | Open Hole/Perforated Interval (MD,Ft) |       | Kick-off Point (MD Ft) | Top of Casing Window (MD Ft) | Date Perfd or Drilled | Date Isolated | Isolation Method | Sacks Cement |
|-----------|-------------------------------------|-----------------|---------------------------------------|-------|------------------------|------------------------------|-----------------------|---------------|------------------|--------------|
| Lateral1  | 20475                               | Perforations    | 11125                                 | 20460 | 10240                  |                              | 06/26/2015            |               |                  |              |
|           | 15303                               |                 |                                       | 15303 |                        |                              |                       |               |                  |              |
| ST 1      | 18629                               |                 | 15087                                 | 18629 |                        |                              |                       |               |                  |              |
| ST 2      | 20478                               |                 | 17858                                 | 20460 |                        |                              |                       |               |                  |              |
|           |                                     |                 |                                       |       |                        |                              |                       |               |                  |              |
|           |                                     |                 |                                       |       |                        |                              |                       |               |                  |              |
|           |                                     |                 |                                       |       |                        |                              |                       |               |                  |              |
|           |                                     |                 |                                       |       |                        |                              |                       |               |                  |              |

**PRODUCTION**

|                                                                                                                                       |                           |                                              |                                    |                           |                             |                             |                                                        |                             |  |
|---------------------------------------------------------------------------------------------------------------------------------------|---------------------------|----------------------------------------------|------------------------------------|---------------------------|-----------------------------|-----------------------------|--------------------------------------------------------|-----------------------------|--|
| Current Producing Open Hole or Perforated Interval(s), This Completion, Top and Bottom, (MD Ft)<br><b>Lateral 1- 11125' to 20460'</b> |                           |                                              |                                    |                           |                             |                             | Name of Zone (If Different from Pool Name)             |                             |  |
| Date Well Completed (SEE INSTRUCTIONS)<br><b>September 4, 2015</b>                                                                    |                           |                                              | Producing Method<br><b>Flowing</b> |                           | Pumping-Size & Type of Pump |                             | Well Status (Producing or Shut-In)<br><b>Producing</b> |                             |  |
| Date of Test<br><b>09/05/2015</b>                                                                                                     | Hours Tested<br><b>24</b> | Choke Size<br><b>34 /64</b>                  | Production for Test                | Oil (Bbls)<br><b>1917</b> | Gas (MCF)<br><b>1528</b>    | Water (Bbls)<br><b>3763</b> | Oil Gravity-API (Corr.)<br><b>42.0 °</b>               | Disposition of Gas          |  |
| Flowing Tubing Pressure (PSI)                                                                                                         |                           | Flowing Casing Pressure (PSI)<br><b>1850</b> |                                    | Calculated 24-Hour Rate   | Oil (Bbls)<br><b>1917</b>   | Gas (MCF)<br><b>1528</b>    | Water (Bbls)<br><b>3763</b>                            | Gas-Oil Ratio<br><b>797</b> |  |

## PLUG BACK INFORMATION

[illegible][illegible]

| Top (Ft) | Bottom (Ft) | Formation | Top (Ft) | Bottom (Ft) | Formation |
|----------|-------------|-----------|----------|-------------|-----------|
|          |             |           |          |             |           |
|          |             |           |          |             |           |
|          |             |           |          |             |           |

| Test Date               | Formation | Top (Ft) | Bottom (Ft) | BH Temp (°F) | CL ppm | H2S ppm | Shut-in 1 (PSIG) | Shut-in 2 (PSIG) |
|-------------------------|-----------|----------|-------------|--------------|--------|---------|------------------|------------------|
| Drill Pipe Recovery     |           |          |             |              |        |         |                  |                  |
| Sample Chamber Recovery |           |          |             |              |        |         |                  |                  |
| Test Date               | Formation | Top (Ft) | Bottom (Ft) | BH Temp (°F) | CL ppm | H2S ppm | Shut-in 1 (PSIG) | Shut-in 2 (PSIG) |
| Drill Pipe Recovery     |           |          |             |              |        |         |                  |                  |
| Sample Chamber Recovery |           |          |             |              |        |         |                  |                  |
| Test Date               | Formation | Top (Ft) | Bottom (Ft) | BH Temp (°F) | CL ppm | H2S ppm | Shut-in 1 (PSIG) | Shut-in 2 (PSIG) |
| Drill Pipe Recovery     |           |          |             |              |        |         |                  |                  |
| Sample Chamber Recovery |           |          |             |              |        |         |                  |                  |
| Test Date               | Formation | Top (Ft) | Bottom (Ft) | BH Temp (°F) | CL ppm | H2S ppm | Shut-in 1 (PSIG) | Shut-in 2 (PSIG) |
| Drill Pipe Recovery     |           |          |             |              |        |         |                  |                  |
| Sample Chamber Recovery |           |          |             |              |        |         |                  |                  |
| Test Date               | Formation | Top (Ft) | Bottom (Ft) | BH Temp (°F) | CL ppm | H2S ppm | Shut-in 1 (PSIG) | Shut-in 2 (PSIG) |
| Drill Pipe Recovery     |           |          |             |              |        |         |                  |                  |
| Sample Chamber Recovery |           |          |             |              |        |         |                  |                  |
| Test Date               | Formation | Top (Ft) | Bottom (Ft) | BH Temp (°F) | CL ppm | H2S ppm | Shut-in 1 (PSIG) | Shut-in 2 (PSIG) |
| Drill Pipe Recovery     |           |          |             |              |        |         |                  |                  |
| Sample Chamber Recovery |           |          |             |              |        |         |                  |                  |



### Well Specific Stimulations

|                                                                                                                                        |                                       |                                |                                                 |                                 |                                                  |                                |
|----------------------------------------------------------------------------------------------------------------------------------------|---------------------------------------|--------------------------------|-------------------------------------------------|---------------------------------|--------------------------------------------------|--------------------------------|
| Date Stimulated<br><b>06/26/2015</b>                                                                                                   | Stimulated Formation<br><b>Bakken</b> | Top (Ft)<br><b>11125</b>       | Bottom (Ft)<br><b>20460</b>                     | Stimulation Stages<br><b>50</b> | Volume<br><b>192526</b>                          | Volume Units<br><b>Barrels</b> |
| Type Treatment<br><b>Sand Frac</b>                                                                                                     | Acid %                                | Lbs Proppant<br><b>9254340</b> | Maximum Treatment Pressure (PSI)<br><b>8844</b> |                                 | Maximum Treatment Rate (BBLS/Min)<br><b>39.0</b> |                                |
| Details<br><b>100 Mesh: 229320</b><br><b>40/70 White: 1221920</b><br><b>20/40 White: 6435200</b><br><b>20/40 Resin Coated: 1367900</b> |                                       |                                |                                                 |                                 |                                                  |                                |

|                 |                      |              |                                  |                    |                                   |              |
|-----------------|----------------------|--------------|----------------------------------|--------------------|-----------------------------------|--------------|
| Date Stimulated | Stimulated Formation | Top (Ft)     | Bottom (Ft)                      | Stimulation Stages | Volume                            | Volume Units |
| Type Treatment  | Acid %               | Lbs Proppant | Maximum Treatment Pressure (PSI) |                    | Maximum Treatment Rate (BBLS/Min) |              |
| Details         |                      |              |                                  |                    |                                   |              |

|                 |                      |              |                                  |                    |                                   |              |
|-----------------|----------------------|--------------|----------------------------------|--------------------|-----------------------------------|--------------|
| Date Stimulated | Stimulated Formation | Top (Ft)     | Bottom (Ft)                      | Stimulation Stages | Volume                            | Volume Units |
| Type Treatment  | Acid %               | Lbs Proppant | Maximum Treatment Pressure (PSI) |                    | Maximum Treatment Rate (BBLS/Min) |              |
| Details         |                      |              |                                  |                    |                                   |              |

|                 |                      |              |                                  |                    |                                   |              |
|-----------------|----------------------|--------------|----------------------------------|--------------------|-----------------------------------|--------------|
| Date Stimulated | Stimulated Formation | Top (Ft)     | Bottom (Ft)                      | Stimulation Stages | Volume                            | Volume Units |
| Type Treatment  | Acid %               | Lbs Proppant | Maximum Treatment Pressure (PSI) |                    | Maximum Treatment Rate (BBLS/Min) |              |
| Details         |                      |              |                                  |                    |                                   |              |

|                 |                      |              |                                  |                    |                                   |              |
|-----------------|----------------------|--------------|----------------------------------|--------------------|-----------------------------------|--------------|
| Date Stimulated | Stimulated Formation | Top (Ft)     | Bottom (Ft)                      | Stimulation Stages | Volume                            | Volume Units |
| Type Treatment  | Acid %               | Lbs Proppant | Maximum Treatment Pressure (PSI) |                    | Maximum Treatment Rate (BBLS/Min) |              |
| Details         |                      |              |                                  |                    |                                   |              |

### ADDITIONAL INFORMATION AND/OR LIST OF ATTACHMENTS

|                                                                                                                                |                                                     |                                       |
|--------------------------------------------------------------------------------------------------------------------------------|-----------------------------------------------------|---------------------------------------|
|                                                                                                                                |                                                     |                                       |
| I hereby swear or affirm that the information provided is true, complete and correct as determined from all available records. | Email Address<br><b>jswenson@oasispetroleum.com</b> | Date<br><b>10/05/2015</b>             |
| Signature<br>                               | Printed Name<br><b>Jennifer Swenson</b>             | Title<br><b>Regulatory Specialist</b> |



**AUTHORIZATION TO PURCHASE AND TRANSPORT OIL FROM LEASE - Form 8**  
INDUSTRIAL COMMISSION OF NORTH DAKOTA  
OIL AND GAS DIVISION  
600 EAST BOULEVARD DEPT 405  
BISMARCK, ND 58505-0840  
SFN 5698 (03-2000)



|                |
|----------------|
| Well File No.  |
| 28652          |
| NDIC CTB No.   |
| To be assigned |

128652

PLEASE READ INSTRUCTIONS BEFORE FILLING OUT FORM.  
PLEASE SUBMIT THE ORIGINAL AND FOUR COPIES.

|                                                             |                 |               |                 |              |                    |
|-------------------------------------------------------------|-----------------|---------------|-----------------|--------------|--------------------|
| Well Name and Number<br><b>KLINE FEDERAL 5300 41-18 10B</b> | Qtr-Qtr<br>LOT4 | Section<br>18 | Township<br>153 | Range<br>100 | County<br>Mckenzie |
|-------------------------------------------------------------|-----------------|---------------|-----------------|--------------|--------------------|

|                                                      |                                    |                |
|------------------------------------------------------|------------------------------------|----------------|
| Operator<br><b>Oasis Petroleum North America LLC</b> | Telephone Number<br>(281) 404-9573 | Field<br>BAKER |
|------------------------------------------------------|------------------------------------|----------------|

|                                    |                 |             |                   |
|------------------------------------|-----------------|-------------|-------------------|
| Address<br>1001 Fannin, Suite 1500 | City<br>Houston | State<br>TX | Zip Code<br>77002 |
|------------------------------------|-----------------|-------------|-------------------|

|                                                                 |                                    |                     |                                     |
|-----------------------------------------------------------------|------------------------------------|---------------------|-------------------------------------|
| Name of First Purchaser<br><b>Oasis Petroleum Marketing LLC</b> | Telephone Number<br>(281) 404-9627 | % Purchased<br>100% | Date Effective<br>September 1, 2015 |
|-----------------------------------------------------------------|------------------------------------|---------------------|-------------------------------------|

|                                                        |                 |             |                   |
|--------------------------------------------------------|-----------------|-------------|-------------------|
| Principal Place of Business<br>1001 Fannin, Suite 1500 | City<br>Houston | State<br>TX | Zip Code<br>77002 |
|--------------------------------------------------------|-----------------|-------------|-------------------|

|               |      |       |          |
|---------------|------|-------|----------|
| Field Address | City | State | Zip Code |
|---------------|------|-------|----------|

|                                         |                                    |                      |                                     |
|-----------------------------------------|------------------------------------|----------------------|-------------------------------------|
| Transporter<br><b>Hiland Crude, LLC</b> | Telephone Number<br>(580) 616-2058 | % Transported<br>75% | Date Effective<br>September 1, 2015 |
|-----------------------------------------|------------------------------------|----------------------|-------------------------------------|

|                          |              |             |                   |
|--------------------------|--------------|-------------|-------------------|
| Address<br>P.O. Box 3886 | City<br>Enid | State<br>OK | Zip Code<br>73702 |
|--------------------------|--------------|-------------|-------------------|

The above named producer authorizes the above named purchaser to purchase the percentage of oil stated above which is produced from the lease designated above until further notice. The oil will be transported by the above named transporter.

|                                                   |               |                   |
|---------------------------------------------------|---------------|-------------------|
| Other First Purchasers Purchasing From This Lease | % Purchased   | Date Effective    |
| Other First Purchasers Purchasing From This Lease | % Purchased   | Date Effective    |
| Other Transporters Transporting From This Lease   | % Transported | Date Effective    |
| <b>Power Crude Transport</b>                      | 25%           | September 1, 2015 |
| Other Transporters Transporting From This Lease   | % Transported | Date Effective    |
|                                                   |               | September 1, 2015 |
| Comments                                          |               |                   |

|                                                                                                                                |                             |                                        |
|--------------------------------------------------------------------------------------------------------------------------------|-----------------------------|----------------------------------------|
| I hereby swear or affirm that the information provided is true, complete and correct as determined from all available records. |                             | Date<br>September 10, 2015             |
| Signature<br><i>Dina Barron</i>                                                                                                | Printed Name<br>Dina Barron | Title<br>Mktg. Contracts Administrator |

|                                   |                               |                              |
|-----------------------------------|-------------------------------|------------------------------|
| Above Signature Witnessed By:     |                               |                              |
| Signature<br><i>Jeremy Harris</i> | Printed Name<br>Jeremy Harris | Title<br>Marketing Scheduler |

FOR STATE USE ONLY

|                                       |
|---------------------------------------|
| Date Approved<br>SEP 18 2015          |
| By<br><i>Ernie Robinson</i>           |
| Title<br>Oil & Gas Production Analyst |



Industrial Commission of North Dakota  
Oil and Gas Division

Well or Facility No  
**28652**

Verbal Approval To Purchase and Transport Oil      Tight Hole      **Yes**

**OPERATOR**

|                                         |                    |                       |
|-----------------------------------------|--------------------|-----------------------|
| Operator                                | Representative     | Rep Phone             |
| <b>OASIS PETROLEUM NORTH AMERICA LL</b> | <b>Todd Hanson</b> | <b>(701) 577-1632</b> |

**WELL INFORMATION**

|                                                     |  |      |                 |       |       |                             |  |
|-----------------------------------------------------|--|------|-----------------|-------|-------|-----------------------------|--|
| Well Name                                           |  |      |                 |       |       | Inspector                   |  |
| KLINE FEDERAL 5300 41-18 10B                        |  |      |                 |       |       | Richard Dunn                |  |
| Well Location                                       |  | QQ   | Sec             | Twp   | Rng   | County                      |  |
|                                                     |  | LOT4 | 18              | 153 N | 100 W | MCKENZIE                    |  |
|                                                     |  |      |                 |       |       | Field                       |  |
|                                                     |  | 500  | Feet From the S |       | Line  | BAKER                       |  |
| Footages                                            |  | 237  | Feet From the W |       | Line  | Pool                        |  |
|                                                     |  |      |                 |       |       | BAKKEN                      |  |
| Date of First Production Through Permanent Wellhead |  |      |                 |       |       | This Is Not The First Sales |  |

**PURCHASER / TRANSPORTER**

|                      |                      |
|----------------------|----------------------|
| Purchaser            | Transporter          |
| <b>Kinder Morgan</b> | <b>Kinder Morgan</b> |

**TANK BATTERY**

|                                   |
|-----------------------------------|
| Single Well Tank Battery Number : |
|-----------------------------------|

**SALES INFORMATION**      This Is Not The First Sales

| ESTIMATED BARRELS TO BE SOLD |      | ACTUAL BARRELS SOLD | DATE |
|------------------------------|------|---------------------|------|
| <b>15000</b>                 | BBLS | BBLS                |      |
|                              | BBLS | BBLS                |      |
|                              | BBLS | BBLS                |      |
|                              | BBLS | BBLS                |      |
|                              | BBLS | BBLS                |      |
|                              | BBLS | BBLS                |      |
|                              | BBLS | BBLS                |      |
|                              | BBLS | BBLS                |      |
|                              | BBLS | BBLS                |      |

**DETAILS**

|                                                                                                                                                                                                                                                                                                                                 |
|---------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|
| Must E-Mail or Call Inspector at 701-770-3554/rsdunn@nd.gov on first date of sales and report amount sold, date sold, and first date of production through the permanent wellhead. Must also forward Forms 6 & 8 to State prior to reaching 15000 Bbl estimate or no later than required time frame for submitting those forms. |
|---------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|

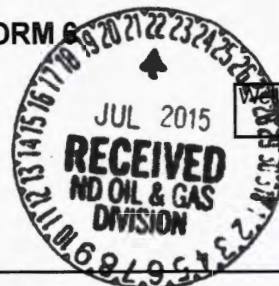
Start Date      **8/25/2015**  
Date Approved      **8/25/2015**  
Approved By      **Richard Dunn**





# WELL COMPLETION OR RECOMPLETION REPORT - FORM 6

INDUSTRIAL COMMISSION OF NORTH DAKOTA  
OIL AND GAS DIVISION  
600 EAST BOULEVARD DEPT 405  
BISMARCK, ND 58505-0840  
SFN 2468 (04-2010)



File No. **28652**

PLEASE READ INSTRUCTIONS BEFORE FILLING OUT FORM.

PLEASE SUBMIT THE ORIGINAL AND ONE COPY.

|                                                             |                                   |                                                           |                                                                                                                                    |
|-------------------------------------------------------------|-----------------------------------|-----------------------------------------------------------|------------------------------------------------------------------------------------------------------------------------------------|
| Designate Type of Completion                                |                                   |                                                           |                                                                                                                                    |
| <input checked="" type="checkbox"/> Oil Well                | <input type="checkbox"/> EOR Well | <input type="checkbox"/> Recompletion                     | <input type="checkbox"/> Deepened Well                                                                                             |
| <input type="checkbox"/> Gas Well                           | <input type="checkbox"/> SWD Well | <input type="checkbox"/> Water Supply Well                | <input type="checkbox"/> Other:                                                                                                    |
| Well Name and Number<br><b>Kline Federal 5300 41-18 10B</b> |                                   | Spacing Unit Description<br><b>Sec. 17/18 T153N R100W</b> |                                                                                                                                    |
| Operator<br><b>Oasis Petroleum North America</b>            |                                   | Telephone Number<br><b>(281) 404-9591</b>                 | Field<br><b>Baker</b>                                                                                                              |
| Address<br><b>1001 Fannin, Suite 1500</b>                   |                                   | Pool<br><b>Bakken</b>                                     |                                                                                                                                    |
| City<br><b>Houston</b>                                      | State<br><b>TX</b>                | Zip Code<br><b>77002</b>                                  | Permit Type<br><input type="checkbox"/> Wildcat <input checked="" type="checkbox"/> Development <input type="checkbox"/> Extension |

## LOCATION OF WELL

|                                                                                                                     |                                           |                                                         |                      |                                  |                                      |                           |
|---------------------------------------------------------------------------------------------------------------------|-------------------------------------------|---------------------------------------------------------|----------------------|----------------------------------|--------------------------------------|---------------------------|
| At Surface<br><b>500 F S L</b>                                                                                      | <b>237 F WL</b>                           | Qtr-Qtr<br><b>SWSW</b>                                  | Section<br><b>18</b> | Township<br><b>153 N</b>         | Range<br><b>100 W</b>                | County<br><b>McKenzie</b> |
| Spud Date<br><b>September 8, 2014</b>                                                                               | Date TD Reached<br><b>January 1, 2015</b> | Drilling Contractor and Rig Number<br><b>Nabors 486</b> |                      | KB Elevation (Ft)<br><b>2082</b> | Graded Elevation (Ft)<br><b>2057</b> |                           |
| Type of Electric and Other Logs Run (See Instructions)<br><b>MWD/GR FROM KOP TO TD; CBL FROM INT. TD TO SURFACE</b> |                                           |                                                         |                      |                                  |                                      |                           |

## CASING & TUBULARS RECORD (Report all strings set in well)

| Well Bore     | String Type  | Size (Inch) | Top Set (MD Ft) | Depth Set (MD Ft) | Hole Size (Inch) | Weight (Lbs/Ft) | Anchor Set (MD Ft) | Packer Set (MD Ft) | Sacks Cement | Top of Cement |
|---------------|--------------|-------------|-----------------|-------------------|------------------|-----------------|--------------------|--------------------|--------------|---------------|
| Surface Hole  | Surface      | 13 3/8      | 0               | 2125              | 17 1/2           | 54.5            |                    |                    | 950          | 0             |
| Vertical Hole | Intermediate | 9 5/8       | 0               | 6125              | 13 1/2           | 36              |                    |                    | 1204         | 2400          |
| Vertical Hole | Intermediate | 7           | 0               | 11125             | 8 3/4            | 32              |                    |                    | 842          |               |
| Lateral1      | Liner        | 4 1/2       | 10190           | 20460             | 6                | 13.5            |                    |                    | 510          | 10190         |
|               |              |             |                 |                   |                  |                 |                    |                    |              |               |
|               |              |             |                 |                   |                  |                 |                    |                    |              |               |
|               |              |             |                 |                   |                  |                 |                    |                    |              |               |

## PERFORATION & OPEN HOLE INTERVALS

| Well Bore | Well Bore TD Drillers Depth (MD Ft) | Completion Type | Open Hole/Perforated Interval (MD,Ft)<br>Top Bottom | Kick-off Point (MD Ft) | Top of Casing Window (MD Ft) | Date Perf'd or Drilled | Date Isolated | Isolation Method | Sacks Cement |
|-----------|-------------------------------------|-----------------|-----------------------------------------------------|------------------------|------------------------------|------------------------|---------------|------------------|--------------|
| Lateral1  | 20475                               | Perforations    |                                                     | 10240                  |                              |                        |               |                  |              |
|           |                                     |                 |                                                     |                        |                              |                        |               |                  |              |
|           |                                     |                 |                                                     |                        |                              |                        |               |                  |              |
|           |                                     |                 |                                                     |                        |                              |                        |               |                  |              |
|           |                                     |                 |                                                     |                        |                              |                        |               |                  |              |
|           |                                     |                 |                                                     |                        |                              |                        |               |                  |              |
|           |                                     |                 |                                                     |                        |                              |                        |               |                  |              |
|           |                                     |                 |                                                     |                        |                              |                        |               |                  |              |

## PRODUCTION

|                                                                                                                      |              |                               |                     |                             |            |                                            |                         |                    |  |
|----------------------------------------------------------------------------------------------------------------------|--------------|-------------------------------|---------------------|-----------------------------|------------|--------------------------------------------|-------------------------|--------------------|--|
| Current Producing Open Hole or Perforated Interval(s), This Completion, Top and Bottom, (MD Ft)<br><b>Lateral 1-</b> |              |                               |                     |                             |            | Name of Zone (If Different from Pool Name) |                         |                    |  |
| Date Well Completed (SEE INSTRUCTIONS)                                                                               |              | Producing Method              |                     | Pumping-Size & Type of Pump |            | Well Status (Producing or Shut-In)         |                         |                    |  |
| Date of Test                                                                                                         | Hours Tested | Choke Size<br><b>164</b>      | Production for Test | Oil (Bbls)                  | Gas (MCF)  | Water (Bbls)                               | Oil Gravity-API (Corr.) | Disposition of Gas |  |
| Flowing Tubing Pressure (PSI)                                                                                        |              | Flowing Casing Pressure (PSI) |                     | Calculated 24-Hour Rate     | Oil (Bbls) | Gas (MCF)                                  | Water (Bbls)            | Gas-Oil Ratio      |  |

## GEOLOGICAL MARKERS

[illegible]

## PLUG BACK INFORMATION

[illegible]

## CORES CUT

| Top (Ft) | Bottom (Ft) | Formation | Top (Ft) | Bottom (Ft) | Formation |
|----------|-------------|-----------|----------|-------------|-----------|
|          |             |           |          |             |           |
|          |             |           |          |             |           |
|          |             |           |          |             |           |

### Drill Stem Test

| Test Date               | Formation | Top (Ft) | Bottom (Ft) | BH Temp (°F) | CL ppm | H2S ppm | Shut-in 1 (PSIG) | Shut-in 2 (PSIG) |
|-------------------------|-----------|----------|-------------|--------------|--------|---------|------------------|------------------|
| Drill Pipe Recovery     |           |          |             |              |        |         |                  |                  |
| Sample Chamber Recovery |           |          |             |              |        |         |                  |                  |
| Test Date               | Formation | Top (Ft) | Bottom (Ft) | BH Temp (°F) | CL ppm | H2S ppm | Shut-in 1 (PSIG) | Shut-in 2 (PSIG) |
| Drill Pipe Recovery     |           |          |             |              |        |         |                  |                  |
| Sample Chamber Recovery |           |          |             |              |        |         |                  |                  |
| Test Date               | Formation | Top (Ft) | Bottom (Ft) | BH Temp (°F) | CL ppm | H2S ppm | Shut-in 1 (PSIG) | Shut-in 2 (PSIG) |
| Drill Pipe Recovery     |           |          |             |              |        |         |                  |                  |
| Sample Chamber Recovery |           |          |             |              |        |         |                  |                  |
| Test Date               | Formation | Top (Ft) | Bottom (Ft) | BH Temp (°F) | CL ppm | H2S ppm | Shut-in 1 (PSIG) | Shut-in 2 (PSIG) |
| Drill Pipe Recovery     |           |          |             |              |        |         |                  |                  |
| Sample Chamber Recovery |           |          |             |              |        |         |                  |                  |
| Test Date               | Formation | Top (Ft) | Bottom (Ft) | BH Temp (°F) | CL ppm | H2S ppm | Shut-in 1 (PSIG) | Shut-in 2 (PSIG) |
| Drill Pipe Recovery     |           |          |             |              |        |         |                  |                  |
| Sample Chamber Recovery |           |          |             |              |        |         |                  |                  |
| Test Date               | Formation | Top (Ft) | Bottom (Ft) | BH Temp (°F) | CL ppm | H2S ppm | Shut-in 1 (PSIG) | Shut-in 2 (PSIG) |
| Drill Pipe Recovery     |           |          |             |              |        |         |                  |                  |
| Sample Chamber Recovery |           |          |             |              |        |         |                  |                  |



**Well Specific Stimulations**

|                                    |                      |              |                                  |                    |                                   |                                |
|------------------------------------|----------------------|--------------|----------------------------------|--------------------|-----------------------------------|--------------------------------|
| Date Stimulated                    | Stimulated Formation | Top (Ft)     | Bottom (Ft)                      | Stimulation Stages | Volume                            | Volume Units<br><b>Barrels</b> |
| Type Treatment<br><b>Sand Frac</b> | Acid %               | Lbs Proppant | Maximum Treatment Pressure (PSI) |                    | Maximum Treatment Rate (BBLS/Min) |                                |
| Details                            |                      |              |                                  |                    |                                   |                                |
| Date Stimulated                    | Stimulated Formation | Top (Ft)     | Bottom (Ft)                      | Stimulation Stages | Volume                            | Volume Units                   |
| Type Treatment                     | Acid %               | Lbs Proppant | Maximum Treatment Pressure (PSI) |                    | Maximum Treatment Rate (BBLS/Min) |                                |
| Details                            |                      |              |                                  |                    |                                   |                                |
| Date Stimulated                    | Stimulated Formation | Top (Ft)     | Bottom (Ft)                      | Stimulation Stages | Volume                            | Volume Units                   |
| Type Treatment                     | Acid %               | Lbs Proppant | Maximum Treatment Pressure (PSI) |                    | Maximum Treatment Rate (BBLS/Min) |                                |
| Details                            |                      |              |                                  |                    |                                   |                                |
| Date Stimulated                    | Stimulated Formation | Top (Ft)     | Bottom (Ft)                      | Stimulation Stages | Volume                            | Volume Units                   |
| Type Treatment                     | Acid %               | Lbs Proppant | Maximum Treatment Pressure (PSI) |                    | Maximum Treatment Rate (BBLS/Min) |                                |
| Details                            |                      |              |                                  |                    |                                   |                                |
| Date Stimulated                    | Stimulated Formation | Top (Ft)     | Bottom (Ft)                      | Stimulation Stages | Volume                            | Volume Units                   |
| Type Treatment                     | Acid %               | Lbs Proppant | Maximum Treatment Pressure (PSI) |                    | Maximum Treatment Rate (BBLS/Min) |                                |
| Details                            |                      |              |                                  |                    |                                   |                                |

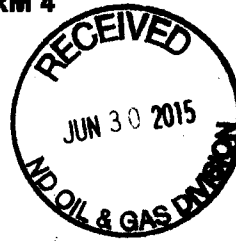
**ADDITIONAL INFORMATION AND/OR LIST OF ATTACHMENTS**

This is a preliminary completion report. A supplemental report will be filed upon first production of the well.

|                                                                                                                                |                             |                       |
|--------------------------------------------------------------------------------------------------------------------------------|-----------------------------|-----------------------|
| I hereby swear or affirm that the information provided is true, complete and correct as determined from all available records. | Email Address               | Date                  |
|                                                                                                                                | jswenson@oasispetroleum.com | 07/21/2015            |
| Signature                                                                                                                      | Printed Name                | Title                 |
|                                             | Jennifer Swenson            | Regulatory Specialist |

**SUNDRY NOTICES AND REPORTS ON WELLS - FORM 4**

INDUSTRIAL COMMISSION OF NORTH DAKOTA  
OIL AND GAS DIVISION  
600 EAST BOULEVARD DEPT 405  
BISMARCK, ND 58505-0840  
SFN 5749 (09-2008)



Well File No.  
**28652**

PLEASE READ INSTRUCTIONS BEFORE FILLING OUT FORM.  
PLEASE SUBMIT THE ORIGINAL AND ONE COPY.

|                                                                                                                                                 |                                                |
|-------------------------------------------------------------------------------------------------------------------------------------------------|------------------------------------------------|
| <input checked="" type="checkbox"/> Notice of Intent                                                                                            | Approximate Start Date<br><b>June 30, 2015</b> |
| <input type="checkbox"/> Report of Work Done                                                                                                    | Date Work Completed                            |
| <input type="checkbox"/> Notice of Intent to Begin a Workover Project that may Qualify for a Tax Exemption Pursuant to NDCC Section 57-51.1-03. | Approximate Start Date                         |

|                                               |                                                   |
|-----------------------------------------------|---------------------------------------------------|
| <input type="checkbox"/> Drilling Prognosis   | <input type="checkbox"/> Spill Report             |
| <input type="checkbox"/> Redrilling or Repair | <input type="checkbox"/> Shooting                 |
| <input type="checkbox"/> Casing or Liner      | <input type="checkbox"/> Acidizing                |
| <input type="checkbox"/> Plug Well            | <input type="checkbox"/> Fracture Treatment       |
| <input type="checkbox"/> Supplemental History | <input type="checkbox"/> Change Production Method |
| <input type="checkbox"/> Temporarily Abandon  | <input type="checkbox"/> Reclamation              |
| <input checked="" type="checkbox"/> Other     | <b>Change well status to CONFIDENTIAL</b>         |

|                                                             |                       |                           |           |              |              |
|-------------------------------------------------------------|-----------------------|---------------------------|-----------|--------------|--------------|
| Well Name and Number<br><b>Kline Federal 5300 41-18 10B</b> |                       |                           |           |              |              |
| Footages                                                    | Qtr-Qtr               | Section                   | Township  | Range        |              |
| <b>500 F S L</b>                                            | <b>237 F W L</b>      | <b>LOT 4</b>              | <b>18</b> | <b>153 N</b> | <b>100 W</b> |
| Field<br><b>Baker</b>                                       | Pool<br><b>Bakken</b> | County<br><b>McKenzie</b> |           |              |              |

| 24-HOUR PRODUCTION RATE |      |       |      |
|-------------------------|------|-------|------|
| Before                  |      | After |      |
| Oil                     | Bbls | Oil   | Bbls |
| Water                   | Bbls | Water | Bbls |
| Gas                     | MCF  | Gas   | MCF  |

|                       |      |       |          |
|-----------------------|------|-------|----------|
| Name of Contractor(s) |      |       |          |
| Address               | City | State | Zip Code |

**DETAILS OF WORK**

Effective immediately, we request **CONFIDENTIAL STATUS** for the above referenced well.

This well has not been completed

*OFF CONFIDENTIAL 12/30/15.*

|                                                     |  |                                         |                          |
|-----------------------------------------------------|--|-----------------------------------------|--------------------------|
| Company<br><b>Oasis Petroleum North America LLC</b> |  | Telephone Number<br><b>281-404-9436</b> |                          |
| Address<br><b>1001 Fannin, Suite 1500</b>           |  |                                         |                          |
| City<br><b>Houston</b>                              |  | State<br><b>TX</b>                      | Zip Code<br><b>77002</b> |
| Signature<br>                                       |  | Printed Name<br><b>Jennifer Swenson</b> |                          |
| Title<br><b>Regulatory Specialist</b>               |  | Date<br><b>June 30, 2015</b>            |                          |
| Email Address<br><b>jswenson@oasispetroleum.com</b> |  |                                         |                          |

| FOR STATE USE ONLY                     |                                              |
|----------------------------------------|----------------------------------------------|
| <input type="checkbox"/> Received      | <input checked="" type="checkbox"/> Approved |
| Date<br><b>7/07/15</b>                 |                                              |
| By<br>                                 |                                              |
| Title<br><b>Engineering Technician</b> |                                              |



Gyrodata Incorporated  
301 Thelma Dr. #413  
Casper, Wyoming 82601

307-472-0182  
Fax: 307-472-0204

### DIRECTIONAL SURVEY CERTIFICATION

RE: Oasis Petroleum (Operator)  
Kline Federal 5300 41-18 10B (Well Name & No.)  
8.19 (Magnetic Declination)

Surveyor: Blake Hunter

I, Kristopher Bertsch, having personal knowledge of all the facts, hereby certify that the attached directional survey run from measured depth of 138 feet to a measured depth of 15212 feet and is projected to 15303 feet, is true and correct as determined from all available records.

  
Signature

MWD Coordinator  
Title

Gyrodata, Inc.  
Company

\*All measurements recorded with a manned MWD Survey Tool.

A Gyrodata Directional Survey

FINAL DEFINITIVE COPY

for

OASIS PETROLEUM

Lease: Kline Federal Well: 5300 41-18 10B, 8 3/4" Case, 6" Hole

Location: Nabors #486, McKenzie County, North Dakota

Job Number: RD0814M177

Run Date: 05 Mar 2015

Surveyor: Blake Hunter

Calculation Method: MINIMUM CURVATURE

Survey Latitude: 48.068944 deg. N Longitude: 103.603319 deg. W

Magnetic Declination: 8.1900 deg

Total Correction: 8.1900 deg

Proposed Well Direction: 90.000 deg

Depth Reference: Rotary Table

Air Gap (RKB to Ground / RKB to MSL): 25.00

Ground Elevation (Ground to MSL / MSL to Mudline): 2057.00

Vertical Section Calculated from Well Head Location

Closure Calculated from Well Head Location

Horizontal Coordinates Calculated from Well Head Location



# A Gyrodata Directional Survey

Oasis Petroleum

Lease: Kline Federal Well: 5300 41-18 10B, 8 3/4" Case, 6" Hole

Location: Nabors #486, McKenzie County, North Dakota

Job Number: RD0814M177

| MEASURED<br>DEPTH<br>feet | INCL<br>deg. | AZIMUTH<br>deg. | DOGLEG<br>SEVERITY<br>deg./<br>100 ft. | VERTICAL<br>DEPTH<br>feet | CLOSURE<br>DIST. feet | AZIMUTH<br>deg. | HORIZONTAL<br>COORDINATES<br>feet |
|---------------------------|--------------|-----------------|----------------------------------------|---------------------------|-----------------------|-----------------|-----------------------------------|
| 0.00                      | 0.00         | 0.00            | 0.00                                   | 0.00                      | 0.0                   | 0.0             | 0.00 N 0.00 E                     |

NO TIE-IN DATA PROVIDED. TIE-IN TO SURFACE.

ALL AZIMUTHS ARE REFERENCED TO TRUE NORTH.

ALL MEASURED DEPTHS AND COORDINATES REFERENCED TO NABORS 486 R.K.B. OF 25 FT.

138-15212 FT. MAGNETIC MEASUREMENT WHILE DRILLING SURVEY.

STRAIGHT LINE PROJECTION TO BIT DEPTH AT 15303 FT. MD.

|         |      |        |      |         |     |      |        |        |
|---------|------|--------|------|---------|-----|------|--------|--------|
| 138.00  | 0.40 | 84.47  | 0.29 | 138.00  | 0.5 | 84.5 | 0.05 N | 0.48 E |
| 232.00  | 0.48 | 53.62  | 0.26 | 232.00  | 1.2 | 74.5 | 0.31 N | 1.12 E |
| 325.00  | 0.31 | 52.74  | 0.18 | 324.99  | 1.8 | 67.0 | 0.69 N | 1.64 E |
| 419.00  | 0.44 | 72.69  | 0.19 | 418.99  | 2.4 | 66.4 | 0.96 N | 2.18 E |
| 515.00  | 0.75 | 86.58  | 0.35 | 514.99  | 3.3 | 70.8 | 1.10 N | 3.16 E |
| 605.00  | 0.35 | 60.13  | 0.52 | 604.98  | 4.2 | 72.3 | 1.28 N | 3.99 E |
| 697.00  | 0.66 | 103.54 | 0.51 | 696.98  | 4.9 | 74.8 | 1.29 N | 4.75 E |
| 789.00  | 0.40 | 66.54  | 0.45 | 788.98  | 5.7 | 76.9 | 1.30 N | 5.56 E |
| 881.00  | 0.31 | 49.84  | 0.15 | 880.97  | 6.2 | 75.3 | 1.58 N | 6.04 E |
| 973.00  | 0.66 | 95.90  | 0.54 | 972.97  | 7.0 | 76.0 | 1.69 N | 6.76 E |
| 1064.00 | 0.35 | 33.32  | 0.65 | 1063.97 | 7.7 | 75.9 | 1.87 N | 7.43 E |
| 1156.00 | 0.22 | 349.64 | 0.27 | 1155.97 | 7.9 | 73.2 | 2.28 N | 7.56 E |
| 1248.00 | 0.48 | 246.89 | 0.62 | 1247.97 | 7.5 | 72.2 | 2.30 N | 7.17 E |
| 1340.00 | 0.35 | 249.53 | 0.14 | 1339.96 | 6.9 | 72.6 | 2.05 N | 6.55 E |

# A Gyrodata Directional Survey

Oasis Petroleum

Lease: Kline Federal Well: 5300 41-18 10B, 8 3/4" Case, 6" Hole

Location: Nabors #486, McKenzie County, North Dakota

Job Number: RD0814M177

| MEASURED<br>DEPTH<br>feet | INCL<br>deg. | AZIMUTH<br>deg. | DOGLEG<br>SEVERITY<br>deg./<br>100 ft. | VERTICAL<br>DEPTH<br>feet | CLOSURE<br>DIST. feet | AZIMUTH<br>deg. | HORIZONTAL<br>COORDINATES<br>feet |        |
|---------------------------|--------------|-----------------|----------------------------------------|---------------------------|-----------------------|-----------------|-----------------------------------|--------|
| 1431.00                   | 0.48         | 266.67          | 0.20                                   | 1430.96                   | 6.2                   | 71.9            | 1.93 N                            | 5.91 E |
| 1526.00                   | 0.35         | 253.84          | 0.17                                   | 1525.96                   | 5.5                   | 70.8            | 1.83 N                            | 5.24 E |
| 1622.00                   | 0.62         | 267.55          | 0.30                                   | 1621.95                   | 4.8                   | 68.8            | 1.72 N                            | 4.44 E |
| 1717.00                   | 0.31         | 285.92          | 0.36                                   | 1716.95                   | 4.1                   | 64.3            | 1.77 N                            | 3.67 E |
| 1811.00                   | 0.62         | 316.77          | 0.41                                   | 1810.95                   | 3.8                   | 54.3            | 2.21 N                            | 3.08 E |
| 1907.00                   | 0.75         | 339.53          | 0.31                                   | 1906.94                   | 4.0                   | 38.3            | 3.18 N                            | 2.51 E |
| 2002.00                   | 0.22         | 357.72          | 0.57                                   | 2001.94                   | 4.6                   | 30.1            | 3.94 N                            | 2.28 E |
| 2071.00                   | 0.57         | 351.48          | 0.51                                   | 2070.94                   | 4.9                   | 26.8            | 4.41 N                            | 2.23 E |
| 2190.00                   | 0.62         | 355.20          | 0.05                                   | 2189.93                   | 6.0                   | 20.3            | 5.64 N                            | 2.08 E |
| 2284.00                   | 0.44         | 349.84          | 0.20                                   | 2283.93                   | 6.8                   | 16.9            | 6.50 N                            | 1.98 E |
| 2379.00                   | 0.48         | 156.39          | 0.96                                   | 2378.92                   | 6.8                   | 17.7            | 6.50 N                            | 2.07 E |
| 2472.00                   | 0.66         | 145.67          | 0.22                                   | 2471.92                   | 6.2                   | 23.9            | 5.70 N                            | 2.53 E |
| 2567.00                   | 0.57         | 153.23          | 0.13                                   | 2566.91                   | 5.7                   | 32.3            | 4.83 N                            | 3.05 E |
| 2662.00                   | 0.48         | 157.01          | 0.10                                   | 2661.91                   | 5.3                   | 40.3            | 4.04 N                            | 3.42 E |
| 2757.00                   | 0.31         | 194.10          | 0.31                                   | 2756.91                   | 4.9                   | 45.8            | 3.42 N                            | 3.51 E |
| 2852.00                   | 0.31         | 178.19          | 0.09                                   | 2851.91                   | 4.5                   | 49.9            | 2.92 N                            | 3.46 E |
| 2945.00                   | 0.35         | 191.02          | 0.09                                   | 2944.91                   | 4.2                   | 55.0            | 2.39 N                            | 3.41 E |
| 3039.00                   | 0.44         | 92.58           | 0.64                                   | 3038.90                   | 4.3                   | 60.7            | 2.09 N                            | 3.72 E |
| 3134.00                   | 0.31         | 108.76          | 0.18                                   | 3133.90                   | 4.8                   | 65.3            | 1.99 N                            | 4.33 E |
| 3227.00                   | 0.40         | 101.02          | 0.11                                   | 3226.90                   | 5.2                   | 69.3            | 1.85 N                            | 4.88 E |
| 3322.00                   | 0.57         | 135.04          | 0.34                                   | 3321.90                   | 5.7                   | 75.4            | 1.45 N                            | 5.54 E |
| 3417.00                   | 0.88         | 92.67           | 0.63                                   | 3416.89                   | 6.7                   | 80.7            | 1.08 N                            | 6.60 E |

# A Gyrodata Directional Survey

Oasis Petroleum

Lease: Kline Federal Well: 5300 41-18 10B, 8 3/4" Case, 6" Hole

Location: Nabors #486, McKenzie County, North Dakota

Job Number: RD0814M177

| MEASURED<br>DEPTH<br>feet | INCL<br>deg. | AZIMUTH<br>deg. | DOGLEG<br>SEVERITY<br>deg./<br>100 ft. | VERTICAL<br>DEPTH<br>feet | CLOSURE<br>DIST. AZIMUTH<br>feet deg. | HORIZONTAL<br>COORDINATES<br>feet |
|---------------------------|--------------|-----------------|----------------------------------------|---------------------------|---------------------------------------|-----------------------------------|
| 3512.00                   | 0.79         | 88.19           | 0.12                                   | 3511.88                   | 8.1 82.4                              | 1.07 N 7.99 E                     |
| 3606.00                   | 0.75         | 83.88           | 0.07                                   | 3605.87                   | 9.3 82.9                              | 1.15 N 9.25 E                     |
| 3701.00                   | 0.70         | 103.92          | 0.27                                   | 3700.86                   | 10.5 84.1                             | 1.08 N 10.43 E                    |
| 3796.00                   | 0.57         | 82.48           | 0.28                                   | 3795.86                   | 11.5 85.0                             | 1.00 N 11.46 E                    |
| 3890.00                   | 0.62         | 78.00           | 0.07                                   | 3889.85                   | 12.5 84.6                             | 1.17 N 12.42 E                    |
| 3984.00                   | 0.57         | 104.98          | 0.30                                   | 3983.85                   | 13.4 85.1                             | 1.15 N 13.37 E                    |
| 4079.00                   | 0.35         | 92.85           | 0.25                                   | 4078.84                   | 14.2 85.9                             | 1.02 N 14.12 E                    |
| 4171.00                   | 0.22         | 101.55          | 0.15                                   | 4170.84                   | 14.6 86.2                             | 0.97 N 14.57 E                    |
| 4266.00                   | 0.31         | 66.31           | 0.19                                   | 4265.84                   | 15.0 86.1                             | 1.03 N 14.98 E                    |
| 4361.00                   | 0.22         | 39.94           | 0.16                                   | 4360.84                   | 15.4 85.2                             | 1.28 N 15.34 E                    |
| 4456.00                   | 0.31         | 70.00           | 0.17                                   | 4455.84                   | 15.8 84.5                             | 1.50 N 15.70 E                    |
| 4550.00                   | 0.26         | 70.44           | 0.05                                   | 4549.84                   | 16.2 84.1                             | 1.66 N 16.14 E                    |
| 4645.00                   | 0.13         | 52.86           | 0.15                                   | 4644.84                   | 16.5 83.7                             | 1.80 N 16.42 E                    |
| 4740.00                   | 0.31         | 355.29          | 0.28                                   | 4739.84                   | 16.6 82.7                             | 2.12 N 16.49 E                    |
| 4835.00                   | 0.31         | 355.64          | 0.00                                   | 4834.84                   | 16.7 80.9                             | 2.63 N 16.45 E                    |
| 4931.00                   | 0.70         | 37.21           | 0.53                                   | 4930.83                   | 17.1 78.7                             | 3.36 N 16.78 E                    |
| 5024.00                   | 0.66         | 34.93           | 0.05                                   | 5023.83                   | 17.9 76.3                             | 4.25 N 17.43 E                    |
| 5119.00                   | 0.70         | 34.23           | 0.04                                   | 5118.82                   | 18.8 74.0                             | 5.18 N 18.07 E                    |
| 5214.00                   | 0.66         | 35.81           | 0.05                                   | 5213.81                   | 19.7 71.9                             | 6.10 N 18.72 E                    |
| 5308.00                   | 0.48         | 68.42           | 0.39                                   | 5307.81                   | 20.5 71.0                             | 6.69 N 19.40 E                    |
| 5403.00                   | 0.35         | 40.91           | 0.25                                   | 5402.81                   | 21.2 70.5                             | 7.05 N 19.96 E                    |
| 5497.00                   | 0.53         | 29.92           | 0.21                                   | 5496.80                   | 21.8 69.4                             | 7.65 N 20.37 E                    |

# A Gyrodata Directional Survey

Oasis Petroleum

Lease: Kline Federal Well: 5300 41-18 10B, 8 3/4" Case, 6" Hole

Location: Nabors #486, McKenzie County, North Dakota

Job Number: RD0814M177

| MEASURED<br>DEPTH<br>feet | INCL<br>deg. | AZIMUTH<br>deg. | DOGLEG<br>SEVERITY<br>deg./<br>100 ft. | VERTICAL<br>DEPTH<br>feet | CLOSURE<br>DIST. feet | AZIMUTH<br>deg. | HORIZONTAL<br>COORDINATES<br>feet |         |
|---------------------------|--------------|-----------------|----------------------------------------|---------------------------|-----------------------|-----------------|-----------------------------------|---------|
| 5592.00                   | 0.31         | 32.64           | 0.23                                   | 5591.80                   | 22.3                  | 68.3            | 8.24 N                            | 20.73 E |
| 5687.00                   | 0.31         | 351.60          | 0.23                                   | 5686.80                   | 22.6                  | 67.3            | 8.71 N                            | 20.83 E |
| 5782.00                   | 0.18         | 358.89          | 0.14                                   | 5781.80                   | 22.7                  | 66.3            | 9.12 N                            | 20.79 E |
| 5876.00                   | 0.18         | 40.29           | 0.14                                   | 5875.80                   | 22.9                  | 65.8            | 9.38 N                            | 20.88 E |
| 5971.00                   | 0.35         | 353.44          | 0.28                                   | 5970.80                   | 23.1                  | 65.0            | 9.78 N                            | 20.94 E |
| 6066.00                   | 0.40         | 27.99           | 0.24                                   | 6065.80                   | 23.5                  | 63.8            | 10.36 N                           | 21.07 E |
| 6090.00                   | 0.35         | 27.37           | 0.21                                   | 6089.80                   | 23.6                  | 63.6            | 10.50 N                           | 21.14 E |
| 6159.00                   | 0.44         | 24.29           | 0.13                                   | 6158.79                   | 24.0                  | 62.9            | 10.93 N                           | 21.34 E |
| 6253.00                   | 0.79         | 16.74           | 0.38                                   | 6252.79                   | 24.7                  | 61.3            | 11.88 N                           | 21.68 E |
| 6348.00                   | 0.84         | 21.48           | 0.09                                   | 6347.78                   | 25.7                  | 59.3            | 13.15 N                           | 22.12 E |
| 6443.00                   | 1.05         | 60.59           | 0.70                                   | 6442.77                   | 27.2                  | 58.4            | 14.23 N                           | 23.14 E |
| 6537.00                   | 1.05         | 27.46           | 0.64                                   | 6536.75                   | 28.8                  | 57.6            | 15.42 N                           | 24.28 E |
| 6632.00                   | 0.48         | 29.66           | 0.60                                   | 6631.74                   | 29.9                  | 56.4            | 16.53 N                           | 24.88 E |
| 6725.00                   | 0.48         | 47.50           | 0.16                                   | 6724.74                   | 30.6                  | 56.0            | 17.14 N                           | 25.36 E |
| 6821.00                   | 0.40         | 177.75          | 0.83                                   | 6820.74                   | 30.8                  | 56.4            | 17.07 N                           | 25.67 E |
| 6914.00                   | 0.35         | 235.32          | 0.39                                   | 6913.74                   | 30.4                  | 56.9            | 16.59 N                           | 25.45 E |
| 7010.00                   | 0.31         | 94.34           | 0.65                                   | 7009.74                   | 30.3                  | 57.2            | 16.40 N                           | 25.47 E |
| 7103.00                   | 0.22         | 165.53          | 0.34                                   | 7102.73                   | 30.4                  | 57.8            | 16.21 N                           | 25.76 E |
| 7197.00                   | 0.48         | 194.98          | 0.33                                   | 7196.73                   | 30.1                  | 58.7            | 15.65 N                           | 25.71 E |
| 7292.00                   | 0.48         | 172.13          | 0.20                                   | 7291.73                   | 29.7                  | 59.9            | 14.87 N                           | 25.66 E |
| 7386.00                   | 0.31         | 205.96          | 0.30                                   | 7385.73                   | 29.3                  | 60.9            | 14.26 N                           | 25.60 E |

# A Gyrodata Directional Survey

Oasis Petroleum

Lease: Kline Federal Well: 5300 41-18 10B, 8 3/4" Case, 6" Hole

Location: Nabors #486, McKenzie County, North Dakota

Job Number: RD0814M177

| MEASURED<br>DEPTH<br>feet | INCL<br>deg. | AZIMUTH<br>deg. | DOGLEG<br>SEVERITY<br>deg./<br>100 ft. | VERTICAL<br>DEPTH<br>feet | CLOSURE<br>DIST. feet | AZIMUTH<br>deg. | HORIZONTAL<br>COORDINATES<br>feet |         |
|---------------------------|--------------|-----------------|----------------------------------------|---------------------------|-----------------------|-----------------|-----------------------------------|---------|
| 7480.00                   | 0.13         | 185.84          | 0.21                                   | 7479.73                   | 29.0                  | 61.3            | 13.92 N                           | 25.48 E |
| 7575.00                   | 0.31         | 163.69          | 0.21                                   | 7574.73                   | 28.9                  | 62.0            | 13.57 N                           | 25.54 E |
| 7669.00                   | 0.31         | 182.67          | 0.11                                   | 7668.72                   | 28.7                  | 63.0            | 13.07 N                           | 25.60 E |
| 7764.00                   | 0.35         | 214.75          | 0.20                                   | 7763.72                   | 28.4                  | 63.7            | 12.57 N                           | 25.42 E |
| 7856.00                   | 0.31         | 194.36          | 0.13                                   | 7855.72                   | 28.0                  | 64.3            | 12.10 N                           | 25.20 E |
| 7951.00                   | 0.35         | 136.00          | 0.34                                   | 7950.72                   | 27.9                  | 65.3            | 11.64 N                           | 25.34 E |
| 8045.00                   | 0.44         | 157.10          | 0.18                                   | 8044.72                   | 28.0                  | 66.6            | 11.11 N                           | 25.68 E |
| 8139.00                   | 0.44         | 132.40          | 0.20                                   | 8138.72                   | 28.1                  | 68.0            | 10.53 N                           | 26.09 E |
| 8234.00                   | 0.26         | 132.58          | 0.19                                   | 8233.71                   | 28.4                  | 69.1            | 10.14 N                           | 26.51 E |
| 8328.00                   | 0.26         | 109.20          | 0.11                                   | 8327.71                   | 28.6                  | 69.7            | 9.92 N                            | 26.87 E |
| 8422.00                   | 0.04         | 9.53            | 0.29                                   | 8421.71                   | 28.8                  | 69.9            | 9.89 N                            | 27.08 E |
| 8516.00                   | 0.62         | 116.75          | 0.67                                   | 8515.71                   | 29.2                  | 70.6            | 9.69 N                            | 27.54 E |
| 8610.00                   | 0.44         | 119.66          | 0.19                                   | 8609.71                   | 29.8                  | 71.8            | 9.28 N                            | 28.31 E |
| 8705.00                   | 0.35         | 104.63          | 0.14                                   | 8704.70                   | 30.3                  | 72.7            | 9.03 N                            | 28.90 E |
| 8800.00                   | 0.35         | 27.46           | 0.46                                   | 8799.70                   | 30.7                  | 72.6            | 9.21 N                            | 29.32 E |
| 8895.00                   | 0.57         | 40.82           | 0.26                                   | 8894.70                   | 31.3                  | 71.7            | 9.83 N                            | 29.76 E |
| 8988.00                   | 0.62         | 46.09           | 0.08                                   | 8987.70                   | 32.2                  | 70.9            | 10.53 N                           | 30.43 E |
| 9082.00                   | 0.66         | 42.58           | 0.06                                   | 9081.69                   | 33.1                  | 70.1            | 11.28 N                           | 31.16 E |
| 9177.00                   | 0.70         | 37.39           | 0.08                                   | 9176.68                   | 34.1                  | 69.2            | 12.14 N                           | 31.88 E |
| 9270.00                   | 0.75         | 28.78           | 0.13                                   | 9269.68                   | 35.1                  | 68.0            | 13.13 N                           | 32.52 E |
| 9365.00                   | 0.66         | 47.85           | 0.26                                   | 9364.67                   | 36.1                  | 67.1            | 14.04 N                           | 33.22 E |
| 9460.00                   | 0.57         | 64.28           | 0.21                                   | 9459.66                   | 37.1                  | 66.8            | 14.61 N                           | 34.06 E |

# A Gyrodata Directional Survey

Oasis Petroleum

Lease: Kline Federal Well: 5300 41-18 10B, 8 3/4" Case, 6" Hole

Location: Nabors #486, McKenzie County, North Dakota

Job Number: RD0814M177

| MEASURED<br>DEPTH<br>feet | INCL<br>deg. | AZIMUTH<br>deg. | DOGLEG<br>SEVERITY<br>deg./<br>100 ft. | VERTICAL<br>DEPTH<br>feet | CLOSURE<br>DIST. feet | AZIMUTH<br>deg. | HORIZONTAL<br>COORDINATES<br>feet |          |
|---------------------------|--------------|-----------------|----------------------------------------|---------------------------|-----------------------|-----------------|-----------------------------------|----------|
| 9553.00                   | 0.57         | 51.54           | 0.14                                   | 9552.66                   | 38.0                  | 66.6            | 15.10 N                           | 34.83 E  |
| 9647.00                   | 0.79         | 90.48           | 0.53                                   | 9646.65                   | 39.0                  | 66.8            | 15.38 N                           | 35.85 E  |
| 9741.00                   | 0.79         | 111.31          | 0.30                                   | 9740.64                   | 40.1                  | 67.8            | 15.14 N                           | 37.10 E  |
| 9834.00                   | 0.88         | 116.67          | 0.13                                   | 9833.63                   | 41.0                  | 69.2            | 14.59 N                           | 38.34 E  |
| 9931.00                   | 1.05         | 100.23          | 0.33                                   | 9930.62                   | 42.3                  | 70.5            | 14.10 N                           | 39.88 E  |
| 10025.00                  | 0.84         | 76.59           | 0.47                                   | 10024.61                  | 43.7                  | 71.2            | 14.10 N                           | 41.39 E  |
| 10120.00                  | 0.75         | 56.64           | 0.30                                   | 10119.60                  | 45.0                  | 71.1            | 14.61 N                           | 42.59 E  |
| 10187.00                  | 0.66         | 66.22           | 0.22                                   | 10186.59                  | 45.8                  | 70.9            | 15.00 N                           | 43.31 E  |
| 10217.00                  | 0.44         | 40.99           | 1.07                                   | 10216.59                  | 46.1                  | 70.8            | 15.16 N                           | 43.54 E  |
| 10248.00                  | 2.68         | 70.70           | 7.45                                   | 10247.58                  | 46.9                  | 70.7            | 15.49 N                           | 44.31 E  |
| 10279.00                  | 7.78         | 74.22           | 16.48                                  | 10278.44                  | 49.8                  | 70.9            | 16.30 N                           | 47.01 E  |
| 10311.00                  | 13.05        | 74.83           | 16.47                                  | 10309.90                  | 55.5                  | 71.3            | 17.84 N                           | 52.59 E  |
| 10343.00                  | 18.46        | 74.57           | 16.91                                  | 10340.69                  | 64.2                  | 71.7            | 20.13 N                           | 60.96 E  |
| 10375.00                  | 20.40        | 71.84           | 6.69                                   | 10370.86                  | 74.8                  | 71.9            | 23.22 N                           | 71.15 E  |
| 10406.00                  | 22.11        | 70.88           | 5.63                                   | 10399.75                  | 86.1                  | 71.8            | 26.81 N                           | 81.79 E  |
| 10438.00                  | 24.53        | 70.79           | 7.56                                   | 10429.14                  | 98.7                  | 71.7            | 30.97 N                           | 93.76 E  |
| 10470.00                  | 25.93        | 73.60           | 5.76                                   | 10458.09                  | 112.4                 | 71.8            | 35.13 N                           | 106.74 E |
| 10502.00                  | 30.11        | 78.00           | 14.56                                  | 10486.33                  | 127.4                 | 72.3            | 38.78 N                           | 121.32 E |
| 10533.00                  | 33.54        | 79.67           | 11.42                                  | 10512.67                  | 143.6                 | 73.0            | 41.93 N                           | 137.35 E |
| 10565.00                  | 37.14        | 79.40           | 11.26                                  | 10538.77                  | 162.0                 | 73.8            | 45.30 N                           | 155.55 E |
| 10597.00                  | 41.14        | 75.09           | 15.11                                  | 10563.59                  | 182.2                 | 74.1            | 49.78 N                           | 175.23 E |
| 10628.00                  | 45.36        | 73.69           | 13.96                                  | 10586.16                  | 203.4                 | 74.2            | 55.51 N                           | 195.68 E |



# A Gyrodata Directional Survey

Oasis Petroleum

Lease: Kline Federal Well: 5300 41-18 10B, 8 3/4" Case, 6" Hole

Location: Nabors #486, McKenzie County, North Dakota

Job Number: RD0814M177

| MEASURED<br>DEPTH<br>feet | INCL<br>deg. | AZIMUTH<br>deg. | DOGLEG<br>SEVERITY<br>deg./<br>100 ft. | VERTICAL<br>DEPTH<br>feet | CLOSURE<br>DIST. feet | AZIMUTH<br>deg. | HORIZONTAL<br>COORDINATES<br>feet |
|---------------------------|--------------|-----------------|----------------------------------------|---------------------------|-----------------------|-----------------|-----------------------------------|
| 10660.00                  | 49.85        | 71.84           | 14.67                                  | 10607.73                  | 227.0                 | 74.0            | 62.52 N 218.24 E                  |
| 10691.00                  | 53.49        | 73.69           | 12.64                                  | 10626.96                  | 251.3                 | 73.9            | 69.72 N 241.46 E                  |
| 10723.00                  | 57.10        | 76.06           | 12.82                                  | 10645.17                  | 277.6                 | 74.0            | 76.57 N 266.85 E                  |
| 10754.00                  | 60.40        | 77.82           | 11.70                                  | 10661.25                  | 304.1                 | 74.2            | 82.55 N 292.67 E                  |
| 10785.00                  | 64.04        | 77.73           | 11.74                                  | 10675.70                  | 331.5                 | 74.5            | 88.35 N 319.47 E                  |
| 10816.00                  | 67.91        | 77.91           | 12.50                                  | 10688.32                  | 359.7                 | 74.8            | 94.33 N 347.14 E                  |
| 10848.00                  | 68.31        | 78.52           | 2.17                                   | 10700.25                  | 389.4                 | 75.1            | 100.39 N 376.20 E                 |
| 10879.00                  | 68.35        | 78.08           | 1.33                                   | 10711.70                  | 418.1                 | 75.3            | 106.23 N 404.41 E                 |
| 10910.00                  | 71.74        | 76.76           | 11.64                                  | 10722.28                  | 447.2                 | 75.4            | 112.58 N 432.85 E                 |
| 10942.00                  | 77.23        | 76.41           | 17.19                                  | 10730.83                  | 478.1                 | 75.5            | 119.73 N 462.83 E                 |
| 10973.00                  | 81.05        | 75.62           | 12.57                                  | 10736.67                  | 508.5                 | 75.5            | 127.09 N 492.37 E                 |
| 11005.00                  | 81.49        | 73.60           | 6.39                                   | 10741.53                  | 540.1                 | 75.5            | 135.48 N 522.86 E                 |
| 11036.00                  | 81.54        | 73.69           | 0.33                                   | 10746.10                  | 570.8                 | 75.4            | 144.12 N 552.28 E                 |
| 11068.00                  | 85.01        | 73.07           | 11.01                                  | 10749.85                  | 602.5                 | 75.3            | 153.21 N 582.73 E                 |
| 11078.00                  | 86.42        | 73.25           | 14.21                                  | 10750.60                  | 612.5                 | 75.2            | 156.10 N 592.27 E                 |
| 11098.00                  | 88.09        | 71.84           | 10.92                                  | 10751.55                  | 632.4                 | 75.2            | 162.09 N 611.33 E                 |
| 11166.00                  | 88.70        | 71.38           | 1.12                                   | 10753.46                  | 700.3                 | 74.8            | 183.53 N 675.83 E                 |
| 11197.00                  | 89.05        | 71.21           | 1.26                                   | 10754.07                  | 731.2                 | 74.7            | 193.47 N 705.18 E                 |
| 11229.00                  | 89.71        | 70.68           | 2.65                                   | 10754.41                  | 763.2                 | 74.5            | 203.92 N 735.43 E                 |
| 11260.00                  | 90.59        | 71.91           | 4.88                                   | 10754.33                  | 794.1                 | 74.4            | 213.86 N 764.79 E                 |
| 11291.00                  | 91.30        | 73.84           | 6.63                                   | 10753.82                  | 825.1                 | 74.3            | 222.99 N 794.41 E                 |

# A Gyrodata Directional Survey

Oasis Petroleum

Lease: Kline Federal Well: 5300 41-18 10B, 8 3/4" Case, 6" Hole

Location: Nabors #486, McKenzie County, North Dakota

Job Number: RD0814M177

| MEASURED<br>DEPTH<br>feet | INCL<br>deg. | AZIMUTH<br>deg. | DOGLEG<br>SEVERITY<br>deg./<br>100 ft. | VERTICAL<br>DEPTH<br>feet | CLOSURE<br>DIST. feet | AZIMUTH<br>deg. | HORIZONTAL<br>COORDINATES<br>feet |           |
|---------------------------|--------------|-----------------|----------------------------------------|---------------------------|-----------------------|-----------------|-----------------------------------|-----------|
| 11322.00                  | 91.96        | 75.07           | 4.50                                   | 10752.94                  | 856.1                 | 74.3            | 231.29 N                          | 824.26 E  |
| 11353.00                  | 92.70        | 76.39           | 4.88                                   | 10751.68                  | 887.1                 | 74.4            | 238.93 N                          | 854.28 E  |
| 11384.00                  | 92.88        | 78.50           | 6.82                                   | 10750.17                  | 918.0                 | 74.5            | 245.66 N                          | 884.50 E  |
| 11415.00                  | 92.88        | 79.38           | 2.84                                   | 10748.61                  | 948.9                 | 74.6            | 251.59 N                          | 914.89 E  |
| 11446.00                  | 92.35        | 80.88           | 5.13                                   | 10747.20                  | 979.7                 | 74.8            | 256.90 N                          | 945.40 E  |
| 11477.00                  | 92.09        | 82.72           | 5.99                                   | 10746.00                  | 1010.4                | 75.0            | 261.32 N                          | 976.06 E  |
| 11508.00                  | 92.18        | 84.39           | 5.39                                   | 10744.84                  | 1041.1                | 75.3            | 264.80 N                          | 1006.84 E |
| 11539.00                  | 91.12        | 86.06           | 6.38                                   | 10743.95                  | 1071.6                | 75.6            | 267.38 N                          | 1037.72 E |
| 11570.00                  | 89.63        | 86.32           | 4.88                                   | 10743.75                  | 1102.1                | 75.8            | 269.44 N                          | 1068.65 E |
| 11602.00                  | 88.53        | 86.76           | 3.70                                   | 10744.26                  | 1133.5                | 76.1            | 271.37 N                          | 1100.58 E |
| 11633.00                  | 89.36        | 87.03           | 2.82                                   | 10744.83                  | 1164.0                | 76.4            | 273.05 N                          | 1131.53 E |
| 11664.00                  | 90.46        | 88.35           | 5.54                                   | 10744.88                  | 1194.4                | 76.7            | 274.30 N                          | 1162.50 E |
| 11695.00                  | 91.08        | 88.70           | 2.30                                   | 10744.46                  | 1224.8                | 77.0            | 275.09 N                          | 1193.49 E |
| 11726.00                  | 91.56        | 88.79           | 1.58                                   | 10743.75                  | 1255.1                | 77.3            | 275.77 N                          | 1224.48 E |
| 11758.00                  | 90.29        | 90.02           | 5.52                                   | 10743.23                  | 1286.4                | 77.6            | 276.11 N                          | 1256.47 E |
| 11820.00                  | 90.02        | 90.81           | 1.35                                   | 10743.07                  | 1347.0                | 78.2            | 275.66 N                          | 1318.47 E |
| 11851.00                  | 90.42        | 90.28           | 2.14                                   | 10742.95                  | 1377.3                | 78.5            | 275.36 N                          | 1349.46 E |
| 11882.00                  | 90.73        | 90.10           | 1.16                                   | 10742.64                  | 1407.6                | 78.7            | 275.26 N                          | 1380.46 E |
| 11913.00                  | 90.77        | 90.37           | 0.88                                   | 10742.23                  | 1438.0                | 79.0            | 275.13 N                          | 1411.46 E |
| 11945.00                  | 89.85        | 90.72           | 3.08                                   | 10742.06                  | 1469.4                | 79.2            | 274.83 N                          | 1443.46 E |
| 12007.00                  | 88.66        | 91.42           | 2.23                                   | 10742.86                  | 1530.1                | 79.7            | 273.67 N                          | 1505.44 E |
| 12069.00                  | 89.05        | 90.19           | 2.08                                   | 10744.10                  | 1591.0                | 80.1            | 272.80 N                          | 1567.42 E |

# A Gyrodata Directional Survey

Oasis Petroleum

Lease: Kline Federal Well: 5300 41-18 10B, 8 3/4" Case, 6" Hole

Location: Nabors #486, McKenzie County, North Dakota

Job Number: RD0814M177

| MEASURED<br>DEPTH<br>feet | INCL<br>deg. | AZIMUTH<br>deg. | DOGLEG<br>SEVERITY<br>deg./<br>100 ft. | VERTICAL<br>DEPTH<br>feet | CLOSURE<br>DIST. feet | AZIMUTH<br>deg. | HORIZONTAL<br>COORDINATES<br>feet |
|---------------------------|--------------|-----------------|----------------------------------------|---------------------------|-----------------------|-----------------|-----------------------------------|
| 12100.00                  | 89.54        | 90.02           | 1.67                                   | 10744.48                  | 1621.5                | 80.3            | 272.74 N 1598.42 E                |
| 12163.00                  | 89.71        | 87.64           | 3.79                                   | 10744.90                  | 1683.8                | 80.6            | 274.03 N 1661.40 E                |
| 12194.00                  | 88.70        | 88.26           | 3.82                                   | 10745.33                  | 1714.6                | 80.8            | 275.14 N 1692.37 E                |
| 12256.00                  | 89.80        | 87.03           | 2.66                                   | 10746.14                  | 1776.2                | 81.0            | 277.68 N 1754.32 E                |
| 12287.00                  | 90.11        | 89.14           | 6.88                                   | 10746.16                  | 1806.9                | 81.1            | 278.72 N 1785.30 E                |
| 12349.00                  | 90.20        | 88.87           | 0.46                                   | 10745.99                  | 1868.4                | 81.4            | 279.80 N 1847.29 E                |
| 12380.00                  | 89.49        | 88.26           | 3.02                                   | 10746.08                  | 1899.1                | 81.5            | 280.57 N 1878.28 E                |
| 12411.00                  | 89.36        | 89.84           | 5.11                                   | 10746.39                  | 1929.8                | 81.6            | 281.09 N 1909.27 E                |
| 12473.00                  | 90.07        | 88.35           | 2.66                                   | 10746.70                  | 1991.3                | 81.9            | 282.07 N 1971.26 E                |
| 12505.00                  | 90.51        | 89.31           | 3.30                                   | 10746.53                  | 2023.1                | 82.0            | 282.72 N 2003.25 E                |
| 12567.00                  | 91.69        | 89.23           | 1.91                                   | 10745.34                  | 2084.6                | 82.2            | 283.51 N 2065.23 E                |
| 12599.00                  | 91.08        | 91.16           | 6.32                                   | 10744.57                  | 2116.3                | 82.3            | 283.40 N 2097.22 E                |
| 12662.00                  | 90.55        | 91.95           | 1.51                                   | 10743.68                  | 2178.5                | 82.6            | 281.69 N 2160.19 E                |
| 12693.00                  | 89.67        | 93.44           | 5.58                                   | 10743.62                  | 2209.0                | 82.7            | 280.23 N 2191.16 E                |
| 12756.00                  | 90.64        | 93.00           | 1.69                                   | 10743.45                  | 2271.0                | 83.0            | 276.69 N 2254.06 E                |
| 12788.00                  | 89.41        | 93.71           | 4.44                                   | 10743.43                  | 2302.5                | 83.1            | 274.82 N 2286.00 E                |
| 12851.00                  | 89.14        | 93.18           | 0.94                                   | 10744.23                  | 2364.5                | 83.4            | 271.04 N 2348.88 E                |
| 12883.00                  | 88.62        | 93.80           | 2.53                                   | 10744.85                  | 2396.0                | 83.6            | 269.09 N 2380.82 E                |
| 12945.00                  | 89.05        | 93.62           | 0.75                                   | 10746.11                  | 2457.0                | 83.8            | 265.08 N 2442.67 E                |
| 13040.00                  | 88.79        | 92.48           | 1.23                                   | 10747.91                  | 2550.8                | 84.1            | 260.02 N 2537.52 E                |
| 13103.00                  | 88.97        | 91.42           | 1.71                                   | 10749.14                  | 2613.2                | 84.3            | 257.88 N 2600.47 E                |
| 13135.00                  | 89.41        | 91.25           | 1.47                                   | 10749.59                  | 2645.0                | 84.4            | 257.14 N 2632.46 E                |

# A Gyrodata Directional Survey

Oasis Petroleum

Lease: Kline Federal Well: 5300 41-18 10B, 8 3/4" Case, 6" Hole

Location: Nabors #486, McKenzie County, North Dakota

Job Number: RD0814M177

| MEASURED<br>DEPTH<br>feet | INCL<br>deg. | AZIMUTH<br>deg. | DOGLEG<br>SEVERITY<br>deg./<br>100 ft. | VERTICAL<br>DEPTH<br>feet | CLOSURE<br>DIST. AZIMUTH<br>feet deg. | HORIZONTAL<br>COORDINATES<br>feet |
|---------------------------|--------------|-----------------|----------------------------------------|---------------------------|---------------------------------------|-----------------------------------|
| 13229.00                  | 87.56        | 91.25           | 1.97                                   | 10752.07                  | 2738.3 84.7                           | 255.09 N 2726.40 E                |
| 13324.00                  | 87.69        | 90.63           | 0.67                                   | 10756.01                  | 2832.7 84.9                           | 253.53 N 2821.30 E                |
| 13355.00                  | 87.74        | 90.46           | 0.57                                   | 10757.25                  | 2863.5 84.9                           | 253.23 N 2852.28 E                |
| 13387.00                  | 88.00        | 90.28           | 0.99                                   | 10758.44                  | 2895.3 85.0                           | 253.03 N 2884.26 E                |
| 13418.00                  | 88.57        | 89.31           | 3.63                                   | 10759.36                  | 2926.2 85.0                           | 253.14 N 2915.24 E                |
| 13481.00                  | 88.97        | 89.40           | 0.65                                   | 10760.72                  | 2989.0 85.1                           | 253.85 N 2978.22 E                |
| 13512.00                  | 89.41        | 89.93           | 2.22                                   | 10761.15                  | 3019.9 85.2                           | 254.03 N 3009.22 E                |
| 13544.00                  | 90.29        | 88.87           | 4.31                                   | 10761.24                  | 3051.8 85.2                           | 254.36 N 3041.22 E                |
| 13607.00                  | 91.08        | 88.87           | 1.25                                   | 10760.49                  | 3114.7 85.3                           | 255.61 N 3104.20 E                |
| 13670.00                  | 90.02        | 88.52           | 1.77                                   | 10759.88                  | 3177.6 85.4                           | 257.04 N 3167.18 E                |
| 13700.00                  | 90.33        | 89.14           | 2.31                                   | 10759.79                  | 3207.5 85.4                           | 257.65 N 3197.17 E                |
| 13732.00                  | 90.42        | 88.61           | 1.68                                   | 10759.58                  | 3239.5 85.4                           | 258.28 N 3229.16 E                |
| 13763.00                  | 89.71        | 89.49           | 3.65                                   | 10759.54                  | 3270.4 85.5                           | 258.80 N 3260.16 E                |
| 13795.00                  | 89.85        | 90.46           | 3.06                                   | 10759.67                  | 3302.3 85.5                           | 258.81 N 3292.16 E                |
| 13858.00                  | 89.05        | 91.86           | 2.56                                   | 10760.27                  | 3365.0 85.6                           | 257.53 N 3355.14 E                |
| 13890.00                  | 88.75        | 91.25           | 2.12                                   | 10760.89                  | 3396.8 85.7                           | 256.67 N 3387.12 E                |
| 13953.00                  | 88.22        | 89.49           | 2.92                                   | 10762.55                  | 3459.6 85.8                           | 256.26 N 3450.10 E                |
| 13984.00                  | 88.53        | 88.87           | 2.24                                   | 10763.43                  | 3490.5 85.8                           | 256.70 N 3481.08 E                |
| 14047.00                  | 89.76        | 91.77           | 5.00                                   | 10764.37                  | 3553.3 85.9                           | 256.35 N 3544.07 E                |
| 14079.00                  | 89.63        | 89.93           | 5.76                                   | 10764.54                  | 3585.2 85.9                           | 255.88 N 3576.06 E                |
| 14110.00                  | 88.92        | 88.79           | 4.33                                   | 10764.93                  | 3616.1 85.9                           | 256.22 N 3607.05 E                |

# A Gyrodata Directional Survey

Oasis Petroleum

Lease: Kline Federal Well: 5300 41-18 10B, 8 3/4" Case, 6" Hole

Location: Nabors #486, McKenzie County, North Dakota

Job Number: RD0814M177

| MEASURED<br>DEPTH<br>feet | INCL<br>deg. | AZIMUTH<br>deg. | DOGLEG<br>SEVERITY<br>deg./<br>100 ft. | VERTICAL<br>DEPTH<br>feet | CLOSURE<br>DIST. feet | AZIMUTH<br>deg. | HORIZONTAL<br>COORDINATES<br>feet |
|---------------------------|--------------|-----------------|----------------------------------------|---------------------------|-----------------------|-----------------|-----------------------------------|
| 14173.00                  | 88.66        | 87.38           | 2.28                                   | 10766.26                  | 3679.1                | 86.0            | 258.33 N 3670.00 E                |
| 14205.00                  | 88.79        | 89.75           | 7.42                                   | 10766.98                  | 3711.0                | 86.0            | 259.13 N 3701.98 E                |
| 14268.00                  | 89.32        | 88.87           | 1.63                                   | 10768.02                  | 3773.9                | 86.1            | 259.89 N 3764.97 E                |
| 14299.00                  | 90.24        | 89.84           | 4.31                                   | 10768.13                  | 3804.9                | 86.1            | 260.24 N 3795.97 E                |
| 14362.00                  | 90.59        | 89.93           | 0.57                                   | 10767.68                  | 3867.7                | 86.1            | 260.36 N 3858.97 E                |
| 14393.00                  | 90.90        | 92.30           | 7.71                                   | 10767.28                  | 3898.6                | 86.2            | 259.76 N 3889.95 E                |
| 14457.00                  | 91.08        | 90.72           | 2.48                                   | 10766.17                  | 3962.3                | 86.3            | 258.07 N 3953.92 E                |
| 14520.00                  | 90.51        | 90.90           | 0.95                                   | 10765.30                  | 4025.1                | 86.3            | 257.18 N 4016.91 E                |
| 14551.00                  | 90.11        | 89.75           | 3.93                                   | 10765.13                  | 4056.1                | 86.4            | 257.01 N 4047.91 E                |
| 14614.00                  | 90.77        | 89.75           | 1.05                                   | 10764.64                  | 4118.9                | 86.4            | 257.28 N 4110.90 E                |
| 14646.00                  | 91.12        | 89.93           | 1.23                                   | 10764.12                  | 4150.9                | 86.4            | 257.37 N 4142.90 E                |
| 14709.00                  | 90.90        | 89.75           | 0.45                                   | 10763.01                  | 4213.8                | 86.5            | 257.55 N 4205.89 E                |
| 14740.00                  | 90.90        | 88.52           | 3.97                                   | 10762.52                  | 4244.7                | 86.5            | 258.01 N 4236.88 E                |
| 14803.00                  | 90.68        | 89.40           | 1.44                                   | 10761.65                  | 4307.7                | 86.6            | 259.16 N 4299.86 E                |
| 14835.00                  | 90.33        | 87.91           | 4.78                                   | 10761.37                  | 4339.6                | 86.6            | 259.91 N 4331.85 E                |
| 14866.00                  | 89.85        | 87.99           | 1.57                                   | 10761.32                  | 4370.6                | 86.6            | 261.02 N 4362.83 E                |
| 14929.00                  | 89.58        | 87.12           | 1.45                                   | 10761.63                  | 4433.6                | 86.6            | 263.71 N 4425.77 E                |
| 14961.00                  | 89.54        | 89.75           | 8.22                                   | 10761.88                  | 4465.6                | 86.6            | 264.58 N 4457.76 E                |
| 15024.00                  | 90.07        | 90.98           | 2.13                                   | 10762.09                  | 4528.5                | 86.7            | 264.18 N 4520.76 E                |
| 15118.00                  | 90.73        | 91.77           | 1.10                                   | 10761.44                  | 4622.2                | 86.8            | 261.92 N 4614.72 E                |
| 15149.00                  | 91.08        | 92.04           | 1.43                                   | 10760.95                  | 4653.0                | 86.8            | 260.89 N 4645.70 E                |
| 15181.00                  | 91.52        | 90.54           | 4.88                                   | 10760.22                  | 4684.9                | 86.8            | 260.17 N 4677.69 E                |

# A Gyrodata Directional Survey

Oasis Petroleum

Lease: Kline Federal Well: 5300 41-18 10B, 8 3/4" Case, 6" Hole

Location: Nabors #486, McKenzie County, North Dakota

Job Number: RD0814M177

| MEASURED<br>DEPTH<br>feet | INCL<br>deg. | AZIMUTH<br>deg. | DOGLEG<br>SEVERITY<br>deg./<br>100 ft. | VERTICAL<br>DEPTH<br>feet | CLOSURE<br>DIST. AZIMUTH<br>feet deg. | HORIZONTAL<br>COORDINATES<br>feet |
|---------------------------|--------------|-----------------|----------------------------------------|---------------------------|---------------------------------------|-----------------------------------|
|---------------------------|--------------|-----------------|----------------------------------------|---------------------------|---------------------------------------|-----------------------------------|

|          |       |       |      |          |             |                    |
|----------|-------|-------|------|----------|-------------|--------------------|
| 15212.00 | 91.91 | 90.98 | 1.90 | 10759.29 | 4715.8 86.8 | 259.76 N 4708.67 E |
|----------|-------|-------|------|----------|-------------|--------------------|

STRAIGHT LINE PROJECTION TO BIT DEPTH AT 15303 FT. MD.

|          |       |       |      |          |             |                    |
|----------|-------|-------|------|----------|-------------|--------------------|
| 15303.00 | 91.91 | 90.98 | 0.00 | 10756.26 | 4806.5 86.9 | 258.21 N 4799.61 E |
|----------|-------|-------|------|----------|-------------|--------------------|

Final Station Closure: Distance: 4806.55 ft Az: 86.92 deg.

WELL PATH





Gyrodata Incorporated  
301 Thelma Dr. #413  
Casper, Wyoming 82601

307-472-0182  
Fax: 307-472-0204

### DIRECTIONAL SURVEY CERTIFICATION

RE: Oasis Petroleum (Operator)  
Kline Federal 5300 41-18 10B ST1 (Well Name & No.)  
8.19 (Magnetic Declination)

Surveyor: Blake Hunter

I, Kristopher Bertsch, having personal knowledge of all the facts, hereby certify that the attached directional survey run from measured depth of 15024 feet to a measured depth of 18551 feet and is projected to 18629 feet, is true and correct as determined from all available records.

  
Signature

MWD Coordinator  
Title

Gyrodata, Inc.  
Company

\*All measurements recorded with a manned MWD Survey Tool.

A Gyrodata Directional Survey

FINAL DEFINITIVE COPY

for

OASIS PETROLEUM

Lease: Kline Federal Well: 5300 41-18 10B ST1, 6" Hole

Location: Nabors #486, McKenzie County, North Dakota

Job Number: RD0814M177ST1

Run Date: 26 Feb 2015

Surveyor: Blake Hunter

Calculation Method: MINIMUM CURVATURE

Survey Latitude: 48.068944 deg. N Longitude: 103.603319 deg. W

Magnetic Declination: 8.1900 deg

Total Correction: 8.1900 deg

Proposed Well Direction: 90.000 deg

Depth Reference: Rotary Table

Air Gap (RKB to Ground / RKB to MSL): 25.00

Ground Elevation (Ground to MSL / MSL to Mudline): 2057.00

Vertical Section Calculated from Well Head Location

Closure Calculated from Well Head Location

Horizontal Coordinates Calculated from Well Head Location

# A Gyrodata Directional Survey

Oasis Petroleum

Lease: Kline Federal Well: 5300 41-18 10B ST1, 6" Hole

Location: Nabors #486, McKenzie County, North Dakota

Job Number: RD0814M177ST1

| MEASURED<br>DEPTH<br>feet | INCL<br>deg. | AZIMUTH<br>deg. | DOGLEG<br>SEVERITY<br>deg./<br>100 ft. | VERTICAL<br>DEPTH<br>feet | CLOSURE<br>DIST. feet | AZIMUTH<br>deg. | HORIZONTAL<br>COORDINATES<br>feet |
|---------------------------|--------------|-----------------|----------------------------------------|---------------------------|-----------------------|-----------------|-----------------------------------|
| 14961.00                  | 89.54        | 89.75           | 0.00                                   | 10761.87                  | 4465.6                | 86.6            | 264.58 N 4457.76 E                |

TIE-ON TO GYRODATA SURVEY OF ORIGINAL WELLBORE AT 14961 FT. MD.

ALL AZIMUTHS ARE REFERENCED TO TRUE NORTH.

ALL MEASURED DEPTHS AND COORDINATES REFERENCED TO NABORS 486 R.K.B. OF 25 FT.

15024-18551 MAGNETIC MEASUREMENT WHILE DRILLING SURVEY.

STRAIGHT LINE PROJECTION TO BIT DEPTH AT 18629 FT. MD.

|          |       |       |      |          |        |      |                    |
|----------|-------|-------|------|----------|--------|------|--------------------|
| 15024.00 | 90.07 | 90.98 | 2.13 | 10762.08 | 4528.5 | 86.7 | 264.18 N 4520.76 E |
| 15087.00 | 87.56 | 92.39 | 4.57 | 10763.39 | 4591.2 | 86.7 | 262.33 N 4583.71 E |
| 15118.00 | 87.30 | 93.09 | 2.41 | 10764.78 | 4622.0 | 86.8 | 260.85 N 4614.64 E |
| 15149.00 | 87.69 | 92.92 | 1.37 | 10766.13 | 4652.8 | 86.8 | 259.22 N 4645.57 E |
| 15181.00 | 87.47 | 91.51 | 4.46 | 10767.48 | 4684.6 | 86.8 | 257.99 N 4677.52 E |
| 15212.00 | 87.16 | 91.77 | 1.30 | 10768.94 | 4715.5 | 86.9 | 257.10 N 4708.47 E |
| 15244.00 | 87.25 | 92.65 | 2.76 | 10770.50 | 4747.3 | 86.9 | 255.87 N 4740.41 E |
| 15275.00 | 88.75 | 91.77 | 5.61 | 10771.58 | 4778.2 | 86.9 | 254.67 N 4771.37 E |
| 15306.00 | 89.05 | 91.25 | 1.94 | 10772.17 | 4809.1 | 87.0 | 253.86 N 4802.35 E |
| 15338.00 | 89.10 | 92.04 | 2.47 | 10772.69 | 4840.9 | 87.0 | 252.94 N 4834.33 E |
| 15369.00 | 89.67 | 91.42 | 2.72 | 10773.02 | 4871.8 | 87.0 | 252.00 N 4865.31 E |
| 15401.00 | 90.02 | 91.25 | 1.22 | 10773.11 | 4903.7 | 87.1 | 251.26 N 4897.31 E |
| 15432.00 | 90.29 | 92.30 | 3.50 | 10773.03 | 4934.6 | 87.1 | 250.30 N 4928.29 E |
| 15495.00 | 89.71 | 89.75 | 4.15 | 10773.03 | 4997.5 | 87.1 | 249.17 N 4991.27 E |

# A Gyrodata Directional Survey

Oasis Petroleum

Lease: Kline Federal Well: 5300 41-18 10B ST1, 6" Hole

Location: Nabors #486, McKenzie County, North Dakota

Job Number: RD0814M177ST1

| MEASURED<br>DEPTH<br>feet | INCL<br>deg. | AZIMUTH<br>deg. | DOGLEG<br>SEVERITY<br>deg./<br>100 ft. | VERTICAL<br>DEPTH<br>feet | CLOSURE<br>DIST. AZIMUTH<br>feet deg. | HORIZONTAL<br>COORDINATES<br>feet |
|---------------------------|--------------|-----------------|----------------------------------------|---------------------------|---------------------------------------|-----------------------------------|
| 15527.00                  | 88.79        | 89.84           | 2.89                                   | 10773.44                  | 5029.5 87.2                           | 249.29 N 5023.27 E                |
| 15590.00                  | 89.32        | 89.49           | 1.01                                   | 10774.48                  | 5092.4 87.2                           | 249.65 N 5086.26 E                |
| 15621.00                  | 89.67        | 88.87           | 2.30                                   | 10774.76                  | 5123.4 87.2                           | 250.10 N 5117.26 E                |
| 15684.00                  | 89.54        | 88.61           | 0.46                                   | 10775.19                  | 5186.3 87.2                           | 251.48 N 5180.24 E                |
| 15716.00                  | 89.54        | 87.73           | 2.75                                   | 10775.45                  | 5218.3 87.2                           | 252.50 N 5212.22 E                |
| 15779.00                  | 89.67        | 88.26           | 0.87                                   | 10775.88                  | 5281.3 87.2                           | 254.71 N 5275.18 E                |
| 15810.00                  | 87.87        | 89.75           | 7.54                                   | 10776.55                  | 5312.3 87.2                           | 255.25 N 5306.17 E                |
| 15873.00                  | 89.05        | 90.54           | 2.25                                   | 10778.24                  | 5375.2 87.3                           | 255.09 N 5369.14 E                |
| 15905.00                  | 91.03        | 90.63           | 6.19                                   | 10778.22                  | 5407.1 87.3                           | 254.76 N 5401.14 E                |
| 15968.00                  | 92.35        | 90.46           | 2.11                                   | 10776.36                  | 5470.0 87.3                           | 254.16 N 5464.11 E                |
| 16000.00                  | 90.81        | 89.93           | 5.09                                   | 10775.48                  | 5502.0 87.4                           | 254.05 N 5496.10 E                |
| 16063.00                  | 91.65        | 89.31           | 1.66                                   | 10774.13                  | 5564.9 87.4                           | 254.47 N 5559.08 E                |
| 16094.00                  | 90.20        | 87.12           | 8.47                                   | 10773.62                  | 5595.9 87.4                           | 255.44 N 5590.06 E                |
| 16126.00                  | 90.29        | 88.17           | 3.29                                   | 10773.49                  | 5627.9 87.4                           | 256.75 N 5622.03 E                |
| 16157.00                  | 89.32        | 86.41           | 6.48                                   | 10773.59                  | 5658.9 87.4                           | 258.22 N 5652.99 E                |
| 16188.00                  | 88.62        | 88.17           | 6.11                                   | 10774.15                  | 5689.9 87.4                           | 259.68 N 5683.95 E                |
| 16251.00                  | 89.27        | 88.61           | 1.25                                   | 10775.31                  | 5752.9 87.4                           | 261.45 N 5746.92 E                |
| 16346.00                  | 89.58        | 88.08           | 0.65                                   | 10776.26                  | 5847.8 87.4                           | 264.19 N 5841.87 E                |
| 16377.00                  | 89.05        | 88.70           | 2.63                                   | 10776.63                  | 5878.8 87.4                           | 265.07 N 5872.86 E                |
| 16440.00                  | 91.52        | 88.87           | 3.93                                   | 10776.32                  | 5941.8 87.4                           | 266.40 N 5935.84 E                |
| 16472.00                  | 89.85        | 90.02           | 6.34                                   | 10775.94                  | 5973.8 87.4                           | 266.71 N 5967.83 E                |
| 16503.00                  | 90.46        | 91.25           | 4.43                                   | 10775.86                  | 6004.7 87.5                           | 266.37 N 5998.83 E                |

# A Gyrodata Directional Survey

Oasis Petroleum

Lease: Kline Federal Well: 5300 41-18 10B ST1, 6" Hole

Location: Nabors #486, McKenzie County, North Dakota

Job Number: RD0814M177ST1

| MEASURED<br>DEPTH<br>feet | INCL<br>deg. | AZIMUTH<br>deg. | DOGLEG<br>SEVERITY<br>deg./<br>100 ft. | VERTICAL<br>DEPTH<br>feet | CLOSURE<br>DIST. AZIMUTH<br>feet deg. | HORIZONTAL<br>COORDINATES<br>feet |
|---------------------------|--------------|-----------------|----------------------------------------|---------------------------|---------------------------------------|-----------------------------------|
| 16535.00                  | 90.42        | 90.63           | 1.94                                   | 10775.61                  | 6036.7 87.5                           | 265.84 N 6030.82 E                |
| 16598.00                  | 88.44        | 91.60           | 3.50                                   | 10776.24                  | 6099.5 87.5                           | 264.62 N 6093.80 E                |
| 16629.00                  | 86.15        | 91.33           | 7.44                                   | 10777.70                  | 6130.4 87.5                           | 263.83 N 6124.76 E                |
| 16724.00                  | 86.37        | 92.48           | 1.23                                   | 10783.90                  | 6225.0 87.6                           | 260.67 N 6219.50 E                |
| 16787.00                  | 87.21        | 92.92           | 1.50                                   | 10787.42                  | 6287.6 87.7                           | 257.71 N 6282.33 E                |
| 16818.00                  | 88.75        | 92.04           | 5.72                                   | 10788.52                  | 6318.5 87.7                           | 256.37 N 6313.28 E                |
| 16851.00                  | 89.14        | 91.33           | 2.45                                   | 10789.12                  | 6351.4 87.7                           | 255.40 N 6346.26 E                |
| 16914.00                  | 89.71        | 91.25           | 0.91                                   | 10789.76                  | 6414.3 87.7                           | 253.98 N 6409.24 E                |
| 16945.00                  | 89.67        | 90.54           | 2.29                                   | 10789.92                  | 6445.2 87.7                           | 253.50 N 6440.24 E                |
| 17008.00                  | 89.71        | 89.75           | 1.26                                   | 10790.26                  | 6508.2 87.8                           | 253.34 N 6503.24 E                |
| 17103.00                  | 89.54        | 88.87           | 0.94                                   | 10790.89                  | 6603.1 87.8                           | 254.48 N 6598.23 E                |
| 17197.00                  | 90.07        | 88.52           | 0.68                                   | 10791.21                  | 6697.1 87.8                           | 256.62 N 6692.20 E                |
| 17228.00                  | 89.89        | 88.17           | 1.27                                   | 10791.22                  | 6728.1 87.8                           | 257.52 N 6723.19 E                |
| 17291.00                  | 90.64        | 88.26           | 1.20                                   | 10790.93                  | 6791.1 87.8                           | 259.48 N 6786.16 E                |
| 17323.00                  | 90.46        | 89.93           | 5.25                                   | 10790.62                  | 6823.1 87.8                           | 259.99 N 6818.15 E                |
| 17354.00                  | 90.73        | 90.28           | 1.43                                   | 10790.30                  | 6854.1 87.8                           | 259.93 N 6849.15 E                |
| 17386.00                  | 90.95        | 90.02           | 1.06                                   | 10789.83                  | 6886.0 87.8                           | 259.85 N 6881.14 E                |
| 17418.00                  | 91.08        | 90.10           | 0.48                                   | 10789.26                  | 6918.0 87.8                           | 259.81 N 6913.14 E                |
| 17449.00                  | 89.01        | 90.90           | 7.16                                   | 10789.24                  | 6949.0 87.9                           | 259.54 N 6944.14 E                |
| 17480.00                  | 89.14        | 91.07           | 0.69                                   | 10789.74                  | 6979.9 87.9                           | 259.01 N 6975.13 E                |
| 17575.00                  | 89.58        | 90.10           | 1.12                                   | 10790.80                  | 7074.8 87.9                           | 258.04 N 7070.11 E                |
| 17607.00                  | 89.89        | 89.66           | 1.68                                   | 10790.95                  | 7106.8 87.9                           | 258.11 N 7102.11 E                |

# A Gyrodata Directional Survey

Oasis Petroleum

Lease: Kline Federal Well: 5300 41-18 10B ST1, 6" Hole

Location: Nabors #486, McKenzie County, North Dakota

Job Number: RD0814M177ST1

| MEASURED<br>DEPTH<br>feet | INCL<br>deg. | AZIMUTH<br>deg. | DOGLEG<br>SEVERITY<br>deg./<br>100 ft. | VERTICAL<br>DEPTH<br>feet | CLOSURE<br>DIST. AZIMUTH<br>feet deg. | HORIZONTAL<br>COORDINATES<br>feet |
|---------------------------|--------------|-----------------|----------------------------------------|---------------------------|---------------------------------------|-----------------------------------|
| 17669.00                  | 90.90        | 89.66           | 1.63                                   | 10790.52                  | 7168.8 87.9                           | 258.47 N 7164.11 E                |
| 17701.00                  | 90.46        | 90.37           | 2.61                                   | 10790.14                  | 7200.7 87.9                           | 258.47 N 7196.11 E                |
| 17764.00                  | 91.03        | 89.84           | 1.24                                   | 10789.32                  | 7263.7 88.0                           | 258.35 N 7259.10 E                |
| 17796.00                  | 89.19        | 89.84           | 5.75                                   | 10789.26                  | 7295.7 88.0                           | 258.44 N 7291.10 E                |
| 17858.00                  | 89.27        | 89.40           | 0.72                                   | 10790.09                  | 7357.6 88.0                           | 258.85 N 7353.09 E                |
| 17952.00                  | 90.81        | 89.14           | 1.66                                   | 10790.03                  | 7451.6 88.0                           | 260.05 N 7447.08 E                |
| 18046.00                  | 90.46        | 89.23           | 0.38                                   | 10788.98                  | 7545.6 88.0                           | 261.39 N 7541.07 E                |
| 18078.00                  | 90.02        | 89.66           | 1.92                                   | 10788.85                  | 7577.6 88.0                           | 261.70 N 7573.07 E                |
| 18140.00                  | 90.37        | 89.49           | 0.63                                   | 10788.64                  | 7639.6 88.0                           | 262.16 N 7635.06 E                |
| 18172.00                  | 88.18        | 89.66           | 6.86                                   | 10789.04                  | 7671.5 88.0                           | 262.39 N 7667.06 E                |
| 18235.00                  | 88.18        | 89.49           | 0.27                                   | 10791.04                  | 7734.5 88.1                           | 262.86 N 7730.02 E                |
| 18266.00                  | 88.53        | 89.49           | 1.13                                   | 10791.93                  | 7765.5 88.1                           | 263.14 N 7761.01 E                |
| 18329.00                  | 89.36        | 89.49           | 1.32                                   | 10793.09                  | 7828.4 88.1                           | 263.70 N 7824.00 E                |
| 18360.00                  | 88.09        | 90.19           | 4.68                                   | 10793.78                  | 7859.4 88.1                           | 263.78 N 7854.99 E                |
| 18423.00                  | 88.48        | 89.93           | 0.74                                   | 10795.67                  | 7922.4 88.1                           | 263.72 N 7917.96 E                |
| 18454.00                  | 88.97        | 89.58           | 1.94                                   | 10796.36                  | 7953.3 88.1                           | 263.85 N 7948.95 E                |
| 18516.00                  | 89.71        | 89.49           | 1.20                                   | 10797.07                  | 8015.3 88.1                           | 264.35 N 8010.95 E                |
| 18551.00                  | 88.04        | 90.28           | 5.28                                   | 10797.76                  | 8050.3 88.1                           | 264.42 N 8045.94 E                |



# A Gyrodata Directional Survey

Oasis Petroleum

Lease: Kline Federal Well: 5300 41-18 10B ST1, 6" Hole

Location: Nabors #486, McKenzie County, North Dakota

Job Number: RD0814M177ST1

| MEASURED<br>DEPTH<br>feet | INCL<br>deg. | AZIMUTH<br>deg. | DOGLEG<br>SEVERITY<br>deg./<br>100 ft. | VERTICAL<br>DEPTH<br>feet | CLOSURE<br>DIST. AZIMUTH<br>feet deg. | HORIZONTAL<br>COORDINATES<br>feet |
|---------------------------|--------------|-----------------|----------------------------------------|---------------------------|---------------------------------------|-----------------------------------|
|---------------------------|--------------|-----------------|----------------------------------------|---------------------------|---------------------------------------|-----------------------------------|

STRAIGHT LINE PROJECTION TO BIT DEPTH AT 18629 FT. MD.

|          |       |       |      |          |             |                    |
|----------|-------|-------|------|----------|-------------|--------------------|
| 18629.00 | 88.04 | 90.28 | 0.00 | 10800.43 | 8128.2 88.1 | 264.04 N 8123.89 E |
|----------|-------|-------|------|----------|-------------|--------------------|

Final Station Closure: Distance: 8128.18 ft Az: 88.14 deg.



Gyrodata Incorporated  
301 Thelma Dr. #413  
Casper, Wyoming 82601

307-472-0182  
Fax: 307-472-0204

## DIRECTIONAL SURVEY CERTIFICATION

RE: Oasis Petroleum (Operator)  
Kline Federal 5300 41-18 10B ST2 (Well Name & No.)  
8.19 (Magnetic Declination)

Surveyor: Blake Hunter

I, Kristopher Bertsch, having personal knowledge of all the facts, hereby certify that the attached directional survey run from measured depth of 17952 feet to a measured depth of 20400 feet and is projected to 20478 feet, is true and correct as determined from all available records.

  
Signature

MWD Coordinator  
Title

Gyrodata, Inc.  
Company

\*All measurements recorded with a manned MWD Survey Tool.

A Gyrodata Directional Survey

FINAL DEFINITIVE COPY

for

OASIS PETROLEUM

Lease: Kline Federal Well: 5300 41-18 10B ST2, 6" Hole

Location: Nabors #486, McKenzie County, North Dakota

Job Number: RD0814M177ST2

Run Date: 26 Feb 2015

Surveyor: Blake Hunter

Calculation Method: MINIMUM CURVATURE

Survey Latitude: 48.068944 deg. N Longitude: 103.603319 deg. W

Magnetic Declination: 8.1900 deg

Total Correction: 8.1900 deg

Proposed Well Direction: 90.000 deg

Depth Reference: Rotary Table

Air Gap (RKB to Ground / RKB to MSL): 25.00

Ground Elevation (Ground to MSL / MSL to Mudline): 2057.00

Vertical Section Calculated from Well Head Location

Closure Calculated from Well Head Location

Horizontal Coordinates Calculated from Well Head Location

# A Gyrodata Directional Survey

Oasis Petroleum

Lease: Kline Federal Well: 5300 41-18 10B ST2, 6" Hole

Location: Nabors #486, McKenzie County, North Dakota

Job Number: RD0814M177ST2

| MEASURED<br>DEPTH<br>feet | INCL<br>deg. | AZIMUTH<br>deg. | DOGLEG<br>SEVERITY<br>deg./<br>100 ft. | VERTICAL<br>DEPTH<br>feet | CLOSURE<br>DIST. AZIMUTH<br>feet deg. | HORIZONTAL<br>COORDINATES<br>feet |
|---------------------------|--------------|-----------------|----------------------------------------|---------------------------|---------------------------------------|-----------------------------------|
|---------------------------|--------------|-----------------|----------------------------------------|---------------------------|---------------------------------------|-----------------------------------|

|          |       |       |      |          |             |                    |
|----------|-------|-------|------|----------|-------------|--------------------|
| 17858.00 | 89.27 | 89.40 | 0.00 | 10790.09 | 7357.6 88.0 | 258.85 N 7353.09 E |
|----------|-------|-------|------|----------|-------------|--------------------|

TIE-ON TO GYRODATA SURVEY OF ST1 AT 17858 FT. MD.

ALL AZIMUTHS ARE REFERENCED TO TRUE NORTH.

ALL MEASURED DEPTHS AND COORDINATES REFERENCED TO NABORS 486 R.K.B. AT 25 FT.

17952-20400 FT. MAGNETIC MEASUREMENT WHILE DRILLING SURVEY.

STRAIGHT LINE PROJECTION TO BIT DEPTH AT 20478 FT. MD.

|          |       |       |      |          |             |                    |
|----------|-------|-------|------|----------|-------------|--------------------|
| 17952.00 | 90.59 | 89.14 | 1.43 | 10790.20 | 7451.6 88.0 | 260.05 N 7447.08 E |
| 17984.00 | 90.81 | 88.87 | 1.09 | 10789.81 | 7483.6 88.0 | 260.60 N 7479.07 E |
| 18015.00 | 89.23 | 90.10 | 6.46 | 10789.80 | 7514.6 88.0 | 260.88 N 7510.07 E |
| 18046.00 | 90.07 | 90.46 | 2.95 | 10789.99 | 7545.6 88.0 | 260.73 N 7541.07 E |
| 18109.00 | 90.42 | 89.84 | 1.13 | 10789.72 | 7608.5 88.0 | 260.57 N 7604.07 E |
| 18140.00 | 90.55 | 90.37 | 1.76 | 10789.46 | 7639.5 88.0 | 260.51 N 7635.07 E |
| 18203.00 | 91.12 | 90.98 | 1.33 | 10788.54 | 7702.4 88.1 | 259.77 N 7698.05 E |
| 18235.00 | 91.65 | 90.98 | 1.66 | 10787.77 | 7734.4 88.1 | 259.22 N 7730.04 E |
| 18297.00 | 92.00 | 91.16 | 0.63 | 10785.79 | 7796.3 88.1 | 258.06 N 7792.00 E |
| 18329.00 | 92.40 | 90.90 | 1.49 | 10784.57 | 7828.2 88.1 | 257.49 N 7823.97 E |
| 18360.00 | 92.57 | 90.90 | 0.55 | 10783.22 | 7859.1 88.1 | 257.00 N 7854.94 E |
| 18423.00 | 91.78 | 90.98 | 1.26 | 10780.83 | 7922.0 88.1 | 255.97 N 7917.88 E |
| 18454.00 | 92.00 | 90.72 | 1.10 | 10779.81 | 7953.0 88.2 | 255.51 N 7948.86 E |
| 18517.00 | 92.84 | 90.63 | 1.34 | 10777.15 | 8015.9 88.2 | 254.77 N 8011.80 E |

# A Gyrodata Directional Survey

Oasis Petroleum

Lease: Kline Federal Well: 5300 41-18 10B ST2, 6" Hole

Location: Nabors #486, McKenzie County, North Dakota

Job Number: RD0814M177ST2

| MEASURED<br>DEPTH<br>feet | INCL<br>deg. | AZIMUTH<br>deg. | DOGLEG<br>SEVERITY<br>deg./<br>100 ft. | VERTICAL<br>DEPTH<br>feet | CLOSURE<br>DIST. AZIMUTH<br>feet deg. | HORIZONTAL<br>COORDINATES<br>feet |
|---------------------------|--------------|-----------------|----------------------------------------|---------------------------|---------------------------------------|-----------------------------------|
| 18548.00                  | 91.30        | 91.16           | 5.25                                   | 10776.03                  | 8046.8 88.2                           | 254.28 N 8042.78 E                |
| 18611.00                  | 90.81        | 91.60           | 1.05                                   | 10774.87                  | 8109.7 88.2                           | 252.77 N 8105.75 E                |
| 18705.00                  | 89.98        | 91.69           | 0.89                                   | 10774.22                  | 8203.5 88.3                           | 250.07 N 8199.70 E                |
| 18768.00                  | 90.90        | 91.07           | 1.76                                   | 10773.74                  | 8266.4 88.3                           | 248.55 N 8262.68 E                |
| 18799.00                  | 91.25        | 90.90           | 1.26                                   | 10773.16                  | 8297.4 88.3                           | 248.02 N 8293.67 E                |
| 18893.00                  | 90.95        | 90.81           | 0.33                                   | 10771.35                  | 8391.3 88.3                           | 246.62 N 8387.65 E                |
| 18987.00                  | 90.77        | 90.90           | 0.21                                   | 10769.94                  | 8485.2 88.3                           | 245.21 N 8481.62 E                |
| 19018.00                  | 90.15        | 91.69           | 3.24                                   | 10769.69                  | 8516.1 88.4                           | 244.51 N 8512.61 E                |
| 19050.00                  | 90.55        | 91.69           | 1.25                                   | 10769.50                  | 8548.1 88.4                           | 243.57 N 8544.60 E                |
| 19081.00                  | 90.95        | 91.42           | 1.56                                   | 10769.09                  | 8579.0 88.4                           | 242.73 N 8575.59 E                |
| 19112.00                  | 91.25        | 91.51           | 1.01                                   | 10768.49                  | 8610.0 88.4                           | 241.93 N 8606.57 E                |
| 19144.00                  | 89.98        | 91.33           | 4.01                                   | 10768.15                  | 8641.9 88.4                           | 241.14 N 8638.56 E                |
| 19175.00                  | 90.02        | 91.60           | 0.88                                   | 10768.15                  | 8672.9 88.4                           | 240.35 N 8669.55 E                |
| 19207.00                  | 88.53        | 91.33           | 4.73                                   | 10768.56                  | 8704.8 88.4                           | 239.53 N 8701.53 E                |
| 19269.00                  | 86.95        | 91.42           | 2.55                                   | 10771.00                  | 8766.7 88.4                           | 238.04 N 8763.47 E                |
| 19300.00                  | 87.25        | 91.51           | 1.01                                   | 10772.57                  | 8797.6 88.5                           | 237.25 N 8794.42 E                |
| 19331.00                  | 87.69        | 91.51           | 1.42                                   | 10773.94                  | 8828.5 88.5                           | 236.44 N 8825.37 E                |
| 19363.00                  | 87.56        | 91.25           | 0.91                                   | 10775.26                  | 8860.5 88.5                           | 235.67 N 8857.34 E                |
| 19426.00                  | 88.48        | 91.07           | 1.49                                   | 10777.44                  | 8923.4 88.5                           | 234.39 N 8920.29 E                |
| 19457.00                  | 89.54        | 90.10           | 4.63                                   | 10777.98                  | 8954.3 88.5                           | 234.08 N 8951.28 E                |
| 19520.00                  | 89.41        | 88.08           | 3.21                                   | 10778.55                  | 9017.3 88.5                           | 235.08 N 9014.27 E                |
| 19551.00                  | 89.41        | 87.29           | 2.55                                   | 10778.87                  | 9048.3 88.5                           | 236.33 N 9045.24 E                |

# A Gyrodata Directional Survey

Oasis Petroleum

Lease: Kline Federal Well: 5300 41-18 10B ST2, 6" Hole

Location: Nabors #486, McKenzie County, North Dakota

Job Number: RD0814M177ST2

| MEASURED<br>DEPTH<br>feet | INCL<br>deg. | AZIMUTH<br>deg. | DOGLEG<br>SEVERITY<br>deg./<br>100 ft. | VERTICAL<br>DEPTH<br>feet | CLOSURE<br>DIST. AZIMUTH<br>feet deg. | HORIZONTAL<br>COORDINATES<br>feet |
|---------------------------|--------------|-----------------|----------------------------------------|---------------------------|---------------------------------------|-----------------------------------|
| 19614.00                  | 89.93        | 87.03           | 0.92                                   | 10779.24                  | 9111.3 88.5                           | 239.45 N 9108.16 E                |
| 19646.00                  | 90.42        | 87.29           | 1.73                                   | 10779.14                  | 9143.3 88.5                           | 241.04 N 9140.12 E                |
| 19740.00                  | 91.16        | 86.32           | 1.30                                   | 10777.84                  | 9237.2 88.5                           | 246.27 N 9233.96 E                |
| 19771.00                  | 91.38        | 86.41           | 0.77                                   | 10777.15                  | 9268.2 88.5                           | 248.24 N 9264.89 E                |
| 19834.00                  | 91.38        | 85.71           | 1.11                                   | 10775.64                  | 9331.1 88.4                           | 252.57 N 9327.73 E                |
| 19865.00                  | 90.33        | 88.17           | 8.63                                   | 10775.17                  | 9362.1 88.4                           | 254.22 N 9358.68 E                |
| 19928.00                  | 90.73        | 87.47           | 1.28                                   | 10774.59                  | 9425.1 88.4                           | 256.62 N 9421.63 E                |
| 19960.00                  | 90.99        | 88.96           | 4.73                                   | 10774.11                  | 9457.1 88.4                           | 257.61 N 9453.61 E                |
| 19991.00                  | 91.30        | 87.73           | 4.09                                   | 10773.49                  | 9488.1 88.4                           | 258.51 N 9484.59 E                |
| 20022.00                  | 91.30        | 88.96           | 3.97                                   | 10772.79                  | 9519.1 88.4                           | 259.40 N 9515.57 E                |
| 20054.00                  | 91.52        | 87.91           | 3.35                                   | 10772.00                  | 9551.1 88.4                           | 260.28 N 9547.54 E                |
| 20117.00                  | 89.63        | 88.70           | 3.25                                   | 10771.37                  | 9614.1 88.4                           | 262.14 N 9610.51 E                |
| 20180.00                  | 88.35        | 91.77           | 5.28                                   | 10772.48                  | 9677.0 88.4                           | 261.88 N 9673.49 E                |
| 20211.00                  | 88.62        | 90.63           | 3.78                                   | 10773.30                  | 9708.0 88.5                           | 261.23 N 9704.47 E                |
| 20305.00                  | 89.19        | 92.13           | 1.71                                   | 10775.10                  | 9801.8 88.5                           | 258.97 N 9798.42 E                |
| 20400.00                  | 89.67        | 90.81           | 1.48                                   | 10776.04                  | 9896.7 88.5                           | 256.53 N 9893.39 E                |

STRAIGHT LINE PROJECTION TO BIT DEPTH AT 20478 FT. MD.

|          |       |       |      |          |             |                    |
|----------|-------|-------|------|----------|-------------|--------------------|
| 20478.00 | 89.67 | 90.81 | 0.00 | 10776.49 | 9974.6 88.5 | 255.43 N 9971.38 E |
|----------|-------|-------|------|----------|-------------|--------------------|

Final Station Closure: Distance: 9974.65 ft Az: 88.53 deg.

# A Gyrodata Directional Survey

Oasis Petroleum

Lease: Kline Federal Well: 5300 41-18 10B ST2, 6" Hole

Location: Nabors #486, McKenzie County, North Dakota

Job Number: RD0814M177ST2

| MEASURED<br>DEPTH<br>feet | INCL<br>deg. | AZIMUTH<br>deg. | DOGLEG<br>SEVERITY<br>deg./<br>100 ft. | VERTICAL<br>DEPTH<br>feet | CLOSURE<br>DIST. AZIMUTH<br>feet deg. | HORIZONTAL<br>COORDINATES<br>feet |
|---------------------------|--------------|-----------------|----------------------------------------|---------------------------|---------------------------------------|-----------------------------------|
|---------------------------|--------------|-----------------|----------------------------------------|---------------------------|---------------------------------------|-----------------------------------|



**SUNDRY NOTICES AND REPORTS ON WELLS - FORM 4**

INDUSTRIAL COMMISSION OF NORTH DAKOTA  
OIL AND GAS DIVISION  
600 EAST BOULEVARD DEPT 405  
BISMARCK, ND 58505-0840  
SFN 5749 (09-2006)



Well File No.

**28652**

PLEASE READ INSTRUCTIONS BEFORE FILLING OUT FORM.  
PLEASE SUBMIT THE ORIGINAL AND ONE COPY.

|                                                                                                                                                 |                                                   |
|-------------------------------------------------------------------------------------------------------------------------------------------------|---------------------------------------------------|
| <input checked="" type="checkbox"/> Notice of Intent                                                                                            | Approximate Start Date<br><b>December 9, 2014</b> |
| <input type="checkbox"/> Report of Work Done                                                                                                    | Date Work Completed                               |
| <input type="checkbox"/> Notice of Intent to Begin a Workover Project that may Qualify for a Tax Exemption Pursuant to NDCC Section 57-51.1-03. | Approximate Start Date                            |

|                                               |                                                   |
|-----------------------------------------------|---------------------------------------------------|
| <input type="checkbox"/> Drilling Prognosis   | <input type="checkbox"/> Spill Report             |
| <input type="checkbox"/> Redrilling or Repair | <input type="checkbox"/> Shooting                 |
| <input type="checkbox"/> Casing or Liner      | <input type="checkbox"/> Acid zing                |
| <input type="checkbox"/> Plug Well            | <input type="checkbox"/> Frac ure Treatment       |
| <input type="checkbox"/> Supplemental History | <input type="checkbox"/> Change Production Method |
| <input type="checkbox"/> Temporarily Abandon  | <input type="checkbox"/> Reclamation              |
| <input type="checkbox"/> Other                | <u>Waiver from tubing/packer requirement</u>      |

|                                                             |                       |                           |                      |                          |                       |
|-------------------------------------------------------------|-----------------------|---------------------------|----------------------|--------------------------|-----------------------|
| Well Name and Number<br><b>Kline Federal 5300 41-18 10B</b> |                       |                           |                      |                          |                       |
| Footages<br><b>500 F S L 237 F W L</b>                      |                       | Qtr-Qtr<br><b>LOT 4</b>   | Section<br><b>18</b> | Township<br><b>153 N</b> | Range<br><b>100 W</b> |
| Field<br><b>BAKER</b>                                       | Pool<br><b>Bakken</b> | County<br><b>MCKENZIE</b> |                      |                          |                       |

| 24-HOUR PRODUCTION RATE |      |       |      |
|-------------------------|------|-------|------|
| Before                  |      | After |      |
| Oil                     | Bbls | Oil   | Bbls |
| Water                   | Bbls | Water | Bbls |
| Gas                     | MCF  | Gas   | MCF  |

|                       |      |       |          |
|-----------------------|------|-------|----------|
| Name of Contractor(s) |      |       |          |
| Address               | City | State | Zip Code |

**DETAILS OF WORK**

Oasis Petroleum North America LLC requests a variance to NDAC 43-02-03-21 for the tubing/packer requirement: Casing, tubing, and cementing requirements during the completion period immediately following the upcoming fracture stimulation.

The following assurances apply:

1. the well is equipped with new 29# and 32# casing at surface with an API burst rating of 11,220 psi;
2. The Frac design will use a safety factor of 0.85 API burst rating to determine the maximum pressure;
3. Damage to the casing during the frac would be detected immediately by monitoring equipment;
4. The casing is exposed to significantly lower rates and pressures during flowback than during the frac job;
5. The frac fluid and formation fluids have very low corrosion and erosion rates;
6. Production equipment will be installed as soon as possible after the well ceases flowing;
7. A 300# gauge will be installed on the surface casing during the flowback period

|                                                     |                                         |                                         |                          |
|-----------------------------------------------------|-----------------------------------------|-----------------------------------------|--------------------------|
| Company<br><b>Oasis Petroleum North America LLC</b> |                                         | Telephone Number<br><b>281-404-9436</b> |                          |
| Address<br><b>1001 Fannin, Suite 1500</b>           |                                         |                                         |                          |
| City<br><b>Houston</b>                              |                                         | State<br><b>TX</b>                      | Zip Code<br><b>77002</b> |
| Signature<br>                                       | Printed Name<br><b>Jennifer Swenson</b> |                                         |                          |
| Title<br><b>Regulatory Assistant</b>                | Date<br><b>December 9, 2014</b>         |                                         |                          |
| Email Address<br><b>jswenson@oasispetroleum.com</b> |                                         |                                         |                          |

**FOR STATE USE ONLY**

|                                    |                                              |
|------------------------------------|----------------------------------------------|
| <input type="checkbox"/> Received  | <input checked="" type="checkbox"/> Approved |
| Date<br><b>December 22, 2014</b>   |                                              |
| By<br>                             |                                              |
| Title<br><b>PETROLEUM ENGINEER</b> |                                              |

**SUNDRY NOTICES AND REPORTS ON WELLS - FORM 4**

INDUSTRIAL COMMISSION OF NORTH DAKOTA  
OIL AND GAS DIVISION  
600 EAST BOULEVARD DEPT 405  
BISMARCK, ND 58505-0840  
SFN 5749 (09-2006)



Well File No.  
**28652**

PLEASE READ INSTRUCTIONS BEFORE FILLING OUT FORM.  
PLEASE SUBMIT THE ORIGINAL AND ONE COPY.

|                                                                                                                                                 |                                                |
|-------------------------------------------------------------------------------------------------------------------------------------------------|------------------------------------------------|
| <input checked="" type="checkbox"/> Notice of Intent                                                                                            | Approximate Start Date<br><b>July 28, 2014</b> |
| <input type="checkbox"/> Report of Work Done                                                                                                    | Date Work Completed                            |
| <input type="checkbox"/> Notice of Intent to Begin a Workover Project that may Qualify for a Tax Exemption Pursuant to NDCC Section 57-51.1-03. | Approximate Start Date                         |

|                                                     |                                                   |
|-----------------------------------------------------|---------------------------------------------------|
| <input type="checkbox"/> Drilling Prognosis         | <input type="checkbox"/> Spill Report             |
| <input type="checkbox"/> Redrilling or Repair       | <input type="checkbox"/> Shooting                 |
| <input checked="" type="checkbox"/> Casing or Liner | <input type="checkbox"/> Acidizing                |
| <input type="checkbox"/> Plug Well                  | <input type="checkbox"/> Fracture Treatment       |
| <input type="checkbox"/> Supplemental History       | <input type="checkbox"/> Change Production Method |
| <input type="checkbox"/> Temporarily Abandon        | <input type="checkbox"/> Reclamation              |
| <input type="checkbox"/> Other                      | <b>Change casing</b>                              |

|                                                             |  |                        |                      |                           |                       |
|-------------------------------------------------------------|--|------------------------|----------------------|---------------------------|-----------------------|
| Well Name and Number<br><b>Kline Federal 5300 41-18 10B</b> |  |                        |                      |                           |                       |
| Footages<br><b>500 F S L 237 F W L</b>                      |  | Qtr-Qtr<br><b>SWSW</b> | Section<br><b>18</b> | Township<br><b>153 N</b>  | Range<br><b>100 W</b> |
| Field<br><b>Baker</b>                                       |  | Pool<br><b>Bakken</b>  |                      | County<br><b>McKenzie</b> |                       |

| 24-HOUR PRODUCTION RATE |      |       |      |
|-------------------------|------|-------|------|
| Before                  |      | After |      |
| Oil                     | Bbls | Oil   | Bbls |
| Water                   | Bbls | Water | Bbls |
| Gas                     | MCF  | Gas   | MCF  |

|                       |      |       |          |
|-----------------------|------|-------|----------|
| Name of Contractor(s) |      |       |          |
| Address               | City | State | Zip Code |

**DETAILS OF WORK**

Oasis Petroleum respectfully requests permission to make the following changes to the above referenced well:

Surface Casing: 13-3/8, 54.5#, 17-1/2" Hole, 2,072' MD  
Dakota Contingency: 9-5/8, 40#, 12-1/4" Hole, 6,101' MD

Attached are revised plats, drill plan, well summary, directional plan and plot

|                                                     |  |                                         |                          |
|-----------------------------------------------------|--|-----------------------------------------|--------------------------|
| Company<br><b>Oasis Petroleum North America LLC</b> |  | Telephone Number<br><b>281-404-9563</b> |                          |
| Address<br><b>1001 Fannin, Suite 1500</b>           |  |                                         |                          |
| City<br><b>Houston</b>                              |  | State<br><b>TX</b>                      | Zip Code<br><b>77002</b> |
| Signature<br><i>Heather McCowan</i>                 |  | Printed Name<br><b>Heather McCowan</b>  |                          |
| Title<br><b>Regulatory Assistant</b>                |  | Date<br><b>July 28, 2014</b>            |                          |
| Email Address<br><b>hmccowan@oasispetroleum.com</b> |  |                                         |                          |

| FOR STATE USE ONLY                            |                                              |
|-----------------------------------------------|----------------------------------------------|
| <input type="checkbox"/> Received             | <input checked="" type="checkbox"/> Approved |
| Date<br><b>8-5-14</b>                         |                                              |
| By<br><i>Anthony Ebele</i>                    |                                              |
| Title<br><b>Petroleum Resource Specialist</b> |                                              |

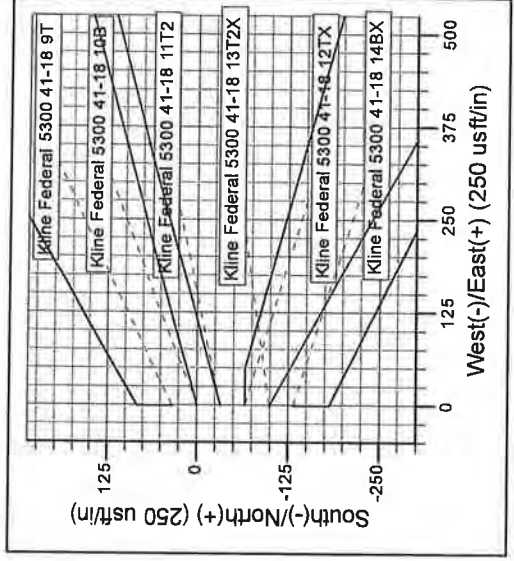
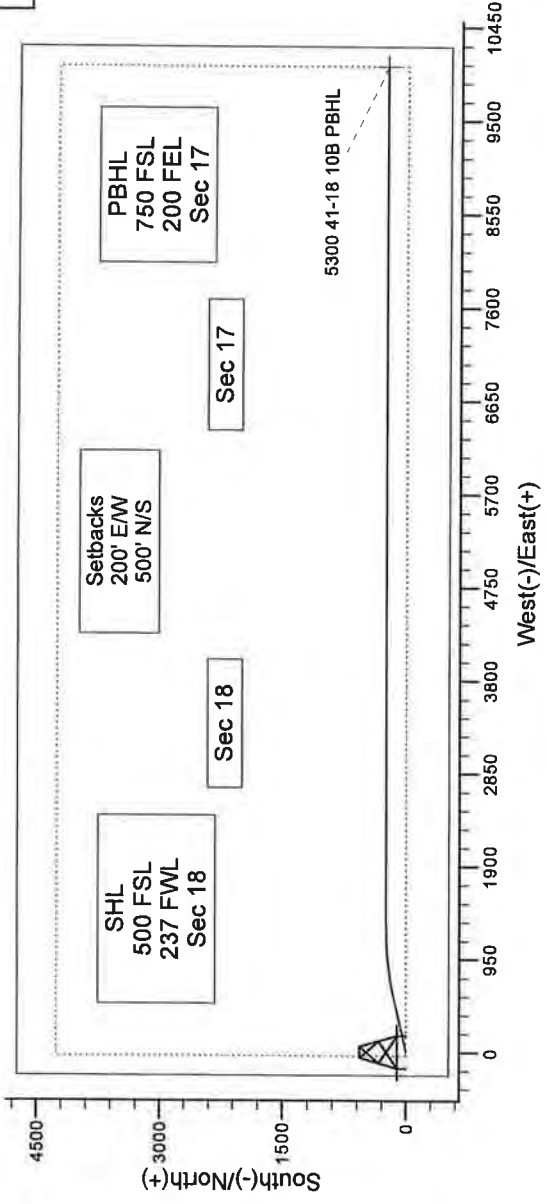




Project: Indian Hills  
Site: 153N-100W-17/18  
Well: Kline Federal 5300 41-18 10B  
Wellbore: Kline Federal 5300 41-18 10B  
Design: Design #1

WELL DETAILS: Kline Federal 5300 41-18 10B

|                      |            |                                  |
|----------------------|------------|----------------------------------|
| Ground Level: 2057.0 | Latitude   | Longitude                        |
| 405202.44            | 1209970.87 | 48° 4' 8.200 N 103° 36' 11.950 W |

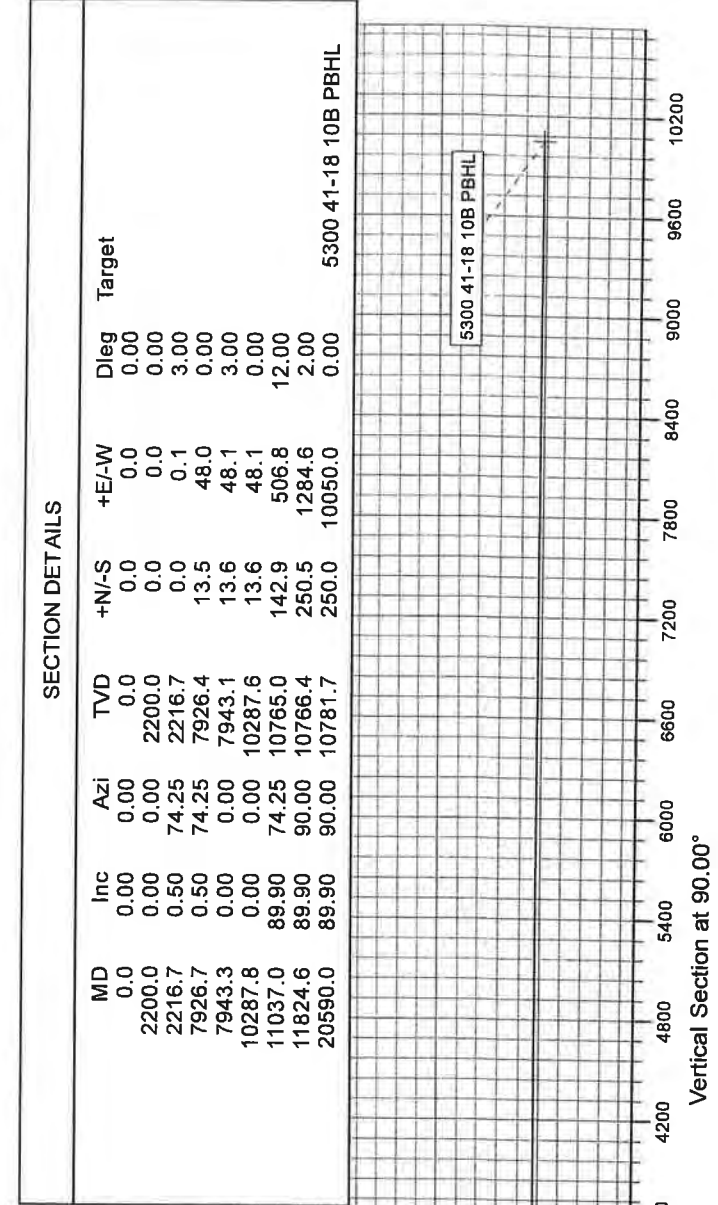


**CASING DETAILS**

| VD     | MD      | Name | Size |
|--------|---------|------|------|
| 0765.0 | 11037.0 | 7"   | 7    |

**SECTION DETAILS**

| MD      | Inc   | Azi   | TVD     | +N/-S | +E/-W   | Dleg  | Target              |
|---------|-------|-------|---------|-------|---------|-------|---------------------|
| 0.0     | 0.00  | 0.00  | 0.0     | 0.0   | 0.0     | 0.00  |                     |
| 2200.0  | 0.00  | 0.00  | 2200.0  | 0.0   | 0.0     | 0.00  |                     |
| 2216.7  | 0.50  | 74.25 | 2216.7  | 0.0   | 0.1     | 3.00  |                     |
| 7926.7  | 0.50  | 74.25 | 7926.4  | 13.5  | 48.0    | 0.00  |                     |
| 7943.3  | 0.00  | 0.00  | 7943.1  | 13.6  | 48.1    | 3.00  |                     |
| 10287.8 | 0.00  | 0.00  | 10287.6 | 13.6  | 48.1    | 0.00  |                     |
| 11037.0 | 89.90 | 74.25 | 10765.0 | 142.9 | 506.8   | 12.00 |                     |
| 11824.6 | 89.90 | 90.00 | 10766.4 | 250.5 | 1284.6  | 2.00  |                     |
| 20590.0 | 89.90 | 90.00 | 10781.7 | 250.0 | 10050.0 | 0.00  | 5300 41-18 10B PBHL |





# Ryan Directional Services Planning Report



|           |                              |                              |                                        |
|-----------|------------------------------|------------------------------|----------------------------------------|
| Database: | EDM 5000.1 Single User Db    | Local Co-ordinate Reference: | Well Kline Federal 5300 41-18 10B      |
| Company:  | Oasis Petroleum              | TVD Reference:               | WELL @ 2082.0usft (Original Well Elev) |
| Project:  | Indian Hills                 | MD Reference:                | WELL @ 2082.0usft (Original Well Elev) |
| Site:     | 153N-100W-17/18              | North Reference:             | True                                   |
| Well:     | Kline Federal 5300 41-18 10B | Survey Calculation Method:   | Minimum Curvature                      |
| Wellbore: | Kline Federal 5300 41-18 10B |                              |                                        |
| Design:   | Design #1                    |                              |                                        |

|             |                            |               |                |
|-------------|----------------------------|---------------|----------------|
| Project     |                            | Indian Hills  |                |
| Map System: | US State Plane 1983        | System Datum: | Mean Sea Level |
| Geo Datum:  | North American Datum 1983  |               |                |
| Map Zone:   | North Dakota Northern Zone |               |                |

|                       |  |                 |  |                   |  |
|-----------------------|--|-----------------|--|-------------------|--|
| Site                  |  | 153N-100W-17/18 |  |                   |  |
| Site Position:        |  | Northing:       |  | 408,962.44 usft   |  |
| From:                 |  | Easting:        |  | 1,210,229.18 usft |  |
| Position Uncertainty: |  | Slot Radius:    |  | Grid Convergence: |  |
| Lat/Long              |  | 0.0 usft        |  | 13-3/16 "         |  |
|                       |  |                 |  | 48° 4' 45.380 N   |  |
|                       |  |                 |  | 103° 36' 10.380 W |  |
|                       |  |                 |  | -2.31             |  |

|                      |                              |               |                     |                   |               |                   |
|----------------------|------------------------------|---------------|---------------------|-------------------|---------------|-------------------|
| Well:                | Kline Federal 5300 41-18 10B |               |                     |                   |               |                   |
| Well Position        | +N-S                         | -3,767.4 usft | Northing:           | 405,202.44 usft   | Latitude:     | 48° 4' 8.200 N    |
|                      | +E-W                         | -106.6 usft   | Easting:            | 1,209,970.87 usft | Longitude:    | 103° 36' 11.950 W |
| Position Uncertainty |                              | 2.0 usft      | Wellhead Elevation: |                   | Ground Level: | 2,057.0 usft      |

|           |            |                              |             |           |                |
|-----------|------------|------------------------------|-------------|-----------|----------------|
| Wellbore  |            | Kline Federal 5300 41-18 10B |             |           |                |
| Magnetics | Model Name | Sample Date                  | Declination | Dip Angle | Field Strength |
|           |            |                              | (°)         | (°)       | (nT)           |
|           | IGRF2010   | 5/19/2014                    | 8.27        | 72.99     | 56,438         |

|                   |                  |           |               |           |
|-------------------|------------------|-----------|---------------|-----------|
| Design:           | Design #1        |           |               |           |
| Audit Notes:      |                  |           |               |           |
| Version:          | Phase:           | PROTOTYPE | Tie On Depth: | 0.0       |
| Vertical Section: | Depth From (TVD) | +N-S      | +E-W          | Direction |
|                   | (usft)           | (usft)    | (usft)        | (°)       |
|                   | 0.0              | 0.0       | 0.0           | 90.00     |

| Measured Depth (usft) | Inclination (°) | Azimuth (°) | Vertical Depth (usft) | +N-S (usft) | +E-W (usft) | Dogleg Rate (°/100usft) | Build Rate (°/100usft) | Turn Rate (°/100usft) | TFO (°) | Target              |
|-----------------------|-----------------|-------------|-----------------------|-------------|-------------|-------------------------|------------------------|-----------------------|---------|---------------------|
| 0.0                   | 0.00            | 0.00        | 0.0                   | 0.0         | 0.0         | 0.00                    | 0.00                   | 0.00                  | 0.00    |                     |
| 2,200.0               | 0.00            | 0.00        | 2,200.0               | 0.0         | 0.0         | 0.00                    | 0.00                   | 0.00                  | 0.00    |                     |
| 2,216.7               | 0.50            | 74.25       | 2,216.7               | 0.0         | 0.1         | 3.00                    | 3.00                   | 0.00                  | 74.25   |                     |
| 7,926.7               | 0.50            | 74.25       | 7,926.4               | 13.5        | 48.0        | 0.00                    | 0.00                   | 0.00                  | 0.00    |                     |
| 7,943.3               | 0.00            | 0.00        | 7,943.1               | 13.6        | 48.1        | 3.00                    | -3.00                  | 0.00                  | 180.00  |                     |
| 10,287.8              | 0.00            | 0.00        | 10,287.6              | 13.6        | 48.1        | 0.00                    | 0.00                   | 0.00                  | 0.00    |                     |
| 11,037.0              | 89.90           | 74.25       | 10,765.0              | 142.9       | 506.8       | 12.00                   | 12.00                  | 0.00                  | 74.25   |                     |
| 11,824.6              | 89.90           | 90.00       | 10,766.4              | 250.5       | 1,284.6     | 2.00                    | 0.00                   | 2.00                  | 90.01   |                     |
| 20,590.0              | 89.90           | 90.00       | 10,781.7              | 250.0       | 10,050.0    | 0.00                    | 0.00                   | 0.00                  | 0.00    | 5300 41-18 10B PBHI |



# Ryan Directional Services Planning Report



|           |                              |                              |                                        |
|-----------|------------------------------|------------------------------|----------------------------------------|
| Database: | EDM 5000.1 Single User Db    | Local Co-ordinate Reference: | Well Kline Federal 5300 41-18 10B      |
| Company:  | Oasis Petroleum              | TVD Reference:               | WELL @ 2082.0usft (Original Well Elev) |
| Project:  | Indian Hills                 | MD Reference:                | WELL @ 2082.0usft (Original Well Elev) |
| Site:     | 153N-100W-17/18              | North Reference:             | True                                   |
| Well:     | Kline Federal 5300 41-18 10B | Survey Calculation Method:   | Minimum Curvature                      |
| Wellbore: | Kline Federal 5300 41-18 10B |                              |                                        |
| Design:   | Design #1                    |                              |                                        |

| Planned Survey        |                 |             |                       |            |            |                         |                         |                        |                       |
|-----------------------|-----------------|-------------|-----------------------|------------|------------|-------------------------|-------------------------|------------------------|-----------------------|
| Measured Depth (usft) | Inclination (°) | Azimuth (°) | Vertical Depth (usft) | N/S (usft) | E/W (usft) | Vertical Section (usft) | Dogleg Rate (°/100usft) | Build Rate (°/100usft) | Turn Rate (°/100usft) |
| 2,200.0               | 0.00            | 0.00        | 2,200.0               | 0.0        | 0.0        | 0.0                     | 0.00                    | 0.00                   | 0.00                  |
| 2,216.7               | 0.50            | 74.25       | 2,216.7               | 0.0        | 0.1        | 0.1                     | 3.00                    | 3.00                   | 0.00                  |
| 2,300.0               | 0.50            | 74.25       | 2,300.0               | 0.2        | 0.8        | 0.8                     | 0.00                    | 0.00                   | 0.00                  |
| 2,400.0               | 0.50            | 74.25       | 2,400.0               | 0.5        | 1.6        | 1.6                     | 0.00                    | 0.00                   | 0.00                  |
| 2,500.0               | 0.50            | 74.25       | 2,500.0               | 0.7        | 2.4        | 2.4                     | 0.00                    | 0.00                   | 0.00                  |
| 2,600.0               | 0.50            | 74.25       | 2,600.0               | 0.9        | 3.3        | 3.3                     | 0.00                    | 0.00                   | 0.00                  |
| 2,700.0               | 0.50            | 74.25       | 2,700.0               | 1.2        | 4.1        | 4.1                     | 0.00                    | 0.00                   | 0.00                  |
| 2,800.0               | 0.50            | 74.25       | 2,800.0               | 1.4        | 5.0        | 5.0                     | 0.00                    | 0.00                   | 0.00                  |
| 2,900.0               | 0.50            | 74.25       | 2,900.0               | 1.6        | 5.8        | 5.8                     | 0.00                    | 0.00                   | 0.00                  |
| 3,000.0               | 0.50            | 74.25       | 3,000.0               | 1.9        | 6.6        | 6.6                     | 0.00                    | 0.00                   | 0.00                  |
| 3,100.0               | 0.50            | 74.25       | 3,100.0               | 2.1        | 7.5        | 7.5                     | 0.00                    | 0.00                   | 0.00                  |
| 3,200.0               | 0.50            | 74.25       | 3,200.0               | 2.3        | 8.3        | 8.3                     | 0.00                    | 0.00                   | 0.00                  |
| 3,300.0               | 0.50            | 74.25       | 3,300.0               | 2.6        | 9.2        | 9.2                     | 0.00                    | 0.00                   | 0.00                  |
| 3,400.0               | 0.50            | 74.25       | 3,400.0               | 2.8        | 10.0       | 10.0                    | 0.00                    | 0.00                   | 0.00                  |
| 3,500.0               | 0.50            | 74.25       | 3,500.0               | 3.1        | 10.8       | 10.8                    | 0.00                    | 0.00                   | 0.00                  |
| 3,600.0               | 0.50            | 74.25       | 3,599.9               | 3.3        | 11.7       | 11.7                    | 0.00                    | 0.00                   | 0.00                  |
| 3,700.0               | 0.50            | 74.25       | 3,699.9               | 3.5        | 12.5       | 12.5                    | 0.00                    | 0.00                   | 0.00                  |
| 3,800.0               | 0.50            | 74.25       | 3,799.9               | 3.8        | 13.4       | 13.4                    | 0.00                    | 0.00                   | 0.00                  |
| 3,900.0               | 0.50            | 74.25       | 3,899.9               | 4.0        | 14.2       | 14.2                    | 0.00                    | 0.00                   | 0.00                  |
| 4,000.0               | 0.50            | 74.25       | 3,999.9               | 4.2        | 15.0       | 15.0                    | 0.00                    | 0.00                   | 0.00                  |
| 4,100.0               | 0.50            | 74.25       | 4,099.9               | 4.5        | 15.9       | 15.9                    | 0.00                    | 0.00                   | 0.00                  |
| 4,200.0               | 0.50            | 74.25       | 4,199.9               | 4.7        | 16.7       | 16.7                    | 0.00                    | 0.00                   | 0.00                  |
| 4,300.0               | 0.50            | 74.25       | 4,299.9               | 5.0        | 17.6       | 17.6                    | 0.00                    | 0.00                   | 0.00                  |
| 4,400.0               | 0.50            | 74.25       | 4,399.9               | 5.2        | 18.4       | 18.4                    | 0.00                    | 0.00                   | 0.00                  |
| 4,500.0               | 0.50            | 74.25       | 4,499.9               | 5.4        | 19.2       | 19.2                    | 0.00                    | 0.00                   | 0.00                  |
| 4,600.0               | 0.50            | 74.25       | 4,599.9               | 5.7        | 20.1       | 20.1                    | 0.00                    | 0.00                   | 0.00                  |
| 4,646.1               | 0.50            | 74.25       | 4,646.0               | 5.8        | 20.5       | 20.5                    | 0.00                    | 0.00                   | 0.00                  |
| Greenhorn             |                 |             |                       |            |            |                         |                         |                        |                       |
| 4,700.0               | 0.50            | 74.25       | 4,699.9               | 5.9        | 20.9       | 20.9                    | 0.00                    | 0.00                   | 0.00                  |
| 4,800.0               | 0.50            | 74.25       | 4,799.9               | 6.1        | 21.8       | 21.8                    | 0.00                    | 0.00                   | 0.00                  |
| 4,900.0               | 0.50            | 74.25       | 4,899.9               | 6.4        | 22.6       | 22.6                    | 0.00                    | 0.00                   | 0.00                  |
| 5,000.0               | 0.50            | 74.25       | 4,999.9               | 6.6        | 23.4       | 23.4                    | 0.00                    | 0.00                   | 0.00                  |
| 5,057.1               | 0.50            | 74.25       | 5,057.0               | 6.7        | 23.9       | 23.9                    | 0.00                    | 0.00                   | 0.00                  |
| Mowry                 |                 |             |                       |            |            |                         |                         |                        |                       |
| 5,100.0               | 0.50            | 74.25       | 5,099.9               | 6.8        | 24.3       | 24.3                    | 0.00                    | 0.00                   | 0.00                  |
| 5,200.0               | 0.50            | 74.25       | 5,199.9               | 7.1        | 25.1       | 25.1                    | 0.00                    | 0.00                   | 0.00                  |
| 5,300.0               | 0.50            | 74.25       | 5,299.9               | 7.3        | 26.0       | 26.0                    | 0.00                    | 0.00                   | 0.00                  |
| 5,400.0               | 0.50            | 74.25       | 5,399.9               | 7.6        | 26.8       | 26.8                    | 0.00                    | 0.00                   | 0.00                  |
| 5,471.1               | 0.50            | 74.25       | 5,471.0               | 7.7        | 27.4       | 27.4                    | 0.00                    | 0.00                   | 0.00                  |
| Dakota                |                 |             |                       |            |            |                         |                         |                        |                       |
| 5,500.0               | 0.50            | 74.25       | 5,499.9               | 7.8        | 27.6       | 27.6                    | 0.00                    | 0.00                   | 0.00                  |
| 5,600.0               | 0.50            | 74.25       | 5,599.9               | 8.0        | 28.5       | 28.5                    | 0.00                    | 0.00                   | 0.00                  |
| 5,700.0               | 0.50            | 74.25       | 5,699.9               | 8.3        | 29.3       | 29.3                    | 0.00                    | 0.00                   | 0.00                  |
| 5,800.0               | 0.50            | 74.25       | 5,799.9               | 8.5        | 30.2       | 30.2                    | 0.00                    | 0.00                   | 0.00                  |
| 5,900.0               | 0.50            | 74.25       | 5,899.9               | 8.7        | 31.0       | 31.0                    | 0.00                    | 0.00                   | 0.00                  |
| 6,000.0               | 0.50            | 74.25       | 5,999.9               | 9.0        | 31.8       | 31.8                    | 0.00                    | 0.00                   | 0.00                  |
| 6,100.0               | 0.50            | 74.25       | 6,099.9               | 9.2        | 32.7       | 32.7                    | 0.00                    | 0.00                   | 0.00                  |
| 6,200.0               | 0.50            | 74.25       | 6,199.8               | 9.5        | 33.5       | 33.5                    | 0.00                    | 0.00                   | 0.00                  |
| 6,300.0               | 0.50            | 74.25       | 6,299.8               | 9.7        | 34.4       | 34.4                    | 0.00                    | 0.00                   | 0.00                  |
| 6,400.0               | 0.50            | 74.25       | 6,399.8               | 9.9        | 35.2       | 35.2                    | 0.00                    | 0.00                   | 0.00                  |
| 6,486.2               | 0.50            | 74.25       | 6,486.0               | 10.1       | 35.9       | 35.9                    | 0.00                    | 0.00                   | 0.00                  |
| Rierson               |                 |             |                       |            |            |                         |                         |                        |                       |
| 6,500.0               | 0.50            | 74.25       | 6,499.8               | 10.2       | 36.0       | 36.0                    | 0.00                    | 0.00                   | 0.00                  |



# Ryan Directional Services Planning Report



|           |                              |                              |                                        |
|-----------|------------------------------|------------------------------|----------------------------------------|
| Database: | EDM 5000.1 Single User Db    | Local Co-ordinate Reference: | Well Kline Federal 5300 41-18 10B      |
| Company:  | Oasis Petroleum              | TVD Reference:               | WELL @ 2082.0usft (Original Well Elev) |
| Project:  | Indian Hills                 | MD Reference:                | WELL @ 2082.0usft (Original Well Elev) |
| Site:     | 153N-100W-17/18              | North Reference:             | True                                   |
| Well:     | Kline Federal 5300 41-18 10B | Survey Calculation Method:   | Minimum Curvature                      |
| Wellbore: | Kline Federal 5300 41-18 10B |                              |                                        |
| Design:   | Design #1                    |                              |                                        |

| Planned Survey                            |                 |             |                       |             |             |                         |                         |                        |                       |
|-------------------------------------------|-----------------|-------------|-----------------------|-------------|-------------|-------------------------|-------------------------|------------------------|-----------------------|
| Measured Depth (usft)                     | Inclination (°) | Azimuth (°) | Vertical Depth (usft) | +N-S (usft) | +E-W (usft) | Vertical Section (usft) | Dogleg Rate (°/100usft) | Build Rate (°/100usft) | Turn Rate (°/100usft) |
| 6,600.0                                   | 0.50            | 74.25       | 6,599.8               | 10.4        | 36.9        | 36.9                    | 0.00                    | 0.00                   | 0.00                  |
| 6,700.0                                   | 0.50            | 74.25       | 6,699.8               | 10.6        | 37.7        | 37.7                    | 0.00                    | 0.00                   | 0.00                  |
| 6,800.0                                   | 0.50            | 74.25       | 6,799.8               | 10.9        | 38.6        | 38.6                    | 0.00                    | 0.00                   | 0.00                  |
| 6,814.2                                   | 0.50            | 74.25       | 6,814.0               | 10.9        | 38.7        | 38.7                    | 0.00                    | 0.00                   | 0.00                  |
| <b>Dunham Salt</b>                        |                 |             |                       |             |             |                         |                         |                        |                       |
| 6,900.0                                   | 0.50            | 74.25       | 6,899.8               | 11.1        | 39.4        | 39.4                    | 0.00                    | 0.00                   | 0.00                  |
| 6,929.2                                   | 0.50            | 74.25       | 6,929.0               | 11.2        | 39.6        | 39.6                    | 0.00                    | 0.00                   | 0.00                  |
| <b>Dunham Salt Base</b>                   |                 |             |                       |             |             |                         |                         |                        |                       |
| 7,000.0                                   | 0.50            | 74.25       | 6,999.8               | 11.4        | 40.2        | 40.2                    | 0.00                    | 0.00                   | 0.00                  |
| 7,024.2                                   | 0.50            | 74.25       | 7,024.0               | 11.4        | 40.4        | 40.4                    | 0.00                    | 0.00                   | 0.00                  |
| <b>Spearfish</b>                          |                 |             |                       |             |             |                         |                         |                        |                       |
| 7,100.0                                   | 0.50            | 74.25       | 7,099.8               | 11.6        | 41.1        | 41.1                    | 0.00                    | 0.00                   | 0.00                  |
| 7,200.0                                   | 0.50            | 74.25       | 7,199.8               | 11.8        | 41.9        | 41.9                    | 0.00                    | 0.00                   | 0.00                  |
| 7,283.2                                   | 0.50            | 74.25       | 7,283.0               | 12.0        | 42.6        | 42.6                    | 0.00                    | 0.00                   | 0.00                  |
| <b>Pine Salt</b>                          |                 |             |                       |             |             |                         |                         |                        |                       |
| 7,300.0                                   | 0.50            | 74.25       | 7,299.8               | 12.1        | 42.8        | 42.8                    | 0.00                    | 0.00                   | 0.00                  |
| 7,318.2                                   | 0.50            | 74.25       | 7,318.0               | 12.1        | 42.9        | 42.9                    | 0.00                    | 0.00                   | 0.00                  |
| <b>Pine Salt Base</b>                     |                 |             |                       |             |             |                         |                         |                        |                       |
| 7,374.2                                   | 0.50            | 74.25       | 7,374.0               | 12.2        | 43.4        | 43.4                    | 0.00                    | 0.00                   | 0.00                  |
| <b>Opeche Salt</b>                        |                 |             |                       |             |             |                         |                         |                        |                       |
| 7,400.0                                   | 0.50            | 74.25       | 7,399.8               | 12.3        | 43.6        | 43.6                    | 0.00                    | 0.00                   | 0.00                  |
| 7,403.2                                   | 0.50            | 74.25       | 7,403.0               | 12.3        | 43.6        | 43.6                    | 0.00                    | 0.00                   | 0.00                  |
| <b>Opeche Salt Base</b>                   |                 |             |                       |             |             |                         |                         |                        |                       |
| 7,500.0                                   | 0.50            | 74.25       | 7,499.8               | 12.5        | 44.4        | 44.4                    | 0.00                    | 0.00                   | 0.00                  |
| 7,600.0                                   | 0.50            | 74.25       | 7,599.8               | 12.8        | 45.3        | 45.3                    | 0.00                    | 0.00                   | 0.00                  |
| 7,605.2                                   | 0.50            | 74.25       | 7,605.0               | 12.8        | 45.3        | 45.3                    | 0.00                    | 0.00                   | 0.00                  |
| <b>Broom Creek (Top of Minnelusa Gp.)</b> |                 |             |                       |             |             |                         |                         |                        |                       |
| 7,684.2                                   | 0.50            | 74.25       | 7,684.0               | 13.0        | 46.0        | 46.0                    | 0.00                    | 0.00                   | 0.00                  |
| <b>Amden</b>                              |                 |             |                       |             |             |                         |                         |                        |                       |
| 7,700.0                                   | 0.50            | 74.25       | 7,699.8               | 13.0        | 46.1        | 46.1                    | 0.00                    | 0.00                   | 0.00                  |
| 7,800.0                                   | 0.50            | 74.25       | 7,799.8               | 13.2        | 47.0        | 47.0                    | 0.00                    | 0.00                   | 0.00                  |
| 7,853.2                                   | 0.50            | 74.25       | 7,853.0               | 13.4        | 47.4        | 47.4                    | 0.00                    | 0.00                   | 0.00                  |
| <b>Tyler</b>                              |                 |             |                       |             |             |                         |                         |                        |                       |
| 7,900.0                                   | 0.50            | 74.25       | 7,899.8               | 13.5        | 47.8        | 47.8                    | 0.00                    | 0.00                   | 0.00                  |
| 7,926.7                                   | 0.50            | 74.25       | 7,926.4               | 13.5        | 48.0        | 48.0                    | 0.00                    | 0.00                   | 0.00                  |
| 7,943.3                                   | 0.00            | 0.00        | 7,943.1               | 13.6        | 48.1        | 48.1                    | 3.00                    | -3.00                  | 0.00                  |
| 8,000.0                                   | 0.00            | 0.00        | 7,999.8               | 13.6        | 48.1        | 48.1                    | 0.00                    | 0.00                   | 0.00                  |
| 8,047.2                                   | 0.00            | 0.00        | 8,047.0               | 13.6        | 48.1        | 48.1                    | 0.00                    | 0.00                   | 0.00                  |
| <b>Otter (Base of Minnelusa Gp.)</b>      |                 |             |                       |             |             |                         |                         |                        |                       |
| 8,100.0                                   | 0.00            | 0.00        | 8,099.8               | 13.6        | 48.1        | 48.1                    | 0.00                    | 0.00                   | 0.00                  |
| 8,200.0                                   | 0.00            | 0.00        | 8,199.8               | 13.6        | 48.1        | 48.1                    | 0.00                    | 0.00                   | 0.00                  |
| 8,300.0                                   | 0.00            | 0.00        | 8,299.8               | 13.6        | 48.1        | 48.1                    | 0.00                    | 0.00                   | 0.00                  |
| 8,390.2                                   | 0.00            | 0.00        | 8,390.0               | 13.6        | 48.1        | 48.1                    | 0.00                    | 0.00                   | 0.00                  |
| <b>Kibbey Lime</b>                        |                 |             |                       |             |             |                         |                         |                        |                       |
| 8,400.0                                   | 0.00            | 0.00        | 8,399.8               | 13.6        | 48.1        | 48.1                    | 0.00                    | 0.00                   | 0.00                  |
| 8,500.0                                   | 0.00            | 0.00        | 8,499.8               | 13.6        | 48.1        | 48.1                    | 0.00                    | 0.00                   | 0.00                  |
| 8,542.2                                   | 0.00            | 0.00        | 8,542.0               | 13.6        | 48.1        | 48.1                    | 0.00                    | 0.00                   | 0.00                  |
| <b>Charles Salt</b>                       |                 |             |                       |             |             |                         |                         |                        |                       |
| 8,600.0                                   | 0.00            | 0.00        | 8,599.8               | 13.6        | 48.1        | 48.1                    | 0.00                    | 0.00                   | 0.00                  |
| 8,700.0                                   | 0.00            | 0.00        | 8,699.8               | 13.6        | 48.1        | 48.1                    | 0.00                    | 0.00                   | 0.00                  |
| 8,800.0                                   | 0.00            | 0.00        | 8,799.8               | 13.6        | 48.1        | 48.1                    | 0.00                    | 0.00                   | 0.00                  |
| 8,900.0                                   | 0.00            | 0.00        | 8,899.8               | 13.6        | 48.1        | 48.1                    | 0.00                    | 0.00                   | 0.00                  |



# Ryan Directional Services Planning Report



|           |                              |                              |                                        |
|-----------|------------------------------|------------------------------|----------------------------------------|
| Database: | EDM 5000.1 Single User Db    | Local Co-ordinate Reference: | Well Kline Federal 5300 41-18 10B      |
| Company:  | Oasis Petroleum              | TVD Reference:               | WELL @ 2082.0usft (Original Well Elev) |
| Project:  | Indian Hills                 | MD Reference:                | WELL @ 2082.0usft (Original Well Elev) |
| Site:     | 153N-100W-1718               | North Reference:             | True                                   |
| Well:     | Kline Federal 5300 41-18 10B | Survey Calculation Method:   | Minimum Curvature                      |
| Wellbore: | Kline Federal 5300 41-18 10B |                              |                                        |
| Design:   | Design #1                    |                              |                                        |

| Planned Survey          |                 |             |                       |             |             |                         |                         |                        |                       |
|-------------------------|-----------------|-------------|-----------------------|-------------|-------------|-------------------------|-------------------------|------------------------|-----------------------|
| Measured Depth (usft)   | Inclination (°) | Azimuth (°) | Vertical Depth (usft) | +N-S (usft) | +E-W (usft) | Vertical Section (usft) | Dogleg Rate (°/100usft) | Build Rate (°/100usft) | Turn Rate (°/100usft) |
| 9,000.0                 | 0.00            | 0.00        | 8,999.8               | 13.6        | 48.1        | 48.1                    | 0.00                    | 0.00                   | 0.00                  |
| 9,100.0                 | 0.00            | 0.00        | 9,099.8               | 13.6        | 48.1        | 48.1                    | 0.00                    | 0.00                   | 0.00                  |
| 9,159.2                 | 0.00            | 0.00        | 9,159.0               | 13.6        | 48.1        | 48.1                    | 0.00                    | 0.00                   | 0.00                  |
| UB                      |                 |             |                       |             |             |                         |                         |                        |                       |
| 9,200.0                 | 0.00            | 0.00        | 9,199.8               | 13.6        | 48.1        | 48.1                    | 0.00                    | 0.00                   | 0.00                  |
| 9,238.2                 | 0.00            | 0.00        | 9,238.0               | 13.6        | 48.1        | 48.1                    | 0.00                    | 0.00                   | 0.00                  |
| Base Last Salt          |                 |             |                       |             |             |                         |                         |                        |                       |
| 9,300.0                 | 0.00            | 0.00        | 9,299.8               | 13.6        | 48.1        | 48.1                    | 0.00                    | 0.00                   | 0.00                  |
| 9,301.2                 | 0.00            | 0.00        | 9,301.0               | 13.6        | 48.1        | 48.1                    | 0.00                    | 0.00                   | 0.00                  |
| Ratcliffe               |                 |             |                       |             |             |                         |                         |                        |                       |
| 9,400.0                 | 0.00            | 0.00        | 9,399.8               | 13.6        | 48.1        | 48.1                    | 0.00                    | 0.00                   | 0.00                  |
| 9,454.2                 | 0.00            | 0.00        | 9,454.0               | 13.6        | 48.1        | 48.1                    | 0.00                    | 0.00                   | 0.00                  |
| Mission Canyon          |                 |             |                       |             |             |                         |                         |                        |                       |
| 9,500.0                 | 0.00            | 0.00        | 9,499.8               | 13.6        | 48.1        | 48.1                    | 0.00                    | 0.00                   | 0.00                  |
| 9,600.0                 | 0.00            | 0.00        | 9,599.8               | 13.6        | 48.1        | 48.1                    | 0.00                    | 0.00                   | 0.00                  |
| 9,700.0                 | 0.00            | 0.00        | 9,699.8               | 13.6        | 48.1        | 48.1                    | 0.00                    | 0.00                   | 0.00                  |
| 9,800.0                 | 0.00            | 0.00        | 9,799.8               | 13.6        | 48.1        | 48.1                    | 0.00                    | 0.00                   | 0.00                  |
| 9,900.0                 | 0.00            | 0.00        | 9,899.8               | 13.6        | 48.1        | 48.1                    | 0.00                    | 0.00                   | 0.00                  |
| 10,000.0                | 0.00            | 0.00        | 9,999.8               | 13.6        | 48.1        | 48.1                    | 0.00                    | 0.00                   | 0.00                  |
| 10,022.2                | 0.00            | 0.00        | 10,022.0              | 13.6        | 48.1        | 48.1                    | 0.00                    | 0.00                   | 0.00                  |
| Lodgepole               |                 |             |                       |             |             |                         |                         |                        |                       |
| 10,100.0                | 0.00            | 0.00        | 10,099.8              | 13.6        | 48.1        | 48.1                    | 0.00                    | 0.00                   | 0.00                  |
| 10,200.0                | 0.00            | 0.00        | 10,199.8              | 13.6        | 48.1        | 48.1                    | 0.00                    | 0.00                   | 0.00                  |
| 10,209.2                | 0.00            | 0.00        | 10,209.0              | 13.6        | 48.1        | 48.1                    | 0.00                    | 0.00                   | 0.00                  |
| Lodgepole Fracture Zone |                 |             |                       |             |             |                         |                         |                        |                       |
| 10,287.8                | 0.00            | 0.00        | 10,287.6              | 13.6        | 48.1        | 48.1                    | 0.00                    | 0.00                   | 0.00                  |
| KOP Build 12°/100'      |                 |             |                       |             |             |                         |                         |                        |                       |
| 10,300.0                | 1.46            | 74.25       | 10,299.8              | 13.6        | 48.2        | 48.2                    | 12.00                   | 12.00                  | 0.00                  |
| 10,325.0                | 4.46            | 74.25       | 10,324.7              | 14.0        | 49.5        | 49.5                    | 12.00                   | 12.00                  | 0.00                  |
| 10,350.0                | 7.46            | 74.25       | 10,349.6              | 14.7        | 52.0        | 52.0                    | 12.00                   | 12.00                  | 0.00                  |
| 10,375.0                | 10.46           | 74.25       | 10,374.3              | 15.7        | 55.7        | 55.7                    | 12.00                   | 12.00                  | 0.00                  |
| 10,400.0                | 13.46           | 74.25       | 10,398.8              | 17.1        | 60.7        | 60.7                    | 12.00                   | 12.00                  | 0.00                  |
| 10,425.0                | 16.46           | 74.25       | 10,422.9              | 18.9        | 66.9        | 66.9                    | 12.00                   | 12.00                  | 0.00                  |
| 10,450.0                | 19.46           | 74.25       | 10,446.7              | 21.0        | 74.4        | 74.4                    | 12.00                   | 12.00                  | 0.00                  |
| 10,475.0                | 22.46           | 74.25       | 10,470.0              | 23.4        | 83.0        | 83.0                    | 12.00                   | 12.00                  | 0.00                  |
| 10,500.0                | 25.46           | 74.25       | 10,492.9              | 26.2        | 92.7        | 92.7                    | 12.00                   | 12.00                  | 0.00                  |
| 10,525.0                | 28.46           | 74.25       | 10,515.1              | 29.2        | 103.6       | 103.6                   | 12.00                   | 12.00                  | 0.00                  |
| 10,550.0                | 31.46           | 74.25       | 10,536.8              | 32.6        | 115.7       | 115.7                   | 12.00                   | 12.00                  | 0.00                  |
| 10,575.0                | 34.46           | 74.25       | 10,557.8              | 36.3        | 128.8       | 128.8                   | 12.00                   | 12.00                  | 0.00                  |
| 10,600.0                | 37.46           | 74.25       | 10,578.0              | 40.3        | 142.9       | 142.9                   | 12.00                   | 12.00                  | 0.00                  |
| 10,625.0                | 40.46           | 74.25       | 10,597.4              | 44.6        | 158.0       | 158.0                   | 12.00                   | 12.00                  | 0.00                  |
| 10,650.0                | 43.46           | 74.25       | 10,616.0              | 49.1        | 174.1       | 174.1                   | 12.00                   | 12.00                  | 0.00                  |
| 10,675.0                | 46.46           | 74.25       | 10,633.7              | 53.9        | 191.1       | 191.1                   | 12.00                   | 12.00                  | 0.00                  |
| 10,700.0                | 49.46           | 74.25       | 10,650.5              | 58.9        | 209.0       | 209.0                   | 12.00                   | 12.00                  | 0.00                  |
| 10,725.0                | 52.46           | 74.25       | 10,666.2              | 64.2        | 227.7       | 227.7                   | 12.00                   | 12.00                  | 0.00                  |
| 10,750.0                | 55.46           | 74.25       | 10,680.9              | 69.7        | 247.1       | 247.1                   | 12.00                   | 12.00                  | 0.00                  |
| 10,775.0                | 58.46           | 74.25       | 10,694.5              | 75.4        | 267.3       | 267.3                   | 12.00                   | 12.00                  | 0.00                  |
| 10,800.0                | 61.46           | 74.25       | 10,707.0              | 81.3        | 288.1       | 288.1                   | 12.00                   | 12.00                  | 0.00                  |
| 10,825.0                | 64.46           | 74.25       | 10,718.4              | 87.3        | 309.5       | 309.5                   | 12.00                   | 12.00                  | 0.00                  |
| 10,843.4                | 66.67           | 74.25       | 10,726.0              | 91.8        | 325.6       | 325.6                   | 12.00                   | 12.00                  | 0.00                  |
| False Bakken            |                 |             |                       |             |             |                         |                         |                        |                       |
| 10,850.0                | 67.46           | 74.25       | 10,728.6              | 93.5        | 331.5       | 331.5                   | 12.00                   | 12.00                  | 0.00                  |
| 10,867.5                | 69.57           | 74.25       | 10,735.0              | 97.9        | 347.2       | 347.2                   | 12.00                   | 12.00                  | 0.00                  |





# Ryan Directional Services Planning Report



|           |                              |                              |                                        |
|-----------|------------------------------|------------------------------|----------------------------------------|
| Database: | EDM 5000.1 Single User Db    | Local Co-ordinate Reference: | Well Kline Federal 5300 41-18 10B      |
| Company:  | Oasis Petroleum              | TVD Reference:               | WELL @ 2082.0usft (Original Well Elev) |
| Project:  | Indian Hills                 | MD Reference:                | WELL @ 2082.0usft (Original Well Elev) |
| Site:     | 153N-100W-17/18              | North Reference:             | True                                   |
| Well:     | Kline Federal 5300 41-18 10B | Survey Calculation Method:   | Minimum Curvature                      |
| Wellbore: | Kline Federal 5300 41-18 10B |                              |                                        |
| Design:   | Design #1                    |                              |                                        |

## Planned Survey

| Measured Depth (usft)                 | Inclination (°) | Azimuth (°) | Vertical Depth (usft) | +N-S (usft) | +E-W (usft) | Vertical Section (usft) | Depth Rate (°/100usft) | Build Rate (°/100usft) | Turn Rate (°/100usft) |
|---------------------------------------|-----------------|-------------|-----------------------|-------------|-------------|-------------------------|------------------------|------------------------|-----------------------|
| <b>Upper Bakken</b>                   |                 |             |                       |             |             |                         |                        |                        |                       |
| 10,875.0                              | 70.46           | 74.25       | 10,737.6              | 99.8        | 354.0       | 354.0                   | 12.00                  | 12.00                  | 0.00                  |
| 10,900.0                              | 73.46           | 74.25       | 10,745.3              | 106.3       | 376.8       | 376.8                   | 12.00                  | 12.00                  | 0.00                  |
| 10,921.7                              | 76.07           | 74.25       | 10,751.0              | 112.0       | 397.0       | 397.0                   | 12.00                  | 12.00                  | 0.00                  |
| <b>Middle Bakken</b>                  |                 |             |                       |             |             |                         |                        |                        |                       |
| 10,925.0                              | 76.46           | 74.25       | 10,751.8              | 112.8       | 400.1       | 400.1                   | 12.00                  | 12.00                  | 0.00                  |
| 10,950.0                              | 79.46           | 74.25       | 10,757.0              | 119.5       | 423.6       | 423.6                   | 12.00                  | 12.00                  | 0.00                  |
| 10,968.3                              | 81.66           | 74.25       | 10,760.0              | 124.4       | 441.0       | 441.0                   | 12.00                  | 12.00                  | 0.00                  |
| <b>Middle Bakken Sand Target</b>      |                 |             |                       |             |             |                         |                        |                        |                       |
| 10,975.0                              | 82.46           | 74.25       | 10,760.9              | 126.2       | 447.4       | 447.4                   | 12.00                  | 12.00                  | 0.00                  |
| 11,000.0                              | 85.46           | 74.25       | 10,763.6              | 132.9       | 471.3       | 471.3                   | 12.00                  | 12.00                  | 0.00                  |
| 11,025.0                              | 88.46           | 74.25       | 10,764.9              | 139.7       | 495.3       | 495.3                   | 12.00                  | 12.00                  | 0.00                  |
| 11,037.0                              | 89.90           | 74.25       | 10,765.0              | 143.0       | 506.9       | 506.9                   | 11.97                  | 11.97                  | 0.00                  |
| <b>EOC - 7"</b>                       |                 |             |                       |             |             |                         |                        |                        |                       |
| 11,100.0                              | 89.90           | 75.51       | 10,765.2              | 159.4       | 567.7       | 567.7                   | 2.00                   | 0.00                   | 2.00                  |
| 11,200.0                              | 89.90           | 77.51       | 10,765.3              | 182.7       | 664.9       | 664.9                   | 2.00                   | 0.00                   | 2.00                  |
| 11,300.0                              | 89.90           | 79.51       | 10,765.5              | 202.6       | 762.9       | 762.9                   | 2.00                   | 0.00                   | 2.00                  |
| 11,400.0                              | 89.90           | 81.51       | 10,765.7              | 219.1       | 861.5       | 861.5                   | 2.00                   | 0.00                   | 2.00                  |
| 11,500.0                              | 89.90           | 83.51       | 10,765.9              | 232.1       | 960.7       | 960.7                   | 2.00                   | 0.00                   | 2.00                  |
| 11,600.0                              | 89.90           | 85.51       | 10,766.0              | 241.7       | 1,060.2     | 1,060.2                 | 2.00                   | 0.00                   | 2.00                  |
| 11,700.0                              | 89.90           | 87.51       | 10,766.2              | 247.8       | 1,160.0     | 1,160.0                 | 2.00                   | 0.00                   | 2.00                  |
| 11,800.0                              | 89.90           | 89.51       | 10,766.4              | 250.4       | 1,260.0     | 1,260.0                 | 2.00                   | 0.00                   | 2.00                  |
| 11,824.6                              | 89.90           | 90.00       | 10,766.4              | 250.5       | 1,284.6     | 1,284.6                 | 2.00                   | 0.00                   | 2.00                  |
| 11,900.0                              | 89.90           | 90.00       | 10,766.6              | 250.5       | 1,360.0     | 1,360.0                 | 0.00                   | 0.00                   | 0.00                  |
| 12,000.0                              | 89.90           | 90.00       | 10,766.7              | 250.5       | 1,460.0     | 1,460.0                 | 0.00                   | 0.00                   | 0.00                  |
| 12,100.0                              | 89.90           | 90.00       | 10,766.9              | 250.5       | 1,560.0     | 1,560.0                 | 0.00                   | 0.00                   | 0.00                  |
| 12,200.0                              | 89.90           | 90.00       | 10,767.1              | 250.5       | 1,660.0     | 1,660.0                 | 0.00                   | 0.00                   | 0.00                  |
| 12,300.0                              | 89.90           | 90.00       | 10,767.3              | 250.5       | 1,760.0     | 1,760.0                 | 0.00                   | 0.00                   | 0.00                  |
| 12,400.0                              | 89.90           | 90.00       | 10,767.4              | 250.5       | 1,860.0     | 1,860.0                 | 0.00                   | 0.00                   | 0.00                  |
| 12,500.0                              | 89.90           | 90.00       | 10,767.6              | 250.5       | 1,960.0     | 1,960.0                 | 0.00                   | 0.00                   | 0.00                  |
| 12,600.0                              | 89.90           | 90.00       | 10,767.8              | 250.5       | 2,060.0     | 2,060.0                 | 0.00                   | 0.00                   | 0.00                  |
| 12,700.0                              | 89.90           | 90.00       | 10,768.0              | 250.4       | 2,160.0     | 2,160.0                 | 0.00                   | 0.00                   | 0.00                  |
| 12,800.0                              | 89.90           | 90.00       | 10,768.1              | 250.4       | 2,260.0     | 2,260.0                 | 0.00                   | 0.00                   | 0.00                  |
| 12,900.0                              | 89.90           | 90.00       | 10,768.3              | 250.4       | 2,360.0     | 2,360.0                 | 0.00                   | 0.00                   | 0.00                  |
| 13,000.0                              | 89.90           | 90.00       | 10,768.5              | 250.4       | 2,460.0     | 2,460.0                 | 0.00                   | 0.00                   | 0.00                  |
| 13,100.0                              | 89.90           | 90.00       | 10,768.7              | 250.4       | 2,560.0     | 2,560.0                 | 0.00                   | 0.00                   | 0.00                  |
| 13,200.0                              | 89.90           | 90.00       | 10,768.8              | 250.4       | 2,660.0     | 2,660.0                 | 0.00                   | 0.00                   | 0.00                  |
| 13,300.0                              | 89.90           | 90.00       | 10,769.0              | 250.4       | 2,760.0     | 2,760.0                 | 0.00                   | 0.00                   | 0.00                  |
| 13,400.0                              | 89.90           | 90.00       | 10,769.2              | 250.4       | 2,860.0     | 2,860.0                 | 0.00                   | 0.00                   | 0.00                  |
| 13,500.0                              | 89.90           | 90.00       | 10,769.3              | 250.4       | 2,960.0     | 2,960.0                 | 0.00                   | 0.00                   | 0.00                  |
| 13,600.0                              | 89.90           | 90.00       | 10,769.5              | 250.4       | 3,060.0     | 3,060.0                 | 0.00                   | 0.00                   | 0.00                  |
| 13,700.0                              | 89.90           | 90.00       | 10,769.7              | 250.4       | 3,160.0     | 3,160.0                 | 0.00                   | 0.00                   | 0.00                  |
| 13,800.0                              | 89.90           | 90.00       | 10,769.9              | 250.4       | 3,260.0     | 3,260.0                 | 0.00                   | 0.00                   | 0.00                  |
| 13,874.6                              | 89.90           | 90.00       | 10,770.0              | 250.4       | 3,334.6     | 3,334.6                 | 0.00                   | 0.00                   | 0.00                  |
| <b>Base Middle Bakken Sand Target</b> |                 |             |                       |             |             |                         |                        |                        |                       |
| 13,900.0                              | 89.90           | 90.00       | 10,770.0              | 250.4       | 3,360.0     | 3,360.0                 | 0.00                   | 0.00                   | 0.00                  |
| 14,000.0                              | 89.90           | 90.00       | 10,770.2              | 250.4       | 3,460.0     | 3,460.0                 | 0.00                   | 0.00                   | 0.00                  |
| 14,100.0                              | 89.90           | 90.00       | 10,770.4              | 250.4       | 3,560.0     | 3,560.0                 | 0.00                   | 0.00                   | 0.00                  |
| 14,200.0                              | 89.90           | 90.00       | 10,770.6              | 250.4       | 3,660.0     | 3,660.0                 | 0.00                   | 0.00                   | 0.00                  |
| 14,300.0                              | 89.90           | 90.00       | 10,770.7              | 250.4       | 3,760.0     | 3,760.0                 | 0.00                   | 0.00                   | 0.00                  |
| 14,400.0                              | 89.90           | 90.00       | 10,770.9              | 250.4       | 3,860.0     | 3,860.0                 | 0.00                   | 0.00                   | 0.00                  |
| 14,500.0                              | 89.90           | 90.00       | 10,771.1              | 250.3       | 3,960.0     | 3,960.0                 | 0.00                   | 0.00                   | 0.00                  |
| 14,600.0                              | 89.90           | 90.00       | 10,771.3              | 250.3       | 4,060.0     | 4,060.0                 | 0.00                   | 0.00                   | 0.00                  |
| 14,700.0                              | 89.90           | 90.00       | 10,771.4              | 250.3       | 4,160.0     | 4,160.0                 | 0.00                   | 0.00                   | 0.00                  |



# Ryan Directional Services Planning Report



|           |                              |                              |                                        |
|-----------|------------------------------|------------------------------|----------------------------------------|
| Database: | EDM 5000.1 Single User Db    | Local Co-ordinate Reference: | Well Kline Federal 5300 41-18 10B      |
| Company:  | Oasis Petroleum              | TVD Reference:               | WELL @ 2082.0usft (Original Well Elev) |
| Project:  | Indian Hills                 | MD Reference:                | WELL @ 2082.0usft (Original Well Elev) |
| Site:     | 153N-100W-17218              | North Reference:             | True                                   |
| Well:     | Kline Federal 5300 41-18 10B | Survey Calculation Method:   | Minimum Curvature                      |
| Wellbore: | Kline Federal 5300 41-18 10B |                              |                                        |
| Design:   | Design #1                    |                              |                                        |

| Planned Survey        |                 |             |                       |            |            |                         |                         |                        |                       |
|-----------------------|-----------------|-------------|-----------------------|------------|------------|-------------------------|-------------------------|------------------------|-----------------------|
| Measured Depth (usft) | Inclination (°) | Azimuth (°) | Vertical Depth (usft) | N-S (usft) | E-W (usft) | Vertical Section (usft) | Dogleg Rate (°/100usft) | Bulld Rate (°/100usft) | Turn Rate (°/100usft) |
| 14,800.0              | 89.90           | 90.00       | 10,771.6              | 250.3      | 4,260.0    | 4,260.0                 | 0.00                    | 0.00                   | 0.00                  |
| 14,900.0              | 89.90           | 90.00       | 10,771.8              | 250.3      | 4,360.0    | 4,360.0                 | 0.00                    | 0.00                   | 0.00                  |
| 15,000.0              | 89.90           | 90.00       | 10,772.0              | 250.3      | 4,460.0    | 4,460.0                 | 0.00                    | 0.00                   | 0.00                  |
| 15,100.0              | 89.90           | 90.00       | 10,772.1              | 250.3      | 4,560.0    | 4,560.0                 | 0.00                    | 0.00                   | 0.00                  |
| 15,200.0              | 89.90           | 90.00       | 10,772.3              | 250.3      | 4,660.0    | 4,660.0                 | 0.00                    | 0.00                   | 0.00                  |
| 15,300.0              | 89.90           | 90.00       | 10,772.5              | 250.3      | 4,760.0    | 4,760.0                 | 0.00                    | 0.00                   | 0.00                  |
| 15,400.0              | 89.90           | 90.00       | 10,772.7              | 250.3      | 4,860.0    | 4,860.0                 | 0.00                    | 0.00                   | 0.00                  |
| 15,500.0              | 89.90           | 90.00       | 10,772.8              | 250.3      | 4,960.0    | 4,960.0                 | 0.00                    | 0.00                   | 0.00                  |
| 15,600.0              | 89.90           | 90.00       | 10,773.0              | 250.3      | 5,060.0    | 5,060.0                 | 0.00                    | 0.00                   | 0.00                  |
| 15,700.0              | 89.90           | 90.00       | 10,773.2              | 250.3      | 5,160.0    | 5,160.0                 | 0.00                    | 0.00                   | 0.00                  |
| 15,800.0              | 89.90           | 90.00       | 10,773.4              | 250.3      | 5,260.0    | 5,260.0                 | 0.00                    | 0.00                   | 0.00                  |
| 15,900.0              | 89.90           | 90.00       | 10,773.5              | 250.3      | 5,360.0    | 5,360.0                 | 0.00                    | 0.00                   | 0.00                  |
| 16,000.0              | 89.90           | 90.00       | 10,773.7              | 250.3      | 5,460.0    | 5,460.0                 | 0.00                    | 0.00                   | 0.00                  |
| 16,100.0              | 89.90           | 90.00       | 10,773.9              | 250.3      | 5,560.0    | 5,560.0                 | 0.00                    | 0.00                   | 0.00                  |
| 16,200.0              | 89.90           | 90.00       | 10,774.1              | 250.3      | 5,660.0    | 5,660.0                 | 0.00                    | 0.00                   | 0.00                  |
| 16,300.0              | 89.90           | 90.00       | 10,774.2              | 250.2      | 5,760.0    | 5,760.0                 | 0.00                    | 0.00                   | 0.00                  |
| 16,400.0              | 89.90           | 90.00       | 10,774.4              | 250.2      | 5,860.0    | 5,860.0                 | 0.00                    | 0.00                   | 0.00                  |
| 16,500.0              | 89.90           | 90.00       | 10,774.6              | 250.2      | 5,960.0    | 5,960.0                 | 0.00                    | 0.00                   | 0.00                  |
| 16,600.0              | 89.90           | 90.00       | 10,774.7              | 250.2      | 6,060.0    | 6,060.0                 | 0.00                    | 0.00                   | 0.00                  |
| 16,700.0              | 89.90           | 90.00       | 10,774.9              | 250.2      | 6,160.0    | 6,160.0                 | 0.00                    | 0.00                   | 0.00                  |
| 16,800.0              | 89.90           | 90.00       | 10,775.1              | 250.2      | 6,260.0    | 6,260.0                 | 0.00                    | 0.00                   | 0.00                  |
| 16,900.0              | 89.90           | 90.00       | 10,775.3              | 250.2      | 6,360.0    | 6,360.0                 | 0.00                    | 0.00                   | 0.00                  |
| 17,000.0              | 89.90           | 90.00       | 10,775.4              | 250.2      | 6,460.0    | 6,460.0                 | 0.00                    | 0.00                   | 0.00                  |
| 17,100.0              | 89.90           | 90.00       | 10,775.6              | 250.2      | 6,560.0    | 6,560.0                 | 0.00                    | 0.00                   | 0.00                  |
| 17,200.0              | 89.90           | 90.00       | 10,775.8              | 250.2      | 6,660.0    | 6,660.0                 | 0.00                    | 0.00                   | 0.00                  |
| 17,300.0              | 89.90           | 90.00       | 10,776.0              | 250.2      | 6,760.0    | 6,760.0                 | 0.00                    | 0.00                   | 0.00                  |
| 17,400.0              | 89.90           | 90.00       | 10,776.1              | 250.2      | 6,860.0    | 6,860.0                 | 0.00                    | 0.00                   | 0.00                  |
| 17,500.0              | 89.90           | 90.00       | 10,776.3              | 250.2      | 6,960.0    | 6,960.0                 | 0.00                    | 0.00                   | 0.00                  |
| 17,600.0              | 89.90           | 90.00       | 10,776.5              | 250.2      | 7,060.0    | 7,060.0                 | 0.00                    | 0.00                   | 0.00                  |
| 17,700.0              | 89.90           | 90.00       | 10,776.7              | 250.2      | 7,160.0    | 7,160.0                 | 0.00                    | 0.00                   | 0.00                  |
| 17,800.0              | 89.90           | 90.00       | 10,776.8              | 250.2      | 7,260.0    | 7,260.0                 | 0.00                    | 0.00                   | 0.00                  |
| 17,900.0              | 89.90           | 90.00       | 10,777.0              | 250.2      | 7,360.0    | 7,360.0                 | 0.00                    | 0.00                   | 0.00                  |
| 18,000.0              | 89.90           | 90.00       | 10,777.2              | 250.1      | 7,460.0    | 7,460.0                 | 0.00                    | 0.00                   | 0.00                  |
| 18,100.0              | 89.90           | 90.00       | 10,777.4              | 250.1      | 7,560.0    | 7,560.0                 | 0.00                    | 0.00                   | 0.00                  |
| 18,200.0              | 89.90           | 90.00       | 10,777.5              | 250.1      | 7,660.0    | 7,660.0                 | 0.00                    | 0.00                   | 0.00                  |
| 18,300.0              | 89.90           | 90.00       | 10,777.7              | 250.1      | 7,760.0    | 7,760.0                 | 0.00                    | 0.00                   | 0.00                  |
| 18,400.0              | 89.90           | 90.00       | 10,777.9              | 250.1      | 7,860.0    | 7,860.0                 | 0.00                    | 0.00                   | 0.00                  |
| 18,500.0              | 89.90           | 90.00       | 10,778.1              | 250.1      | 7,960.0    | 7,960.0                 | 0.00                    | 0.00                   | 0.00                  |
| 18,600.0              | 89.90           | 90.00       | 10,778.2              | 250.1      | 8,060.0    | 8,060.0                 | 0.00                    | 0.00                   | 0.00                  |
| 18,700.0              | 89.90           | 90.00       | 10,778.4              | 250.1      | 8,160.0    | 8,160.0                 | 0.00                    | 0.00                   | 0.00                  |
| 18,800.0              | 89.90           | 90.00       | 10,778.6              | 250.1      | 8,260.0    | 8,260.0                 | 0.00                    | 0.00                   | 0.00                  |
| 18,900.0              | 89.90           | 90.00       | 10,778.8              | 250.1      | 8,360.0    | 8,360.0                 | 0.00                    | 0.00                   | 0.00                  |
| 19,000.0              | 89.90           | 90.00       | 10,778.9              | 250.1      | 8,460.0    | 8,460.0                 | 0.00                    | 0.00                   | 0.00                  |
| 19,100.0              | 89.90           | 90.00       | 10,779.1              | 250.1      | 8,560.0    | 8,560.0                 | 0.00                    | 0.00                   | 0.00                  |
| 19,200.0              | 89.90           | 90.00       | 10,779.3              | 250.1      | 8,660.0    | 8,660.0                 | 0.00                    | 0.00                   | 0.00                  |
| 19,300.0              | 89.90           | 90.00       | 10,779.5              | 250.1      | 8,760.0    | 8,760.0                 | 0.00                    | 0.00                   | 0.00                  |
| 19,400.0              | 89.90           | 90.00       | 10,779.6              | 250.1      | 8,860.0    | 8,860.0                 | 0.00                    | 0.00                   | 0.00                  |
| 19,500.0              | 89.90           | 90.00       | 10,779.8              | 250.1      | 8,960.0    | 8,960.0                 | 0.00                    | 0.00                   | 0.00                  |
| 19,600.0              | 89.90           | 90.00       | 10,780.0              | 250.1      | 9,060.0    | 9,060.0                 | 0.00                    | 0.00                   | 0.00                  |
| 19,700.0              | 89.90           | 90.00       | 10,780.1              | 250.1      | 9,160.0    | 9,160.0                 | 0.00                    | 0.00                   | 0.00                  |
| 19,800.0              | 89.90           | 90.00       | 10,780.3              | 250.0      | 9,260.0    | 9,260.0                 | 0.00                    | 0.00                   | 0.00                  |
| 19,900.0              | 89.90           | 90.00       | 10,780.5              | 250.0      | 9,360.0    | 9,360.0                 | 0.00                    | 0.00                   | 0.00                  |
| 20,000.0              | 89.90           | 90.00       | 10,780.7              | 250.0      | 9,460.0    | 9,460.0                 | 0.00                    | 0.00                   | 0.00                  |
| 20,100.0              | 89.90           | 90.00       | 10,780.8              | 250.0      | 9,560.0    | 9,560.0                 | 0.00                    | 0.00                   | 0.00                  |



# Ryan Directional Services Planning Report



|           |                              |                              |                                        |
|-----------|------------------------------|------------------------------|----------------------------------------|
| Database: | EDM 5000.1 Single User Db    | Local Co-ordinate Reference: | Well Kline Federal 5300 41-18 10B      |
| Company:  | Oasis Petroleum              | TVD Reference:               | WELL @ 2082.0usft (Original Well Elev) |
| Project:  | Indian Hills                 | MD Reference:                | WELL @ 2082.0usft (Original Well Elev) |
| Site:     | 153N-100W-17/18              | North Reference:             | True                                   |
| Well:     | Kline Federal 5300 41-18 10B | Survey Calculation Method:   | Minimum Curvature                      |
| Wellbore: | Kline Federal 5300 41-18 10B |                              |                                        |
| Design:   | Design #1                    |                              |                                        |

| Planned Survey        |                 |             |                       |             |             |                         |                         |                        |                       |
|-----------------------|-----------------|-------------|-----------------------|-------------|-------------|-------------------------|-------------------------|------------------------|-----------------------|
| Measured Depth (usft) | Inclination (°) | Azimuth (°) | Vertical Depth (usft) | +N-S (usft) | +E-W (usft) | Vertical Section (usft) | Dogleg Rate (°/100usft) | Build Rate (°/100usft) | Turn Rate (°/100usft) |
| 20,200.0              | 89.90           | 90.00       | 10,781.0              | 250.0       | 9,660.0     | 9,660.0                 | 0.00                    | 0.00                   | 0.00                  |
| 20,300.0              | 89.90           | 90.00       | 10,781.2              | 250.0       | 9,760.0     | 9,760.0                 | 0.00                    | 0.00                   | 0.00                  |
| 20,400.0              | 89.90           | 90.00       | 10,781.4              | 250.0       | 9,860.0     | 9,860.0                 | 0.00                    | 0.00                   | 0.00                  |
| 20,500.0              | 89.90           | 90.00       | 10,781.5              | 250.0       | 9,960.0     | 9,960.0                 | 0.00                    | 0.00                   | 0.00                  |
| 20,590.0              | 89.90           | 90.00       | 10,781.7              | 250.0       | 10,050.0    | 10,050.0                | 0.00                    | 0.00                   | 0.00                  |
| 5300 41-18 10B PBHL   |                 |             |                       |             |             |                         |                         |                        |                       |

| Design Targets                                              |               |              |            |             |             |                 |                |                 |                   |
|-------------------------------------------------------------|---------------|--------------|------------|-------------|-------------|-----------------|----------------|-----------------|-------------------|
| Target Name                                                 | Dip Angle (°) | Dip Dir. (°) | TVD (usft) | +N-S (usft) | +E-W (usft) | Northing (usft) | Easting (usft) | Latitude        | Longitude         |
| 5300 41-18 10B PBHL<br>- plan hits target center<br>- Point | 0.00          | 0.00         | 10,781.7   | 250.0       | 10,050.0    | 405,047.29      | 1,220,022.78   | 48° 4' 10.641 N | 103° 33' 43.967 W |

| Casing Points         |                       |      |                     |                   |  |  |
|-----------------------|-----------------------|------|---------------------|-------------------|--|--|
| Measured Depth (usft) | Vertical Depth (usft) | Name | Casing Diameter (") | Hole Diameter (") |  |  |
| 11,037.0              | 10,765.0              | 7"   | 7                   | 8-3/4             |  |  |



# Ryan Directional Services Planning Report



|           |                              |                              |                                        |
|-----------|------------------------------|------------------------------|----------------------------------------|
| Database: | EDM 5000.1 Single User Db    | Local Co-ordinate Reference: | Well Kline Federal 5300 41-18 10B      |
| Company:  | Oasis Petroleum              | TVD Reference:               | WELL @ 2082.0usft (Original Well Elev) |
| Project:  | Indian Hills                 | MB Reference:                | WELL @ 2082.0usft (Original Well Elev) |
| Site:     | 153N-100W-17/18              | North Reference:             | True                                   |
| Well:     | Kline Federal 5300 41-18 10B | Survey Calculation Method:   | Minimum Curvature                      |
| Wellbore: | Kline Federal 5300 41-18 10B |                              |                                        |
| Design:   | Design #1                    |                              |                                        |

| Formations            |                       |                                    |           |         |                   |  |
|-----------------------|-----------------------|------------------------------------|-----------|---------|-------------------|--|
| Measured Depth (usft) | Vertical Depth (usft) | Name                               | Lithology | Dip (°) | Dip Direction (°) |  |
| 1,972.0               | 1,972.0               | Pierre                             |           |         |                   |  |
| 4,646.1               | 4,646.0               | Greenhorn                          |           |         |                   |  |
| 5,057.1               | 5,057.0               | Mowry                              |           |         |                   |  |
| 5,471.1               | 5,471.0               | Dakota                             |           |         |                   |  |
| 6,486.2               | 6,486.0               | Rierdon                            |           |         |                   |  |
| 6,814.2               | 6,814.0               | Dunham Salt                        |           |         |                   |  |
| 6,929.2               | 6,929.0               | Dunham Salt Base                   |           |         |                   |  |
| 7,024.2               | 7,024.0               | Spearfish                          |           |         |                   |  |
| 7,283.2               | 7,283.0               | Pine Salt                          |           |         |                   |  |
| 7,318.2               | 7,318.0               | Pine Salt Base                     |           |         |                   |  |
| 7,374.2               | 7,374.0               | Opeche Salt                        |           |         |                   |  |
| 7,403.2               | 7,403.0               | Opeche Salt Base                   |           |         |                   |  |
| 7,605.2               | 7,605.0               | Broom Creek (Top of Minnelusa Gp.) |           |         |                   |  |
| 7,684.2               | 7,684.0               | Amsden                             |           |         |                   |  |
| 7,853.2               | 7,853.0               | Tyler                              |           |         |                   |  |
| 8,047.2               | 8,047.0               | Otter (Base of Minnelusa Gp.)      |           |         |                   |  |
| 8,390.2               | 8,390.0               | Kibbey Lime                        |           |         |                   |  |
| 8,542.2               | 8,542.0               | Charles Salt                       |           |         |                   |  |
| 9,159.2               | 9,159.0               | UB                                 |           |         |                   |  |
| 9,238.2               | 9,238.0               | Base Last Salt                     |           |         |                   |  |
| 9,301.2               | 9,301.0               | Ratcliffe                          |           |         |                   |  |
| 9,454.2               | 9,454.0               | Mission Canyon                     |           |         |                   |  |
| 10,022.2              | 10,022.0              | Lodgepole                          |           |         |                   |  |
| 10,209.2              | 10,209.0              | Lodgepole Fracture Zone            |           |         |                   |  |
| 10,843.4              | 10,726.0              | False Bakken                       |           |         |                   |  |
| 10,867.5              | 10,735.0              | Upper Bakken                       |           |         |                   |  |
| 10,921.7              | 10,751.0              | Middle Bakken                      |           |         |                   |  |
| 10,968.3              | 10,760.0              | Middle Bakken Sand Target          |           |         |                   |  |
| 13,874.6              | 10,770.0              | Base Middle Bakken Sand Target     |           |         |                   |  |

| Plan Annotations      |                       |                   |            |                    |
|-----------------------|-----------------------|-------------------|------------|--------------------|
| Measured Depth (usft) | Vertical Depth (usft) | Local Coordinates |            |                    |
|                       |                       | N-S (usft)        | E-W (usft) | Comment            |
| 10,287.8              | 10,287.6              | 13.6              | 48.1       | KOP Build 12°/100' |
| 11,037.0              | 10,765.0              | 143.0             | 506.9      | EOC                |

**Oasis Petroleum  
Well Summary  
Kline Federal 5300 41-18 10B  
Section 18 T153N R100W  
McKenzie County, ND**

**SURFACE CASING AND CEMENT DESIGN**

| Size    | Interval     | Weight | Grade | Coupling | I.D.    | Drift   | Make-up Torque (ft-lbs) |         |       |
|---------|--------------|--------|-------|----------|---------|---------|-------------------------|---------|-------|
|         |              |        |       |          |         |         | Minimum                 | Optimum | Max   |
| 13-3/8" | 0' to 2,072' | 54.5   | J-55  | STC      | 12.615" | 12.459" | 4,100                   | 5,470   | 6,840 |

| Interval     | Description                    | Collapse    | Burst       | Tension      |
|--------------|--------------------------------|-------------|-------------|--------------|
|              |                                | (psi) a     | (psi) b     | (1000 lbs) c |
| 0' to 2,072' | 13-3/8", 54.5#, J-55, STC, 8rd | 1130 / 1.17 | 2730 / 2.82 | 514 / 2.61   |

**API Rating & Safety Factor**

- a) Based on full casing evacuation with 9 ppg fluid on backside (2,072' setting depth).
- b) Burst pressure based on 9 ppg fluid with no fluid on backside (2,072' setting depth).
- c) Based on string weight in 9 ppg fluid at 2,072' TVD plus 100k# overpull. (Buoyed weight equals 97k lbs.)

Cement volumes are based on 13-3/8" casing set in 17-1/2" hole with 50% excess to circulate cement back to surface.  
Mix and pump the following slurry.

**Pre-flush (Spacer):**        **20 bbls** fresh water

**Lead Slurry:**            **635 sks** (328 bbls) 2.9 yield conventional system with 94 lb/sk cement, .25 lb/sk D130 Lost Circulation Control Agent, 2% CaCl<sub>2</sub>, 4% D079 Extender and 2% D053 Expanding Agent.

**Tail Slurry:**            **349 sks** (72 bbls) 1.16 yield conventional system with 94 lb/sk cement, .25% CaCl<sub>2</sub> and 0.25 lb/sk Lost Circulation Control Agent

**Oasis Petroleum  
Well Summary  
Kline Federal 5300 31-18 10B  
Section 18 T153N R100W  
McKenzie County, ND**

**CONTINGENCY SURFACE CASING AND CEMENT DESIGN**

|        |              |        |        |          |        |       | Make-up Torque (ft-lbs) |         |       |
|--------|--------------|--------|--------|----------|--------|-------|-------------------------|---------|-------|
| Size   | Interval     | Weight | Grade  | Coupling | I.D.   | Drift | Minimum                 | Optimum | Max   |
| 9-5/8" | 0' to 6,101' | 40     | HCL-80 | LTC      | 8.835" | 8.75" | 5,450                   | 7,270   | 9,090 |

| Interval     | Description                   | Collapse    | Burst       | Tension      |
|--------------|-------------------------------|-------------|-------------|--------------|
|              |                               | (psi) a     | (psi) b     | (1000 lbs) c |
| 0' to 6,101' | 9-5/8", 40#, HCL-80, LTC, 8rd | 4230 / 2.13 | 5750 / 3.72 | 837 / 2.78   |

**API Rating & Safety Factor**

- a) Collapse pressure based on 11.5 ppg fluid on the backside and 9 ppg fluid inside of casing.
- b) Burst pressure calculated from a gas kick coming from the production zone (Bakken Pool) at 9,000 psi and a subsequent breakdown at the 9-5/8" shoe, based on a 13.5#/ft fracture gradient. Backup of 9 ppg fluid.
- c) Yield based on string weight in 10 ppg fluid, (207k lbs buoyed weight) plus 100k lbs overpull.

Cement volumes are based on 9-5/8" casing set in 12-1/4" hole with 10% excess in OH and 0% excess inside surface casing. TOC at surface.

**Pre-flush (Spacer):**      **20 bbls** Chem wash

**Lead Slurry:**            **598 sks** (309 bbls) Conventional system with 75 lb/sk cement, 0.5 lb/sk lost circulation, 10% expanding agent, 2% extender, 2% CaCl<sub>2</sub>, 0.2% anti-foam and 0.4% fluid loss agent.

**Tail Slurry:**             **349 sks** (72 bbls) Conventional system with 94 lb/sk cement, 0.3% anti-settling agent, 0.3% fluid loss agent, 0.3 lb/sk lost circulation control agent, 0.2% anti-foam and 0.1% retarder.



**Oasis Petroleum  
Well Summary  
Kline Federal 5300 41-18 10B  
Section 18 T153N R100W  
McKenzie County, ND**

INTERMEDIATE CASING AND CEMENT DESIGN

| Size | Interval          | Weight | Grade   | Coupling | I.D.   | Drift    | Make-up Torque (ft-lbs) |         |       |
|------|-------------------|--------|---------|----------|--------|----------|-------------------------|---------|-------|
|      |                   |        |         |          |        |          | Minimum                 | Optimum | Max   |
| 7"   | 0' – 6,664'       | 29     | P-110   | LTC      | 6.184" | 6.059"   | 5,980                   | 7,970   | 8,770 |
| 7"   | 6,664' – 10,288'  | 32     | HCP-110 | LTC      | 6.094" | 6.000*** | 6,730                   | 8,970   | 9,870 |
| 7"   | 10,288' – 11,037' | 29     | P-110   | LTC      | 6.184" | 6.059"   | 5,980                   | 7,970   | 8,770 |

\*\*\*Special drift

| Interval          | Length | Description                | Collapse       | Burst        | Tension      |
|-------------------|--------|----------------------------|----------------|--------------|--------------|
|                   |        |                            | (psi) a        | (psi) b      | (1000 lbs) c |
| 0' – 6,664'       | 6,664' | 7", 29#, P-110, LTC, 8rd   | 8530 / 2.46*   | 11220 / 1.19 | 797 / 2.09   |
| 6,664' – 10,288'  | 3,624' | 7", 32#, HCP-110, LTC, 8rd | 11820 / 2.20*  | 12460 / 1.29 |              |
| 6,664' – 10,288'  | 3,624' | 7", 32#, HCP-110, LTC, 8rd | 11820 / 1.07** | 12460 / 1.29 |              |
| 10,288' – 11,037' | 749'   | 7", 29 lb, P-110, LTC, 8rd | 8530 / 1.52*   | 11220 / 1.16 |              |

API Rating & Safety Factor

- a) \*Assume full casing evacuation with 10 ppg fluid on backside. \*\*Assume full casing evacuation with 1.2 psi/ft equivalent fluid gradient across salt intervals.
- b) Burst pressure based on 9000 psig max press for stimulation plus 10.2 ppg fluid in casing and 9 ppg fluid on backside-to 10,765' TVD.
- c) Based on string weight in 10 ppg fluid, (280k lbs buoyed weight) plus 100k

Cement volumes are estimates based on 7" casing set in an 8-3/4" hole with 30% excess.

**Pre-flush (Spacer):**      **50 bbls Saltwater**  
**40 bbls Weighted MudPush Express**

**Lead Slurry:**              **157 sks** (72 bbls) 2.21 yield conventional system with 47 lb/sk cement, 37 lb/sk D035 extender, 3.0% KCl, 3.0% D154 extender, 0.3% D208 viscosifier, 0.07% retarder, 0.2% anti-foam, 0.5 lb/sk, D130 LCM.

**Tail Slurry:**                **608 sks** (168 bbls) 1.54 yield conventional system with 94 lb/sk cement, 3.0% KCl, 35.0% Silica, 0.5% retarder, 0.2% fluid loss, 0.2% anti-foam and 0.5 lb/sk LCM.

**Oasis Petroleum  
Well Summary  
Kline Federal 5300 41-18 10B  
Section 18 T153N R100W  
McKenzie County, ND**

**PRODUCTION LINER**

| Size   | Interval          | Weight | Grade | Coupling | I.D.  | Drift  | Estimated Torque |
|--------|-------------------|--------|-------|----------|-------|--------|------------------|
| 4-1/2" | 10,238' – 20,590' | 13.5   | P-110 | BTC      | 3.92" | 3.795" | 4,500            |

| Interval          | Description                      | Collapse     | Burst        | Tension      |
|-------------------|----------------------------------|--------------|--------------|--------------|
|                   |                                  | (psi) a      | (psi) b      | (1000 lbs) c |
| 10,238' – 20,590' | 4-1/2", 13.5 lb, P-110, BTC, 8rd | 10680 / 2.00 | 12410 / 1.28 | 443 / 2.01   |

**API Rating & Safety Factor**

- a) Based on full casing evacuation with 9.5 ppg fluid on backside @ 10,782' TVD.  
Burst pressure based on 9000 psi treating pressure with 10.2 ppg internal fluid gradient and 9 ppg external
- b) fluid gradient @ 10,782' TVD.
- c) Based on string weight in 9.5 ppg fluid (Buoyed weight: 119k lbs.) plus 100k lbs overpull.

**Oasis Petroleum does not use Diesel Fuel, as defined by the US EPA in the list below, in our hydraulic fracture operations.**

**68334-30-5 (Primary Name: Fuels, diesel)  
68476-34-6 (Primary Name: Fuels, diesel, No. 2)  
68476-30-2 (Primary Name: Fuel oil No. 2)  
68476-31-3 (Primary Name: Fuel oil, No. 4)  
8008-20-6 (Primary Name: Kerosene)**

| DRILLING PLAN                                                                                                                      |                              |                      |                                         |          |                                 |                                                                                                                                                                                                                                     |                          |                |  |
|------------------------------------------------------------------------------------------------------------------------------------|------------------------------|----------------------|-----------------------------------------|----------|---------------------------------|-------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|--------------------------|----------------|--|
| OPERATOR                                                                                                                           | Oasis Petroleum              |                      |                                         |          | COUNTY/STATE                    | McKenzie Co., ND                                                                                                                                                                                                                    |                          |                |  |
| WELL NAME                                                                                                                          | Kline Federal 5300 41-18 108 |                      |                                         |          | RIG                             | 0                                                                                                                                                                                                                                   |                          |                |  |
| WELL TYPE                                                                                                                          | Horizontal Middle Bakken     |                      |                                         |          |                                 |                                                                                                                                                                                                                                     |                          |                |  |
| LOCATION                                                                                                                           | SWNW 18-153N-100W            |                      | Surface Location (survey plat): 500' fs |          | 237' fs                         |                                                                                                                                                                                                                                     |                          |                |  |
| EST. T.D.                                                                                                                          | 20,590'                      |                      |                                         |          | GROUND ELEV:                    | 2057 Finished Pad Elev.                                                                                                                                                                                                             |                          | Sub Height: 25 |  |
|                                                                                                                                    | TOTAL LATERAL                |                      | 9,553'                                  |          | KB ELEV:                        | 2082                                                                                                                                                                                                                                |                          |                |  |
| PROGNOSIS:                                                                                                                         | Based on 2,082' KB(est)      |                      |                                         |          | LOGS:                           | <b>Type</b><br>OH Logs: Triple Combo KOP to Kibby (or min run of 1800' whichever is greater); GR/Res to BSC;<br>GR to surf; CND through the Dakota<br>CBL/GR: Above top of cement/GR to base of casing<br>MWD GR: KOP to lateral TD |                          |                |  |
| MARKER                                                                                                                             | DEPTH (Surf Loc)             | DATUM (Surf Loc)     |                                         |          |                                 |                                                                                                                                                                                                                                     |                          |                |  |
| Pierre                                                                                                                             | NDIC MAP                     | 1,972                | 110                                     |          |                                 |                                                                                                                                                                                                                                     |                          |                |  |
| Greenhorn                                                                                                                          |                              | 4,646                | (2,564)                                 |          |                                 |                                                                                                                                                                                                                                     |                          |                |  |
| Mowry                                                                                                                              |                              | 5,057                | (2,975)                                 |          |                                 |                                                                                                                                                                                                                                     |                          |                |  |
| Dakota                                                                                                                             |                              | 5,471                | (3,389)                                 |          |                                 |                                                                                                                                                                                                                                     |                          |                |  |
| Rierdon                                                                                                                            |                              | 6,486                | (4,404)                                 |          |                                 |                                                                                                                                                                                                                                     |                          |                |  |
| Dunham Salt                                                                                                                        |                              | 6,814                | (4,732)                                 |          |                                 |                                                                                                                                                                                                                                     |                          |                |  |
| Dunham Salt Base                                                                                                                   |                              | 6,929                | (4,847)                                 |          |                                 |                                                                                                                                                                                                                                     |                          |                |  |
| Spearfish                                                                                                                          |                              | 7,024                | (4,942)                                 |          |                                 |                                                                                                                                                                                                                                     |                          |                |  |
| Pine Salt                                                                                                                          |                              | 7,283                | (5,201)                                 |          |                                 |                                                                                                                                                                                                                                     |                          |                |  |
| Pine Salt Base                                                                                                                     |                              | 7,318                | (5,238)                                 |          |                                 |                                                                                                                                                                                                                                     |                          |                |  |
| Opeche Salt                                                                                                                        |                              | 7,374                | (5,292)                                 |          |                                 |                                                                                                                                                                                                                                     |                          |                |  |
| Opeche Salt Base                                                                                                                   |                              | 7,403                | (5,321)                                 |          |                                 |                                                                                                                                                                                                                                     |                          |                |  |
| Broom Creek (Top of Minnelusa Gp.)                                                                                                 |                              | 7,605                | (5,523)                                 |          |                                 |                                                                                                                                                                                                                                     |                          |                |  |
| Amsden                                                                                                                             |                              | 7,684                | (5,602)                                 |          |                                 |                                                                                                                                                                                                                                     |                          |                |  |
| Tyler                                                                                                                              |                              | 7,853                | (5,771)                                 |          |                                 |                                                                                                                                                                                                                                     |                          |                |  |
| Otter (Base of Minnelusa Gp.)                                                                                                      |                              | 8,047                | (5,965)                                 |          |                                 |                                                                                                                                                                                                                                     |                          |                |  |
| Kibbey Lime                                                                                                                        |                              | 8,390                | (6,308)                                 |          |                                 |                                                                                                                                                                                                                                     |                          |                |  |
| Charles Salt                                                                                                                       |                              | 8,542                | (6,460)                                 |          |                                 |                                                                                                                                                                                                                                     |                          |                |  |
| UB                                                                                                                                 |                              | 9,159                | (7,077)                                 |          |                                 |                                                                                                                                                                                                                                     |                          |                |  |
| Base Last Salt                                                                                                                     |                              | 9,238                | (7,156)                                 |          |                                 |                                                                                                                                                                                                                                     |                          |                |  |
| Ratcliffe                                                                                                                          |                              | 9,301                | (7,219)                                 |          |                                 |                                                                                                                                                                                                                                     |                          |                |  |
| Mission Canyon                                                                                                                     |                              | 9,454                | (7,372)                                 |          |                                 |                                                                                                                                                                                                                                     |                          |                |  |
| Lodgepole                                                                                                                          |                              | 10,022               | (7,940)                                 |          |                                 |                                                                                                                                                                                                                                     |                          |                |  |
| Lodgepole Fracture Zone                                                                                                            |                              | 10,209               | (8,127)                                 |          |                                 |                                                                                                                                                                                                                                     |                          |                |  |
| False Bakken                                                                                                                       |                              | 10,726               | (8,644)                                 |          |                                 |                                                                                                                                                                                                                                     |                          |                |  |
| Upper Bakken                                                                                                                       |                              | 10,735               | (8,653)                                 |          |                                 |                                                                                                                                                                                                                                     |                          |                |  |
| Middle Bakken                                                                                                                      |                              | 10,751               | (8,669)                                 |          |                                 |                                                                                                                                                                                                                                     |                          |                |  |
| Middle Bakken Sand Target                                                                                                          |                              | 10,760               | (8,678)                                 |          |                                 |                                                                                                                                                                                                                                     |                          |                |  |
| Base Middle Bakken Sand Target                                                                                                     |                              | 10,770               | (8,688)                                 |          |                                 |                                                                                                                                                                                                                                     |                          |                |  |
| Lower Bakken                                                                                                                       |                              | 10,785               | (8,703)                                 |          |                                 |                                                                                                                                                                                                                                     |                          |                |  |
| Three Forks                                                                                                                        |                              | 10,813               | (8,731)                                 |          |                                 |                                                                                                                                                                                                                                     |                          |                |  |
| Dip Rate:                                                                                                                          | -0.1                         |                      |                                         |          |                                 |                                                                                                                                                                                                                                     |                          |                |  |
| Max. Anticipated BHP:                                                                                                              | 4673                         |                      |                                         |          | Surface Formation: Glacial till |                                                                                                                                                                                                                                     |                          |                |  |
| MUD:                                                                                                                               | Interval                     | Type                 | WT                                      | Vis      | WL                              | Remarks                                                                                                                                                                                                                             |                          |                |  |
| Surface:                                                                                                                           | 0' - 2,072'                  | PW/Gel - Lime Sweeps | 8.4-9.0                                 | 28-32    | NC                              | Circ Mud Tanks                                                                                                                                                                                                                      |                          |                |  |
| Intermediate:                                                                                                                      | 2,072' - 11,037'             | Invert               | 9.5-10.4                                | 40-50    | 30+HHP                          | Circ Mud Tanks                                                                                                                                                                                                                      |                          |                |  |
| Lateral:                                                                                                                           | 11,037' - 20,590'            | Salt Water           | 9.8-10.2                                | 28-32    | NC                              | Circ Mud Tanks                                                                                                                                                                                                                      |                          |                |  |
| CASING:                                                                                                                            | Size                         | Wt ppf               | Hole                                    | Depth    | Cement                          | WOC                                                                                                                                                                                                                                 | Remarks                  |                |  |
| Surface:                                                                                                                           | 13-3/8"                      | 54.5#                | 17-1/2"                                 | 2,072'   | To Surface                      | 12                                                                                                                                                                                                                                  | 100' into Pierre         |                |  |
| Dakota Contingency:                                                                                                                | 9-5/8"                       | 40#                  | 12-1/4"                                 | 6,101'   | To Surface                      | 12                                                                                                                                                                                                                                  | Below Dakota             |                |  |
| Intermediate:                                                                                                                      | 7"                           | 29/32#               | 8-3/4"                                  | 11,037'  | 3971                            | 24                                                                                                                                                                                                                                  | 1500' above Dakota       |                |  |
| Production Liner:                                                                                                                  | 4.5"                         | 13.5#                | 6"                                      | 20,590'  | TOL @ 10,238'                   |                                                                                                                                                                                                                                     | 50' above KOP            |                |  |
| PROBABLE PLUGS, IF REQ'D:                                                                                                          |                              |                      |                                         |          |                                 |                                                                                                                                                                                                                                     |                          |                |  |
| OTHER:                                                                                                                             | MD                           | TVD                  | FNU/FSL                                 | FEL/FWL  | S-T-R                           | AZI                                                                                                                                                                                                                                 |                          |                |  |
| Surface:                                                                                                                           | 2,072                        | 2,072                | 500' FSL                                | 237' FWL | SEC 18-T153N-R100W              |                                                                                                                                                                                                                                     | Survey Company:          |                |  |
| KOP:                                                                                                                               | 10,288'                      | 10,288'              | 514' FSL                                | 285' FWL | SEC 18-T153N-R100W              |                                                                                                                                                                                                                                     | Build Rate: 12 deg /100' |                |  |
| EOC:                                                                                                                               | 11,037'                      | 10,765'              | 643' FSL                                | 744' FWL | SEC 18-T153N-R100W              | 74.25                                                                                                                                                                                                                               |                          |                |  |
| Casing Point:                                                                                                                      | 11,037'                      | 10,765'              | 643' FSL                                | 744' FWL | SEC 18-T153N-R100W              | 74.25                                                                                                                                                                                                                               |                          |                |  |
| Middle Bakken Lateral TD:                                                                                                          | 20,590'                      | 10,782'              | 750' FSL                                | 200' FEL | SEC 17-T153N-R100W              | 90.00                                                                                                                                                                                                                               |                          |                |  |
| Comments:                                                                                                                          |                              |                      |                                         |          |                                 |                                                                                                                                                                                                                                     |                          |                |  |
| Request a Sundry for an Open Hole Log Waiver                                                                                       |                              |                      |                                         |          |                                 |                                                                                                                                                                                                                                     |                          |                |  |
| Exception well: Oasis Petroleum's Kline Federal 5300 11-18H (153N 100W 18 NW NW)                                                   |                              |                      |                                         |          |                                 |                                                                                                                                                                                                                                     |                          |                |  |
| 35 packers, 35 sleeves, no frac string                                                                                             |                              |                      |                                         |          |                                 |                                                                                                                                                                                                                                     |                          |                |  |
| Oasis Petroleum does not use Diesel Fuel, as defined by the US EPA in the list below, in our hydraulic fracture operations.        |                              |                      |                                         |          |                                 |                                                                                                                                                                                                                                     |                          |                |  |
| 68334-30-5 (Primary Name: Fuels, diesel) 68476-34-6 (Primary Name: Fuels, diesel, No. 2) 68476-30-2 (Primary Name: Fuel oil No. 2) |                              |                      |                                         |          |                                 |                                                                                                                                                                                                                                     |                          |                |  |
| 68476-31-3 (Primary Name: Fuel oil, No. 4) 8008-20-6 (Primary Name: Kerosene)                                                      |                              |                      |                                         |          |                                 |                                                                                                                                                                                                                                     |                          |                |  |
|                                                 |                              |                      |                                         |          |                                 |                                                                                                                                                                                                                                     |                          |                |  |
| Geology: M. Steed 4/3/2014                                                                                                         |                              |                      |                                         |          | Engineering: hbader rpm 7/18/14 |                                                                                                                                                                                                                                     |                          |                |  |



# Oil and Gas Division

Lynn D. Helms - Director

Bruce E. Hicks - Assistant Director

## Department of Mineral Resources

Lynn D. Helms - Director

## North Dakota Industrial Commission

www.dmr.nd.gov/oilgas

28652

BRANDI TERRY  
OASIS PETROLEUM NORTH AMERICA LLC  
1001 FANNIN STE 1500  
HOUSTON, TX 77002 USA

Date: 6/23/2014

**RE: CORES AND SAMPLES**

Well Name: **KLINE FEDERAL 5300 41-18 10B** Well File No.: **28652**  
Location: **LOT4 18-153-100** County: **MCKENZIE**  
Permit Type: **Development - HORIZONTAL**  
Field: **BAKER** Target Horizon: **BAKKEN**

Dear BRANDI TERRY:

North Dakota Century Code Section 38-08-04 provides for the preservation of cores and samples and their shipment to the State Geologist when requested. The following is required on the above referenced well:

- 1) All cores, core chips and samples must be submitted to the State Geologist as provided for under North Dakota Century Code: Section 38-08-04 and North Dakota Administrative Code: Section 43-02-03-38.1.
- 2) Samples: The Operator is to begin collecting sample drill cuttings no lower than the:  
**Base of the Last Charles Salt**
  - Sample cuttings shall be collected at:
    - o 30' maximum intervals through all vertical and build sections.
    - o 100' maximum intervals through any horizontal sections.
  - Samples must be washed, dried, placed in standard sample envelopes (3" x 4.5"), packed in the correct order into standard sample boxes (3.5" x 5.25" x 15.25").
  - Samples boxes are to be carefully identified with a label that indicates the operator, well name, well file number, American Petroleum Institute (API) number, location and depth of samples; and forwarded in to the state core and sample library within 30 days of the completion of drilling operations.
- 3) Cores: Any cores cut shall be preserved in correct order, boxed in standard core boxes (4.5", 4.5", 35.75"), and the entire core forwarded to the state core and samples library within 180 days of completion of drilling operations. Any extension of time must have approval on a Form 4 Sundry Notice.

All cores, core chips, and samples must be shipped, prepaid, to the state core and samples library at the following address:

**ND Geological Survey Core Library  
2835 Campus Road, Stop 8156  
Grand Forks, ND 58202**

North Dakota Century Code Section 38-08-16 allows for a civil penalty for any violation of Chapter 38 08 not to exceed \$12,500 for each offense, and each day's violation is a separate offense.

Sincerely

  
Stephen Fried  
Geologist

**SUNDRY NOTICES AND REPORTS ON WELLS - FORM 4**

INDUSTRIAL COMMISSION OF NORTH DAKOTA  
OIL AND GAS DIVISION  
600 EAST BOULEVARD DEPT 405  
BISMARCK, ND 58505-0840  
SFN 5749 (09-2006)



Well File No.

28652

PLEASE READ INSTRUCTIONS BEFORE FILLING OUT FORM.  
PLEASE SUBMIT THE ORIGINAL AND ONE COPY.

|                                                                                                                                                 |                                                  |
|-------------------------------------------------------------------------------------------------------------------------------------------------|--------------------------------------------------|
| <input checked="" type="checkbox"/> Notice of Intent                                                                                            | Approximate Start Date<br><b>October 1, 2014</b> |
| <input type="checkbox"/> Report of Work Done                                                                                                    | Date Work Completed                              |
| <input type="checkbox"/> Notice of Intent to Begin a Workover Project that may Qualify for a Tax Exemption Pursuant to NDCC Section 57-51.1-03. | Approximate Start Date                           |

|                                               |                                                   |
|-----------------------------------------------|---------------------------------------------------|
| <input type="checkbox"/> Drilling Prognosis   | <input type="checkbox"/> Spill Report             |
| <input type="checkbox"/> Redrilling or Repair | <input type="checkbox"/> Shooting                 |
| <input type="checkbox"/> Casing or Liner      | <input type="checkbox"/> Acidizing                |
| <input type="checkbox"/> Plug Well            | <input type="checkbox"/> Fracture Treatment       |
| <input type="checkbox"/> Supplemental History | <input type="checkbox"/> Change Production Method |
| <input type="checkbox"/> Temporarily Abandon  | <input type="checkbox"/> Reclamation              |
| <input checked="" type="checkbox"/> Other     | <b>Waiver to rule Rule 43-02-03-31</b>            |

|                                                             |               |              |                 |              |              |
|-------------------------------------------------------------|---------------|--------------|-----------------|--------------|--------------|
| Well Name and Number<br><b>Kline Federal 5300 41-18 10B</b> |               |              |                 |              |              |
| Footages                                                    |               | Qtr-Qtr      | Section         | Township     | Range        |
| <b>500 F S L 237 F W L</b>                                  |               | <u>Lot 4</u> | <b>18</b>       | <b>153 N</b> | <b>100 W</b> |
| Field                                                       | Pool          |              | County          |              |              |
|                                                             | <b>Bakken</b> |              | <b>McKenzie</b> |              |              |

| 24-HOUR PRODUCTION RATE |      |       |      |
|-------------------------|------|-------|------|
| Before                  |      | After |      |
| Oil                     | Bbls | Oil   | Bbls |
| Water                   | Bbls | Water | Bbls |
| Gas                     | MCF  | Gas   | MCF  |

|                       |      |       |          |
|-----------------------|------|-------|----------|
| Name of Contractor(s) |      |       |          |
| Address               | City | State | Zip Code |

**DETAILS OF WORK**

Oasis Petroleum respectfully requests a waiver to Rule 43-02-03-31 in regards to running open hole logs for the above referenced well. Justification for this request is as follows:

#20275  
The Oasis Petroleum's Kline Federal 5300 11-18H (153N 100W 18 NW NW) located within a mile of the subject well

If this exception is approved, Oasis Petroleum will run a CBL on the intermediate string, and we will also run GR to surface. Oasis Petroleum will also submit two digital copies of each cased hole log and a copy of the mud log containing MWD gamma ray.

|                                                     |  |                                         |                          |
|-----------------------------------------------------|--|-----------------------------------------|--------------------------|
| Company<br><b>Oasis Petroleum North America LLC</b> |  | Telephone Number<br><b>281-404-9563</b> |                          |
| Address<br><b>1001 Fannin, Suite 1500</b>           |  |                                         |                          |
| City<br><b>Houston</b>                              |  | State<br><b>TX</b>                      | Zip Code<br><b>77002</b> |
| Signature<br><i>Heather McCowan</i>                 |  | Printed Name<br><b>Heather McCowan</b>  |                          |
| Title<br><b>Regulatory Assistant</b>                |  | Date<br><b>May 29, 2014</b>             |                          |
| Email Address<br><b>hmccowan@oasispetroleum.com</b> |  |                                         |                          |

| FOR STATE USE ONLY                          |                                              |
|---------------------------------------------|----------------------------------------------|
| <input type="checkbox"/> Received           | <input checked="" type="checkbox"/> Approved |
| Date<br><u>6-17-2014</u>                    |                                              |
| By<br><i>Stephen Fried</i>                  |                                              |
| Title<br><b>Stephen Fried<br/>Geologist</b> |                                              |



**SUNDRY NOTICES AND REPORTS ON WELLS - FORM 4**

INDUSTRIAL COMMISSION OF NORTH DAKOTA  
OIL AND GAS DIVISION  
600 EAST BOULEVARD DEPT 405  
BISMARCK, ND 58505-0840  
SFN 5749 (09-2006)

Well File No.  
28652

PLEASE READ INSTRUCTIONS BEFORE FILLING OUT FORM.  
PLEASE SUBMIT THE ORIGINAL AND ONE COPY.

|                                                                                                                                                 |                                                  |                                               |                                                   |
|-------------------------------------------------------------------------------------------------------------------------------------------------|--------------------------------------------------|-----------------------------------------------|---------------------------------------------------|
| <input checked="" type="checkbox"/> Notice of Intent                                                                                            | Approximate Start Date<br><b>October 1, 2014</b> | <input type="checkbox"/> Drilling Prognosis   | <input type="checkbox"/> Spill Report             |
| <input type="checkbox"/> Report of Work Done                                                                                                    | Date Work Completed                              | <input type="checkbox"/> Redrilling or Repair | <input type="checkbox"/> Shooting                 |
| <input type="checkbox"/> Notice of Intent to Begin a Workover Project that may Qualify for a Tax Exemption Pursuant to NDCC Section 57-51.1-03. | Approximate Start Date                           | <input type="checkbox"/> Casing or Liner      | <input type="checkbox"/> Acidizing                |
|                                                                                                                                                 |                                                  | <input type="checkbox"/> Plug Well            | <input type="checkbox"/> Fracture Treatment       |
|                                                                                                                                                 |                                                  | <input type="checkbox"/> Supplemental History | <input type="checkbox"/> Change Production Method |
|                                                                                                                                                 |                                                  | <input type="checkbox"/> Temporarily Abandon  | <input type="checkbox"/> Reclamation              |
|                                                                                                                                                 |                                                  | <input checked="" type="checkbox"/> Other     | <b>Suspension of Drilling</b>                     |

|                                                             |  |               |           |                 |              |  |  |
|-------------------------------------------------------------|--|---------------|-----------|-----------------|--------------|--|--|
| Well Name and Number<br><b>Kline Federal 5300 41-18 10B</b> |  |               |           |                 |              |  |  |
| Footages                                                    |  | Qtr-Qtr       | Section   | Township        | Range        |  |  |
| <b>500 F S L 237 F WL</b>                                   |  | <b>SWSW</b>   | <b>18</b> | <b>153 N</b>    | <b>100 W</b> |  |  |
| Field                                                       |  | Pool          |           | County          |              |  |  |
| <b>Bakken</b>                                               |  | <b>Bakken</b> |           | <b>McKenzie</b> |              |  |  |

| 24-HOUR PRODUCTION RATE |      |       |      |
|-------------------------|------|-------|------|
| Before                  |      | After |      |
| Oil                     | Bbls | Oil   | Bbls |
| Water                   | Bbls | Water | Bbls |
| Gas                     | MCF  | Gas   | MCF  |

|                                                          |      |       |          |
|----------------------------------------------------------|------|-------|----------|
| Name of Contractor(s)<br><b>Advanced Energy Services</b> |      |       |          |
| Address                                                  | City | State | Zip Code |

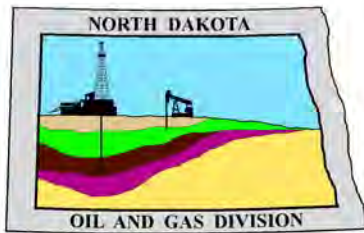
**DETAILS OF WORK**

Oasis Petroleum North America LLC requests permission for suspension of drilling for up to 90 days for the referenced well under NDAC 43-02-03-55. Oasis Petroleum North America LLC intends to drill the surface hole with freshwater based drilling mud and set surface casing with a small drilling rig and move off within 3 to 5 days. The casing will be set at a depth pre-approved by the NDIC per the Application for Permit to Drill NDAC 43-02-03-21. No saltwater will be used in the drilling and cementing operations of the surface casing. Once the surface casing is cemented, a plug or mechanical seal will be placed at the top of the casing to prevent any foreign matter from getting into the well. A rig capable of drilling to TD will move onto the location within the 90 days previously outlined to complete the drilling and casing plan as per the APD. The undersigned states that this request for suspension of drilling operations in accordance with the Subsection 4 of Section 43-02-03-55 of the NDAC, is being requested to take advantage of the cost savings and time savings of using an initial rig that is smaller than the rig necessary to drill a well to total depth but is not intended to alter or extend the terms and conditions of, or suspend any obligation under, any oil and gas lease with acreage in or under the spacing or drilling unit for the above-referenced well. Oasis Petroleum North America LLC understands NDAC 43-02-03-31 requirements regarding confidentiality pertaining to this permit. The drilling pit will be fenced immediately after construction if the well pad is located in a pasture (NDAC 43-02-03-19 & 19.1). Oasis Petroleum North America LLC will plug and abandon the well and reclaim the well site if the well is not drilled by the larger rotary rig within 90 days after spudding the well with the smaller drilling rig.

Notify NDIC inspector Richard Dunn at 701-770-3554 with spud and TD info.

|                                                     |  |                                           |                          |
|-----------------------------------------------------|--|-------------------------------------------|--------------------------|
| Company<br><b>Oasis Petroleum North America LLC</b> |  | Telephone Number<br><b>(281) 404-9563</b> |                          |
| Address<br><b>1001 Fannin, Suite 1500</b>           |  |                                           |                          |
| City<br><b>Houston</b>                              |  | State<br><b>TX</b>                        | Zip Code<br><b>77002</b> |
| Signature<br><i>Heather McCowan</i>                 |  | Printed Name<br><b>Heather McCowan</b>    |                          |
| Title<br><b>Regulatory Assistant</b>                |  | Date<br><b>May 29, 2014</b>               |                          |
| Email Address<br><b>hmccowan@oasispetroleum.com</b> |  |                                           |                          |

|                                               |                                              |
|-----------------------------------------------|----------------------------------------------|
| FOR STATE USE ONLY                            |                                              |
| <input type="checkbox"/> Received             | <input checked="" type="checkbox"/> Approved |
| Date<br><b>6/17/14</b>                        |                                              |
| By<br><b>Nathaniel Erbele</b>                 |                                              |
| Title<br><b>Petroleum Resource Specialist</b> |                                              |



# Oil and Gas Division

Lynn D. Helms - Director

Bruce E. Hicks - Assistant Director

## Department of Mineral Resources

Lynn D. Helms - Director

## North Dakota Industrial Commission

[www.oilgas.nd.gov](http://www.oilgas.nd.gov)

June 17, 2014

Heather McCowan  
Regulatory Assistant  
OASIS PETROLEUM NORTH AMERICA LLC  
1001 Fannin Suite 1500  
Houston, TX 77002

**RE: HORIZONTAL WELL  
KLINE FEDERAL 5300 41-18 10B  
LOT4 Section 18-153N-100W  
McKenzie County  
Well File # 28652**

Dear Heather:

Pursuant to Commission Order No. 23752, approval to drill the above captioned well is hereby given. The approval is granted on the condition that all portions of the well bore not isolated by cement, be no closer than the **500' setback** from the north & south boundaries and **200' setback** from the east & west boundaries within the 1280 acre spacing unit consisting of Sections 18 & 17 T153N R100W.

**PERMIT STIPULATIONS: Due to the proximity of Lake Sakakawea to the well site, a dike is required surrounding the entire location. Effective June 1, 2014, a covered leak-proof container (with placard) for filter sock disposal must be maintained on the well site beginning when the well is spud, and must remain on-site during clean-out, completion, and flow-back whenever filtration operations are conducted. OASIS PETRO NO AMER must contact NDIC Field Inspector Richard Dunn at 701-770-3554 prior to location construction.**

### Drilling pit

NDAC 43-02-03-19.4 states that "a pit may be utilized to bury drill cuttings and solids generated during well drilling and completion operations, providing the pit can be constructed, used and reclaimed in a manner that will prevent pollution of the land surface and freshwaters. Reserve and circulation of mud system through earthen pits are prohibited. All pits shall be inspected by an authorized representative of the director prior to lining and use. Drill cuttings and solids must be stabilized in a manner approved by the director prior to placement in a cuttings pit."

### Form 1 Changes & Hard Lines

Any changes, shortening of casing point or lengthening at Total Depth must have prior approval by the NDIC. The proposed directional plan is at a legal location. Based on the azimuth of the proposed lateral the maximum legal coordinate from the well head is: 10052' east.

### Location Construction Commencement (Three Day Waiting Period)

Operators shall not commence operations on a drill site until the 3rd business day following publication of the approved drilling permit on the NDIC - OGD Daily Activity Report. If circumstances require operations to commence before the 3rd business day following publication on the Daily Activity Report, the waiting period may be waived by the Director. Application for a waiver must be by sworn affidavit providing the information necessary to evaluate the extenuating circumstances, the factors of NDAC 43-02-03-16.2 (1), (a)-(f), and any other information that would allow the Director to conclude that in the event another owner seeks revocation of the drilling permit, the applicant should retain the permit.



Heather McCowan

June 17, 2014

Page 2

### **Permit Fee & Notification**

Payment was received in the amount of \$100 via credit card .The permit fee has been received. It is requested that notification be given immediately upon the spudding of the well. This information should be relayed to the Oil & Gas Division, Bismarck, via telephone. The following information must be included: Well name, legal location, permit number, drilling contractor, company representative, date and time of spudding. Office hours are 8:00 a.m. to 12:00 p.m. and 1:00 p.m. to 5:00 p.m. Central Time. Our telephone number is (701) 328-8020, leave a message if after hours or on the weekend.

### **Survey Requirements for Horizontal, Horizontal Re-entry, and Directional Wells**

NDAC Section 43-02-03-25 (Deviation Tests and Directional Surveys) states in part (that) the survey contractor shall file a certified copy of all surveys with the director free of charge within thirty days of completion. Surveys must be submitted as one electronic copy, or in a form approved by the director. However, the director may require the directional survey to be filed immediately after completion if the survey is needed to conduct the operation of the director's office in a timely manner. Certified surveys must be submitted via email in one adobe document, with a certification cover page to [certsurvey@nd.gov](mailto:certsurvey@nd.gov).

Survey points shall be of such frequency to accurately determine the entire location of the well bore.

Specifically, the Horizontal and Directional well survey frequency is 100 feet in the vertical, 30 feet in the curve (or when sliding) and 90 feet in the lateral.

### **Surface casing cement**

Tail cement utilized on surface casing must have a minimum compressive strength of 500 psi within 12 hours, and tail cement utilized on production casing must have a minimum compressive strength of 500 psi before drilling the plug or initiating tests.

### **Logs**

NDAC Section 43-02-03-31 requires the running of (1) a suite of open hole logs from which formation tops and porosity zones can be determined, (2) a Gamma Ray Log run from total depth to ground level elevation of the well bore, and (3) a log from which the presence and quality of cement can be determined (Standard CBL or Ultrasonic cement evaluation log) in every well in which production or intermediate casing has been set, this log must be run prior to completing the well.

All logs run must be submitted free of charge, as one digital TIFF (tagged image file format) copy and one digital LAS (log ASCII) formatted copy. Digital logs may be submitted on a standard CD, DVD, or attached to an email sent to [digitallogs@nd.gov](mailto:digitallogs@nd.gov)

Thank you for your cooperation.

Sincerely,

Nathaniel Erbele

Petroleum Resource Specialist



# APPLICATION FOR PERMIT TO DRILL HORIZONTAL WELL - FORM 1H

INDUSTRIAL COMMISSION OF NORTH DAKOTA  
OIL AND GAS DIVISION  
600 EAST BOULEVARD DEPT 405  
BISMARCK, ND 58505-0840  
SFN 54269 (08-2005)

PLEASE READ INSTRUCTIONS BEFORE FILLING OUT FORM.  
PLEASE SUBMIT THE ORIGINAL AND ONE COPY.

|                                                      |                                      |                                                          |                                             |
|------------------------------------------------------|--------------------------------------|----------------------------------------------------------|---------------------------------------------|
| Type of Work<br><b>New Location</b>                  | Type of Well<br><b>Oil &amp; Gas</b> | Approximate Date Work Will Start<br><b>10 / 1 / 2013</b> | Confidential Status<br><b>No</b>            |
| Operator<br><b>OASIS PETROLEUM NORTH AMERICA LLC</b> |                                      |                                                          | Telephone Number<br><b>281-404-9591</b>     |
| Address<br><b>1001 Fannin Suite 1500</b>             |                                      | City<br><b>Houston</b>                                   | State<br><b>TX</b> Zip Code<br><b>77002</b> |

☒ Notice has been provided to the owner of any permanently occupied dwelling within 1,320 feet.

☒ This well is not located within five hundred feet of an occupied dwelling.

## WELL INFORMATION (If more than one lateral proposed, enter data for additional laterals on page 2)

|                                                                                                    |                                                          |                                                                                                         |                                                                                                    |                                                      |                                                                                                                 |                                  |                                     |
|----------------------------------------------------------------------------------------------------|----------------------------------------------------------|---------------------------------------------------------------------------------------------------------|----------------------------------------------------------------------------------------------------|------------------------------------------------------|-----------------------------------------------------------------------------------------------------------------|----------------------------------|-------------------------------------|
| Well Name<br><b>KLINE FEDERAL</b>                                                                  |                                                          |                                                                                                         |                                                                                                    | Well Number<br><b>5300 41-18 10B</b>                 |                                                                                                                 |                                  |                                     |
| Surface Footages<br><b>500 F S L 237 F W L</b>                                                     |                                                          | Qtr-Qtr<br><b>LOT4</b>                                                                                  | Section<br><b>18</b>                                                                               | Township<br><b>153 N</b>                             | Range<br><b>100 W</b>                                                                                           | County<br><b>McKenzie</b>        |                                     |
| Longstring Casing Point Footages<br><b>642 F S L 743 F W L</b>                                     |                                                          | Qtr-Qtr<br><b>LOT4</b>                                                                                  | Section<br><b>18</b>                                                                               | Township<br><b>153 N</b>                             | Range<br><b>100 W</b>                                                                                           | County<br><b>McKenzie</b>        |                                     |
| Longstring Casing Point Coordinates From Well Head<br><b>142 N From WH 506 E From WH</b>           |                                                          | Azimuth<br><b>74.25 °</b>                                                                               | Longstring Total Depth<br><b>11037 Feet MD 10765 Feet TVD</b>                                      |                                                      |                                                                                                                 |                                  |                                     |
| Bottom Hole Footages From Nearest Section Line<br><b>750 F S L 202 F E L</b>                       |                                                          | Qtr-Qtr<br><b>LOT4</b>                                                                                  | Section<br><b>17</b>                                                                               | Township<br><b>153 N</b>                             | Range<br><b>100 W</b>                                                                                           | County<br><b>McKenzie</b>        |                                     |
| Bottom Hole Coordinates From Well Head<br><b>250 N From WH 10050 E From WH</b>                     |                                                          | KOP Lateral 1<br><b>10288 Feet MD</b>                                                                   | Azimuth Lateral 1<br><b>90 °</b>                                                                   |                                                      | Estimated Total Depth Lateral 1<br><b>20590 Feet MD 10781 Feet TVD</b>                                          |                                  |                                     |
| Latitude of Well Head<br><b>48 ° 04 ' 8.20 "</b>                                                   | Longitude of Well Head<br><b>-103 ° 36 ' 11.95 "</b>     | NAD Reference<br><b>NAD83</b>                                                                           | Description of (Subject to NDIC Approval)<br><b>Spacing Unit: Sections 18 &amp; 17 T153N R100W</b> |                                                      |                                                                                                                 |                                  |                                     |
| Ground Elevation<br><b>2055 Feet Above S.L.</b>                                                    | Acres in Spacing/Drilling Unit<br><b>1280</b>            | Spacing/Drilling Unit Setback Requirement<br><b>500 Feet N/S 200 Feet E/W</b>                           |                                                                                                    |                                                      | Industrial Commission Order<br><b>23752</b>                                                                     |                                  |                                     |
| North Line of Spacing/Drilling Unit<br><b>10544 Feet</b>                                           | South Line of Spacing/Drilling Unit<br><b>10489 Feet</b> | East Line of Spacing/Drilling Unit<br><b>5244 Feet</b>                                                  |                                                                                                    |                                                      | West Line of Spacing/Drilling Unit<br><b>5256 Feet</b>                                                          |                                  |                                     |
| Objective Horizons<br><b>Bakken</b>                                                                |                                                          |                                                                                                         |                                                                                                    |                                                      | Pierre Shale Top<br><b>1972</b>                                                                                 |                                  |                                     |
| Proposed Surface Casing                                                                            | Size<br><b>9 - 5/8 "</b>                                 | Weight<br><b>36 Lb./Ft.</b>                                                                             | Depth<br><b>2072 Feet</b>                                                                          | Cement Volume<br><b>749 Sacks</b>                    | <b>NOTE: Surface hole must be drilled with fresh water and surface casing must be cemented back to surface.</b> |                                  |                                     |
| Proposed Longstring Casing                                                                         | Size<br><b>7 - "</b>                                     | Weight(s)<br><b>29/32 Lb./Ft.</b>                                                                       | Longstring Total Depth<br><b>11037 Feet MD 10765 Feet TVD</b>                                      |                                                      | Cement Volume<br><b>752 Sacks</b>                                                                               | Cement Top<br><b>3971 Feet</b>   | Top Dakota Sand<br><b>5471 Feet</b> |
| Base Last Charles Salt (If Applicable)<br><b>9238 Feet</b>                                         |                                                          | <b>NOTE: Intermediate or longstring casing string must be cemented above the top Dakota Group Sand.</b> |                                                                                                    |                                                      |                                                                                                                 |                                  |                                     |
| Proposed Logs<br><b>Triple Combo: KOP to Kibby GR/Res to BSC GR to surf CND through the Dakota</b> |                                                          |                                                                                                         |                                                                                                    |                                                      |                                                                                                                 |                                  |                                     |
| Drilling Mud Type (Vertical Hole - Below Surface Casing)<br><b>Invert</b>                          |                                                          |                                                                                                         |                                                                                                    | Drilling Mud Type (Lateral)<br><b>Salt Water Gel</b> |                                                                                                                 |                                  |                                     |
| Survey Type in Vertical Portion of Well<br><b>MWD Every 100 Feet</b>                               |                                                          | Survey Frequency: Build Section<br><b>30 Feet</b>                                                       |                                                                                                    | Survey Frequency: Lateral<br><b>90 Feet</b>          |                                                                                                                 | Survey Contractor<br><b>Ryan</b> |                                     |

**NOTE: A Gamma Ray log must be run to ground surface and a CBL must be run on intermediate or longstring casing string if set.**

Surveys are required at least every 30 feet in the build section and every 90 feet in the lateral section of a horizontal well. Measurement inaccuracies are not considered when determining compliance with the spacing/drilling unit boundary setback requirement except in the following scenarios: 1) When the angle between the well bore and the respective boundary is 10 degrees or less; or 2) If Industry standard methods and equipment are not utilized. Consult the applicable field order for exceptions.

If measurement inaccuracies are required to be considered, a 2° MWD measurement inaccuracy will be applied to the horizontal portion of the well bore. This measurement inaccuracy is applied to the well bore from KOP to TD.

**REQUIRED ATTACHMENTS:** Certified surveyor's plat, horizontal section plat, estimated geological tops, proposed mud/cementing plan, directional plot/plan, \$100 fee.  
See Page 2 for Comments section and signature block.

**COMMENTS, ADDITIONAL INFORMATION, AND/OR LIST OF ATTACHMENTS**

**Documents forwarded by email: Drill plan with drilling fluids, Well Summary with casing/cement plans, Directional Plan & Plot, Plats**

**Lateral 2**

|                                                  |                   |                                        |          |                                |       |         |
|--------------------------------------------------|-------------------|----------------------------------------|----------|--------------------------------|-------|---------|
| KOP Lateral 2                                    | Azimuth Lateral 2 | Estimated Total Depth Lateral 2        |          | KOP Coordinates From Well Head |       |         |
| Feet MD                                          | °                 | Feet MD                                | Feet TVD | From WH                        |       | From WH |
| Formation Entry Point Coordinates From Well Head |                   | Bottom Hole Coordinates From Well Head |          |                                |       |         |
| From WH                                          |                   | From WH                                |          | From WH                        |       | From WH |
| KOP Footages From Nearest Section Line           |                   | Qtr-Qtr                                | Section  | Township                       | Range | County  |
| F                                                | L                 |                                        |          | N                              | W     |         |
| Bottom Hole Footages From Nearest Section Line   |                   | Qtr-Qtr                                | Section  | Township                       | Range | County  |
| F                                                | L                 |                                        |          | N                              | W     |         |

**Lateral 3**

|                                                                                                          |                        |                                                                                |         |                                                                        |                                |
|----------------------------------------------------------------------------------------------------------|------------------------|--------------------------------------------------------------------------------|---------|------------------------------------------------------------------------|--------------------------------|
| KOP Lateral 3<br>Feet MD                                                                                 | Azimuth Lateral 3<br>° | Estimated Total Depth Lateral 3<br>Feet MD                      Feet TVD       |         | KOP Coordinates From Well Head<br>From WH                      From WH |                                |
| Formation Entry Point Coordinates From Well Head<br>From WH                      From WH                 |                        | Bottom Hole Coordinates From Well Head<br>From WH                      From WH |         |                                                                        |                                |
| KOP Footages From Nearest Section Line<br>F              L                      F              L         |                        | Qtr-Qtr                                                                        | Section | Township<br>N                                                          | Range<br>W              County |
| Bottom Hole Footages From Nearest Section Line<br>F              L                      F              L |                        | Qtr-Qtr                                                                        | Section | Township<br>N                                                          | Range<br>W              County |

**Lateral 4**

|                                                  |                   |                                        |          |                                |         |        |
|--------------------------------------------------|-------------------|----------------------------------------|----------|--------------------------------|---------|--------|
| KOP Lateral 4                                    | Azimuth Lateral 4 | Estimated Total Depth Lateral 4        |          | KOP Coordinates From Well Head |         |        |
| Feet MD                                          | °                 | Feet MD                                | Feet TVD | From WH                        | From WH |        |
| Formation Entry Point Coordinates From Well Head |                   | Bottom Hole Coordinates From Well Head |          |                                |         |        |
| From WH                                          | From WH           | From WH                                | From WH  |                                |         |        |
| KOP Footages From Nearest Section Line           |                   | Qtr-Qtr                                | Section  | Township                       | Range   | County |
| F                                                | L                 | F                                      | L        | N                              | W       |        |
| Bottom Hole Footages From Nearest Section Line   |                   | Qtr-Qtr                                | Section  | Township                       | Range   | County |
| F                                                | L                 | F                                      | L        | N                              | W       |        |

**Lateral 5**

|                                                                                                          |                        |                                                                                |         |                                                                        |            |        |
|----------------------------------------------------------------------------------------------------------|------------------------|--------------------------------------------------------------------------------|---------|------------------------------------------------------------------------|------------|--------|
| KOP Lateral 5<br>Feet MD                                                                                 | Azimuth Lateral 5<br>° | Estimated Total Depth Lateral 5<br>Feet MD                      Feet TVD       |         | KOP Coordinates From Well Head<br>From WH                      From WH |            |        |
| Formation Entry Point Coordinates From Well Head<br>From WH                      From WH                 |                        | Bottom Hole Coordinates From Well Head<br>From WH                      From WH |         |                                                                        |            |        |
| KOP Footages From Nearest Section Line<br>F              L                      F              L         |                        | Qtr-Qtr                                                                        | Section | Township<br>N                                                          | Range<br>W | County |
| Bottom Hole Footages From Nearest Section Line<br>F              L                      F              L |                        | Qtr-Qtr                                                                        | Section | Township<br>N                                                          | Range<br>W | County |

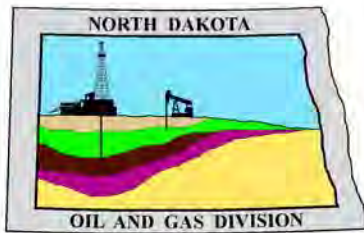
|                                                                                                                           |                                        |                                      |
|---------------------------------------------------------------------------------------------------------------------------|----------------------------------------|--------------------------------------|
| I hereby swear or affirm the information provided is true, complete and correct as determined from all available records. |                                        | Date<br><b>5 / 29 / 2014</b>         |
| <b>ePermit</b>                                                                                                            | Printed Name<br><b>Heather McCowan</b> | Title<br><b>Regulatory Assistant</b> |
|                                                                                                                           |                                        |                                      |

**FOR STATE USE ONLY**

|                                        |                                       |
|----------------------------------------|---------------------------------------|
| Permit and File Number<br><b>28652</b> | API Number<br><b>33 - 053 - 06026</b> |
| Field<br><b>BAKER</b>                  |                                       |
| Pool<br><b>BAKKEN</b>                  | Permit Type<br><b>DEVELOPMENT</b>     |

**FOR STATE USE ONLY**

|                                               |
|-----------------------------------------------|
| Date Approved<br><b>6 / 17 / 2014</b>         |
| By<br><b>Nathaniel Erbele</b>                 |
| Title<br><b>Petroleum Resource Specialist</b> |



# Oil and Gas Division

Lynn D. Helms - Director

Bruce E. Hicks - Assistant Director

## Department of Mineral Resources

Lynn D. Helms - Director

## North Dakota Industrial Commission

[www.oilgas.nd.gov](http://www.oilgas.nd.gov)

April 9, 2014

**RE: Filter Socks and Other Filter Media  
Leakproof Container Required  
Oil and Gas Wells**

Dear Operator,

North Dakota Administrative Code Section 43-02-03-19.2 states in part that all waste material associated with exploration or production of oil and gas must be properly disposed of in an authorized facility in accord with all applicable local, state, and federal laws and regulations.

Filtration systems are commonly used during oil and gas operations in North Dakota. The Commission is very concerned about the proper disposal of used filters (including filter socks) used by the oil and gas industry.

Effective June 1, 2014, a container must be maintained on each well drilled in North Dakota beginning when the well is spud and must remain on-site during clean-out, completion, and flow-back whenever filtration operations are conducted. The on-site container must be used to store filters until they can be properly disposed of in an authorized facility. Such containers must be:

- leakproof to prevent any fluids from escaping the container
- covered to prevent precipitation from entering the container
- placard to indicate only filters are to be placed in the container

If the operator will not utilize a filtration system, a waiver to the container requirement will be considered, but only upon the operator submitting a Sundry Notice (Form 4) justifying their request.

As previously stated in our March 13, 2014 letter, North Dakota Administrative Code Section 33-20-02.1-01 states in part that every person who transports solid waste (which includes oil and gas exploration and production wastes) is required to have a valid permit issued by the North Dakota Department of Health, Division of Waste Management. Please contact the Division of Waste Management at (701) 328-5166 with any questions on the solid waste program. Note oil and gas exploration and production wastes include produced water, drilling mud, invert mud, tank bottom sediment, pipe scale, filters, and fly ash.

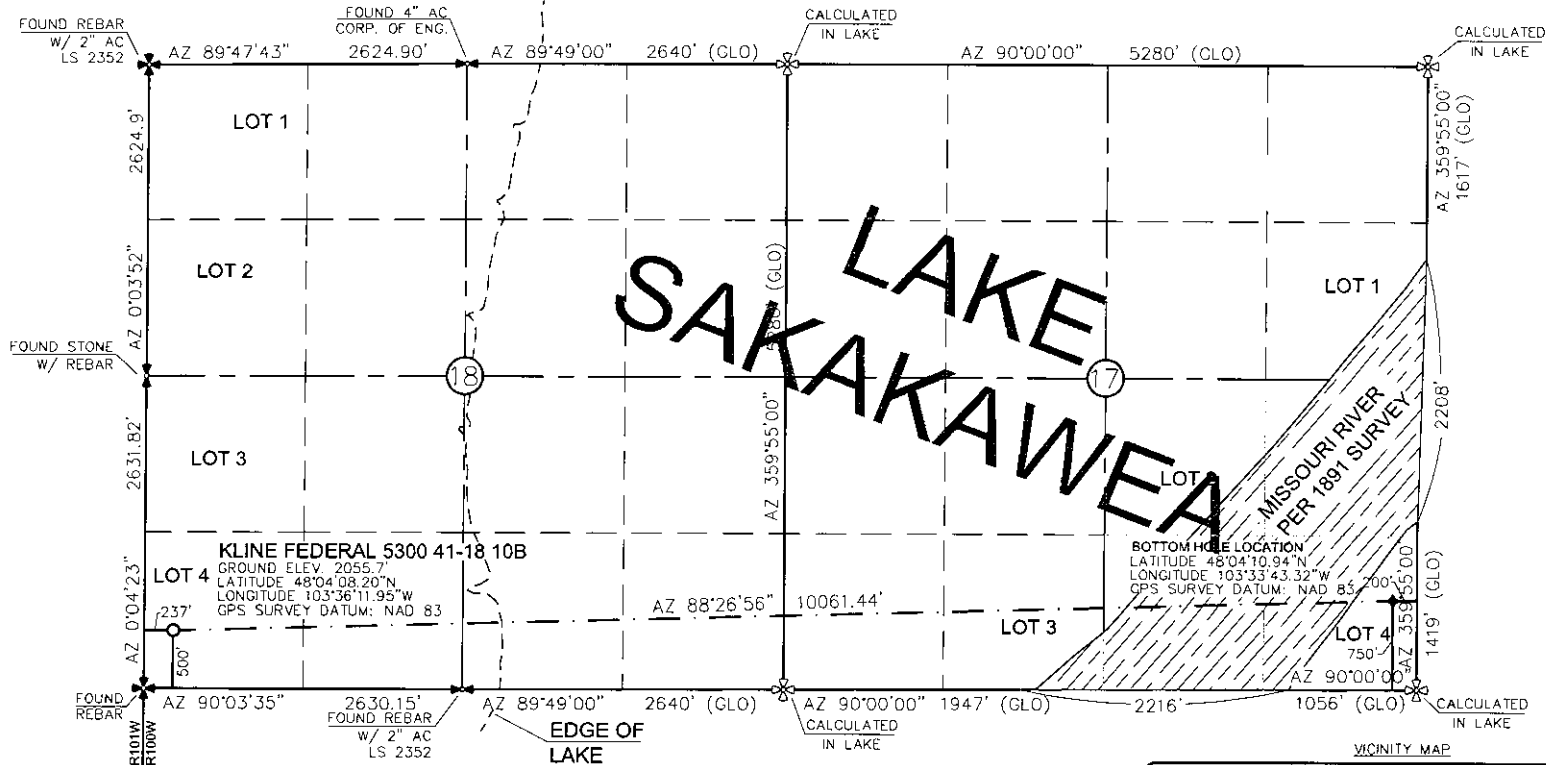
Thank you for your cooperation.

Sincerely,

*Bruce E. Hicks*

Assistant Director

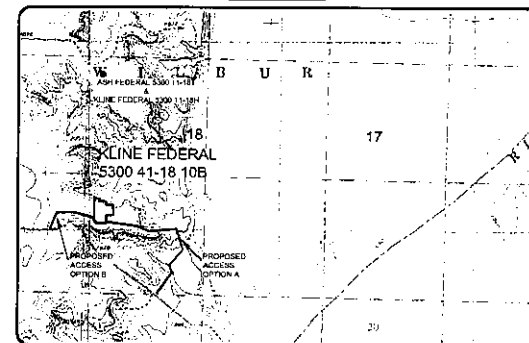
**WELL LOCATION PLAT**  
 OASIS PETROLEUM NORTH AMERICA, LLC  
 1001 FANNIN, SUITE 1500, HOUSTON, TX 77002  
 "KLINE FEDERAL 5300 41-18 10B"  
 500 FEET FROM SOUTH LINE AND 237 FEET FROM WEST LINE  
 SECTION 18, T153N, R100W, 5th P.M., MCKENZIE COUNTY, NORTH DAKOTA



THIS DOCUMENT WAS ORIGINALLY ISSUED AND SEALED BY ROBERT L. PROCIVE, PLS, REGISTRATION NUMBER 2884 ON 4/15/14 AND THE ORIGINAL DOCUMENTS ARE STORED AT THE OFFICES OF INTERSTATE ENGINEERING, INC.

STAKED ON 4/15/14  
 VERTICAL CONTROL DATUM WAS BASED UPON CONTROL POINT 4 WITH AN ELEVATION OF 2090.8'  
 THIS SURVEY AND PLAT IS BEING PROVIDED AT THE REQUEST OF ERIC BAYES OF OASIS PETROLEUM. I CERTIFY THAT THIS PLAT CORRECTLY REPRESENTS WORK PERFORMED BY ME OR UNDER MY SUPERVISION AND IS TRUE AND CORRECT TO THE BEST OF MY KNOWLEDGE AND BELIEF.

*Robert L. Procive*  
 ROBERT L. PROCIVE 2884LS

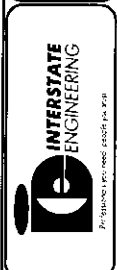


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| By | Date | Checked |
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|-----------------------------------------------------------------------------------------------------------------------|---------------------------------------------------------------------------------------------------|
| OASIS PETROLEUM NORTH AMERICA, LLC<br>WELL LOCATION PLAT<br>SECTION 18, T153N, R100W<br>MCKENZIE COUNTY, NORTH DAKOTA | Project No.: 514-02-01-04<br>Drawn By: [Signature]<br>Checked By: [Signature]<br>Date: APRIL 2014 |
|-----------------------------------------------------------------------------------------------------------------------|---------------------------------------------------------------------------------------------------|

Interstate Engineering, Inc.  
 P.O. Box 648  
 425 East Main Street  
 Spearfish, SD 57583  
 Phone: (605) 433-6817  
 Fax: (605) 433-6818  
 www.interstateeng.com  
 Other: When I have been used in a project, I have been used in a project.



1/8  
 SHEET NO.

| DRILLING PLAN                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                             |                |                              |  |                     |                                 |                  |                   |                |                    |           |       |         |       |       |         |        |       |         |         |       |         |             |       |         |                  |       |         |           |       |         |           |       |         |                |       |         |             |       |         |                  |       |         |                                    |       |         |        |       |         |       |       |         |                               |       |         |             |       |         |              |       |         |    |       |         |                |       |         |           |       |         |                |       |         |           |        |         |                         |        |         |              |        |         |              |        |         |               |        |         |                           |        |         |                                |        |         |              |        |         |             |        |         |                                                                                                                                  |  |  |  |  |  |
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| OPERATOR                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                  |                | Oasis Petroleum              |  |                     | COUNTY/STATE                    |                  | McKenzie Co., ND  |                |                    |           |       |         |       |       |         |        |       |         |         |       |         |             |       |         |                  |       |         |           |       |         |           |       |         |                |       |         |             |       |         |                  |       |         |                                    |       |         |        |       |         |       |       |         |                               |       |         |             |       |         |              |       |         |    |       |         |                |       |         |           |       |         |                |       |         |           |        |         |                         |        |         |              |        |         |              |        |         |               |        |         |                           |        |         |                                |        |         |              |        |         |             |        |         |                                                                                                                                  |  |  |  |  |  |
| WELL NAME                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                 |                | Kline Federal 5300 41-18 10B |  |                     | RIG                             |                  | 0                 |                |                    |           |       |         |       |       |         |        |       |         |         |       |         |             |       |         |                  |       |         |           |       |         |           |       |         |                |       |         |             |       |         |                  |       |         |                                    |       |         |        |       |         |       |       |         |                               |       |         |             |       |         |              |       |         |    |       |         |                |       |         |           |       |         |                |       |         |           |        |         |                         |        |         |              |        |         |              |        |         |               |        |         |                           |        |         |                                |        |         |              |        |         |             |        |         |                                                                                                                                  |  |  |  |  |  |
| WELL TYPE                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                 |                | Horizontal Middle Bakken     |  |                     |                                 |                  |                   |                |                    |           |       |         |       |       |         |        |       |         |         |       |         |             |       |         |                  |       |         |           |       |         |           |       |         |                |       |         |             |       |         |                  |       |         |                                    |       |         |        |       |         |       |       |         |                               |       |         |             |       |         |              |       |         |    |       |         |                |       |         |           |       |         |                |       |         |           |        |         |                         |        |         |              |        |         |              |        |         |               |        |         |                           |        |         |                                |        |         |              |        |         |             |        |         |                                                                                                                                  |  |  |  |  |  |
| LOCATION                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                  |                | SWNW 18-153N-100W            |  |                     | Surface Location (survey plat): |                  | 500' fsl 237' fwl |                |                    |           |       |         |       |       |         |        |       |         |         |       |         |             |       |         |                  |       |         |           |       |         |           |       |         |                |       |         |             |       |         |                  |       |         |                                    |       |         |        |       |         |       |       |         |                               |       |         |             |       |         |              |       |         |    |       |         |                |       |         |           |       |         |                |       |         |           |        |         |                         |        |         |              |        |         |              |        |         |               |        |         |                           |        |         |                                |        |         |              |        |         |             |        |         |                                                                                                                                  |  |  |  |  |  |
| EST. T.D.                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                 |                | 20,590'                      |  |                     | GROUND ELEV:                    |                  | 2057              |                | Finished Pad Elev. |           |       |         |       |       |         |        |       |         |         |       |         |             |       |         |                  |       |         |           |       |         |           |       |         |                |       |         |             |       |         |                  |       |         |                                    |       |         |        |       |         |       |       |         |                               |       |         |             |       |         |              |       |         |    |       |         |                |       |         |           |       |         |                |       |         |           |        |         |                         |        |         |              |        |         |              |        |         |               |        |         |                           |        |         |                                |        |         |              |        |         |             |        |         |                                                                                                                                  |  |  |  |  |  |
|                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                           |                | TOTAL LATERA 9,553'          |  |                     | KB ELEV:                        |                  | 2082              |                |                    |           |       |         |       |       |         |        |       |         |         |       |         |             |       |         |                  |       |         |           |       |         |           |       |         |                |       |         |             |       |         |                  |       |         |                                    |       |         |        |       |         |       |       |         |                               |       |         |             |       |         |              |       |         |    |       |         |                |       |         |           |       |         |                |       |         |           |        |         |                         |        |         |              |        |         |              |        |         |               |        |         |                           |        |         |                                |        |         |              |        |         |             |        |         |                                                                                                                                  |  |  |  |  |  |
| PROGNOSIS: Based on 2,082' KB(est)                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                        |                |                              |  | LOGS: Type Interval |                                 |                  |                   |                |                    |           |       |         |       |       |         |        |       |         |         |       |         |             |       |         |                  |       |         |           |       |         |           |       |         |                |       |         |             |       |         |                  |       |         |                                    |       |         |        |       |         |       |       |         |                               |       |         |             |       |         |              |       |         |    |       |         |                |       |         |           |       |         |                |       |         |           |        |         |                         |        |         |              |        |         |              |        |         |               |        |         |                           |        |         |                                |        |         |              |        |         |             |        |         |                                                                                                                                  |  |  |  |  |  |
| <table><tr><th>MARKER</th><th>DEPTH (Surf Loc)</th><th>DATUM (Surf Loc)</th></tr><tr><td>Pierre</td><td>NDIC MAP 1,972</td><td>110</td></tr><tr><td>Greenhorn</td><td>4,646</td><td>(2,564)</td></tr><tr><td>Mowry</td><td>5,057</td><td>(2,975)</td></tr><tr><td>Dakota</td><td>5,471</td><td>(3,389)</td></tr><tr><td>Rierdon</td><td>6,486</td><td>(4,404)</td></tr><tr><td>Dunham Salt</td><td>6,814</td><td>(4,732)</td></tr><tr><td>Dunham Salt Base</td><td>6,929</td><td>(4,847)</td></tr><tr><td>Spearfish</td><td>7,024</td><td>(4,942)</td></tr><tr><td>Pine Salt</td><td>7,283</td><td>(5,201)</td></tr><tr><td>Pine Salt Base</td><td>7,318</td><td>(5,236)</td></tr><tr><td>Opeche Salt</td><td>7,374</td><td>(5,292)</td></tr><tr><td>Opeche Salt Base</td><td>7,403</td><td>(5,321)</td></tr><tr><td>Broom Creek (Top of Minnelusa Gp.)</td><td>7,605</td><td>(5,523)</td></tr><tr><td>Amsden</td><td>7,684</td><td>(5,602)</td></tr><tr><td>Tyler</td><td>7,853</td><td>(5,771)</td></tr><tr><td>Otter (Base of Minnelusa Gp.)</td><td>8,047</td><td>(5,965)</td></tr><tr><td>Kibbey Lime</td><td>8,390</td><td>(6,308)</td></tr><tr><td>Charles Salt</td><td>8,542</td><td>(6,460)</td></tr><tr><td>UB</td><td>9,159</td><td>(7,077)</td></tr><tr><td>Base Last Salt</td><td>9,238</td><td>(7,156)</td></tr><tr><td>Ratcliffe</td><td>9,301</td><td>(7,219)</td></tr><tr><td>Mission Canyon</td><td>9,454</td><td>(7,372)</td></tr><tr><td>Lodgepole</td><td>10,022</td><td>(7,940)</td></tr><tr><td>Lodgepole Fracture Zone</td><td>10,209</td><td>(8,127)</td></tr><tr><td>False Bakken</td><td>10,726</td><td>(8,644)</td></tr><tr><td>Upper Bakken</td><td>10,735</td><td>(8,653)</td></tr><tr><td>Middle Bakken</td><td>10,751</td><td>(8,669)</td></tr><tr><td>Middle Bakken Sand Target</td><td>10,760</td><td>(8,678)</td></tr><tr><td>Base Middle Bakken Sand Target</td><td>10,770</td><td>(8,688)</td></tr><tr><td>Lower Bakken</td><td>10,785</td><td>(8,703)</td></tr><tr><td>Three Forks</td><td>10,813</td><td>(8,731)</td></tr></table> |                |                              |  | MARKER              | DEPTH (Surf Loc)                | DATUM (Surf Loc) | Pierre            | NDIC MAP 1,972 | 110                | Greenhorn | 4,646 | (2,564) | Mowry | 5,057 | (2,975) | Dakota | 5,471 | (3,389) | Rierdon | 6,486 | (4,404) | Dunham Salt | 6,814 | (4,732) | Dunham Salt Base | 6,929 | (4,847) | Spearfish | 7,024 | (4,942) | Pine Salt | 7,283 | (5,201) | Pine Salt Base | 7,318 | (5,236) | Opeche Salt | 7,374 | (5,292) | Opeche Salt Base | 7,403 | (5,321) | Broom Creek (Top of Minnelusa Gp.) | 7,605 | (5,523) | Amsden | 7,684 | (5,602) | Tyler | 7,853 | (5,771) | Otter (Base of Minnelusa Gp.) | 8,047 | (5,965) | Kibbey Lime | 8,390 | (6,308) | Charles Salt | 8,542 | (6,460) | UB | 9,159 | (7,077) | Base Last Salt | 9,238 | (7,156) | Ratcliffe | 9,301 | (7,219) | Mission Canyon | 9,454 | (7,372) | Lodgepole | 10,022 | (7,940) | Lodgepole Fracture Zone | 10,209 | (8,127) | False Bakken | 10,726 | (8,644) | Upper Bakken | 10,735 | (8,653) | Middle Bakken | 10,751 | (8,669) | Middle Bakken Sand Target | 10,760 | (8,678) | Base Middle Bakken Sand Target | 10,770 | (8,688) | Lower Bakken | 10,785 | (8,703) | Three Forks | 10,813 | (8,731) | OH Logs: Triple Combo KOP to Kibby (or min run of 1800' whichever is greater); GR/Res to BSC; GR to surf; CND through the Dakota |  |  |  |  |  |
|                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                           |                |                              |  | MARKER              | DEPTH (Surf Loc)                | DATUM (Surf Loc) |                   |                |                    |           |       |         |       |       |         |        |       |         |         |       |         |             |       |         |                  |       |         |           |       |         |           |       |         |                |       |         |             |       |         |                  |       |         |                                    |       |         |        |       |         |       |       |         |                               |       |         |             |       |         |              |       |         |    |       |         |                |       |         |           |       |         |                |       |         |           |        |         |                         |        |         |              |        |         |              |        |         |               |        |         |                           |        |         |                                |        |         |              |        |         |             |        |         |                                                                                                                                  |  |  |  |  |  |
| Pierre                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                    | NDIC MAP 1,972 | 110                          |  |                     |                                 |                  |                   |                |                    |           |       |         |       |       |         |        |       |         |         |       |         |             |       |         |                  |       |         |           |       |         |           |       |         |                |       |         |             |       |         |                  |       |         |                                    |       |         |        |       |         |       |       |         |                               |       |         |             |       |         |              |       |         |    |       |         |                |       |         |           |       |         |                |       |         |           |        |         |                         |        |         |              |        |         |              |        |         |               |        |         |                           |        |         |                                |        |         |              |        |         |             |        |         |                                                                                                                                  |  |  |  |  |  |
| Greenhorn                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                 | 4,646          | (2,564)                      |  |                     |                                 |                  |                   |                |                    |           |       |         |       |       |         |        |       |         |         |       |         |             |       |         |                  |       |         |           |       |         |           |       |         |                |       |         |             |       |         |                  |       |         |                                    |       |         |        |       |         |       |       |         |                               |       |         |             |       |         |              |       |         |    |       |         |                |       |         |           |       |         |                |       |         |           |        |         |                         |        |         |              |        |         |              |        |         |               |        |         |                           |        |         |                                |        |         |              |        |         |             |        |         |                                                                                                                                  |  |  |  |  |  |
| Mowry                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                     | 5,057          | (2,975)                      |  |                     |                                 |                  |                   |                |                    |           |       |         |       |       |         |        |       |         |         |       |         |             |       |         |                  |       |         |           |       |         |           |       |         |                |       |         |             |       |         |                  |       |         |                                    |       |         |        |       |         |       |       |         |                               |       |         |             |       |         |              |       |         |    |       |         |                |       |         |           |       |         |                |       |         |           |        |         |                         |        |         |              |        |         |              |        |         |               |        |         |                           |        |         |                                |        |         |              |        |         |             |        |         |                                                                                                                                  |  |  |  |  |  |
| Dakota                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                    | 5,471          | (3,389)                      |  |                     |                                 |                  |                   |                |                    |           |       |         |       |       |         |        |       |         |         |       |         |             |       |         |                  |       |         |           |       |         |           |       |         |                |       |         |             |       |         |                  |       |         |                                    |       |         |        |       |         |       |       |         |                               |       |         |             |       |         |              |       |         |    |       |         |                |       |         |           |       |         |                |       |         |           |        |         |                         |        |         |              |        |         |              |        |         |               |        |         |                           |        |         |                                |        |         |              |        |         |             |        |         |                                                                                                                                  |  |  |  |  |  |
| Rierdon                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                   | 6,486          | (4,404)                      |  |                     |                                 |                  |                   |                |                    |           |       |         |       |       |         |        |       |         |         |       |         |             |       |         |                  |       |         |           |       |         |           |       |         |                |       |         |             |       |         |                  |       |         |                                    |       |         |        |       |         |       |       |         |                               |       |         |             |       |         |              |       |         |    |       |         |                |       |         |           |       |         |                |       |         |           |        |         |                         |        |         |              |        |         |              |        |         |               |        |         |                           |        |         |                                |        |         |              |        |         |             |        |         |                                                                                                                                  |  |  |  |  |  |
| Dunham Salt                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                               | 6,814          | (4,732)                      |  |                     |                                 |                  |                   |                |                    |           |       |         |       |       |         |        |       |         |         |       |         |             |       |         |                  |       |         |           |       |         |           |       |         |                |       |         |             |       |         |                  |       |         |                                    |       |         |        |       |         |       |       |         |                               |       |         |             |       |         |              |       |         |    |       |         |                |       |         |           |       |         |                |       |         |           |        |         |                         |        |         |              |        |         |              |        |         |               |        |         |                           |        |         |                                |        |         |              |        |         |             |        |         |                                                                                                                                  |  |  |  |  |  |
| Dunham Salt Base                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                          | 6,929          | (4,847)                      |  |                     |                                 |                  |                   |                |                    |           |       |         |       |       |         |        |       |         |         |       |         |             |       |         |                  |       |         |           |       |         |           |       |         |                |       |         |             |       |         |                  |       |         |                                    |       |         |        |       |         |       |       |         |                               |       |         |             |       |         |              |       |         |    |       |         |                |       |         |           |       |         |                |       |         |           |        |         |                         |        |         |              |        |         |              |        |         |               |        |         |                           |        |         |                                |        |         |              |        |         |             |        |         |                                                                                                                                  |  |  |  |  |  |
| Spearfish                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                 | 7,024          | (4,942)                      |  |                     |                                 |                  |                   |                |                    |           |       |         |       |       |         |        |       |         |         |       |         |             |       |         |                  |       |         |           |       |         |           |       |         |                |       |         |             |       |         |                  |       |         |                                    |       |         |        |       |         |       |       |         |                               |       |         |             |       |         |              |       |         |    |       |         |                |       |         |           |       |         |                |       |         |           |        |         |                         |        |         |              |        |         |              |        |         |               |        |         |                           |        |         |                                |        |         |              |        |         |             |        |         |                                                                                                                                  |  |  |  |  |  |
| Pine Salt                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                 | 7,283          | (5,201)                      |  |                     |                                 |                  |                   |                |                    |           |       |         |       |       |         |        |       |         |         |       |         |             |       |         |                  |       |         |           |       |         |           |       |         |                |       |         |             |       |         |                  |       |         |                                    |       |         |        |       |         |       |       |         |                               |       |         |             |       |         |              |       |         |    |       |         |                |       |         |           |       |         |                |       |         |           |        |         |                         |        |         |              |        |         |              |        |         |               |        |         |                           |        |         |                                |        |         |              |        |         |             |        |         |                                                                                                                                  |  |  |  |  |  |
| Pine Salt Base                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                            | 7,318          | (5,236)                      |  |                     |                                 |                  |                   |                |                    |           |       |         |       |       |         |        |       |         |         |       |         |             |       |         |                  |       |         |           |       |         |           |       |         |                |       |         |             |       |         |                  |       |         |                                    |       |         |        |       |         |       |       |         |                               |       |         |             |       |         |              |       |         |    |       |         |                |       |         |           |       |         |                |       |         |           |        |         |                         |        |         |              |        |         |              |        |         |               |        |         |                           |        |         |                                |        |         |              |        |         |             |        |         |                                                                                                                                  |  |  |  |  |  |
| Opeche Salt                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                               | 7,374          | (5,292)                      |  |                     |                                 |                  |                   |                |                    |           |       |         |       |       |         |        |       |         |         |       |         |             |       |         |                  |       |         |           |       |         |           |       |         |                |       |         |             |       |         |                  |       |         |                                    |       |         |        |       |         |       |       |         |                               |       |         |             |       |         |              |       |         |    |       |         |                |       |         |           |       |         |                |       |         |           |        |         |                         |        |         |              |        |         |              |        |         |               |        |         |                           |        |         |                                |        |         |              |        |         |             |        |         |                                                                                                                                  |  |  |  |  |  |
| Opeche Salt Base                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                          | 7,403          | (5,321)                      |  |                     |                                 |                  |                   |                |                    |           |       |         |       |       |         |        |       |         |         |       |         |             |       |         |                  |       |         |           |       |         |           |       |         |                |       |         |             |       |         |                  |       |         |                                    |       |         |        |       |         |       |       |         |                               |       |         |             |       |         |              |       |         |    |       |         |                |       |         |           |       |         |                |       |         |           |        |         |                         |        |         |              |        |         |              |        |         |               |        |         |                           |        |         |                                |        |         |              |        |         |             |        |         |                                                                                                                                  |  |  |  |  |  |
| Broom Creek (Top of Minnelusa Gp.)                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                        | 7,605          | (5,523)                      |  |                     |                                 |                  |                   |                |                    |           |       |         |       |       |         |        |       |         |         |       |         |             |       |         |                  |       |         |           |       |         |           |       |         |                |       |         |             |       |         |                  |       |         |                                    |       |         |        |       |         |       |       |         |                               |       |         |             |       |         |              |       |         |    |       |         |                |       |         |           |       |         |                |       |         |           |        |         |                         |        |         |              |        |         |              |        |         |               |        |         |                           |        |         |                                |        |         |              |        |         |             |        |         |                                                                                                                                  |  |  |  |  |  |
| Amsden                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                    | 7,684          | (5,602)                      |  |                     |                                 |                  |                   |                |                    |           |       |         |       |       |         |        |       |         |         |       |         |             |       |         |                  |       |         |           |       |         |           |       |         |                |       |         |             |       |         |                  |       |         |                                    |       |         |        |       |         |       |       |         |                               |       |         |             |       |         |              |       |         |    |       |         |                |       |         |           |       |         |                |       |         |           |        |         |                         |        |         |              |        |         |              |        |         |               |        |         |                           |        |         |                                |        |         |              |        |         |             |        |         |                                                                                                                                  |  |  |  |  |  |
| Tyler                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                     | 7,853          | (5,771)                      |  |                     |                                 |                  |                   |                |                    |           |       |         |       |       |         |        |       |         |         |       |         |             |       |         |                  |       |         |           |       |         |           |       |         |                |       |         |             |       |         |                  |       |         |                                    |       |         |        |       |         |       |       |         |                               |       |         |             |       |         |              |       |         |    |       |         |                |       |         |           |       |         |                |       |         |           |        |         |                         |        |         |              |        |         |              |        |         |               |        |         |                           |        |         |                                |        |         |              |        |         |             |        |         |                                                                                                                                  |  |  |  |  |  |
| Otter (Base of Minnelusa Gp.)                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                             | 8,047          | (5,965)                      |  |                     |                                 |                  |                   |                |                    |           |       |         |       |       |         |        |       |         |         |       |         |             |       |         |                  |       |         |           |       |         |           |       |         |                |       |         |             |       |         |                  |       |         |                                    |       |         |        |       |         |       |       |         |                               |       |         |             |       |         |              |       |         |    |       |         |                |       |         |           |       |         |                |       |         |           |        |         |                         |        |         |              |        |         |              |        |         |               |        |         |                           |        |         |                                |        |         |              |        |         |             |        |         |                                                                                                                                  |  |  |  |  |  |
| Kibbey Lime                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                               | 8,390          | (6,308)                      |  |                     |                                 |                  |                   |                |                    |           |       |         |       |       |         |        |       |         |         |       |         |             |       |         |                  |       |         |           |       |         |           |       |         |                |       |         |             |       |         |                  |       |         |                                    |       |         |        |       |         |       |       |         |                               |       |         |             |       |         |              |       |         |    |       |         |                |       |         |           |       |         |                |       |         |           |        |         |                         |        |         |              |        |         |              |        |         |               |        |         |                           |        |         |                                |        |         |              |        |         |             |        |         |                                                                                                                                  |  |  |  |  |  |
| Charles Salt                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                              | 8,542          | (6,460)                      |  |                     |                                 |                  |                   |                |                    |           |       |         |       |       |         |        |       |         |         |       |         |             |       |         |                  |       |         |           |       |         |           |       |         |                |       |         |             |       |         |                  |       |         |                                    |       |         |        |       |         |       |       |         |                               |       |         |             |       |         |              |       |         |    |       |         |                |       |         |           |       |         |                |       |         |           |        |         |                         |        |         |              |        |         |              |        |         |               |        |         |                           |        |         |                                |        |         |              |        |         |             |        |         |                                                                                                                                  |  |  |  |  |  |
| UB                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                        | 9,159          | (7,077)                      |  |                     |                                 |                  |                   |                |                    |           |       |         |       |       |         |        |       |         |         |       |         |             |       |         |                  |       |         |           |       |         |           |       |         |                |       |         |             |       |         |                  |       |         |                                    |       |         |        |       |         |       |       |         |                               |       |         |             |       |         |              |       |         |    |       |         |                |       |         |           |       |         |                |       |         |           |        |         |                         |        |         |              |        |         |              |        |         |               |        |         |                           |        |         |                                |        |         |              |        |         |             |        |         |                                                                                                                                  |  |  |  |  |  |
| Base Last Salt                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                            | 9,238          | (7,156)                      |  |                     |                                 |                  |                   |                |                    |           |       |         |       |       |         |        |       |         |         |       |         |             |       |         |                  |       |         |           |       |         |           |       |         |                |       |         |             |       |         |                  |       |         |                                    |       |         |        |       |         |       |       |         |                               |       |         |             |       |         |              |       |         |    |       |         |                |       |         |           |       |         |                |       |         |           |        |         |                         |        |         |              |        |         |              |        |         |               |        |         |                           |        |         |                                |        |         |              |        |         |             |        |         |                                                                                                                                  |  |  |  |  |  |
| Ratcliffe                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                 | 9,301          | (7,219)                      |  |                     |                                 |                  |                   |                |                    |           |       |         |       |       |         |        |       |         |         |       |         |             |       |         |                  |       |         |           |       |         |           |       |         |                |       |         |             |       |         |                  |       |         |                                    |       |         |        |       |         |       |       |         |                               |       |         |             |       |         |              |       |         |    |       |         |                |       |         |           |       |         |                |       |         |           |        |         |                         |        |         |              |        |         |              |        |         |               |        |         |                           |        |         |                                |        |         |              |        |         |             |        |         |                                                                                                                                  |  |  |  |  |  |
| Mission Canyon                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                            | 9,454          | (7,372)                      |  |                     |                                 |                  |                   |                |                    |           |       |         |       |       |         |        |       |         |         |       |         |             |       |         |                  |       |         |           |       |         |           |       |         |                |       |         |             |       |         |                  |       |         |                                    |       |         |        |       |         |       |       |         |                               |       |         |             |       |         |              |       |         |    |       |         |                |       |         |           |       |         |                |       |         |           |        |         |                         |        |         |              |        |         |              |        |         |               |        |         |                           |        |         |                                |        |         |              |        |         |             |        |         |                                                                                                                                  |  |  |  |  |  |
| Lodgepole                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                 | 10,022         | (7,940)                      |  |                     |                                 |                  |                   |                |                    |           |       |         |       |       |         |        |       |         |         |       |         |             |       |         |                  |       |         |           |       |         |           |       |         |                |       |         |             |       |         |                  |       |         |                                    |       |         |        |       |         |       |       |         |                               |       |         |             |       |         |              |       |         |    |       |         |                |       |         |           |       |         |                |       |         |           |        |         |                         |        |         |              |        |         |              |        |         |               |        |         |                           |        |         |                                |        |         |              |        |         |             |        |         |                                                                                                                                  |  |  |  |  |  |
| Lodgepole Fracture Zone                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                   | 10,209         | (8,127)                      |  |                     |                                 |                  |                   |                |                    |           |       |         |       |       |         |        |       |         |         |       |         |             |       |         |                  |       |         |           |       |         |           |       |         |                |       |         |             |       |         |                  |       |         |                                    |       |         |        |       |         |       |       |         |                               |       |         |             |       |         |              |       |         |    |       |         |                |       |         |           |       |         |                |       |         |           |        |         |                         |        |         |              |        |         |              |        |         |               |        |         |                           |        |         |                                |        |         |              |        |         |             |        |         |                                                                                                                                  |  |  |  |  |  |
| False Bakken                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                              | 10,726         | (8,644)                      |  |                     |                                 |                  |                   |                |                    |           |       |         |       |       |         |        |       |         |         |       |         |             |       |         |                  |       |         |           |       |         |           |       |         |                |       |         |             |       |         |                  |       |         |                                    |       |         |        |       |         |       |       |         |                               |       |         |             |       |         |              |       |         |    |       |         |                |       |         |           |       |         |                |       |         |           |        |         |                         |        |         |              |        |         |              |        |         |               |        |         |                           |        |         |                                |        |         |              |        |         |             |        |         |                                                                                                                                  |  |  |  |  |  |
| Upper Bakken                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                              | 10,735         | (8,653)                      |  |                     |                                 |                  |                   |                |                    |           |       |         |       |       |         |        |       |         |         |       |         |             |       |         |                  |       |         |           |       |         |           |       |         |                |       |         |             |       |         |                  |       |         |                                    |       |         |        |       |         |       |       |         |                               |       |         |             |       |         |              |       |         |    |       |         |                |       |         |           |       |         |                |       |         |           |        |         |                         |        |         |              |        |         |              |        |         |               |        |         |                           |        |         |                                |        |         |              |        |         |             |        |         |                                                                                                                                  |  |  |  |  |  |
| Middle Bakken                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                             | 10,751         | (8,669)                      |  |                     |                                 |                  |                   |                |                    |           |       |         |       |       |         |        |       |         |         |       |         |             |       |         |                  |       |         |           |       |         |           |       |         |                |       |         |             |       |         |                  |       |         |                                    |       |         |        |       |         |       |       |         |                               |       |         |             |       |         |              |       |         |    |       |         |                |       |         |           |       |         |                |       |         |           |        |         |                         |        |         |              |        |         |              |        |         |               |        |         |                           |        |         |                                |        |         |              |        |         |             |        |         |                                                                                                                                  |  |  |  |  |  |
| Middle Bakken Sand Target                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                 | 10,760         | (8,678)                      |  |                     |                                 |                  |                   |                |                    |           |       |         |       |       |         |        |       |         |         |       |         |             |       |         |                  |       |         |           |       |         |           |       |         |                |       |         |             |       |         |                  |       |         |                                    |       |         |        |       |         |       |       |         |                               |       |         |             |       |         |              |       |         |    |       |         |                |       |         |           |       |         |                |       |         |           |        |         |                         |        |         |              |        |         |              |        |         |               |        |         |                           |        |         |                                |        |         |              |        |         |             |        |         |                                                                                                                                  |  |  |  |  |  |
| Base Middle Bakken Sand Target                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                            | 10,770         | (8,688)                      |  |                     |                                 |                  |                   |                |                    |           |       |         |       |       |         |        |       |         |         |       |         |             |       |         |                  |       |         |           |       |         |           |       |         |                |       |         |             |       |         |                  |       |         |                                    |       |         |        |       |         |       |       |         |                               |       |         |             |       |         |              |       |         |    |       |         |                |       |         |           |       |         |                |       |         |           |        |         |                         |        |         |              |        |         |              |        |         |               |        |         |                           |        |         |                                |        |         |              |        |         |             |        |         |                                                                                                                                  |  |  |  |  |  |
| Lower Bakken                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                              | 10,785         | (8,703)                      |  |                     |                                 |                  |                   |                |                    |           |       |         |       |       |         |        |       |         |         |       |         |             |       |         |                  |       |         |           |       |         |           |       |         |                |       |         |             |       |         |                  |       |         |                                    |       |         |        |       |         |       |       |         |                               |       |         |             |       |         |              |       |         |    |       |         |                |       |         |           |       |         |                |       |         |           |        |         |                         |        |         |              |        |         |              |        |         |               |        |         |                           |        |         |                                |        |         |              |        |         |             |        |         |                                                                                                                                  |  |  |  |  |  |
| Three Forks                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                               | 10,813         | (8,731)                      |  |                     |                                 |                  |                   |                |                    |           |       |         |       |       |         |        |       |         |         |       |         |             |       |         |                  |       |         |           |       |         |           |       |         |                |       |         |             |       |         |                  |       |         |                                    |       |         |        |       |         |       |       |         |                               |       |         |             |       |         |              |       |         |    |       |         |                |       |         |           |       |         |                |       |         |           |        |         |                         |        |         |              |        |         |              |        |         |               |        |         |                           |        |         |                                |        |         |              |        |         |             |        |         |                                                                                                                                  |  |  |  |  |  |
| CBL/GR: Above top of cement/GR to base of casing                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                          |                |                              |  |                     |                                 |                  |                   |                |                    |           |       |         |       |       |         |        |       |         |         |       |         |             |       |         |                  |       |         |           |       |         |           |       |         |                |       |         |             |       |         |                  |       |         |                                    |       |         |        |       |         |       |       |         |                               |       |         |             |       |         |              |       |         |    |       |         |                |       |         |           |       |         |                |       |         |           |        |         |                         |        |         |              |        |         |              |        |         |               |        |         |                           |        |         |                                |        |         |              |        |         |             |        |         |                                                                                                                                  |  |  |  |  |  |
| MWD GR: KOP to lateral TD                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                 |                |                              |  |                     |                                 |                  |                   |                |                    |           |       |         |       |       |         |        |       |         |         |       |         |             |       |         |                  |       |         |           |       |         |           |       |         |                |       |         |             |       |         |                  |       |         |                                    |       |         |        |       |         |       |       |         |                               |       |         |             |       |         |              |       |         |    |       |         |                |       |         |           |       |         |                |       |         |           |        |         |                         |        |         |              |        |         |              |        |         |               |        |         |                           |        |         |                                |        |         |              |        |         |             |        |         |                                                                                                                                  |  |  |  |  |  |

 DEVIATION: | | | ||  | | | | Surf: | | 3 deg. max., 1 deg / 100'; srvy every 500' | |
|  | | | | Prod: | | 5 deg. max., 1 deg / 100'; srvy every 100' | |
|  | | | | DST'S: None planned | | | | | |
|  | | | | CORES: None planned | | | | | |
|  | | | | MUDLOGGING: | | | | | |
|  | | | | Two-Man: | | 8,342' | |
|  | | | | ~200' above the Charles (Kibbey) to Casing point; Casing point to TD | | | | | |
|  | | | | 30' samples at direction of wellsite geologist; 10' through target @ curve land | | | | | |
|  | | | | BOP: 11" 5000 psi blind, pipe & annular | | | | | |
| Dip Rate: -0.1 | | | |  | | | | | |
| Max. Anticipated BHP: 4673 | | | | Surface Formation: Glacial till | | | | | |
| MUD: | | Interval | | Type | | WT | | Vis | |
| Surface: | | 0' - 2,072' | | FW/Gel - Lime Sweeps | | 8.4-9.0 | | 28-32 | |
| Intermediate: | | 2,072' - 11,037' | | Invert | | 9.5-10.4 | | 40-50 | |
| Lateral: | | 11,037' - 20,590' | | Salt Water | | 9.8-10.2 | | 28-32 | |
| CASING: | | Size | | Wt ppf | | Hole | | Depth | |
| Surface: | | 9-5/8" | | 36# | | 13-1/2" | | 2,072' | |
| Intermediate: | | 7" | | 29/32# | | 8-3/4" | | 11,037' | |
| Production Liner: | | 4.5" | | 11.6# | | 6" | | 20,590' | |
|  | |  | |  | | TOL @ 10,238' | |  | |
| PROBABLE PLUGS, IF REQ'D: | | | | | | | | | |
| OTHER: | | MD | | TVD | | FNL/FSL | | FEL/FWL | |
| Surface: | | 2,072 | | 2,072 | | 500' FSL | | 237' FWL | |
| KOP: | | 10,288' | | 10,288' | | 514' FSL | | 285' FWL | |
| EOC: | | 11,037' | | 10,765' | | 643' FSL | | 744' FWL | |
| Casing Point: | | 11,037' | | 10,765' | | 643' FSL | | 744' FWL | |
| Middle Bakken Lateral TD: | | 20,590' | | 10,782' | | 750' FSL | | 200' FEL | |
|  | |  | |  | |  | | S-T-R | |
|  | |  | |  | |  | | SEC 18-T153N-R100W | |
|  | |  | |  | |  | | SEC 18-T153N-R100W | |
|  | |  | |  | |  | | SEC 18-T153N-R100W | |
|  | |  | |  | |  | | SEC 18-T153N-R100W | |
|  | |  | |  | |  | | SEC 17-T153N-R100W | |
|  | |  | |  | |  | | AZI | |
|  | |  | |  | |  | | Survey Company: | |
|  | |  | |  | |  | | Build Rate: 12 deg /100' | |
| Comments: | | | | | | | | | |
| Request a Sundry for an Open Hole Log Waiver | | | | | | | | | |
| Exception well: Oasis Petroleum's Kline Federal 5300 11-18H (153N 100W 18 NW NW) | | | | | | | | | |
| 35 packers, 35 sleeves, no frac string | | | | | | | | | |
| Oasis Petroleum does not use Diesel Fuel, as defined by the US EPA in the list below, in our hydraulic fracture operations. | | | | | | | | | |
| 68334-30-5 (Primary Name: Fuels, diesel) 68476-34-6 (Primary Name: Fuels, diesel, No. 2)68476-30-2 (Primary Name: Fuel oil No. 2) | | | | | | | | | |
| 68476-31-3 (Primary Name: Fuel oil, No. 4)8008-20-6 (Primary Name: Kerosene) | | | | | | | | | |
|  | | | | | | | | | |
| Geology: M. Steed 4/3/2014 | | | | | Engineering: hbadler rpm | | | | |



**Oasis Petroleum  
Well Summary  
Kline Federal 5300 41-18 10B  
Section 18 T153N R100W  
McKenzie County, ND**

**SURFACE CASING AND CEMENT DESIGN**

| Size   | Interval     | Weight | Grade | Coupling | I.D.   | Drift  | Make-up Torque (ft-lbs) |         |      |
|--------|--------------|--------|-------|----------|--------|--------|-------------------------|---------|------|
|        |              |        |       |          |        |        | Minimum                 | Optimum | Max  |
| 9-5/8" | 0' to 2,072' | 36     | J-55  | LTC      | 8.921" | 8.765" | 3400                    | 4530    | 5660 |

| Interval     | Description                 | Collapse    | Burst       | Tension      | Cost per ft |
|--------------|-----------------------------|-------------|-------------|--------------|-------------|
|              |                             | (psi) a     | (psi) b     | (1000 lbs) c |             |
| 0' to 2,072' | 9-5/8", 36#, J-55, LTC, 8rd | 2020 / 2.08 | 3520 / 3.63 | 453 / 2.76   |             |

**API Rating & Safety Factor**

- a) Based on full casing evacuation with 9 ppg fluid on backside (2,072' setting depth).
- b) Burst pressure based on 9 ppg fluid with no fluid on backside (2,072' setting depth).
- c) Based on string weight in 9 ppg fluid at 2,072' TVD plus 100k# overpull. (Buoyed weight equals 64k lbs.)

Cement volumes are based on 9-5/8" casing set in 13-1/2" hole with 60% excess to circulate cement back to surface. Mix and pump the following slurry.

**Pre-flush (Spacer):**      **20 bbls** fresh water

**Lead Slurry:**              **449 sks** (232 bbls) Conventional system with 94 lb/sk cement, 4% extender, 2% expanding agent, 2% CaCl<sub>2</sub> and 0.25 lb/sk lost circulation control agent

**Tail Slurry:**                **300 sks** (62 bbls) Conventional system with 94 lb/sk cement, 3% NaCl, and 0.25 lb/sk lost circulation control agent

**Oasis Petroleum  
Well Summary  
Kline Federal 5300 41-18 10B  
Section 18 T153N R100W  
McKenzie County, ND**

**INTERMEDIATE CASING AND CEMENT DESIGN**

| Size | Interval          | Weight | Grade   | Coupling | I.D.   | Drift    | Make-up Torque (ft-lbs) |         |       |
|------|-------------------|--------|---------|----------|--------|----------|-------------------------|---------|-------|
|      |                   |        |         |          |        |          | Minimum                 | Optimum | Max   |
| 7"   | 0' – 6,614'       | 29     | P-110   | LTC      | 6.184" | 6.059"   | 5,980                   | 7,970   | 8,770 |
| 7"   | 6,614' – 10,288'  | 32     | HCP-110 | LTC      | 6.094" | 6.000*** | 6,730                   | 8,970   | 9,870 |
| 7"   | 10,288' – 11,037' | 29     | P-110   | LTC      | 6.184" | 6.059"   | 5,980                   | 7,970   | 8,770 |

\*\*\*Special drift

| Interval          | Length | Description                | Collapse       | Burst        | Tension      |
|-------------------|--------|----------------------------|----------------|--------------|--------------|
|                   |        |                            | (psi) a        | (psi) b      | (1000 lbs) c |
| 0' – 6,614'       | 6,614' | 7", 29#, P-110, LTC, 8rd   | 8530 / 2.48*   | 11220 / 1.19 | 797 / 2.10   |
| 6,614' – 10,288'  | 3,674' | 7", 32#, HCP-110, LTC, 8rd | 11820 / 2.21*  | 12460 / 1.29 |              |
| 6,614' – 10,288'  | 3,674' | 7", 32#, HCP-110, LTC, 8rd | 11820 / 1.07** | 12460 / 1.29 |              |
| 10,288' – 11,037' | 749'   | 7", 29 lb, P-110, LTC, 8rd | 8530 / 1.53*   | 11220 / 1.16 |              |

**API Rating & Safety Factor**

- a) \*Assume full casing evacuation with 10 ppg fluid on backside  
 \*\*Assume full casing evacuation with 1.2 psi/ft equivalent fluid gradient across salt intervals. (Bottom of last salt 9,238' TVD)
- b) Burst pressure based on 9000 psig max press for stimulation plus 10.2 ppg fluid in casing and 9 ppg fluid on backside-to 10,765' TVD.
- c) Based on string weight in 10 ppg fluid, (280k lbs buoyed weight) plus 100k

Cement volumes are estimates based on 7" casing set in an 8-3/4" hole with 30% excess.

**Pre-flush (Spacer):**      **170 bbls** Saltwater  
                                  **20 bbls** CW8 System  
                                  **10 bbls** Fresh Water

**Lead Slurry:**              **177 sks** (82 bbls) Conventional system with 47 lb/sk cement, 10% NaCl, 34 lb/sk extender, 10% D020 extender, 1% D079 extender, 1% anti-settling agent, 1% fluid loss agent, 0.2% anti-foam agent, 0.7% retarder, 0.125 lb/sk lost circulation control agent, and 0.3% dispersant

**Tail Slurry:**                **575 sks** (168 bbls) Conventional system with 94 lb/sk cement, 10% NaCl, 35% Silica, 0.2% fluid loss agent, 0.8% dispersant, 0.125 lb/sk lost circulation control agent and 0.3% retarder

**Oasis Petroleum  
Well Summary  
Kline Federal 5300 41-18 10B  
Section 18 T153N R100W  
McKenzie County, ND**

**PRODUCTION LINER**

|             |                   |               |              |                 |             |              | <b>Make-up Torque (ft-lbs)</b> |                |            |
|-------------|-------------------|---------------|--------------|-----------------|-------------|--------------|--------------------------------|----------------|------------|
| <b>Size</b> | <b>Interval</b>   | <b>Weight</b> | <b>Grade</b> | <b>Coupling</b> | <b>I.D.</b> | <b>Drift</b> | <b>Minimum</b>                 | <b>Optimum</b> | <b>Max</b> |
| 4-1/2"      | 10,238' – 20,590' | 11.6          | P-110        | BTC             | 4.000"      | 3.875"       |                                |                |            |

| <b>Interval</b>   | <b>Description</b>               | <b>Collapse</b> | <b>Burst</b>   | <b>Tension</b>      | <b>Cost per ft</b> |
|-------------------|----------------------------------|-----------------|----------------|---------------------|--------------------|
|                   |                                  | (psi) <b>a</b>  | (psi) <b>b</b> | (1000 lbs) <b>c</b> |                    |
| 10,238' – 20,590' | 4-1/2", 11.6 lb, P-110, BTC, 8rd | 7560 / 1.42     | 10690 / 1.11   | 385 / 1.90          |                    |

**API Rating & Safety Factor**

- a) Based on full casing evacuation with 9.5 ppg fluid on backside @ 10,765' TVD.  
Burst pressure based on 9000 psi treating pressure with 10.2 ppg internal fluid gradient and 9 ppg external
- b) fluid gradient @ 10,765' TVD.
- c) Based on string weight in 9.5 ppg fluid (Buoyed weight: 102k lbs.) plus 100k lbs overpull.



# **Oasis Petroleum**

**Indian Hills**

**153N-100W-17/18**

**Kline Federal 5300 41-18 10B**

**Kline Federal 5300 41-18 10B**

**Plan: Design #1**

## **Standard Planning Report**

**23 May, 2014**



**RYAN DIRECTIONAL  
SERVICES**

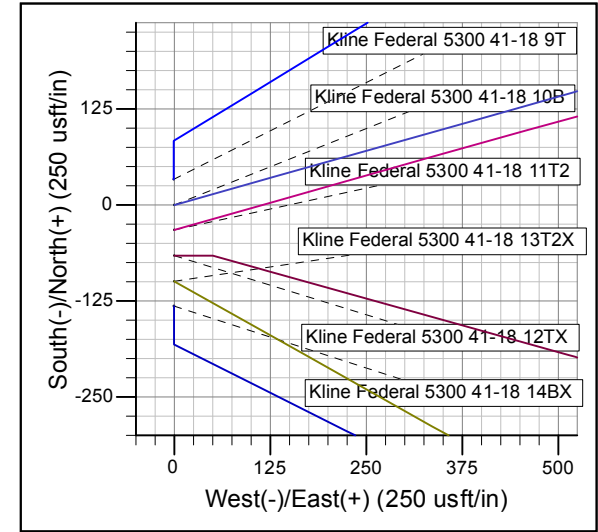
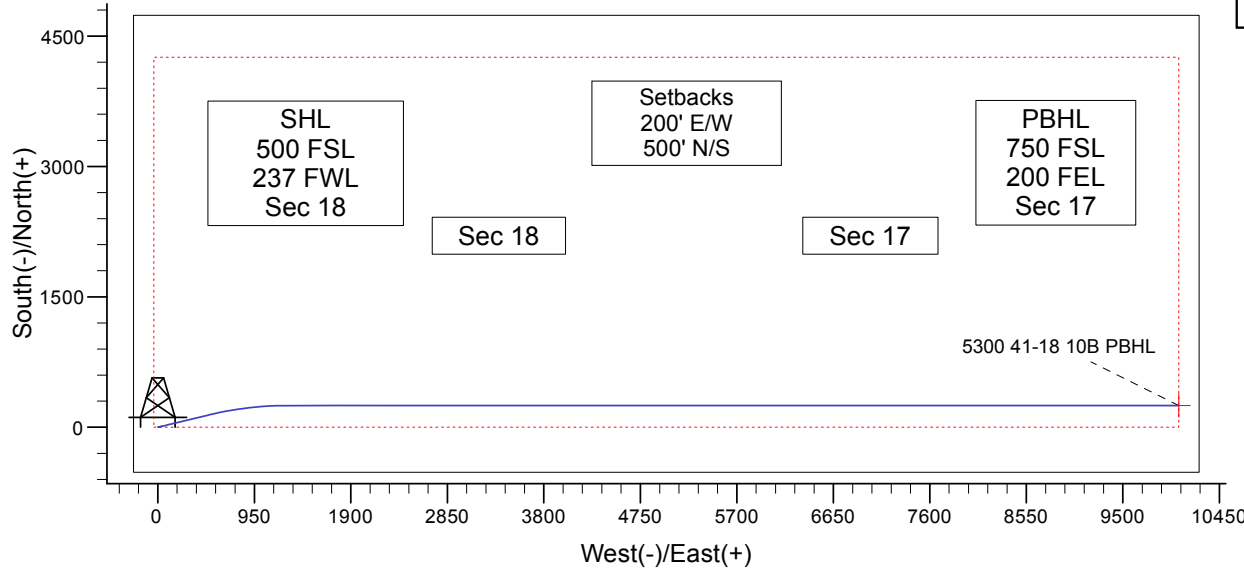
**A NABORS COMPANY**

Project: Indian Hills  
 Site: 153N-100W-17/18  
 Well: Kline Federal 5300 41-18 10B  
 Wellbore: Kline Federal 5300 41-18 10B  
 Design: Design #1



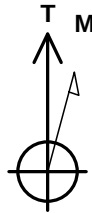
# WELL DETAILS: Kline Federal 5300 41-18 10B

Ground Level: 2057.0  
 Northing 405202.44 Easting 1209970.87 Latitude 48° 4' 8.200 N Longitude 103° 36' 11.950 W



## CASING DETAILS

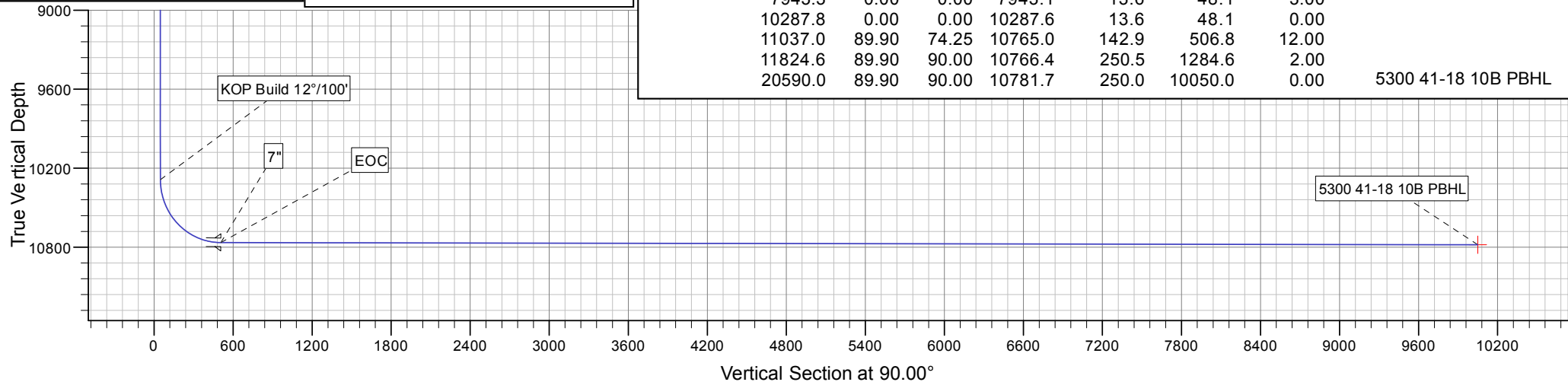
| TVD     | MD      | Name | Size |
|---------|---------|------|------|
| 10765.0 | 11037.0 | 7"   | 7"   |



Azimuths to True North  
 Magnetic North: 8.27°  
 Magnetic Field  
 Strength: 56438.3snT  
 Dip Angle: 72.99°  
 Date: 5/19/2014  
 Model: IGRF2010

## SECTION DETAILS

| MD      | Inc   | Azi   | TVD     | +N/-S | +E/-W   | Dleg  | Target              |
|---------|-------|-------|---------|-------|---------|-------|---------------------|
| 0.0     | 0.00  | 0.00  | 0.0     | 0.0   | 0.0     | 0.00  |                     |
| 2200.0  | 0.00  | 0.00  | 2200.0  | 0.0   | 0.0     | 0.00  |                     |
| 2216.7  | 0.50  | 74.25 | 2216.7  | 0.0   | 0.1     | 3.00  |                     |
| 7926.7  | 0.50  | 74.25 | 7926.4  | 13.5  | 48.0    | 0.00  |                     |
| 7943.3  | 0.00  | 0.00  | 7943.1  | 13.6  | 48.1    | 3.00  |                     |
| 10287.8 | 0.00  | 0.00  | 10287.6 | 13.6  | 48.1    | 0.00  |                     |
| 11037.0 | 89.90 | 74.25 | 10765.0 | 142.9 | 506.8   | 12.00 |                     |
| 11824.6 | 89.90 | 90.00 | 10766.4 | 250.5 | 1284.6  | 2.00  |                     |
| 20590.0 | 89.90 | 90.00 | 10781.7 | 250.0 | 10050.0 | 0.00  | 5300 41-18 10B PBHL |





# Ryan Directional Services

ÄÄ!!"#%&'()\*



|           |                                  |                              |                                      |
|-----------|----------------------------------|------------------------------|--------------------------------------|
| Database: | 789\$/111:3\$;!#Ä&\$<=&)\$8>     | Local Co-ordinate Reference: | + &ÄÄ\$,Ä"!&\$-&.&YÄ\$23435\$316     |
| Company:  | E "=\$Ä&*)(Ä&BH                  | TVD Reference:               | +7?? @\$A15A:1B=C*\$DE)"#1879\$7Ä&FG |
| Project:  | !!."!\$J"ÄÄ=                     | MD Reference:                | +7?? @\$A15A:1B=C*\$DE)"#1879\$7Ä&FG |
| Site:     | 3/OL4311+43MN35                  | North Reference:             | K)B&                                 |
| Well:     | ,Ä"!&\$-&.&) Ä\$/011\$23435\$316 | Survey Calculation Method:   | 9"!HBH\$OB)F *B)&                    |
| Wellbore: | ,Ä"!&\$-&.&) Ä\$/011\$23435\$316 |                              |                                      |
| Design:   | 8&="#!\$P3                       |                              |                                      |

|             |                                 |               |                |
|-------------|---------------------------------|---------------|----------------|
| Project     | !!."!\$J"ÄÄ=                    |               |                |
| Map System: | <,\$;* *&\$ÄÄ I&\$3Q50          | System Datum: | 9&!\$;&\$?&F&Ä |
| Geo Datum:  | L()*(R\$SH)"T I\$8 *BH\$3Q50    |               |                |
| Map Zone:   | L()*(R\$8 U(* \$L()*(R&)!\$V(!& |               |                |

|                       |                 |                   |                    |
|-----------------------|-----------------|-------------------|--------------------|
| Site                  | 3/OL4311+43MN35 |                   |                    |
| Site Position:        |                 | Northing:         | 215ZQXA:22\$B=C*   |
| From:                 | ? *N?(!#        | Easting:          | 3ZA31ZAAQ:35\$B=C* |
| Position Uncertainty: | 1:1\$B=C*       | Slot Radius:      | 3040N3X\$Y         |
|                       |                 | Grid Convergence: | 4A:03\$W           |

|                      |                                  |                     |               |
|----------------------|----------------------------------|---------------------|---------------|
| Well                 | ,Ä"!&\$-&.&) Ä\$/011\$23435\$316 |                     |               |
| Well Position        | +N/-S                            | 40ZMXM:2\$B=C*      | Northing:     |
|                      | +E/-W                            | 431X:X\$B=C*        | Easting:      |
| Position Uncertainty | A:1\$B=C*                        | Wellhead Elevation: | Ground Level: |

|           |                                  |             |                |
|-----------|----------------------------------|-------------|----------------|
| Wellbore  | ,Ä"!&\$-&.&) Ä\$/011\$23435\$316 |             |                |
| Magnetics | Model Name                       | Sample Date | Declination    |
|           | I\%-A131                         | /N3QNA132   | 5:AM           |
|           |                                  |             | Dip Angle      |
|           |                                  |             | MA:QQ          |
|           |                                  |             | Field Strength |
|           |                                  |             | /XZ205         |

|                   |                  |           |               |
|-------------------|------------------|-----------|---------------|
| Design            | 8&="#!\$P3       |           |               |
| Audit Notes:      |                  |           |               |
| Version:          | Phase:           | Ä%EKEK)Ä7 | Tie On Depth: |
| Vertical Section: | Depth From (TVD) | +N/-S     | +E/-W         |
|                   | (usft)           | (usft)    | (usft)        |
|                   | 1:1              | 1:1       | 1:1           |
|                   |                  |           | Direction     |
|                   |                  |           | (°)           |
|                   |                  |           | Q1:11         |

|               |             |         |          |        |          |             |             |             |        |                       |
|---------------|-------------|---------|----------|--------|----------|-------------|-------------|-------------|--------|-----------------------|
| Plan Sections |             |         |          |        |          |             |             |             |        |                       |
| Measured      | Inclination | Azimuth | Vertical | +N/-S  | +E/-W    | Dogleg      | Build       | Turn        | TFO    | Target                |
| Depth         | (°)         | (°)     | Depth    | (usft) | (usft)   | Rate        | Rate        | Rate        | (°)    |                       |
| (usft)        |             |         | (usft)   |        |          | (°/100usft) | (°/100usft) | (°/100usft) |        |                       |
| 1:1           | 1:11        | 1:11    | 1:1      | 1:1    | 1:1      | 1:11        | 1:11        | 1:11        | 1:11   |                       |
| AZA11:1       | 1:11        | 1:11    | AZA11:1  | 1:1    | 1:1      | 1:11        | 1:11        | 1:11        | 1:11   |                       |
| AZA3X:M       | 1:/1        | M2:A/   | AZA3X:M  | 1:1    | 1:3      | 0:11        | 0:11        | 1:11        | M2:A/  |                       |
| MZQAX:M       | 1:/1        | M2:A/   | MZQAX:2  | 30:/   | 25:1     | 1:11        | 1:11        | 1:11        | 1:11   |                       |
| MZQ20:0       | 1:11        | 1:11    | MZQ20:3  | 30:X   | 25:3     | 0:11        | 40:11       | 1:11        | 351:11 |                       |
| 31ZA5M:5      | 1:11        | 1:11    | 31ZA5M:X | 30:X   | 25:3     | 1:11        | 1:11        | 1:11        | 1:11   |                       |
| 33Z10M:1      | 5Q:Q1       | M2:A/   | 31ZMX/:1 | 32A:Q  | /1X:5    | 3A:11       | 3A:11       | 1:11        | M2:A/  |                       |
| 33Z5A2:X      | 5Q:Q1       | Q1:11   | 31ZMXX:2 | A/1:/  | 3ZA52:X  | A:11        | 1:11        | A:11        | Q1:13  |                       |
| A1Z/Q1:1      | 5Q:Q1       | Q1:11   | 31ZM53:M | A/1:1  | 31Z1/1:1 | 1:11        | 1:11        | 1:11        | 1:11   | /011\$23435\$316\$Ä6J |



|                  |                                 |                                     |                                       |
|------------------|---------------------------------|-------------------------------------|---------------------------------------|
| <b>Database:</b> | 789\$/111:3\$;!#Ä&\$<=&)\$8>    | <b>Local Co-ordinate Reference:</b> | + &ÄÄ\$,Ä"!&\$-&.)YÄ\$23435\$316      |
| <b>Company:</b>  | E ="=\$Ä&*)(Ä&BH                | <b>TVD Reference:</b>               | +7?? \$@A15A:1B=C*\$SDE)"#1879\$7Ä&FG |
| <b>Project:</b>  | !!." !J"ÄÄ=                     | <b>MD Reference:</b>                | +7?? \$@A15A:1B=C*\$SDE)"#1879\$7Ä&FG |
| <b>Site:</b>     | 3/OL4311+43MN35                 | <b>North Reference:</b>             | K)B&                                  |
| <b>Well:</b>     | ,Ä"!&\$-&.) Ä\$/011\$23435\$316 | <b>Survey Calculation Method:</b>   | 9"!HBH\$OB)F *B)&                     |
| <b>Wellbore:</b> | ,Ä"!&\$-&.) Ä\$/011\$23435\$316 |                                     |                                       |
| <b>Design:</b>   | 8&="#!\$P3                      |                                     |                                       |

| Planned Survey        |                 |             |                       |              |              |                         |                         |                        |                       |  |
|-----------------------|-----------------|-------------|-----------------------|--------------|--------------|-------------------------|-------------------------|------------------------|-----------------------|--|
| Measured Depth (usft) | Inclination (°) | Azimuth (°) | Vertical Depth (usft) | +N/-S (usft) | +E/-W (usft) | Vertical Section (usft) | Dogleg Rate (°/100usft) | Build Rate (°/100usft) | Turn Rate (°/100usft) |  |
| AZA11:1               | 1:11            | 1:11        | AZA11:1               | 1:1          | 1:1          | 1:1                     | 1:11                    | 1:11                   | 1:11                  |  |
| AZA3X:M               | 1:/             | M2:A/       | AZA3X:M               | 1:1          | 1:3          | 1:3                     | 0:11                    | 0:11                   | 1:11                  |  |
| AZ011:1               | 1:/             | M2:A/       | AZ011:1               | 1:A          | 1:5          | 1:5                     | 1:11                    | 1:11                   | 1:11                  |  |
| AZ211:1               | 1:/             | M2:A/       | AZ211:1               | 1:/          | 3:X          | 3:X                     | 1:11                    | 1:11                   | 1:11                  |  |
| AZ/11:1               | 1:/             | M2:A/       | AZ/11:1               | 1:M          | A:2          | A:2                     | 1:11                    | 1:11                   | 1:11                  |  |
| AZX11:1               | 1:/             | M2:A/       | AZX11:1               | 1:Q          | 0:0          | 0:0                     | 1:11                    | 1:11                   | 1:11                  |  |
| AZM11:1               | 1:/             | M2:A/       | AZM11:1               | 3:A          | 2:3          | 2:3                     | 1:11                    | 1:11                   | 1:11                  |  |
| AZ511:1               | 1:/             | M2:A/       | AZ511:1               | 3:2          | /:1          | /:1                     | 1:11                    | 1:11                   | 1:11                  |  |
| AZQ11:1               | 1:/             | M2:A/       | AZQ11:1               | 3:X          | /:5          | /:5                     | 1:11                    | 1:11                   | 1:11                  |  |
| OZ111:1               | 1:/             | M2:A/       | OZ111:1               | 3:Q          | X:X          | X:X                     | 1:11                    | 1:11                   | 1:11                  |  |
| OZ311:1               | 1:/             | M2:A/       | OZ311:1               | A:3          | M:/          | M:/                     | 1:11                    | 1:11                   | 1:11                  |  |
| OZA11:1               | 1:/             | M2:A/       | OZA11:1               | A:0          | 5:0          | 5:0                     | 1:11                    | 1:11                   | 1:11                  |  |
| OZ011:1               | 1:/             | M2:A/       | OZ011:1               | A:X          | Q:A          | Q:A                     | 1:11                    | 1:11                   | 1:11                  |  |
| OZ211:1               | 1:/             | M2:A/       | OZ211:1               | A:5          | 31:1         | 31:1                    | 1:11                    | 1:11                   | 1:11                  |  |
| OZ/11:1               | 1:/             | M2:A/       | OZ/11:1               | 0:3          | 31:5         | 31:5                    | 1:11                    | 1:11                   | 1:11                  |  |
| OZX11:1               | 1:/             | M2:A/       | OZ/QQ:Q               | 0:0          | 33:M         | 33:M                    | 1:11                    | 1:11                   | 1:11                  |  |
| OZM11:1               | 1:/             | M2:A/       | OZXQQ:Q               | 0:/          | 3A:/         | 3A:/                    | 1:11                    | 1:11                   | 1:11                  |  |
| OZ511:1               | 1:/             | M2:A/       | OZMQQ:Q               | 0:5          | 30:2         | 30:2                    | 1:11                    | 1:11                   | 1:11                  |  |
| OZQ11:1               | 1:/             | M2:A/       | OZ5QQ:Q               | 2:1          | 32:A         | 32:A                    | 1:11                    | 1:11                   | 1:11                  |  |
| 2Z111:1               | 1:/             | M2:A/       | OZQQQ:Q               | 2:A          | 3:/          | 3:/                     | 1:11                    | 1:11                   | 1:11                  |  |
| 2Z311:1               | 1:/             | M2:A/       | 2Z1QQ:Q               | 2:/          | 3:/Q         | 3:/Q                    | 1:11                    | 1:11                   | 1:11                  |  |
| 2ZA11:1               | 1:/             | M2:A/       | 2Z3QQ:Q               | 2:M          | 3X:M         | 3X:M                    | 1:11                    | 1:11                   | 1:11                  |  |
| 2Z011:1               | 1:/             | M2:A/       | 2ZAQQ:Q               | /:1          | 3M:X         | 3M:X                    | 1:11                    | 1:11                   | 1:11                  |  |
| 2Z211:1               | 1:/             | M2:A/       | 2Z0QQ:Q               | /:A          | 35:2         | 35:2                    | 1:11                    | 1:11                   | 1:11                  |  |
| 2Z/11:1               | 1:/             | M2:A/       | 2Z2QQ:Q               | /:2          | 3Q:A         | 3Q:A                    | 1:11                    | 1:11                   | 1:11                  |  |
| 2ZX11:1               | 1:/             | M2:A/       | 2Z/QQ:Q               | /:M          | A1:3         | A1:3                    | 1:11                    | 1:11                   | 1:11                  |  |
| 2ZX2X:3               | 1:/             | M2:A/       | 2ZX2X:1               | /:5          | A1:/         | A1:/                    | 1:11                    | 1:11                   | 1:11                  |  |
| Greenhorn             |                 |             |                       |              |              |                         |                         |                        |                       |  |
| 2ZM11:1               | 1:/             | M2:A/       | 2ZXQQ:Q               | /:Q          | A1:Q         | A1:Q                    | 1:11                    | 1:11                   | 1:11                  |  |
| 2Z511:1               | 1:/             | M2:A/       | 2ZMQQ:Q               | X:3          | A3:5         | A3:5                    | 1:11                    | 1:11                   | 1:11                  |  |
| 2ZQ11:1               | 1:/             | M2:A/       | 2Z5QQ:Q               | X:2          | AA:X         | AA:X                    | 1:11                    | 1:11                   | 1:11                  |  |
| /Z111:1               | 1:/             | M2:A/       | 2ZQQQ:Q               | X:X          | A0:2         | A0:2                    | 1:11                    | 1:11                   | 1:11                  |  |
| /Z1/M:3               | 1:/             | M2:A/       | /Z1/M:1               | X:M          | A0:Q         | A0:Q                    | 1:11                    | 1:11                   | 1:11                  |  |
| Mowry                 |                 |             |                       |              |              |                         |                         |                        |                       |  |
| /Z311:1               | 1:/             | M2:A/       | /Z1QQ:Q               | X:5          | A2:0         | A2:0                    | 1:11                    | 1:11                   | 1:11                  |  |
| /ZA11:1               | 1:/             | M2:A/       | /Z3QQ:Q               | M:3          | A:/3         | A:/3                    | 1:11                    | 1:11                   | 1:11                  |  |
| /Z011:1               | 1:/             | M2:A/       | /ZAQQ:Q               | M:0          | AX:1         | AX:1                    | 1:11                    | 1:11                   | 1:11                  |  |
| /Z211:1               | 1:/             | M2:A/       | /Z0QQ:Q               | M:X          | AX:5         | AX:5                    | 1:11                    | 1:11                   | 1:11                  |  |
| /Z2M3:3               | 1:/             | M2:A/       | /Z2M3:1               | M:M          | AM:2         | AM:2                    | 1:11                    | 1:11                   | 1:11                  |  |
| Dakota                |                 |             |                       |              |              |                         |                         |                        |                       |  |
| /Z/11:1               | 1:/             | M2:A/       | /Z2QQ:Q               | M:5          | AM:X         | AM:X                    | 1:11                    | 1:11                   | 1:11                  |  |
| /ZX11:1               | 1:/             | M2:A/       | /Z/QQ:Q               | 5:1          | A5:/         | A5:/                    | 1:11                    | 1:11                   | 1:11                  |  |
| /ZM11:1               | 1:/             | M2:A/       | /ZXQQ:Q               | 5:0          | AQ:0         | AQ:0                    | 1:11                    | 1:11                   | 1:11                  |  |
| /Z511:1               | 1:/             | M2:A/       | /ZMQQ:Q               | 5:/          | 01:A         | 01:A                    | 1:11                    | 1:11                   | 1:11                  |  |
| /ZQ11:1               | 1:/             | M2:A/       | /Z5QQ:Q               | 5:M          | 03:1         | 03:1                    | 1:11                    | 1:11                   | 1:11                  |  |
| XZ111:1               | 1:/             | M2:A/       | /ZQQQ:Q               | Q:1          | 03:5         | 03:5                    | 1:11                    | 1:11                   | 1:11                  |  |
| XZ311:1               | 1:/             | M2:A/       | XZ1QQ:Q               | Q:A          | 0A:M         | 0A:M                    | 1:11                    | 1:11                   | 1:11                  |  |
| XZA11:1               | 1:/             | M2:A/       | XZ3QQ:5               | Q:/          | 00:/         | 00:/                    | 1:11                    | 1:11                   | 1:11                  |  |
| XZ011:1               | 1:/             | M2:A/       | XZAQQ:5               | Q:M          | 02:2         | 02:2                    | 1:11                    | 1:11                   | 1:11                  |  |
| XZ211:1               | 1:/             | M2:A/       | XZ0QQ:5               | Q:Q          | 0:/A         | 0:/A                    | 1:11                    | 1:11                   | 1:11                  |  |
| XZ25X:A               | 1:/             | M2:A/       | XZ25X:1               | 31:3         | 0:/Q         | 0:/Q                    | 1:11                    | 1:11                   | 1:11                  |  |
| Rierdon               |                 |             |                       |              |              |                         |                         |                        |                       |  |
| XZ/11:1               | 1:/             | M2:A/       | XZ2QQ:5               | 31:A         | 0X:1         | 0X:1                    | 1:11                    | 1:11                   | 1:11                  |  |

|                  |                                 |                                     |                                     |
|------------------|---------------------------------|-------------------------------------|-------------------------------------|
| <b>Database:</b> | 789\$/111:3\$;!#Ä&\$<=&)\$8>    | <b>Local Co-ordinate Reference:</b> | + &ÄÄ\$,Ä"!&\$-&.)YÄ\$23435\$316    |
| <b>Company:</b>  | E ="=\$Ä&*)(Ä&BH                | <b>TVD Reference:</b>               | +7?? \$@A15A:1B=C*\$DE)"#189\$7Ä&FG |
| <b>Project:</b>  | !!." !J"ÄÄ=                     | <b>MD Reference:</b>                | +7?? \$@A15A:1B=C*\$DE)"#189\$7Ä&FG |
| <b>Site:</b>     | 3/OL4311+43MN35                 | <b>North Reference:</b>             | K)B&                                |
| <b>Well:</b>     | ,Ä"!&\$-&.) Ä\$/011\$23435\$316 | <b>Survey Calculation Method:</b>   | 9"!HBH\$OB)F *B)&                   |
| <b>Wellbore:</b> | ,Ä"!&\$-&.) Ä\$/011\$23435\$316 |                                     |                                     |
| <b>Design:</b>   | 8&="#!\$P3                      |                                     |                                     |

| Planned Survey                     |                    |                |                             |                 |                 |                               |                               |                              |                             |  |
|------------------------------------|--------------------|----------------|-----------------------------|-----------------|-----------------|-------------------------------|-------------------------------|------------------------------|-----------------------------|--|
| Measured<br>Depth<br>(usft)        | Inclination<br>(°) | Azimuth<br>(°) | Vertical<br>Depth<br>(usft) | +N/-S<br>(usft) | +E/-W<br>(usft) | Vertical<br>Section<br>(usft) | Dogleg<br>Rate<br>(°/100usft) | Build<br>Rate<br>(°/100usft) | Turn<br>Rate<br>(°/100usft) |  |
| XZX11:1                            | 1:1                | M2:A/          | XZ/QQ:5                     | 31:2            | 0X:Q            | 0X:Q                          | 1:11                          | 1:11                         | 1:11                        |  |
| XZM11:1                            | 1:1                | M2:A/          | XZXQQ:5                     | 31:X            | 0M:M            | 0M:M                          | 1:11                          | 1:11                         | 1:11                        |  |
| XZ511:1                            | 1:1                | M2:A/          | XZMQQ:5                     | 31:Q            | 05:X            | 05:X                          | 1:11                          | 1:11                         | 1:11                        |  |
| XZ532:A                            | 1:1                | M2:A/          | XZ532:1                     | 31:Q            | 05:M            | 05:M                          | 1:11                          | 1:11                         | 1:11                        |  |
| Dunham Salt                        |                    |                |                             |                 |                 |                               |                               |                              |                             |  |
| XZQ11:1                            | 1:1                | M2:A/          | XZ5QQ:5                     | 33:3            | 0Q:2            | 0Q:2                          | 1:11                          | 1:11                         | 1:11                        |  |
| XZQAQ:A                            | 1:1                | M2:A/          | XZQAQ:1                     | 33:A            | 0Q:X            | 0Q:X                          | 1:11                          | 1:11                         | 1:11                        |  |
| Dunham Salt Base                   |                    |                |                             |                 |                 |                               |                               |                              |                             |  |
| MZ111:1                            | 1:1                | M2:A/          | XZQQQ:5                     | 33:2            | 21:A            | 21:A                          | 1:11                          | 1:11                         | 1:11                        |  |
| MZ1A2:A                            | 1:1                | M2:A/          | MZ1A2:1                     | 33:2            | 21:2            | 21:2                          | 1:11                          | 1:11                         | 1:11                        |  |
| Spearfish                          |                    |                |                             |                 |                 |                               |                               |                              |                             |  |
| MZ311:1                            | 1:1                | M2:A/          | MZ1QQ:5                     | 33:X            | 23:3            | 23:3                          | 1:11                          | 1:11                         | 1:11                        |  |
| MZA11:1                            | 1:1                | M2:A/          | MZ3QQ:5                     | 33:5            | 23:Q            | 23:Q                          | 1:11                          | 1:11                         | 1:11                        |  |
| MZA50:A                            | 1:1                | M2:A/          | MZA50:1                     | 3A:1            | 2A:X            | 2A:X                          | 1:11                          | 1:11                         | 1:11                        |  |
| Pine Salt                          |                    |                |                             |                 |                 |                               |                               |                              |                             |  |
| MZ011:1                            | 1:1                | M2:A/          | MZAQQ:5                     | 3A:3            | 2A:5            | 2A:5                          | 1:11                          | 1:11                         | 1:11                        |  |
| MZ035:A                            | 1:1                | M2:A/          | MZ035:1                     | 3A:3            | 2A:Q            | 2A:Q                          | 1:11                          | 1:11                         | 1:11                        |  |
| Pine Salt Base                     |                    |                |                             |                 |                 |                               |                               |                              |                             |  |
| MZ0M2:A                            | 1:1                | M2:A/          | MZ0M2:1                     | 3A:A            | 20:2            | 20:2                          | 1:11                          | 1:11                         | 1:11                        |  |
| Opeche Salt                        |                    |                |                             |                 |                 |                               |                               |                              |                             |  |
| MZ211:1                            | 1:1                | M2:A/          | MZ0QQ:5                     | 3A:0            | 20:X            | 20:X                          | 1:11                          | 1:11                         | 1:11                        |  |
| MZ210:A                            | 1:1                | M2:A/          | MZ210:1                     | 3A:0            | 20:X            | 20:X                          | 1:11                          | 1:11                         | 1:11                        |  |
| Opeche Salt Base                   |                    |                |                             |                 |                 |                               |                               |                              |                             |  |
| MZ/11:1                            | 1:1                | M2:A/          | MZ2QQ:5                     | 3A:/            | 22:2            | 22:2                          | 1:11                          | 1:11                         | 1:11                        |  |
| MZX11:1                            | 1:1                | M2:A/          | MZ/QQ:5                     | 3A:5            | 2/:0            | 2/:0                          | 1:11                          | 1:11                         | 1:11                        |  |
| MZX1/:A                            | 1:1                | M2:A/          | MZX1/:1                     | 3A:5            | 2/:0            | 2/:0                          | 1:11                          | 1:11                         | 1:11                        |  |
| Broom Creek (Top of Minnelusa Gp.) |                    |                |                             |                 |                 |                               |                               |                              |                             |  |
| MZX52:A                            | 1:1                | M2:A/          | MZX52:1                     | 30:1            | 2X:1            | 2X:1                          | 1:11                          | 1:11                         | 1:11                        |  |
| Amsden                             |                    |                |                             |                 |                 |                               |                               |                              |                             |  |
| MZM11:1                            | 1:1                | M2:A/          | MZXQQ:5                     | 30:1            | 2X:3            | 2X:3                          | 1:11                          | 1:11                         | 1:11                        |  |
| MZ511:1                            | 1:1                | M2:A/          | MZMQQ:5                     | 30:A            | 2M:1            | 2M:1                          | 1:11                          | 1:11                         | 1:11                        |  |
| MZ5/0:A                            | 1:1                | M2:A/          | MZ5/0:1                     | 30:2            | 2M:2            | 2M:2                          | 1:11                          | 1:11                         | 1:11                        |  |
| Tyler                              |                    |                |                             |                 |                 |                               |                               |                              |                             |  |
| MZQ11:1                            | 1:1                | M2:A/          | MZ5QQ:5                     | 30:/            | 2M:5            | 2M:5                          | 1:11                          | 1:11                         | 1:11                        |  |
| MZQAX:M                            | 1:1                | M2:A/          | MZQAX:2                     | 30:/            | 25:1            | 25:1                          | 1:11                          | 1:11                         | 1:11                        |  |
| MZQ20:0                            | 1:11               | 1:11           | MZQ20:3                     | 30:X            | 25:3            | 25:3                          | 0:11                          | 40:11                        | 1:11                        |  |
| 5Z111:1                            | 1:11               | 1:11           | MZQQQ:5                     | 30:X            | 25:3            | 25:3                          | 1:11                          | 1:11                         | 1:11                        |  |
| 5Z12M:A                            | 1:11               | 1:11           | 5Z12M:1                     | 30:X            | 25:3            | 25:3                          | 1:11                          | 1:11                         | 1:11                        |  |
| Otter (Base of Minnelusa Gp.)      |                    |                |                             |                 |                 |                               |                               |                              |                             |  |
| 5Z311:1                            | 1:11               | 1:11           | 5Z1QQ:5                     | 30:X            | 25:3            | 25:3                          | 1:11                          | 1:11                         | 1:11                        |  |
| 5ZA11:1                            | 1:11               | 1:11           | 5Z3QQ:5                     | 30:X            | 25:3            | 25:3                          | 1:11                          | 1:11                         | 1:11                        |  |
| 5Z011:1                            | 1:11               | 1:11           | 5ZAQQ:5                     | 30:X            | 25:3            | 25:3                          | 1:11                          | 1:11                         | 1:11                        |  |
| 5Z0Q1:A                            | 1:11               | 1:11           | 5Z0Q1:1                     | 30:X            | 25:3            | 25:3                          | 1:11                          | 1:11                         | 1:11                        |  |
| Kibbey Lime                        |                    |                |                             |                 |                 |                               |                               |                              |                             |  |
| 5Z211:1                            | 1:11               | 1:11           | 5Z0QQ:5                     | 30:X            | 25:3            | 25:3                          | 1:11                          | 1:11                         | 1:11                        |  |
| 5Z/11:1                            | 1:11               | 1:11           | 5Z2QQ:5                     | 30:X            | 25:3            | 25:3                          | 1:11                          | 1:11                         | 1:11                        |  |
| 5Z/2A:A                            | 1:11               | 1:11           | 5Z/2A:1                     | 30:X            | 25:3            | 25:3                          | 1:11                          | 1:11                         | 1:11                        |  |
| Charles Salt                       |                    |                |                             |                 |                 |                               |                               |                              |                             |  |
| 5ZX11:1                            | 1:11               | 1:11           | 5Z/QQ:5                     | 30:X            | 25:3            | 25:3                          | 1:11                          | 1:11                         | 1:11                        |  |
| 5ZM11:1                            | 1:11               | 1:11           | 5ZXQQ:5                     | 30:X            | 25:3            | 25:3                          | 1:11                          | 1:11                         | 1:11                        |  |
| 5Z511:1                            | 1:11               | 1:11           | 5ZMQQ:5                     | 30:X            | 25:3            | 25:3                          | 1:11                          | 1:11                         | 1:11                        |  |
| 5ZQ11:1                            | 1:11               | 1:11           | 5Z5QQ:5                     | 30:X            | 25:3            | 25:3                          | 1:11                          | 1:11                         | 1:11                        |  |

|                  |                                  |                                     |                                      |
|------------------|----------------------------------|-------------------------------------|--------------------------------------|
| <b>Database:</b> | 789\$/111:3\$;"!#Ä&\$<=&)\$8>    | <b>Local Co-ordinate Reference:</b> | + &ÄÄ\$,Ä"!&\$-&.&YÄ\$23435\$316     |
| <b>Company:</b>  | E ="=\$Ä&*)(Ä&BH                 | <b>TVD Reference:</b>               | +7?? \$@A15A:1B=C*\$DE)"#1879\$7Ä&FG |
| <b>Project:</b>  | !!." !J"ÄÄ=                      | <b>MD Reference:</b>                | +7?? \$@A15A:1B=C*\$DE)"#1879\$7Ä&FG |
| <b>Site:</b>     | 3/OL4311+43MN35                  | <b>North Reference:</b>             | K)B&                                 |
| <b>Well:</b>     | ,Ä"!&\$-&.&) Ä\$/011\$23435\$316 | <b>Survey Calculation Method:</b>   | 9"!HBH\$OB)F *B)&                    |
| <b>Wellbore:</b> | ,Ä"!&\$-&.&) Ä\$/011\$23435\$316 |                                     |                                      |
| <b>Design:</b>   | 8&="#!\$P3                       |                                     |                                      |

| Planned Survey                 |                 |             |                       |              |              |                         |                         |                        |                       |  |
|--------------------------------|-----------------|-------------|-----------------------|--------------|--------------|-------------------------|-------------------------|------------------------|-----------------------|--|
| Measured Depth (usft)          | Inclination (°) | Azimuth (°) | Vertical Depth (usft) | +N/-S (usft) | +E/-W (usft) | Vertical Section (usft) | Dogleg Rate (°/100usft) | Build Rate (°/100usft) | Turn Rate (°/100usft) |  |
| QZ111:1                        | 1:11            | 1:11        | 5ZQQQ:5               | 30:X         | 25:3         | 25:3                    | 1:11                    | 1:11                   | 1:11                  |  |
| QZ311:1                        | 1:11            | 1:11        | QZ1QQ:5               | 30:X         | 25:3         | 25:3                    | 1:11                    | 1:11                   | 1:11                  |  |
| QZ3/Q:A                        | 1:11            | 1:11        | QZ3/Q:1               | 30:X         | 25:3         | 25:3                    | 1:11                    | 1:11                   | 1:11                  |  |
| <b>UB</b>                      |                 |             |                       |              |              |                         |                         |                        |                       |  |
| QZA11:1                        | 1:11            | 1:11        | QZ3QQ:5               | 30:X         | 25:3         | 25:3                    | 1:11                    | 1:11                   | 1:11                  |  |
| QZA05:A                        | 1:11            | 1:11        | QZA05:1               | 30:X         | 25:3         | 25:3                    | 1:11                    | 1:11                   | 1:11                  |  |
| <b>Base Last Salt</b>          |                 |             |                       |              |              |                         |                         |                        |                       |  |
| QZ011:1                        | 1:11            | 1:11        | QZAQQ:5               | 30:X         | 25:3         | 25:3                    | 1:11                    | 1:11                   | 1:11                  |  |
| QZ013:A                        | 1:11            | 1:11        | QZ013:1               | 30:X         | 25:3         | 25:3                    | 1:11                    | 1:11                   | 1:11                  |  |
| <b>Ratcliffe</b>               |                 |             |                       |              |              |                         |                         |                        |                       |  |
| QZ211:1                        | 1:11            | 1:11        | QZ0QQ:5               | 30:X         | 25:3         | 25:3                    | 1:11                    | 1:11                   | 1:11                  |  |
| QZ2/2:A                        | 1:11            | 1:11        | QZ2/2:1               | 30:X         | 25:3         | 25:3                    | 1:11                    | 1:11                   | 1:11                  |  |
| <b>Mission Canyon</b>          |                 |             |                       |              |              |                         |                         |                        |                       |  |
| QZ/11:1                        | 1:11            | 1:11        | QZ2QQ:5               | 30:X         | 25:3         | 25:3                    | 1:11                    | 1:11                   | 1:11                  |  |
| QZX11:1                        | 1:11            | 1:11        | QZ/QQ:5               | 30:X         | 25:3         | 25:3                    | 1:11                    | 1:11                   | 1:11                  |  |
| QZM11:1                        | 1:11            | 1:11        | QZXQQ:5               | 30:X         | 25:3         | 25:3                    | 1:11                    | 1:11                   | 1:11                  |  |
| QZ511:1                        | 1:11            | 1:11        | QZMQQ:5               | 30:X         | 25:3         | 25:3                    | 1:11                    | 1:11                   | 1:11                  |  |
| QZQ11:1                        | 1:11            | 1:11        | QZ5QQ:5               | 30:X         | 25:3         | 25:3                    | 1:11                    | 1:11                   | 1:11                  |  |
| 31Z111:1                       | 1:11            | 1:11        | QZQQQ:5               | 30:X         | 25:3         | 25:3                    | 1:11                    | 1:11                   | 1:11                  |  |
| 31Z1AA:A                       | 1:11            | 1:11        | 31Z1AA:1              | 30:X         | 25:3         | 25:3                    | 1:11                    | 1:11                   | 1:11                  |  |
| <b>Lodgepole</b>               |                 |             |                       |              |              |                         |                         |                        |                       |  |
| 31Z311:1                       | 1:11            | 1:11        | 31Z1QQ:5              | 30:X         | 25:3         | 25:3                    | 1:11                    | 1:11                   | 1:11                  |  |
| 31ZA11:1                       | 1:11            | 1:11        | 31Z3QQ:5              | 30:X         | 25:3         | 25:3                    | 1:11                    | 1:11                   | 1:11                  |  |
| 31ZA1Q:A                       | 1:11            | 1:11        | 31ZA1Q:1              | 30:X         | 25:3         | 25:3                    | 1:11                    | 1:11                   | 1:11                  |  |
| <b>Lodgepole Fracture Zone</b> |                 |             |                       |              |              |                         |                         |                        |                       |  |
| 31ZA5M:5                       | 1:11            | 1:11        | 31ZA5M:X              | 30:X         | 25:3         | 25:3                    | 1:11                    | 1:11                   | 1:11                  |  |
| <b>KOP Build 12°/100'</b>      |                 |             |                       |              |              |                         |                         |                        |                       |  |
| 31Z011:1                       | 3:2X            | M2:A/       | 31ZAQQ:5              | 30:X         | 25:A         | 25:A                    | 3A:11                   | 3A:11                  | 1:11                  |  |
| 31Z0A:1                        | 2:2X            | M2:A/       | 31Z0A2:M              | 32:1         | 2Q:/         | 2Q:/                    | 3A:11                   | 3A:11                  | 1:11                  |  |
| 31Z0/1:1                       | M:2X            | M2:A/       | 31Z02Q:X              | 32:M         | /A:1         | /A:1                    | 3A:11                   | 3A:11                  | 1:11                  |  |
| 31Z0M/:1                       | 31:2X           | M2:A/       | 31Z0M2:0              | 3/:M         | //:M         | //:M                    | 3A:11                   | 3A:11                  | 1:11                  |  |
| 31Z211:1                       | 30:2X           | M2:A/       | 31Z0Q5:5              | 3M:3         | X1:M         | X1:M                    | 3A:11                   | 3A:11                  | 1:11                  |  |
| 31Z2A/:1                       | 3X:2X           | M2:A/       | 31Z2AA:Q              | 35:Q         | XX:Q         | XX:Q                    | 3A:11                   | 3A:11                  | 1:11                  |  |
| 31Z2/1:1                       | 3Q:2X           | M2:A/       | 31Z22X:M              | A3:1         | M2:2         | M2:2                    | 3A:11                   | 3A:11                  | 1:11                  |  |
| 31Z2M/:1                       | AA:2X           | M2:A/       | 31Z2M1:1              | A0:2         | 50:1         | 50:1                    | 3A:11                   | 3A:11                  | 1:11                  |  |
| 31Z/11:1                       | A/:2X           | M2:A/       | 31Z2QA:Q              | AX:A         | QA:M         | QA:M                    | 3A:11                   | 3A:11                  | 1:11                  |  |
| 31Z/A/:1                       | A5:2X           | M2:A/       | 31Z/3/:3              | AQ:A         | 310:X        | 310:X                   | 3A:11                   | 3A:11                  | 1:11                  |  |
| 31Z//1:1                       | 03:2X           | M2:A/       | 31Z/0X:5              | 0A:X         | 33/:M        | 33/:M                   | 3A:11                   | 3A:11                  | 1:11                  |  |
| 31Z/M/:1                       | 02:2X           | M2:A/       | 31Z//M:5              | 0X:0         | 3A5:5        | 3A5:5                   | 3A:11                   | 3A:11                  | 1:11                  |  |
| 31ZX11:1                       | 0M:2X           | M2:A/       | 31Z/M5:1              | 21:0         | 32A:Q        | 32A:Q                   | 3A:11                   | 3A:11                  | 1:11                  |  |
| 31ZXA/:1                       | 21:2X           | M2:A/       | 31Z/QM:2              | 22:X         | 3/5:1        | 3/5:1                   | 3A:11                   | 3A:11                  | 1:11                  |  |
| 31ZX/1:1                       | 20:2X           | M2:A/       | 31ZX3X:1              | 2Q:3         | 3M2:3        | 3M2:3                   | 3A:11                   | 3A:11                  | 1:11                  |  |
| 31ZXM/:1                       | 2X:2X           | M2:A/       | 31ZX00:M              | /0:Q         | 3Q3:3        | 3Q3:3                   | 3A:11                   | 3A:11                  | 1:11                  |  |
| 31ZM11:1                       | 2Q:2X           | M2:A/       | 31ZX/1:/              | /5:Q         | A1Q:1        | A1Q:1                   | 3A:11                   | 3A:11                  | 1:11                  |  |
| 31ZMA/:1                       | /A:2X           | M2:A/       | 31ZXXX:A              | X2:A         | AAM:M        | AAM:M                   | 3A:11                   | 3A:11                  | 1:11                  |  |
| 31ZM/1:1                       | //:2X           | M2:A/       | 31ZX51:Q              | XQ:M         | A2M:3        | A2M:3                   | 3A:11                   | 3A:11                  | 1:11                  |  |
| 31ZMM/:1                       | /5:2X           | M2:A/       | 31ZXQ2:/              | M/:2         | AXM:0        | AXM:0                   | 3A:11                   | 3A:11                  | 1:11                  |  |
| 31Z511:1                       | X3:2X           | M2:A/       | 31ZM1M:1              | 53:0         | A55:3        | A55:3                   | 3A:11                   | 3A:11                  | 1:11                  |  |
| 31Z5A/:1                       | X2:2X           | M2:A/       | 31ZM35:2              | 5M:0         | 01Q:/        | 01Q:/                   | 3A:11                   | 3A:11                  | 1:11                  |  |
| 31Z520:2                       | XX:XM           | M2:A/       | 31ZMAX:1              | Q3:5         | 0A/:X        | 0A/:X                   | 3A:11                   | 3A:11                  | 1:11                  |  |
| <b>False Bakken</b>            |                 |             |                       |              |              |                         |                         |                        |                       |  |
| 31Z5/1:1                       | XM:2X           | M2:A/       | 31ZMA5:X              | Q0:/         | 003:/        | 003:/                   | 3A:11                   | 3A:11                  | 1:11                  |  |
| 31Z5XM:/                       | XQ:/M           | M2:A/       | 31ZM0/:1              | QM:Q         | 02M:A        | 02M:A                   | 3A:11                   | 3A:11                  | 1:11                  |  |

|           |                                  |                              |                                      |
|-----------|----------------------------------|------------------------------|--------------------------------------|
| Database: | 789\$/111.3\$;"!#Ä&\$<=&)\$8>    | Local Co-ordinate Reference: | + &ÄÄ\$,Ä"!&\$-&.&YÄ\$23435\$316     |
| Company:  | E ="=\$Ä&*)(Ä&BH                 | TVD Reference:               | +7?? \$@A15A:1B=C*\$SDE)"#189\$7Ä&FG |
| Project:  | !!." !J"ÄÄ=                      | MD Reference:                | +7?? \$@A15A:1B=C*\$SDE)"#189\$7Ä&FG |
| Site:     | 3/OL4311+43MN35                  | North Reference:             | K)B&                                 |
| Well:     | ,Ä"!&\$-&.&) Ä\$/011\$23435\$316 | Survey Calculation Method:   | 9"!HBH\$OB)F *B)&                    |
| Wellbore: | ,Ä"!&\$-&.&) Ä\$/011\$23435\$316 |                              |                                      |
| Design:   | 8&="#!\$P3                       |                              |                                      |

| Planned Survey                 |                    |                |                             |                 |                 |                               |                               |                              |                             |  |
|--------------------------------|--------------------|----------------|-----------------------------|-----------------|-----------------|-------------------------------|-------------------------------|------------------------------|-----------------------------|--|
| Measured<br>Depth<br>(usft)    | Inclination<br>(°) | Azimuth<br>(°) | Vertical<br>Depth<br>(usft) | +N/-S<br>(usft) | +E/-W<br>(usft) | Vertical<br>Section<br>(usft) | Dogleg<br>Rate<br>(°/100usft) | Build<br>Rate<br>(°/100usft) | Turn<br>Rate<br>(°/100usft) |  |
| Upper Bakken                   |                    |                |                             |                 |                 |                               |                               |                              |                             |  |
| 31Z5M/:1                       | M1:2X              | M2:A/          | 31ZM0M:X                    | QQ:5            | 0/2:1           | 0/2:1                         | 3A:11                         | 3A:11                        | 1:11                        |  |
| 31ZQ11:1                       | M0:2X              | M2:A/          | 31ZM2/:0                    | 31X:0           | 0MX:5           | 0MX:5                         | 3A:11                         | 3A:11                        | 1:11                        |  |
| 31ZQA3:M                       | MX:1M              | M2:A/          | 31ZM/3:1                    | 33A:1           | 0QM:1           | 0QM:1                         | 3A:11                         | 3A:11                        | 1:11                        |  |
| Middle Bakken                  |                    |                |                             |                 |                 |                               |                               |                              |                             |  |
| 31ZQA/:1                       | MX:2X              | M2:A/          | 31ZM/3:5                    | 33A:5           | 211:3           | 211:3                         | 3A:11                         | 3A:11                        | 1:11                        |  |
| 31ZQ/1:1                       | MQ:2X              | M2:A/          | 31ZM/M:1                    | 33Q:/           | 2A0:X           | 2A0:X                         | 3A:11                         | 3A:11                        | 1:11                        |  |
| 31ZQX5:0                       | 53:XX              | M2:A/          | 31ZMX1:1                    | 3A2:2           | 223:1           | 223:1                         | 3A:11                         | 3A:11                        | 1:11                        |  |
| Middle Bakken Sand Target      |                    |                |                             |                 |                 |                               |                               |                              |                             |  |
| 31ZQM/:1                       | 5A:2X              | M2:A/          | 31ZMX1:Q                    | 3AX:A           | 22M:2           | 22M:2                         | 3A:11                         | 3A:11                        | 1:11                        |  |
| 33Z111:1                       | 5/:2X              | M2:A/          | 31ZMX0:X                    | 30A:Q           | 2M3:0           | 2M3:0                         | 3A:11                         | 3A:11                        | 1:11                        |  |
| 33Z1A/:1                       | 55:2X              | M2:A/          | 31ZMX2:Q                    | 30Q:M           | 2Q/:0           | 2Q/:0                         | 3A:11                         | 3A:11                        | 1:11                        |  |
| 33Z10M:1                       | 5Q:Q1              | M2:A/          | 31ZMX/:1                    | 320:1           | /1X:Q           | /1X:Q                         | 33:QM                         | 33:QM                        | 1:11                        |  |
| EOC - 7"                       |                    |                |                             |                 |                 |                               |                               |                              |                             |  |
| 33Z311:1                       | 5Q:Q1              | M/:/3          | 31ZMX/:A                    | 3/Q:2           | /XM:M           | /XM:M                         | A:11                          | 1:11                         | A:11                        |  |
| 33ZA11:1                       | 5Q:Q1              | MM:/3          | 31ZMX/:0                    | 35A:M           | XX2:Q           | XX2:Q                         | A:11                          | 1:11                         | A:11                        |  |
| 33Z011:1                       | 5Q:Q1              | MQ:/3          | 31ZMX/:/                    | A1A:X           | MXA:Q           | MXA:Q                         | A:11                          | 1:11                         | A:11                        |  |
| 33Z211:1                       | 5Q:Q1              | 53:/3          | 31ZMX/:M                    | A3Q:3           | 5X3:/           | 5X3:/                         | A:11                          | 1:11                         | A:11                        |  |
| 33Z/11:1                       | 5Q:Q1              | 50:/3          | 31ZMX/:Q                    | A0A:3           | QX1:M           | QX1:M                         | A:11                          | 1:11                         | A:11                        |  |
| 33ZX11:1                       | 5Q:Q1              | 5/:/3          | 31ZMXX:1                    | A23:M           | 3Z1X1:A         | 3Z1X1:A                       | A:11                          | 1:11                         | A:11                        |  |
| 33ZM11:1                       | 5Q:Q1              | 5M:/3          | 31ZMXXX:A                   | A2M:5           | 3Z3X1:1         | 3Z3X1:1                       | A:11                          | 1:11                         | A:11                        |  |
| 33Z511:1                       | 5Q:Q1              | 5Q:/3          | 31ZMXX:2                    | A/1:2           | 3ZAX1:1         | 3ZAX1:1                       | A:11                          | 1:11                         | A:11                        |  |
| 33Z5A2:X                       | 5Q:Q1              | Q1:11          | 31ZMXX:2                    | A/1:/           | 3ZA52:X         | 3ZA52:X                       | A:11                          | 1:11                         | A:11                        |  |
| 33ZQ11:1                       | 5Q:Q1              | Q1:11          | 31ZMXX:X                    | A/1:/           | 3Z0X1:1         | 3Z0X1:1                       | 1:11                          | 1:11                         | 1:11                        |  |
| 3AZ111:1                       | 5Q:Q1              | Q1:11          | 31ZMXX:M                    | A/1:/           | 3Z2X1:1         | 3Z2X1:1                       | 1:11                          | 1:11                         | 1:11                        |  |
| 3AZ311:1                       | 5Q:Q1              | Q1:11          | 31ZMXX:Q                    | A/1:/           | 3Z/X1:1         | 3Z/X1:1                       | 1:11                          | 1:11                         | 1:11                        |  |
| 3AZA11:1                       | 5Q:Q1              | Q1:11          | 31ZMXM:3                    | A/1:/           | 3ZXX1:1         | 3ZXX1:1                       | 1:11                          | 1:11                         | 1:11                        |  |
| 3AZ011:1                       | 5Q:Q1              | Q1:11          | 31ZMXM:0                    | A/1:/           | 3ZMX1:1         | 3ZMX1:1                       | 1:11                          | 1:11                         | 1:11                        |  |
| 3AZ211:1                       | 5Q:Q1              | Q1:11          | 31ZMXM:2                    | A/1:/           | 3Z5X1:1         | 3Z5X1:1                       | 1:11                          | 1:11                         | 1:11                        |  |
| 3AZ/11:1                       | 5Q:Q1              | Q1:11          | 31ZMXM:X                    | A/1:/           | 3ZQX1:1         | 3ZQX1:1                       | 1:11                          | 1:11                         | 1:11                        |  |
| 3AZX11:1                       | 5Q:Q1              | Q1:11          | 31ZMXM:5                    | A/1:/           | AZ1X1:1         | AZ1X1:1                       | 1:11                          | 1:11                         | 1:11                        |  |
| 3AZM11:1                       | 5Q:Q1              | Q1:11          | 31ZMX5:1                    | A/1:2           | AZ3X1:1         | AZ3X1:1                       | 1:11                          | 1:11                         | 1:11                        |  |
| 3AZ511:1                       | 5Q:Q1              | Q1:11          | 31ZMX5:3                    | A/1:2           | AZAX1:1         | AZAX1:1                       | 1:11                          | 1:11                         | 1:11                        |  |
| 3AZQ11:1                       | 5Q:Q1              | Q1:11          | 31ZMX5:0                    | A/1:2           | AZ0X1:1         | AZ0X1:1                       | 1:11                          | 1:11                         | 1:11                        |  |
| 30Z111:1                       | 5Q:Q1              | Q1:11          | 31ZMX5:/                    | A/1:2           | AZ2X1:1         | AZ2X1:1                       | 1:11                          | 1:11                         | 1:11                        |  |
| 30Z311:1                       | 5Q:Q1              | Q1:11          | 31ZMX5:M                    | A/1:2           | AZ/X1:1         | AZ/X1:1                       | 1:11                          | 1:11                         | 1:11                        |  |
| 30ZA11:1                       | 5Q:Q1              | Q1:11          | 31ZMX5:5                    | A/1:2           | AZXX1:1         | AZXX1:1                       | 1:11                          | 1:11                         | 1:11                        |  |
| 30Z011:1                       | 5Q:Q1              | Q1:11          | 31ZMXQ:1                    | A/1:2           | AZMX1:1         | AZMX1:1                       | 1:11                          | 1:11                         | 1:11                        |  |
| 30Z211:1                       | 5Q:Q1              | Q1:11          | 31ZMXQ:A                    | A/1:2           | AZ5X1:1         | AZ5X1:1                       | 1:11                          | 1:11                         | 1:11                        |  |
| 30Z/11:1                       | 5Q:Q1              | Q1:11          | 31ZMXQ:0                    | A/1:2           | AZQX1:1         | AZQX1:1                       | 1:11                          | 1:11                         | 1:11                        |  |
| 30ZX11:1                       | 5Q:Q1              | Q1:11          | 31ZMXQ:/                    | A/1:2           | 0Z1X1:1         | 0Z1X1:1                       | 1:11                          | 1:11                         | 1:11                        |  |
| 30ZM11:1                       | 5Q:Q1              | Q1:11          | 31ZMXQ:M                    | A/1:2           | 0Z3X1:1         | 0Z3X1:1                       | 1:11                          | 1:11                         | 1:11                        |  |
| 30Z511:1                       | 5Q:Q1              | Q1:11          | 31ZMXQ:Q                    | A/1:2           | 0ZAX1:1         | 0ZAX1:1                       | 1:11                          | 1:11                         | 1:11                        |  |
| 30Z5M2:X                       | 5Q:Q1              | Q1:11          | 31ZMM1:1                    | A/1:2           | 0Z002:X         | 0Z002:X                       | 1:11                          | 1:11                         | 1:11                        |  |
| Base Middle Bakken Sand Target |                    |                |                             |                 |                 |                               |                               |                              |                             |  |
| 30ZQ11:1                       | 5Q:Q1              | Q1:11          | 31ZMM1:1                    | A/1:2           | 0Z0X1:1         | 0Z0X1:1                       | 1:11                          | 1:11                         | 1:11                        |  |
| 32Z111:1                       | 5Q:Q1              | Q1:11          | 31ZMM1:A                    | A/1:2           | 0Z2X1:1         | 0Z2X1:1                       | 1:11                          | 1:11                         | 1:11                        |  |
| 32Z311:1                       | 5Q:Q1              | Q1:11          | 31ZMM1:2                    | A/1:2           | 0Z/X1:1         | 0Z/X1:1                       | 1:11                          | 1:11                         | 1:11                        |  |
| 32ZA11:1                       | 5Q:Q1              | Q1:11          | 31ZMM1:X                    | A/1:2           | 0ZXX1:1         | 0ZXX1:1                       | 1:11                          | 1:11                         | 1:11                        |  |
| 32Z011:1                       | 5Q:Q1              | Q1:11          | 31ZMM1:M                    | A/1:2           | 0ZMX1:1         | 0ZMX1:1                       | 1:11                          | 1:11                         | 1:11                        |  |
| 32Z211:1                       | 5Q:Q1              | Q1:11          | 31ZMM1:Q                    | A/1:2           | 0Z5X1:1         | 0Z5X1:1                       | 1:11                          | 1:11                         | 1:11                        |  |
| 32Z/11:1                       | 5Q:Q1              | Q1:11          | 31ZMM3:3                    | A/1:0           | 0ZQX1:1         | 0ZQX1:1                       | 1:11                          | 1:11                         | 1:11                        |  |
| 32ZX11:1                       | 5Q:Q1              | Q1:11          | 31ZMM3:0                    | A/1:0           | 2Z1X1:1         | 2Z1X1:1                       | 1:11                          | 1:11                         | 1:11                        |  |
| 32ZM11:1                       | 5Q:Q1              | Q1:11          | 31ZMM3:2                    | A/1:0           | 2Z3X1:1         | 2Z3X1:1                       | 1:11                          | 1:11                         | 1:11                        |  |

|                  |                                  |                                     |                                      |
|------------------|----------------------------------|-------------------------------------|--------------------------------------|
| <b>Database:</b> | 789\$/111:3\$;"#Ä&\$<=&)\$8>     | <b>Local Co-ordinate Reference:</b> | + &ÄÄ\$.Ä"!&\$-&.&YÄ\$23435\$316     |
| <b>Company:</b>  | E ="=\$Ä&*)(Ä&BH                 | <b>TVD Reference:</b>               | +7?? \$@A15A:1B=C*\$SDE)"#187\$7Ä&FG |
| <b>Project:</b>  | !!."!\$J"ÄÄ=                     | <b>MD Reference:</b>                | +7?? \$@A15A:1B=C*\$SDE)"#187\$7Ä&FG |
| <b>Site:</b>     | 3/OL4311+43MN35                  | <b>North Reference:</b>             | K)B&                                 |
| <b>Well:</b>     | ,Ä"!&\$-&.&) Ä\$/011\$23435\$316 | <b>Survey Calculation Method:</b>   | 9"!HBH\$OB)F *B)&                    |
| <b>Wellbore:</b> | ,Ä"!&\$-&.&) Ä\$/011\$23435\$316 |                                     |                                      |
| <b>Design:</b>   | 8&="#!\$P3                       |                                     |                                      |

| Planned Survey        |                 |             |                       |              |              |                         |                         |                        |                       |
|-----------------------|-----------------|-------------|-----------------------|--------------|--------------|-------------------------|-------------------------|------------------------|-----------------------|
| Measured Depth (usft) | Inclination (°) | Azimuth (°) | Vertical Depth (usft) | +N/-S (usft) | +E/-W (usft) | Vertical Section (usft) | Dogleg Rate (°/100usft) | Build Rate (°/100usft) | Turn Rate (°/100usft) |
| 32Z511:1              | 5Q:Q1           | Q1:11       | 31ZMM3:X              | A/1:0        | 2ZAX1:1      | 2ZAX1:1                 | 1:11                    | 1:11                   | 1:11                  |
| 32ZQ11:1              | 5Q:Q1           | Q1:11       | 31ZMM3:5              | A/1:0        | 2Z0X1:1      | 2Z0X1:1                 | 1:11                    | 1:11                   | 1:11                  |
| 3/Z111:1              | 5Q:Q1           | Q1:11       | 31ZMMA:1              | A/1:0        | 2Z2X1:1      | 2Z2X1:1                 | 1:11                    | 1:11                   | 1:11                  |
| 3/Z311:1              | 5Q:Q1           | Q1:11       | 31ZMMA:3              | A/1:0        | 2Z/X1:1      | 2Z/X1:1                 | 1:11                    | 1:11                   | 1:11                  |
| 3/ZA11:1              | 5Q:Q1           | Q1:11       | 31ZMMA:0              | A/1:0        | 2ZXX1:1      | 2ZXX1:1                 | 1:11                    | 1:11                   | 1:11                  |
| 3/Z011:1              | 5Q:Q1           | Q1:11       | 31ZMMA:/              | A/1:0        | 2ZMX1:1      | 2ZMX1:1                 | 1:11                    | 1:11                   | 1:11                  |
| 3/Z211:1              | 5Q:Q1           | Q1:11       | 31ZMMA:M              | A/1:0        | 2Z5X1:1      | 2Z5X1:1                 | 1:11                    | 1:11                   | 1:11                  |
| 3/Z/11:1              | 5Q:Q1           | Q1:11       | 31ZMMA:5              | A/1:0        | 2ZQX1:1      | 2ZQX1:1                 | 1:11                    | 1:11                   | 1:11                  |
| 3/ZX11:1              | 5Q:Q1           | Q1:11       | 31ZMM0:1              | A/1:0        | /Z1X1:1      | /Z1X1:1                 | 1:11                    | 1:11                   | 1:11                  |
| 3/ZM11:1              | 5Q:Q1           | Q1:11       | 31ZMM0:A              | A/1:0        | /Z3X1:1      | /Z3X1:1                 | 1:11                    | 1:11                   | 1:11                  |
| 3/Z511:1              | 5Q:Q1           | Q1:11       | 31ZMM0:2              | A/1:0        | /ZAX1:1      | /ZAX1:1                 | 1:11                    | 1:11                   | 1:11                  |
| 3/ZQ11:1              | 5Q:Q1           | Q1:11       | 31ZMM0:/              | A/1:0        | /Z0X1:1      | /Z0X1:1                 | 1:11                    | 1:11                   | 1:11                  |
| 3XZ111:1              | 5Q:Q1           | Q1:11       | 31ZMM0:M              | A/1:0        | /Z2X1:1      | /Z2X1:1                 | 1:11                    | 1:11                   | 1:11                  |
| 3XZ311:1              | 5Q:Q1           | Q1:11       | 31ZMM0:Q              | A/1:0        | /Z/X1:1      | /Z/X1:1                 | 1:11                    | 1:11                   | 1:11                  |
| 3XZA11:1              | 5Q:Q1           | Q1:11       | 31ZMM2:3              | A/1:0        | /ZXX1:1      | /ZXX1:1                 | 1:11                    | 1:11                   | 1:11                  |
| 3XZ011:1              | 5Q:Q1           | Q1:11       | 31ZMM2:A              | A/1:A        | /ZMX1:1      | /ZMX1:1                 | 1:11                    | 1:11                   | 1:11                  |
| 3XZ211:1              | 5Q:Q1           | Q1:11       | 31ZMM2:2              | A/1:A        | /Z5X1:1      | /Z5X1:1                 | 1:11                    | 1:11                   | 1:11                  |
| 3XZ/11:1              | 5Q:Q1           | Q1:11       | 31ZMM2:X              | A/1:A        | /ZQX1:1      | /ZQX1:1                 | 1:11                    | 1:11                   | 1:11                  |
| 3XZX11:1              | 5Q:Q1           | Q1:11       | 31ZMM2:M              | A/1:A        | XZ1X1:1      | XZ1X1:1                 | 1:11                    | 1:11                   | 1:11                  |
| 3XZM11:1              | 5Q:Q1           | Q1:11       | 31ZMM2:Q              | A/1:A        | XZ3X1:1      | XZ3X1:1                 | 1:11                    | 1:11                   | 1:11                  |
| 3XZ511:1              | 5Q:Q1           | Q1:11       | 31ZMM:/3              | A/1:A        | XZAX1:1      | XZAX1:1                 | 1:11                    | 1:11                   | 1:11                  |
| 3XZQ11:1              | 5Q:Q1           | Q1:11       | 31ZMM:/0              | A/1:A        | XZ0X1:1      | XZ0X1:1                 | 1:11                    | 1:11                   | 1:11                  |
| 3MZ111:1              | 5Q:Q1           | Q1:11       | 31ZMM:/2              | A/1:A        | XZ2X1:1      | XZ2X1:1                 | 1:11                    | 1:11                   | 1:11                  |
| 3MZ311:1              | 5Q:Q1           | Q1:11       | 31ZMM:/X              | A/1:A        | XZ/X1:1      | XZ/X1:1                 | 1:11                    | 1:11                   | 1:11                  |
| 3MZA11:1              | 5Q:Q1           | Q1:11       | 31ZMM:/5              | A/1:A        | XZXX1:1      | XZXX1:1                 | 1:11                    | 1:11                   | 1:11                  |
| 3MZ011:1              | 5Q:Q1           | Q1:11       | 31ZMMX:1              | A/1:A        | XZMX1:1      | XZMX1:1                 | 1:11                    | 1:11                   | 1:11                  |
| 3MZ211:1              | 5Q:Q1           | Q1:11       | 31ZMMX:3              | A/1:A        | XZ5X1:1      | XZ5X1:1                 | 1:11                    | 1:11                   | 1:11                  |
| 3MZ/11:1              | 5Q:Q1           | Q1:11       | 31ZMMX:0              | A/1:A        | XZQX1:1      | XZQX1:1                 | 1:11                    | 1:11                   | 1:11                  |
| 3MZX11:1              | 5Q:Q1           | Q1:11       | 31ZMMX:/              | A/1:A        | MZ1X1:1      | MZ1X1:1                 | 1:11                    | 1:11                   | 1:11                  |
| 3MZM11:1              | 5Q:Q1           | Q1:11       | 31ZMMX:M              | A/1:A        | MZ3X1:1      | MZ3X1:1                 | 1:11                    | 1:11                   | 1:11                  |
| 3MZ511:1              | 5Q:Q1           | Q1:11       | 31ZMMX:5              | A/1:A        | MZAX1:1      | MZAX1:1                 | 1:11                    | 1:11                   | 1:11                  |
| 3MZQ11:1              | 5Q:Q1           | Q1:11       | 31ZMMM:1              | A/1:A        | MZ0X1:1      | MZ0X1:1                 | 1:11                    | 1:11                   | 1:11                  |
| 35Z111:1              | 5Q:Q1           | Q1:11       | 31ZMMM:A              | A/1:3        | MZ2X1:1      | MZ2X1:1                 | 1:11                    | 1:11                   | 1:11                  |
| 35Z311:1              | 5Q:Q1           | Q1:11       | 31ZMMM:2              | A/1:3        | MZ/X1:1      | MZ/X1:1                 | 1:11                    | 1:11                   | 1:11                  |
| 35ZA11:1              | 5Q:Q1           | Q1:11       | 31ZMMM:/              | A/1:3        | MZXX1:1      | MZXX1:1                 | 1:11                    | 1:11                   | 1:11                  |
| 35Z011:1              | 5Q:Q1           | Q1:11       | 31ZMMM:M              | A/1:3        | MZMX1:1      | MZMX1:1                 | 1:11                    | 1:11                   | 1:11                  |
| 35Z211:1              | 5Q:Q1           | Q1:11       | 31ZMMM:Q              | A/1:3        | MZ5X1:1      | MZ5X1:1                 | 1:11                    | 1:11                   | 1:11                  |
| 35Z/11:1              | 5Q:Q1           | Q1:11       | 31ZMM5:3              | A/1:3        | MZQX1:1      | MZQX1:1                 | 1:11                    | 1:11                   | 1:11                  |
| 35ZX11:1              | 5Q:Q1           | Q1:11       | 31ZMM5:A              | A/1:3        | 5Z1X1:1      | 5Z1X1:1                 | 1:11                    | 1:11                   | 1:11                  |
| 35ZM11:1              | 5Q:Q1           | Q1:11       | 31ZMM5:2              | A/1:3        | 5Z3X1:1      | 5Z3X1:1                 | 1:11                    | 1:11                   | 1:11                  |
| 35Z511:1              | 5Q:Q1           | Q1:11       | 31ZMM5:X              | A/1:3        | 5ZAX1:1      | 5ZAX1:1                 | 1:11                    | 1:11                   | 1:11                  |
| 35ZQ11:1              | 5Q:Q1           | Q1:11       | 31ZMM5:5              | A/1:3        | 5Z0X1:1      | 5Z0X1:1                 | 1:11                    | 1:11                   | 1:11                  |
| 3QZ111:1              | 5Q:Q1           | Q1:11       | 31ZMM5:Q              | A/1:3        | 5Z2X1:1      | 5Z2X1:1                 | 1:11                    | 1:11                   | 1:11                  |
| 3QZ311:1              | 5Q:Q1           | Q1:11       | 31ZMMQ:3              | A/1:3        | 5Z/X1:1      | 5Z/X1:1                 | 1:11                    | 1:11                   | 1:11                  |
| 3QZA11:1              | 5Q:Q1           | Q1:11       | 31ZMMQ:0              | A/1:3        | 5ZXX1:1      | 5ZXX1:1                 | 1:11                    | 1:11                   | 1:11                  |
| 3QZ011:1              | 5Q:Q1           | Q1:11       | 31ZMMQ:/              | A/1:3        | 5ZMX1:1      | 5ZMX1:1                 | 1:11                    | 1:11                   | 1:11                  |
| 3QZ211:1              | 5Q:Q1           | Q1:11       | 31ZMMQ:X              | A/1:3        | 5Z5X1:1      | 5Z5X1:1                 | 1:11                    | 1:11                   | 1:11                  |
| 3QZ/11:1              | 5Q:Q1           | Q1:11       | 31ZMMQ:5              | A/1:3        | 5ZQX1:1      | 5ZQX1:1                 | 1:11                    | 1:11                   | 1:11                  |
| 3QZX11:1              | 5Q:Q1           | Q1:11       | 31ZM51:1              | A/1:3        | QZ1X1:1      | QZ1X1:1                 | 1:11                    | 1:11                   | 1:11                  |
| 3QZM11:1              | 5Q:Q1           | Q1:11       | 31ZM51:3              | A/1:3        | QZ3X1:1      | QZ3X1:1                 | 1:11                    | 1:11                   | 1:11                  |
| 3QZ511:1              | 5Q:Q1           | Q1:11       | 31ZM51:0              | A/1:1        | QZAX1:1      | QZAX1:1                 | 1:11                    | 1:11                   | 1:11                  |
| 3QZQ11:1              | 5Q:Q1           | Q1:11       | 31ZM51:/              | A/1:1        | QZ0X1:1      | QZ0X1:1                 | 1:11                    | 1:11                   | 1:11                  |
| A1Z111:1              | 5Q:Q1           | Q1:11       | 31ZM51:M              | A/1:1        | QZ2X1:1      | QZ2X1:1                 | 1:11                    | 1:11                   | 1:11                  |
| A1Z311:1              | 5Q:Q1           | Q1:11       | 31ZM51:5              | A/1:1        | QZ/X1:1      | QZ/X1:1                 | 1:11                    | 1:11                   | 1:11                  |



|           |                                 |                              |                                      |
|-----------|---------------------------------|------------------------------|--------------------------------------|
| Database: | 789\$/111:3\$;"!#Ä&\$<=&)\$8>   | Local Co-ordinate Reference: | + &ÄÄ\$,Ä"!&\$-&.)YÄ\$23435\$316     |
| Company:  | E="\$Ä&*)(Ä&BH                  | TVD Reference:               | +7?? @\$A15A:1B=C*\$DE)"#1879\$7Ä&FG |
| Project:  | !!."!\$J"ÄÄ=                    | MD Reference:                | +7?? @\$A15A:1B=C*\$DE)"#1879\$7Ä&FG |
| Site:     | 3/OL4311+43MN35                 | North Reference:             | K)B&                                 |
| Well:     | ,Ä"!&\$-&.) Ä\$/011\$23435\$316 | Survey Calculation Method:   | 9"!HBH\$OB)F *B)&                    |
| Wellbore: | ,Ä"!&\$-&.) Ä\$/011\$23435\$316 |                              |                                      |
| Design:   | 8&="#!\$P3                      |                              |                                      |

| Planned Survey        |                 |             |                       |              |              |                         |                         |                        |                       |  |
|-----------------------|-----------------|-------------|-----------------------|--------------|--------------|-------------------------|-------------------------|------------------------|-----------------------|--|
| Measured Depth (usft) | Inclination (°) | Azimuth (°) | Vertical Depth (usft) | +N/-S (usft) | +E/-W (usft) | Vertical Section (usft) | Dogleg Rate (°/100usft) | Build Rate (°/100usft) | Turn Rate (°/100usft) |  |
| A1ZA11:1              | 5Q:Q1           | Q1:11       | 31ZM53:1              | A/1:1        | QZXX1:1      | QZXX1:1                 | 1:11                    | 1:11                   | 1:11                  |  |
| A1Z011:1              | 5Q:Q1           | Q1:11       | 31ZM53:A              | A/1:1        | QZMX1:1      | QZMX1:1                 | 1:11                    | 1:11                   | 1:11                  |  |
| A1Z211:1              | 5Q:Q1           | Q1:11       | 31ZM53:2              | A/1:1        | QZ5X1:1      | QZ5X1:1                 | 1:11                    | 1:11                   | 1:11                  |  |
| A1Z/11:1              | 5Q:Q1           | Q1:11       | 31ZM53:/              | A/1:1        | QZQX1:1      | QZQX1:1                 | 1:11                    | 1:11                   | 1:11                  |  |
| A1Z/Q1:1              | 5Q:Q1           | Q1:11       | 31ZM53:M              | A/1:1        | 31Z1/1:1     | 31Z1/1:1                | 1:11                    | 1:11                   | 1:11                  |  |
| 5300 41-18 10B PBHL   |                 |             |                       |              |              |                         |                         |                        |                       |  |

| Design Targets                                                    |               |              |            |              |              |                 |                |                                         |  |  |
|-------------------------------------------------------------------|---------------|--------------|------------|--------------|--------------|-----------------|----------------|-----------------------------------------|--|--|
| Target Name                                                       |               |              |            |              |              |                 |                |                                         |  |  |
| - hit/miss target                                                 | Dip Angle (°) | Dip Dir. (°) | TVD (usft) | +N/-S (usft) | +E/-W (usft) | Northing (usft) | Easting (usft) | Latitude                                |  |  |
| - Shape                                                           |               |              |            |              |              |                 |                | Longitude                               |  |  |
| /011\$23435\$316\$Ä6J?<br>4\$Ä!\$R"*=\$#&*\$T&!*\$&)<br>4\$Ä("!*" | 1:11          | 1:11         | 31ZM53:M   | A/1:1        | 31Z1/1:1     | 21/Z12M:AQ      | 3ZAA1Z1AA:M5   | 25W\$2[\$31:X23\$L 310W\$00[\$20:QXM\$+ |  |  |

| Casing Points         |                       |      |  |                     |                   |
|-----------------------|-----------------------|------|--|---------------------|-------------------|
| Measured Depth (usft) | Vertical Depth (usft) | Name |  | Casing Diameter (") | Hole Diameter (") |
| 33Z10M:1              | 31ZMX/:1              | MY   |  | M                   | 5012              |

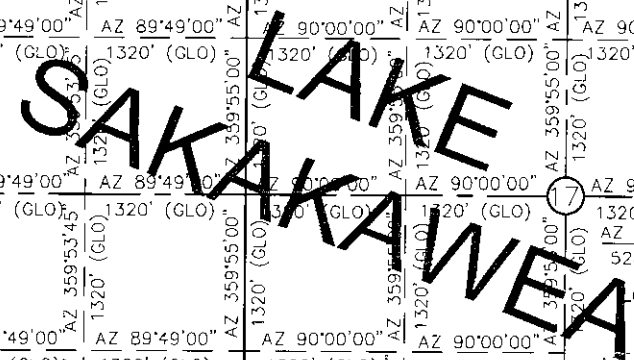


|                  |                                  |                                     |                                     |
|------------------|----------------------------------|-------------------------------------|-------------------------------------|
| <b>Database:</b> | 789\$/111:3\$;!#Ä&\$<=&)\$8>     | <b>Local Co-ordinate Reference:</b> | + &ÄÄ\$,Ä"!&\$-&.&YÄ\$23435\$316    |
| <b>Company:</b>  | E ="=\$Ä&*)(Ä&BH                 | <b>TVD Reference:</b>               | +7?? @\$A15A:1B=C*\$DE)"#187\$7Ä&FG |
| <b>Project:</b>  | !!." !\$J"ÄÄ=                    | <b>MD Reference:</b>                | +7?? @\$A15A:1B=C*\$DE)"#187\$7Ä&FG |
| <b>Site:</b>     | 3/OL4311+43MN35                  | <b>North Reference:</b>             | K)B&                                |
| <b>Well:</b>     | ,Ä"!&\$-&.&) Ä\$/011\$23435\$316 | <b>Survey Calculation Method:</b>   | 9"!HBH\$OB)F *B)&                   |
| <b>Wellbore:</b> | ,Ä"!&\$-&.&) Ä\$/011\$23435\$316 |                                     |                                     |
| <b>Design:</b>   | 8&="#!\$P3                       |                                     |                                     |

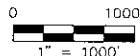
| Formations            |                                             |      |           |         |                   |  |
|-----------------------|---------------------------------------------|------|-----------|---------|-------------------|--|
| Measured Depth (usft) | Vertical Depth (usft)                       | Name | Lithology | Dip (°) | Dip Direction (°) |  |
| 3ZQMA:1               | 3ZQMA:1 Ä"&)&                               |      |           |         |                   |  |
| 2ZX2X:3               | 2ZX2X:1 \)&!R(!                             |      |           |         |                   |  |
| /Z1/M:3               | /Z1/M:1 9(^)_                               |      |           |         |                   |  |
| /Z2M3:3               | /Z2M3:1 8 U(*                               |      |           |         |                   |  |
| XZ25X:A               | XZ25X:1 %"&).(!                             |      |           |         |                   |  |
| XZ532:A               | XZ532:1 8B!R H\$; Ä*                        |      |           |         |                   |  |
| XZQAQ:A               | XZQAQ:18B!R H\$; Ä*\$6 =&                   |      |           |         |                   |  |
| MZ1A2:A               | MZ1A2:1 ;'& )C"=R                           |      |           |         |                   |  |
| MZA50:A               | MZA50:1 Ä"!&\$; Ä*                          |      |           |         |                   |  |
| MZ035:A               | MZ035:1 Ä"!&\$; Ä*\$6 =&                    |      |           |         |                   |  |
| MZ0M2:A               | MZ0M2:1 E'&TR&\$; Ä*                        |      |           |         |                   |  |
| MZ210:A               | MZ210:1 E'&TR&\$; Ä*\$6 =&                  |      |           |         |                   |  |
| MZX1/:A               | MZX1/:1 6)(H\$O)&&U\$D!\$(C\$9"!&ÄB= \$!':G |      |           |         |                   |  |
| MZX52:A               | MZX52:1 SH=.&!                              |      |           |         |                   |  |
| MZ5/0:A               | MZ5/0:1 K_Ä&)                               |      |           |         |                   |  |
| 5Z12M:A               | 5Z12M:1 E**&)\$D6 =&\$(C\$9"!&ÄB= \$!':G    |      |           |         |                   |  |
| 5Z0Q1:A               | 5Z0Q1:1 ,">&_?"H&                           |      |           |         |                   |  |
| 5Z/2A:A               | 5Z/2A:1 OR )Ä&=\$; Ä*                       |      |           |         |                   |  |
| QZ3/Q:A               | QZ3/Q:1 <6                                  |      |           |         |                   |  |
| QZA05:A               | QZA05:1 6 =&\$? =*\$; Ä*                    |      |           |         |                   |  |
| QZ013:A               | QZ013:1 % *TÄ&Ä&                            |      |           |         |                   |  |
| QZ2/2:A               | QZ2/2:1 9"=="(!\$O !_(!                     |      |           |         |                   |  |
| 31Z1AA:A              | 31Z1AA:1 ?(.#&'(Ä&                          |      |           |         |                   |  |
| 31ZA1Q:A              | 31ZA1Q:1 ?(.#&'(Ä&\$-) T*B)&\$V(!&          |      |           |         |                   |  |
| 31Z520:2              | 31ZMAX:1 - Ä=&\$6 UU&!                      |      |           |         |                   |  |
| 31Z5XM:/              | 31ZM0/:1 <"&)\$6 UU&!                       |      |           |         |                   |  |
| 31ZQA3:M              | 31ZM/3:1 9"..Ä&\$6 UU&!                     |      |           |         |                   |  |
| 31ZQX5:0              | 31ZMX1:1 9"..Ä&\$6 UU&!\$; !.\$ K)#&*       |      |           |         |                   |  |
| 30Z5M2:X              | 31ZMM1:16 =&\$9"..Ä&\$6 UU&!\$; !.\$ K)#&*  |      |           |         |                   |  |

| Plan Annotations      |                       |                   |              |                       |  |
|-----------------------|-----------------------|-------------------|--------------|-----------------------|--|
| Measured Depth (usft) | Vertical Depth (usft) | Local Coordinates |              |                       |  |
|                       |                       | +N/-S (usft)      | +E/-W (usft) | Comment               |  |
| 31ZA5M:5              | 31ZA5M:X              | 30:X              | 25:3         | ,EÄ \$6B"Ä.\$3AWN311[ |  |
| 33Z10M:1              | 31ZMX/:1              | 320:1             | /1X:Q        | 7EO                   |  |

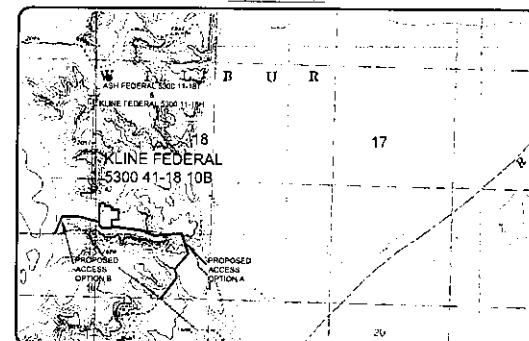
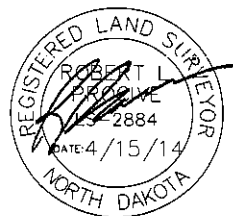
SECTIONS 17 & 18, T153N, R100W, 5th P.M., MCKENZIE COUNTY, NORTH DAKOTA



ALL AZIMUTHS ARE BASED ON G.P.S. OBSERVATIONS. THE ORIGINAL SURVEY OF THIS AREA FOR THE GENERAL LAND OFFICE (G.L.O.) WAS 1891. THE CORNERS FOUND ARE AS INDICATED AND ALL OTHERS ARE COMPUTED FROM THOSE CORNERS FOUND AND BASED ON G.L.O. DATA. THE MAPPING ANGLE FOR THIS AREA IS APPROXIMATELY 0°03'.



-   - MONUMENT - RECOVERED  
  - MONUMENT - NOT RECOVERED



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|                                                                                                                            |                            |
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| OASIS PETROLEUM NORTH AMERICA, LLC<br>SECTION BREAKDOWN<br>SECTIONS 17 & 18, T153N, R100W<br>MCKENZIE COUNTY, NORTH DAKOTA |                            |
| Drawn By: _____                                                                                                            | Project No.: SPC-09-081 D4 |
| Book: _____                                                                                                                | Date: APRIL 2014           |
| Created By: _____                                                                                                          |                            |

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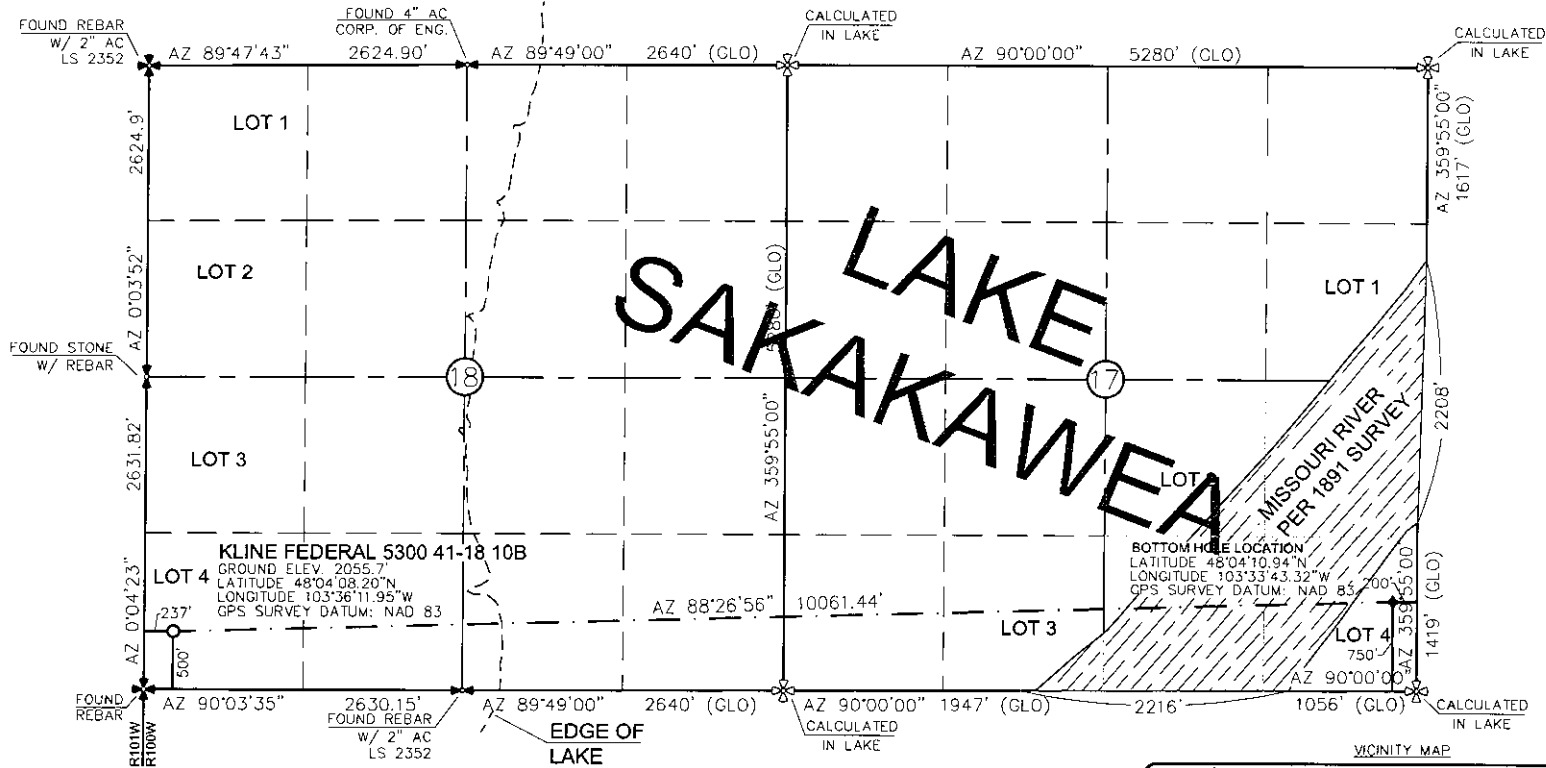


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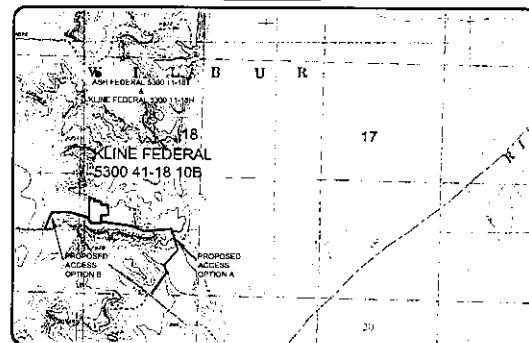
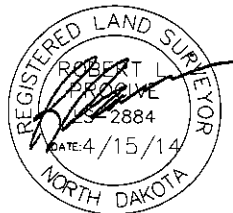
500 FEET FROM SOUTH LINE AND 237 FEET FROM WEST LINE  
SECTION 18, T153N, R100W, 5th P.M., MCKENZIE COUNTY, NORTH DAKOTA



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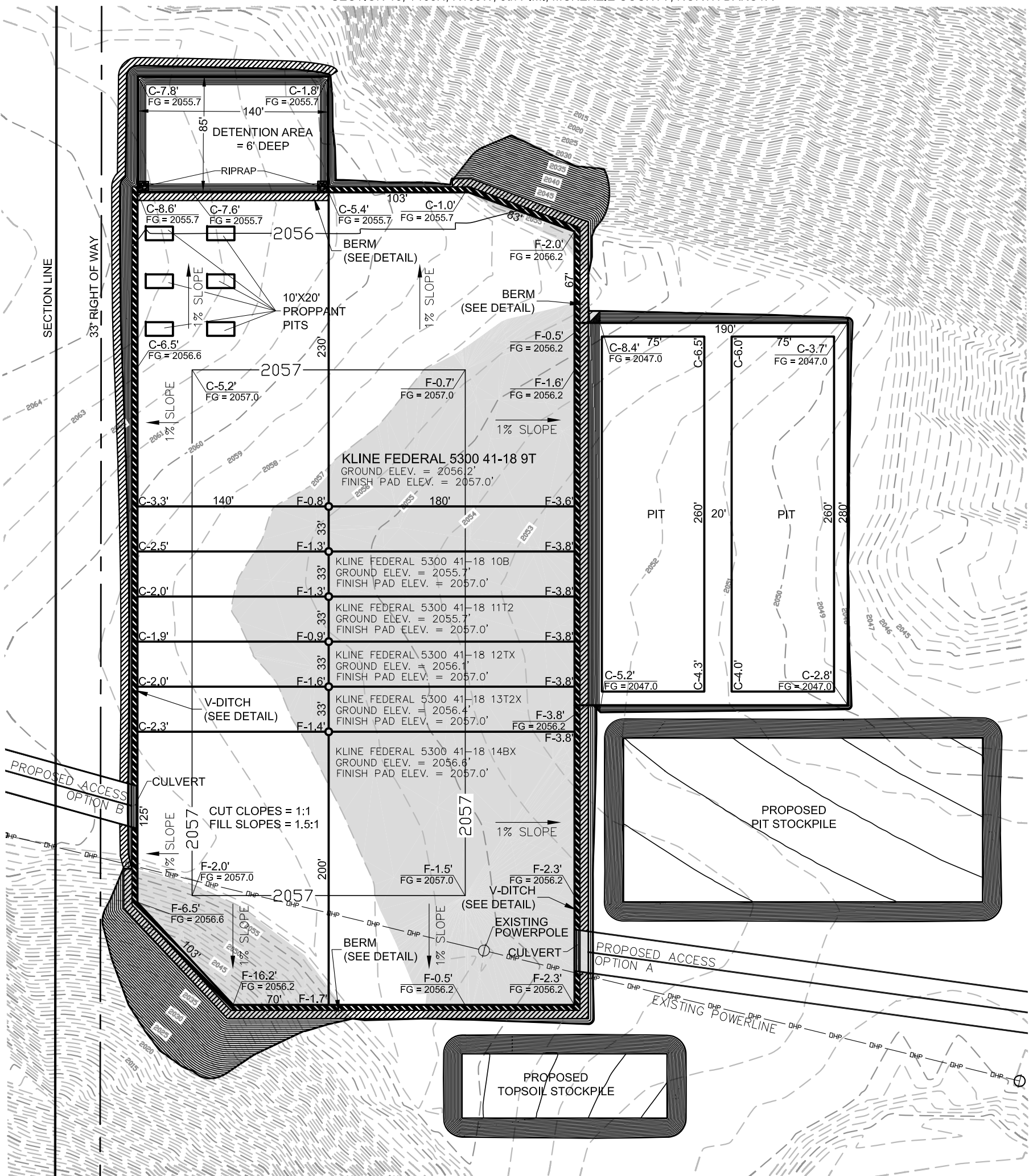
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"KLINE FEDERAL 5300 41-18 9T"  
533 FEET FROM SOUTH LINE AND 237 FEET FROM WEST LINE  
SECTION 18, T153N, R100W, 5th P.M., MCKENZIE COUNTY, NORTH DAKOTA

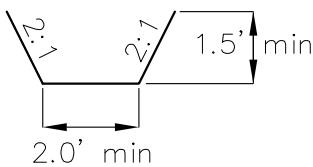


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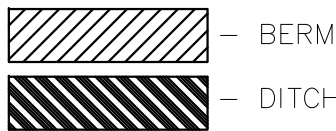
NOTE: Pad dimensions shown are to usable area, the v-ditch and berm areas shall be built to the outside of the pad dimensions.



V-DITCH DETAIL

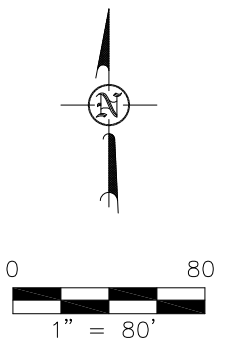


————— Proposed Contours  
----- Original Contours



NOTE: All utilities shown are preliminary only, a complete utilities location is recommended before construction.

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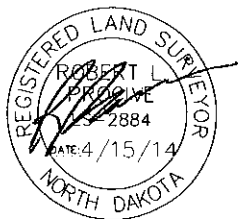
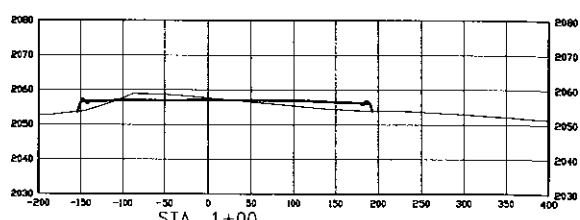
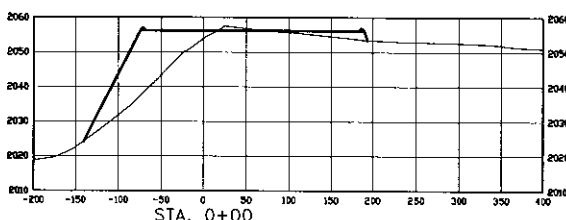
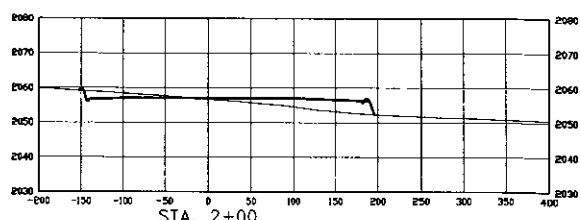
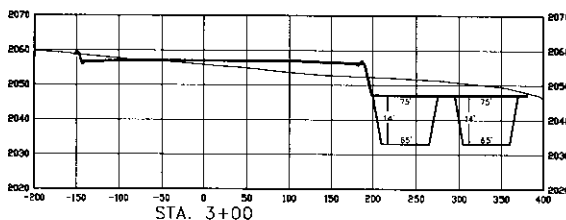
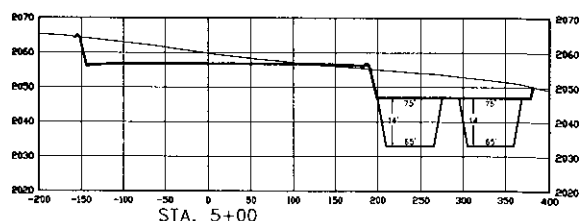
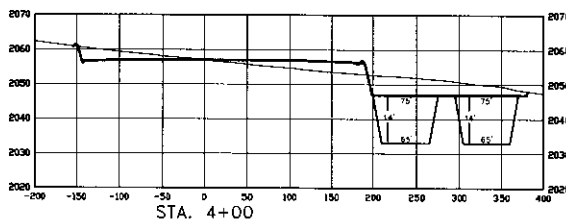
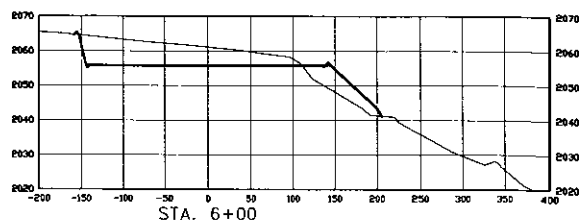
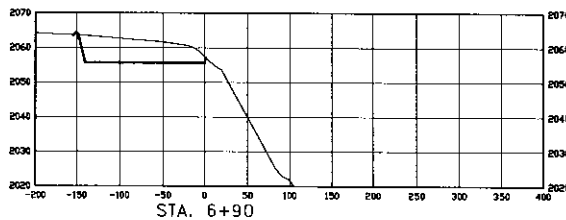
OASIS PETROLEUM NORTH AMERICA, LLC  
PAD LAYOUT  
SECTION 18, T153N, R100W  
MCKENZIE COUNTY, NORTH DAKOTA

Drawn By: B.H.H. Project No.: S14-09-081.05  
Checked By: D.D.K. Date: APRIL 2014

| Revision No. | Date | By | Description |
|--------------|------|----|-------------|
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© 2014 S14-09-081.05 Oasis Petroleum - Kline Federal 5300 41-18 9T  
Location: CAD Kline Federal 5300 41-18 9T.dwg - 4/15/2014 3:59 PM joan schmierer

**CROSS SECTIONS**  
 OASIS PETROLEUM NORTH AMERICA, LLC  
 1001 FANNIN, SUITE 1500, HOUSTON, TX 77002  
 "KLINE FEDERAL 5300 41-18 108"  
 500 FEET FROM SOUTH LINE AND 237 FEET FROM WEST LINE  
 SECTION 18, T153N, R100W, 5th P.M., MCKENZIE COUNTY, NORTH DAKOTA



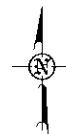
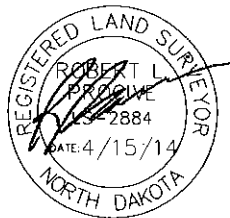
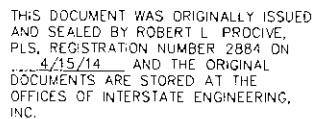
THIS DOCUMENT WAS ORIGINALLY ISSUED  
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SCALE  
 HORIZ 1"=140'  
 VERT 1"=35'

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| <div style="font-size: 2em; font-weight: bold;">7/8</div> <div style="font-size: 0.8em;">SHEET NO.</div>                                                                                                             | <div style="font-weight: bold; font-size: 0.8em;">INTERSTATE<br/>ENGINEERING</div> <div style="font-size: 0.6em;">Professional engineering services</div> | Interstate Engineering, Inc.<br>P.O. Box 648<br>425 East Main Street<br>Sidney, Montana 59270<br>Ph: (406) 433-5617<br>Fax: (406) 433-5618<br>www.interstateeng.com<br><small>Other offices in Minneapolis, North Dakota and South Dakota</small> | OASIS PETROLEUM NORTH AMERICA, LLC<br>CROSS SECTIONS<br>SECTION 18, T153N, R100W<br>MCKENZIE COUNTY, NORTH DAKOTA | <table border="1" style="width: 100%;"> <thead> <tr> <th>Revision No.</th> <th>Date</th> <th>By</th> <th>Description</th> </tr> </thead> <tbody> <tr><td> </td><td> </td><td> </td><td> </td></tr> <tr><td> </td><td> </td><td> </td><td> </td></tr> <tr><td> </td><td> </td><td> </td><td> </td></tr> </tbody> </table> | Revision No.                                                                                                                                                                                                  | Date | By | Description |  |  |  |  |  |  |  |  |  |  |  |  |
|----------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|-----------------------------------------------------------------------------------------------------------------------------------------------------------|---------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|-------------------------------------------------------------------------------------------------------------------|--------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|---------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|------|----|-------------|--|--|--|--|--|--|--|--|--|--|--|--|
|                                                                                                                                                                                                                      |                                                                                                                                                           | Revision No.                                                                                                                                                                                                                                      | Date                                                                                                              | By                                                                                                                                                                                                                                                                                                                       | Description                                                                                                                                                                                                   |      |    |             |  |  |  |  |  |  |  |  |  |  |  |  |
|                                                                                                                                                                                                                      |                                                                                                                                                           |                                                                                                                                                                                                                                                   |                                                                                                                   |                                                                                                                                                                                                                                                                                                                          |                                                                                                                                                                                                               |      |    |             |  |  |  |  |  |  |  |  |  |  |  |  |
|                                                                                                                                                                                                                      |                                                                                                                                                           |                                                                                                                                                                                                                                                   |                                                                                                                   |                                                                                                                                                                                                                                                                                                                          |                                                                                                                                                                                                               |      |    |             |  |  |  |  |  |  |  |  |  |  |  |  |
|                                                                                                                                                                                                                      |                                                                                                                                                           |                                                                                                                                                                                                                                                   |                                                                                                                   |                                                                                                                                                                                                                                                                                                                          |                                                                                                                                                                                                               |      |    |             |  |  |  |  |  |  |  |  |  |  |  |  |
| <table border="1" style="width: 100%;"> <tr> <td>Drawn By: <u>B.H.M.</u></td> <td>Project No.: <u>514-0908104</u></td> </tr> <tr> <td>Checked By: <u>B.L.P.</u></td> <td>Date: <u>APRIL 2014</u></td> </tr> </table> | Drawn By: <u>B.H.M.</u>                                                                                                                                   | Project No.: <u>514-0908104</u>                                                                                                                                                                                                                   | Checked By: <u>B.L.P.</u>                                                                                         | Date: <u>APRIL 2014</u>                                                                                                                                                                                                                                                                                                  | <small>© 2014 INTERSTATE ENGINEERING, INC. All rights reserved. This drawing is the property of Interstate Engineering, Inc. and is not to be reproduced without written permission. 11/15/14 1:10 PM</small> |      |    |             |  |  |  |  |  |  |  |  |  |  |  |  |
| Drawn By: <u>B.H.M.</u>                                                                                                                                                                                              | Project No.: <u>514-0908104</u>                                                                                                                           |                                                                                                                                                                                                                                                   |                                                                                                                   |                                                                                                                                                                                                                                                                                                                          |                                                                                                                                                                                                               |      |    |             |  |  |  |  |  |  |  |  |  |  |  |  |
| Checked By: <u>B.L.P.</u>                                                                                                                                                                                            | Date: <u>APRIL 2014</u>                                                                                                                                   |                                                                                                                                                                                                                                                   |                                                                                                                   |                                                                                                                                                                                                                                                                                                                          |                                                                                                                                                                                                               |      |    |             |  |  |  |  |  |  |  |  |  |  |  |  |

OASIS PETROLEUM NORTH AMERICA, LLC  
1001 FANNIN, SUITE 1500, HOUSTON, TX 77002  
"KLINE FEDERAL 5300 41-18 10B"  
500 FEET FROM SOUTH LINE AND 237 FEET FROM WEST LINE  
SECTION 18, T153N, R100W, 5th P.M., MCKENZIE COUNTY, NORTH DAKOTA



4/8



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OASIS PETROLEUM NORTH AMERICA, LLC  
ACCESS APPROACH  
SECTION 18, T153N, R100W  
MCKENZIE COUNTY, NORTH DAKOTA

|                           |                                   |
|---------------------------|-----------------------------------|
| Drawn By: <u>B.H.J.</u>   | Project No.: <u>S14-05-061.04</u> |
| Checked By: <u>R.L.P.</u> | Date: <u>APRIL 2014</u>           |

[illegible]



OASIS PETROLEUM NORTH AMERICA, LLC  
KLINE FEDERAL 5300 41-18 10B  
500' FSL/237' FWL  
QUAD LOCATION MAP  
SECTION 18, T153N, R100W  
MCKENZIE COUNTY, NORTH DAKOTA

ASH FEDERAL 5300 11-18T  
&  
KLINE FEDERAL 5300 11-18H

18  
KLINE FEDERAL  
5300 41-18 10B

PROPOSED  
ACCESS  
OPTION B

PROPOSED  
ACCESS  
OPTION A

Indian Creek Bay  
Recreation Area

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OASIS PETROLEUM NORTH AMERICA, LLC  
QUAD LOCATION MAP  
SECTION 18, T153N, R100W  
MCKENZIE COUNTY, NORTH DAKOTA

|                           |                                   |
|---------------------------|-----------------------------------|
| Drawn By: <u>B.H.H.</u>   | Project No.: <u>S14-09-081-04</u> |
| Checked By: <u>R.L.P.</u> | Date: <u>APRIL 2014</u>           |

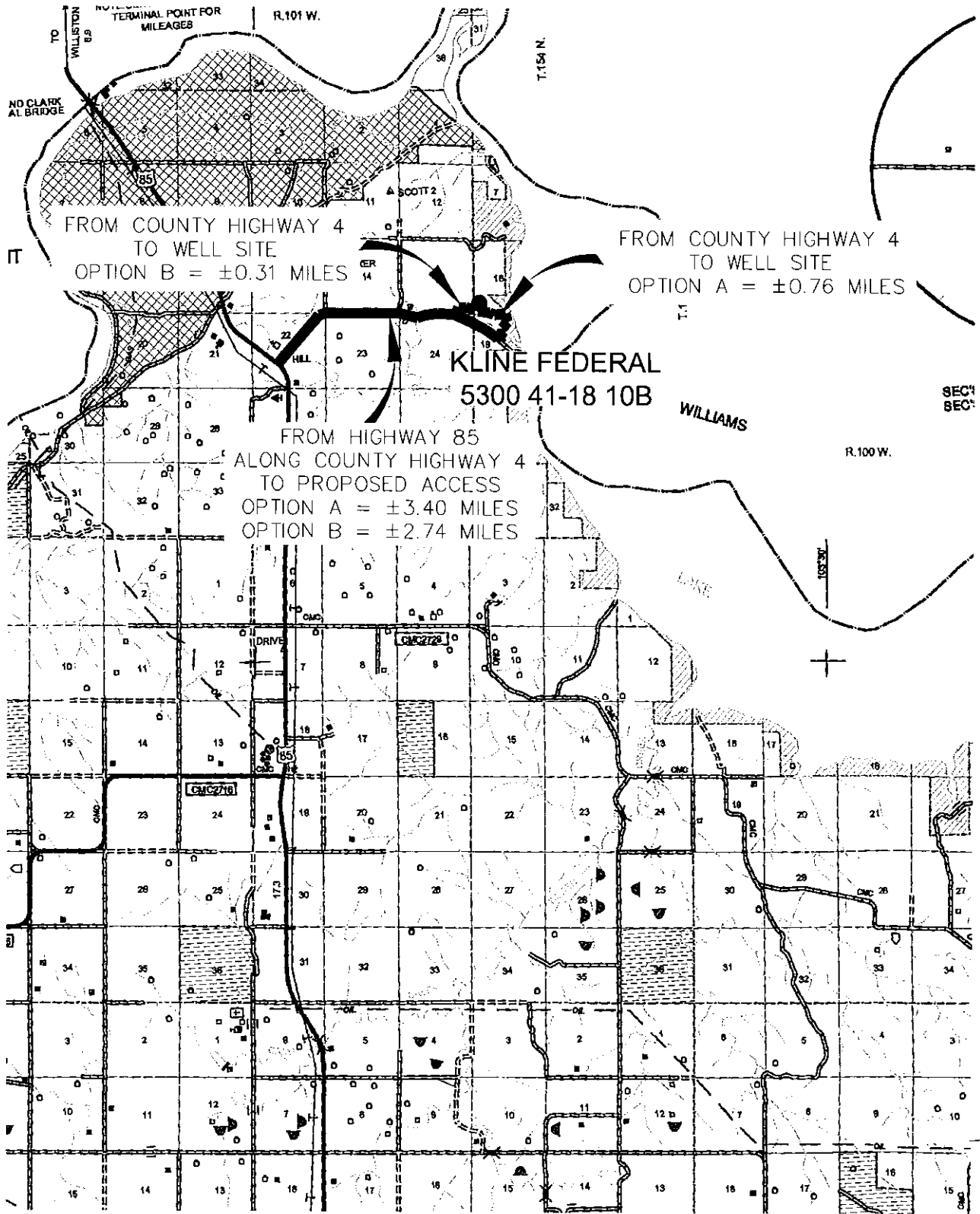
[illegible]

# COUNTY ROAD MAP

OASIS PETROLEUM NORTH AMERICA, LLC  
1001 FANNIN, SUITE 1500, HOUSTON, TX 77002

"KLINE FEDERAL 5300 41-18 10B"

500 FEET FROM SOUTH LINE AND 237 FEET FROM WEST LINE  
SECTION 18, T153N, R100W, 5th P.M., MCKENZIE COUNTY, NORTH DAKOTA



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SCALE: 1" = 2 MILE

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OASIS PETROLEUM NORTH AMERICA, LLC  
COUNTY ROAD MAP  
SECTION 18, T153N, R100W  
MCKENZIE COUNTY, NORTH DAKOTA

Drawn By: B.H.H. Project No.: S14-09-081.04  
Checked By: R.L.P. Date: APRIL 2014

| Revision No. | Date | By | Description |
|--------------|------|----|-------------|
|              |      |    |             |
|              |      |    |             |
|              |      |    |             |
|              |      |    |             |





5/29/2014

Mineral Resources Permit Manager  
North Dakota Industrial Commission  
600 East Boulevard Avenue Dept. 405  
Bismarck, ND 58505-0840

RE: **Kline Federal 5300 41-18 9T**  
**Kline Federal 5300 41-18 10B**  
**Kline Federal 5300 41-18 11T2**  
**Kline Federal 5300 41-18 12TX**  
**Kline Federal 5300 41-18 13T2X**  
**Kline Federal 5300 41-18 14BX**  
Request for a legal street address

Dear NDIC:

Oasis Petroleum has requested a physical street address for the Kline Federal 5300 41-18 9T, Kline Federal 5300 41-18 10B, Kline Federal 5300 41-18 11T2, Kline Federal 5300 41-18 12TX, Kline Federal 5300 41-18 13T2X, and Kline Federal 5300 41-18 14BX. The request was made to Aaron Chisolm, in McKenzie County. Upon receiving a legal street address, Oasis will submit the address to the NDIC on a Sundry Notice (form 4) pursuant to 43-02-03-28.

Thank you for your consideration.

Respectfully,

A handwritten signature in black ink, appearing to read "Heather McCowan". The signature is written in a cursive, flowing style.

Heather McCowan  
Regulatory Assistant  
Oasis Petroleum North America, LLC

Surface Damage Affidavit

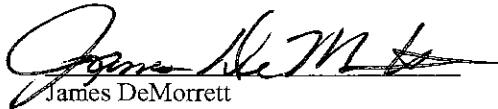
STATE OF NORTH DAKOTA     )  
                                          ) ss.  
COUNTY OF WILLIAMS         )

James DeMorrett, being duly sworn, states as follows:

The below described lands are not located in a wellhead protection/source water protection area per the website provided by the NDIC

and the adjacent landowner is fully aware that Oasis Petroleum will be drilling and operating multiple wells on spacing unit from a surface location in Section 18: Lot 4 a/k/a SW¼SW¼, Township 153 North, Range 100 West, 5<sup>th</sup> P.M., McKenzie County, North Dakota. (totaling 6.52 acres). The adjacent surface landowner is fully aware that the above referenced location will consist of the well bores and pumping units only and will not have cuttings pits, storage tanks or flare pits.

Dated this 16<sup>th</sup> day of June, 2014

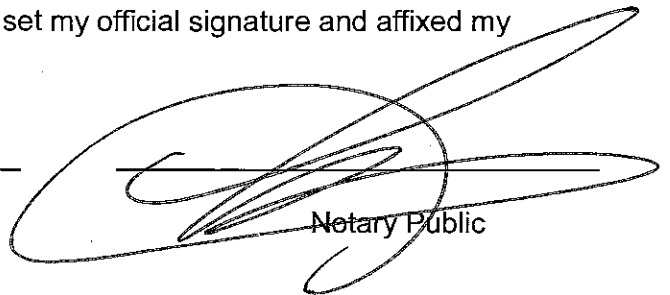
  
James DeMorrett

STATE OF NORTH DAKOTA     }  
                                          } ss.  
COUNTY OF WILLIAMS         }

BE IT REMEMBERED, That on this 16 day of June, 2014 before me, a Notary Public, in and for said County and State, personally appeared James DeMorrett, to me known to be the identical persons described in and who executed the within and foregoing instrument and acknowledged to me that he executed the same as his free and voluntary act and deed for the uses and purposes therein set forth.

IN WITNESS WHEREOF, I have hereunto set my official signature and affixed my notarial seal, the day and year first above written.

My commission expires: Nov 18, 2017

  
Notary Public

